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NATOPS PROGRAM TECHNICAL PUBLICATIONS AND PRODUCTS; STYLE, FORMAT, AND COMMON TECHNICAL CONTENT

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1. SCOPE

1.1 Scope. This specification covers general style, format, and technical content requirements for preparation of Naval Air Training and Operating Procedures Standardization (NATOPS) publications and products.

1.1.1 Use of sample figures. The figures contained in this specification are examples intended to illustrate style, format, and sample content. These figures should not be used as a source of technical data or for determining exact technical content and scale requirements.

1.2 NATOPS publications.

NATOPS Flight Manual
NATOPS Manual
NATOPS Flight Manual Supplements
NATOPS Partial Flight Manual
NATOPS Pilot Pocket Checklist (PCL) and Aircrew Pocket Checklist (APCL)
NATOPS Card Checklist (CCL)
NATOPS Servicing Checklist (SCL)
NATOPS Functional Checkflight Checklist (FCFCL)
NATOPS Passenger Information Card (PIC)
NATOPS Ditching and Bailout Placards (DBPs)
NATOPS Takeoff and Landing Data (TOLD) Cards

1.3 NATOPS publication development. A NATOPS publication may exist in any one of the following stages:

1.3.1 Draft NATOPS publication. A NATOPS publication in its initial stage of development may be generated as a draft. The title of the publication should be annotated with the word “Draft.” Draft NATOPS publications may be generated as single-sided pages, and should follow standard NATOPS one-column format. Parts not applicable to a specific manual or for which information is to be supplied at a later date should have a statement to that effect on the part page. Chapters for which information is to be supplied at a later date should have a statement to that effect on the first page of the chapter. These draft publications are not professionally printed and are duplicated only in very limited quantities for use during the initial development and testing of an aircraft. A draft NATOPS manual does not contain a letter of promulgation. Draft NATOPS publications may be changed at will by the assigned NATOPS Model Manager. Changes to these publications are not subject to the approval processes contained in OPNAVINST 3710.7-series.

1.3.2 Preliminary NATOPS publication. A PRELIMINARY NATOPS Flight Manual is a version of a NATOPS Flight Manual typically used for an aircraft during its initial production and Fleet introduction (see 6.7c.). The word “PRELIMINARY” appears in the title of a Preliminary publication to clearly identify its developmental nature. Parts not applicable to a specific manual or for which a chapter is to be provided at a later date should have a statement to that effect on the part page. Chapters for which information is to be supplied at a later date should have a statement to that effect on the first page of the chapter. A Preliminary NATOPS Flight Manual does not contain a Letter of Promulgation, and recommended changes are not subject to Advisory Group review when processed. A Preliminary NATOPS publication makes technical information and operational procedures available for test, verification, or training purposes in

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advance of the promulgated NATOPS publication. A Preliminary NATOPS publication is professionally formatted for paper or electronic media distribution and resembles a mature NATOPS publication.

1.3.2.1 Updating a preliminary NATOPS publication. The assigned NATOPS Model Manager may use streamlined, approval procedures detailed in OPNAVINST 3710.7-series to approve changes to the NATOPS manual and to provide flightcrews with the latest information more rapidly. The Model Manager is authorized to make procedural changes without using the formal NATOPS change process. He/she should maintain complete records of these changes and should ensure that all users are promptly informed of the changes. The Model Manager should consult with Commander Naval Air Systems Command (AIR-4.0P) to obtain Interim Change numbers for interim changes to Preliminary NATOPS publications. Technical changes to a Preliminary NATOPS must be reviewed and approved by AIR-4.0P.

1.3.2.2 Obtaining promulgation. The simplest way to upgrade a Preliminary NATOPS Flight Manual to a promulgated NATOPS Flight Manual is to conduct a NATOPS review conference. Following the conference, the proposed Flight Manual should be forwarded to AIR-4.0P via the Cognizant Command for final review and inclusion of a Letter of Promulgation. As a rule, the officially promulgated Flight Manual should be available to the users before the first aircraft deploy operationally.

1.3.3 Promulgated NATOPS publication. A promulgated NATOPS manual is a mature document that contains, as a minimum, all of the informational elements required by this specification. A NATOPS Flight Manual (NFM) or NATOPS Manual (NM) is certified as a complete and promulgated publication when approval is granted for a letter of promulgation (LOP) from Commander Naval Air Systems Command to be incorporated within it. A NATOPS checklist publication achieves its promulgation through the approval for incorporation of a LOP in the NFM or NM from which it is derived. A promulgated NATOPS publication is issued as a professionally prepared paper or electronic document. All changes to the contents of promulgated NATOPS publications are subject to the formal review processes detailed in OPNAVINST 3710.7-series.

1.4 Types of NATOPS publications.

1.4.1 NATOPS manual publications.

1.4.1.1 NATOPS flight manual. A NATOPS flight manual (NFM) is written for a specific, piloted aircraft or unmanned air vehicle (UAV) and contains standardized ground and flight operating procedures, training requirements, and technical data necessary for the safe and effective operation of the aircraft or UAV. NATOPS flight manuals do not include tactical doctrine. They are compiled and kept current by fleet reviews. Each NFM contains a letter of promulgation (LOP) signed by the officer delegated such authority by the Chief of Naval Operations. NATOPS flight manuals are normally unclassified publications. Classified subject matter may be placed in a NATOPS flight manual supplement in order to maintain an unclassified NATOPS flight manual. When present, classified material should be presented in accordance with the current SECNAV Instruction 5510.36-series.

1.4.1.2 NATOPS manual (general series). A NATOPS Manual is different from a NATOPS flight manual in that it contains general system descriptions and procedures or reference information for aircraft-related operations and evolutions (e.g., CV NATOPS Manual, NATOPS Instrument Flight Manual, NATOPS Air-to-Air Refueling Manual, etc.) and is normally not for

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just one model of aircraft or UAV. As such, a general series NATOPS manual is subject to the same standardized formatting requirements as NATOPS flight manuals, but its chapters are organized appropriately for the subject addressed by the publication title.

1.4.1.3 NATOPS flight manual supplement. A NATOPS flight manual supplement is one that contains some of the information mandated for inclusion in the NATOPS flight manual, but for convenience is printed under separate cover (see 6.7g.). As such, the NATOPS supplement is an extension of the basic NATOPS flight manual. Technical content of a NATOPS supplement should follow the same order of arrangement specified for formal NATOPS flight manuals, with no requirement that all parts or chapters of parts be present. All other requirements for format and content and arrangement of front matter and indices should apply. Common types of NATOPS manual supplements include the following:

1.4.1.3.1 Systems or weapon systems supplement. A system or weapon systems supplement is published when there is extensive system or weapon systems information or when systems or weapon systems differ between aircraft.

1.4.1.3.2 Aircrew operator supplement. An aircrew operator supplement is published when aircraft systems operating instructions concern only one or several of the aircrew members.

1.4.1.3.3 Special mission supplement. A special mission supplement provides general information and checklists for special flight missions not included in the basic mission profiles presented in the NATOPS flight manual.

1.4.1.3.4 Aircraft performance data supplement. An aircraft performance data supplement is published when aircraft performance data are applicable to several aircraft NATOPS flight manuals or when the volume of the performance data is sufficiently large enough to be bound as a separate publication to reduce the size of the NATOPS flight manual.

1.4.1.4 NATOPS partial flight manual. A NATOPS partial manual is intended for use with a NATOPS flight manual, some parts of which are replaced by the partial manual. A NATOPS partial flight manual provides operating instructions for aircraft that differ from the basic aircraft series by model or through modification. The format and method of reproduction are the same as specified for the NATOPS flight manual. Content should be only the additional information necessary to cover the differences between the standard aircraft and those modified. The difference data should be presented under a paragraph heading corresponding to that contained in the NATOPS flight manual. If the data are not related to a specific system or paragraph heading, new headings may be established. Each chapter should contain an introductory statement, as appropriate, that refers to the basic publication and states, “Except for the following, all other systems descriptions, operating procedures, and/or limitations are covered in NAVAIR _____.” If normal or emergency procedures are changed from the formal flight manual, the entire procedure, chapter, or part (if the part is not subdivided into chapters) should be included in the partial flight manual. Specific approval to vary from this provision must be granted by NAVAIR 4.0P or CNAF N455.

1.4.1.5 Commercial Derivative Aircraft (CDA) NATOPS flight manuals. CDA NATOPS Flight manuals have been prepared for some Navy and Marine Corps aircraft that are configured and utilized in a manner similar to their commercial passenger and cargo transport counterparts. Because the Original Equipment Manufacturer’s (OEM) flight manuals and checklist publications are updated by the aircraft manufacturer on a regular basis and the information in them

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reflects the experience derived from a significantly greater number of aircraft than those operated by the Naval Service alone, some Naval aircraft communities have opted to use the commercially-available publications in lieu of expending the resources necessary to maintain a separate set of NATOPS publication. CDA NATOPS flight manuals are organized in the traditional NFM format and contained only the information not found in the commercially available OEM flight manuals and checklist publications. Each part and chapter in a CDA NFM either contains the information found in a traditional NFM or points to the OEM publication that contains it. As a result, a CDA NFM contains a relatively small number of pages and is a relative shell when compared to the number of pages it would have as a standalone NFM. The procuring activity (program office) is responsible for maintaining access to current OEM publications for the aircraft community throughout the life cycle of the aircraft if a CDA NFM is used.

1.4.2 NATOPS checklist publications. A NATOPS aircraft checklist publication contains information that is extracted directly from the related NATOPS flight manual and normally placed in the checklist in an abbreviated form.

1.4.2.1 NATOPS pocket checklists. A NATOPS aircraft pocket checklist (pilot, aircrew or as specified), published in handbook form, contains normal procedures, emergency procedures, special procedures, and reference data for use in aircraft in which a NATOPS aircraft flight manual cannot be carried and readily used in the cockpit.

1.4.2.2 NATOPS card checklist. A NATOPS aircraft card checklist, normally published on a one-sheet card, contains normal and emergency procedures for use in aircraft in which a NATOPS flight manual can be carried and readily used in the cockpit.

1.4.2.3 NATOPS servicing checklist. A NATOPS aircraft servicing checklist contains servicing information in handbook form from the aircraft NATOPS flight manual or maintenance publications for use by the aircrew in servicing the aircraft for flight at locations away from the home base.

1.4.2.4 NATOPS functional checkflight checklist. A NATOPS aircraft functional checkflight checklist is published in index card form for reference and one-time use in recording aircraft flight data during a functional checkflight.

1.4.3 Other NATOPS products. Other published NATOPS products that contain information extracted from the relative aircraft NATOPS flight manual include:

1.4.3.1 NATOPS ditching and bailout placards. NATOPS aircraft ditching and bailout placards comprise a set of adhesive-backed placards for posting in the aircraft. Each placard is posted at a crew station and provides parachute location, raft location, evacuation route, and responsibilities during a ditching or bailout situation for the aircrewman at that particular station.

1.4.3.2 NATOPS takeoff and landing data cards. NATOPS aircraft takeoff and landing data (TOLD) cards are a set of reference cards that contain precalculated runway distance and airspeed requirements for selected takeoff or landing weights and configurations. TOLD cards are referred to by the pilots to determine airspeed and runway length requirements for the current ambient conditions.

1.4.3.3 NATOPS passenger information card. A NATOPS aircraft passenger information card contains information for use by passengers during aircraft emergency situations.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they should meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-3013	Glossary of Definitions, Ground Rules, and Mission Profiles to Define Air Vehicle Performance Capability
MIL-STD-38784	Technical Manuals; General Style and Format Requirements

DEPARTMENT OF DEFENSE HANDBOOK

MIL-HDBK-274	Electrical Grounding for Aircraft Safety

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE PUBLICATION

JOINT PUB 1-02	Department of Defense Dictionary of Military and Associated Terms

(Copies of JP 1-02 are available online at http://www.dtic.mil/doctrine/jel/doddict/.)

DEPARTMENT OF THE NAVY PUBLICATION

NTRP 1-02	Navy Supplement to the DoD Dictionary of Military and Associated Terms

(Copies of the NTRP are available online at http://www.nwdc.navy.mil or from the Commander, Navy Warfare Development Command, 686 Cushing Road, Newport, RI 02841-1207.)

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NAVAL AIR SYSTEMS COMMAND SPECIFICATION

SD-8706C	General Specification for Design Examinations, Engineering, Aircraft Weapon Systems

(Copies of SD-8706C are available by contacting NATOPS@navy.mil or AIR 4.0P, Commander Naval Air Systems Command, Bldg. 460, Room 228, 22244 Cedar Point Road, Patuxent River, MD 20670–1163.)

UNITED STATES GOVERNMENT PRINTING OFFICE

United States Government Printing Office (GPO) Style Manual

(Copies of the GPO Style Manual are available online at http://access.gpo.gov/ or from the Superintendent of Documents, U.S. Government Printing Office, 732 North Capitol St., N.W., Washington, DC 20402-0002.)

DEPARTMENT OF DEFENSE DIRECTIVES

DoD 5220.22-M	National Industrial Security Program Operating Manual (NISPOM) available at http://www.dss.mil
DoDD 5230.24	Distribution Statements on Technical Documents

(Copies of Department of Defense publications are available online at http://www.dtic.mil or from the contracting officer.)

DEPARTMENT OF THE NAVY INSTRUCTIONS

SECNAVINST 5510.36	Department of the Navy (DON) Information Security Program (ISP) Instruction
OPNAVINST 3710.7	NATOPS General Flight and Operating Instructions
COMNAVAIRFORINST 4790.2	The Naval Aviation Maintenance Program (NAMP)
NAVSUPINST 5600.26	Bar Coding of all Stock Numbered Technical Manuals

(Copies of COMNAVAIRFORINST 4790.2 are available online at http://logistics.navair.navy.mil/4790/affected.cfm or from the contracting officer. Copies of the other instructions are available online at http://doni.daps.dla.mil or from the contracting officer.)

NAVAL AIR SYSTEMS COMMAND MANUALS

NAVAIR 00-25-100	Naval Air Systems Command Technical Manual Program
NAVAIR 00-80R-14-1	NATOPS U.S. Navy Aircraft Emergency Rescue Information Manual
NAVAIR 00-80T-110	NATOPS Air-to-Air Refueling Manual
NAVAIR 00-80T-122	Helicopter Operating Procedures for Air-Capable Ships NATOPS Manual
NAVAIR 01-1B-40	Technical Manual, Weight and Balance Data
NAVAIR 01-1B-50	Technical Manual, USN/USMC Aircraft Weight and Balance Control

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(Copies of NAVAIR publications are available online at https://www.natec.navy.mil; or from the Commanding Officer, Naval Air Technical Data and Engineering Service Command, P.O. Box 357031, San Diego, CA 92135-7031; or may be obtained from the contracting officer.)

2.3 Non-Government publications. The following industry documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Y14.38	Abbreviations and Acronyms (DoD adopted)

(Copies of these documents are available from http://www.asme.org or ASME Information Central Orders/Inquiries, P.O. Box 2300, Fairfield, NJ 07007-2300.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE).

IEEE 91 or 91A	Graphic Symbols for Logic Functions (DoD adopted)
IEEE 315 or 315A	Graphic Symbols for Electrical and Electronics Diagrams

(Copies of these documents are available from http://www.ieee.org or from IEEE Customer Service Center, P.O. Box 1331, 445 Hoes Lane, Piscataway, NJ 08855-1331.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Technical content requirements for NATOPS products. The technical content of all NATOPS products shall comply with the following:

3.1.1 Copyrights and advertising. Copyrighted material shall not be included in any publication prepared in accordance with this specification without written permission of the copyright owner (see 6.2). Proprietary legends shall not be shown. The manual shall not contain advertising matter. All material prepared in accordance with this specification shall be Government property.

3.1.2 Security classification. The classification authority shall be in accordance with appropriate OPNAV 5513 Series Instruction for Department of the Navy classification guidance. If classification is based on multiple sources, the guidelines provided in SECNAVINST 5510.36 apply. In the latter case, the words “Multiple Sources” shall appear on the DERIVED FROM line. Downgrading and declassification data shall appear only once in a classified publication — on the lower left of the title page of a new or revised publication or on the cover sheet of a change to a classified publication. If a major component of a publication is used separately (e.g., an appendix consisting of plastic templates), it shall be marked as a separate document with classification warning notice(s) and declassification data.

3.1.2.1 Title page classification. The title page of a classified publication shall show the highest level of classification of any material contained therein. Consider each volume of a multi-volume publication separately for this purpose. The classification markings (SECRET or CONFIDENTIAL) shall be placed at the top and bottom of the title page (see figure 1) and back cover. These markings shall be not greater than 24-point Arial/Helvetica bold or equivalent

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typeface. Except for the title page, change cover sheet, and back cover, all other elements of a classified publication shall be treated as individual chapters and pages marked according to the highest level of their contents. Treat change cover sheets as title pages for classification. Covers and title pages shall also carry classification warning notices, if applicable (see 3.1.2.5).

Note: An unclassified publication shall not be marked UNCLASSIFIED.

For purposes of page classification, unless the title page contains classified material, the page shall also be marked at the bottom center, "(This page is UNCLASSIFIED.)."

3.1.2.2 Publication title classification. All titles shall be unclassified. Whenever the title of a classified publication is used in reference to that publication, it shall be followed by "(U)."

Note: This rule does not apply in unclassified publications.

3.1.2.3 Page classification. In a classified publication, the pages in a chapter or appendix (except blank pages) shall be marked according to the highest level of classification within that chapter or appendix. Include classification warning notice if appropriate (see 3.1.2.5). Each page shall be marked at center top and bottom. The top marking shall be one line above any other marginal copy (see 3.3.2.2). The bottom marking shall be one line below the page number. If a chapter or appendix in a classified publication is wholly unclassified, each page within that chapter or appendix shall be marked UNCLASSIFIED; no other classification markings are necessary. For page classification markings, 14-point Arial bold or equivalent shall be used.

3.1.2.4 Table of contents, list of illustrations, and index classification. Unless they contain classified entries, the table of contents, list of illustrations, and index shall be marked "UNCLASSIFIED." If an entry is classified, the classification of the entry shall be marked, and the table of contents, list of illustrations, and index shall each be marked with its highest classification. When classified entries are present, place the following statement in the lower left of the first page of the applicable document:

"ALL UNMARKED ENTRIES IN THE TABLE OF CONTENTS/LIST OF ILLUSTRATIONS/INDEX ARE UNCLASSIFIED."

3.1.2.5 Marking symbols. Classifications and warning notices are marked on individual pages of a classified publication. Pages that contain restricted data (RD) or formerly restricted data (FRD) shall be marked SECRET/RD or SECRET/FRD as appropriate. The entire chapter shall be so marked, if applicable. Individual paragraphs and other subordinate elements shall be identified by the use of marking symbols. Marking symbols shall be used within the text of a classified chapter or appendix or any other subdivision of these sections (e.g., heading, title, or listing) instead of fully written classification and warning notices. The symbols most commonly used are:

SYMBOL — MEANING

(U) — Unclassified
(C) — Confidential
(S) — Secret
(C-RD) — Confidential-Restricted Data
(S-RD) — Secret-Restricted Data
(C-FRD) — Confidential-Formerly Restricted Data
(S-FRD) — Secret-Formerly Restricted Data
(U-O) — U.S. Only

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*Note: Publication data that require a higher or more restrictive classification or warning notice (e.g., Top Secret or sensitive compartmented information) shall not be included in NATOPS publication*s.

3.1.2.6 Chapter and appendix title classification. In a classified chapter or appendix, the classification of its title shall be indicated with the appropriate marking symbol (see 3.1.2.5) in parentheses following the title (see figure 2).

3.1.2.7 Paragraph classification. In a classified chapter or appendix, each numbered paragraph that contains run-in text shall be marked with an appropriate classification symbol (see 3.1.2.5) following the paragraph number, but preceding the heading. Precede each unnumbered paragraph with the appropriate marking symbol for that paragraph. Lines of a paragraph that are carried over to another column or page do not require separate marking.

a. When the entire contents of a chapter or appendix in a classified publication are unclassified, it is not necessary to mark individual paragraphs. Warnings, cautions, and notes shall be marked with the appropriate marking symbol preceding the text. A sample of paragraph marking is shown on figure 2.

3.1.2.8 Stand-alone paragraph heading classification. In a classified chapter or appendix, every effort shall be made to avoid classified stand-alone headings. If this is not possible, the stand-alone heading shall carry a classification marking symbol following the number, but preceding the heading (see figure 2).

3.1.2.9 Listing classification. A listing shall be annotated with a classification marking symbol when necessary if it expresses a complete thought. A list of components is not a complete thought; do not give this list a marking symbol (see figure 2).

3.1.2.10 Figure classification. For classified publications, the figure classification markings shall be centered 1 pica below the illustration and 1 pica above the figure title. Set in 10-point Arial uppercase letters or equivalent typeface (see figure 3). Designate an unclassified figure in the same way. It is not necessary to mark figures in an unclassified chapter.

3.1.2.11 Figure title classification. Every effort shall be made to create figure titles that are both descriptive and unclassified. Unclassified figure titles in classified chapters shall be marked (U). The appropriate classification marking symbol shall be placed between the figure number and its title (see figure 3). It is not necessary to mark figure titles in an unclassified chapter.

3.1.3 Changes. A Change shall be prepared when only a portion of the total pages of the publication is affected, as determined by the procuring activity (see 6.2). When pages are added only to the end of a chapter or to the back of the publication, a Change is normally prepared, regardless of the number of pages involved. A Change shall be prepared so that its pages can be substituted for existing pages of the publication or can be added to the publication. The Change Pages shall match as closely as possible the type style and size used in the basic publication.

3.1.3.1 Change number. On each page containing changes or additions, the words CHANGE (number) shall be placed at the bottom of the page in the right corner for right-hand pages and in the left-hand corner for left-hand pages on the same line as the page number. This also applies to added pages (e.g., 1-2a, 1-2b). Backup pages shall retain existing corner markings.

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3.1.3.2 Numbering changed pages, paragraphs, steps, and figures.

3.1.3.2.1 Added pages. When new material that necessitates inserting additional pages prior to the end of a chapter is added, the inserted pages shall follow an even page and shall be numbered as that page number with consecutive letter suffixes (e.g., pages 2-4a through 2-4c). When new pages are added to the end of a chapter, 3.1.3.2.7 applies.

3.1.3.2.2 Added figures. Figures added between existing figures shall be identified by the preceding figure number plus consecutive letter suffixes. For example, 2-3a, 2-3b, and 2-3c would be assigned to three new figures inserted between existing figures 2-3 and 2-4. When new figures are added to the end of a chapter, 3.1.3.2.7 applies. In both cases, the new figures shall be added in the List of Illustrations.

3.1.3.2.3 Deleted figures. When a Change deletes a figure without substitution, the space occupied by the deleted figure can be used for text, if necessary. A phrase such as, “Figure 4-3 deleted by Change 2” shall be placed at the bottom of the page. When this Change results in a blank page, 3.1.3.2.4 applies. The statement of deletion shall appear on the page until the first revision or until the chapter is revised or rearranged. A modified statement of deletion shall also appear in the List of Illustrations next to the appropriate figure number (e.g., Figure 4-3 deleted by Change 2).

3.1.3.2.4 Deleted pages. When page number continuity is broken by the deletion of a page, and a blank page (or a missing leaf) results, a statement indicating the deletion shall be placed in the bottom margin of the preceding page or top margin of the following page (e.g., “Page 2-17, including figure 2-2, deleted by Change 1”). A modified statement shall also be placed in the List of Effective Pages (e.g., 2-17, Deleted by Change 2). These statements shall remain in both places until the situation is corrected by a Change or Revision.

3.1.3.2.5 Paragraphs. When a new paragraph is added between existing paragraphs during a Change, the new paragraph shall inherit the number of the preceding paragraph with an upper-case letter “A” added (e.g., paragraph 1.3.12.2.4A). To eliminate the requirement to renumber subsequent paragraphs of a chapter when a paragraph is deleted without replacement during a Change, the Deleted paragraph number shall be retained with an explanatory statement inserted (e.g., “Paragraph 1.3.12.2.4 Deleted by Change 3”). When new paragraphs are added to the end of a chapter, 3.1.3.2.7 applies.

3.1.3.2.6 Listings and procedural steps. When adding a listing and/or procedural steps between existing listings or procedural steps, the entire listing or procedure shall be revised.

3.1.3.2.7 Changes at end of chapter. Pages, paragraphs and figures added at the end of a chapter shall be numbered consecutively, starting with the next available sequential numbers. Do not assign letter suffixes to these pages, paragraphs and figures as would be done with new pages, paragraphs and figures inserted within a chapter. Place the change number at the bottom of these pages in the right-hand corner on right-hand pages and in the left-hand corner on left-hand pages.

3.1.3.2.8 Change symbol. Changes to text and figures (including new material or added pages) shall be indicated by a black vertical line (change bar) beside lines of affected text in the nearest outside margin. The change bar shall be 6.5 points in width and shall be set 6 points from

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the applicable text or figure frame (see figure 4). If a chapter is completely changed or new, place a vertical line in the margin by only the chapter number and title. Delete previous change bars on a page when the page is subsequently changed or the publication is revised; change bars shall reflect current changes only.

a. For single-column formatted pages or full-page art, add new change bars on the right side of right-hand pages and on the left side of left-hand pages.

b. On new art, extend the change bar to the bottom of the figure title.

c. On turn pages, add change bars parallel to the right or left border of the turn-page art.

d. Change symbols are not required for:

(1) Table of contents, list of illustrations, alphabetical index, and list of effective pages.

(2) Blank space resulting from deletion of text or a figure.

(3) Correction of minor inaccuracies such as spelling, punctuation, and relocation of material, unless such correction changes the meaning of material.

(4) On new/revised art when the figure width does not fill the full column or page, ensure that the change bar is affixed in the margin and not inside the 20- or 42-pica image area.

3.1.3.3 Errata. An erratum shall be issued to a basic publication or its latest change to correct an error in printing or to make an administrative correction. Errors in printing include reprinting to correct misprinted pages or pages inadvertently dropped. Administrative corrections include correcting a page number in the change notice or the list of effective pages. An erratum does not contain a signature block, shall be unclassified, and may direct the insertion of pen-and-ink changes. An erratum shall not be issued to a classified NATOPS publication, nor be used to reclassify material or to change the substantive content of a publication; a Change shall be issued for these purposes. An erratum shall be incorporated in the next Revision or Change to the publication (see figure 5). Generally, an erratum contains:

a. Day, month, and year of issue of the erratum. The first erratum shall be identified as "ERRATUM" and dated 1 day after the date of the Revision or Change that it corrects. The second erratum, if needed, shall be dated 2 days after the date of the Revision or Change that it corrects, and so on.

b. A title line including the words "ERRATUM TO" the publication number and Revision date. If the erratum is to a Change, include also the Change number and Change date.

c. A list of the pages contained in the erratum.

d. Brief statement of the purpose(s) for which the erratum is issued.

e. Precise description(s) of the changes to be made.

f. Instructions for recording incorporation of the erratum on the Record of Changes page (use the form, "Err to publication no. [original or change no.], date of issue of erratum")

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g. Instructions for destroying superseded material after the erratum has been entered in the basic publication.

h. The issuing authority for the erratum.

i. All replacement pages of an erratum shall identify the page status in the lower right corner of right-hand pages and in the lower left-hand corner of left-hand pages as "Erratum to ORIGINAL (or CHANGE NUMBER)."

3.1.3.4 Interim changes. An interim change (IC) is a Change to a NATOPS publication issued via a relatively rapid means of dissemination, such as a naval message. An IC is usually initiated by an urgent change recommendation (UCR), which may be submitted by any user of a NATOPS publication. AIR-4.0P issues ICs and reviews and approves each IC for release to the fleet. ICs, which are released between formal NATOPS publication reviews, are numbered consecutively and subsequently incorporated into the respective NATOPS publication or product. Each time the respective NATOPS publication is updated, the Interim Change Summary page shall be annotated to reflect all ICs issued since the publication's last printing. The current trend for issuing ICs is via change pages downloaded from AIR-4.0P's Airworthiness Website's NATOPS link at https://airworthiness.navair.navy.mil; however, simple ICs may still be issued as pen-and-ink changes via direction from AIR-4.0P.

3.1.4 Revisions. A Revision is a second or subsequent edition of a publication that supersedes the preceding edition. A Revision shall incorporate all previously issued Changes and any outstanding interim changes to the existing publication. Preparation of a Revision shall be approved in advance by the procuring activity. The procuring activity shall decide whether a Change or Revision will be prepared by weighing the advantages of a completely updated publication against the increased costs and time to prepare, print, and distribute a Revision.

3.1.4.1 Renumbering and removal. All paragraphs, illustrations, and pages shall be renumbered, as necessary, to eliminate all number suffixes and to establish correct sequence. All partial pages shall be eliminated.

3.1.4.2 Supersedure notice. When a publication supersedes an existing publication(s), a supersedure notice shall be placed on the title page immediately below the title of the publication. The supersedure notice shall include the NAVAIR number and revision date of each wholly or partially superseded publication.

3.1.4.3 Numbering revisions. A NATOPS Manual normally retains the same publication number as the superseded edition.

3.1.4.4 Change symbol. Changes to text and figures (including new material or added pages) shall be indicated as specified in 3.1.3.2.8.

3.1.5 Emergency procedures border. Each page (including entire chapters) of a NATOPS publication addressing emergency procedures shall be identified by an emergency border. This border, consisting of alternating black and white 9-point lines at a 45° angle, shall "bleed" off three sides of the page for flight manuals and general series manuals (top, right, and bottom of right-hand pages; and top, left, and bottom of left-hand pages). Dimensions are ¼ inch (top and bottom) and 1/8 inch (left edge for left-hand pages, right edge for right-hand pages). See figure 6

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for a sample NATOPS flight manual emergency border page. NATOPS pilot pocket checklists and aircrew pocket checklists have emergency borders on the left side of odd numbered pages only and on the left and top edges on even-numbered pages. PCL and APCL emergency borders are ½ inch in depth. Card checklists have emergency borders on all four sides. NATOPS ditching and bailout placards have a solid red, reflective border on all four sides.

3.1.6 Nomenclature. The names of systems, equipment, circuit breakers, switches, handles, knobs, and dials shall be as listed in the Illustrated Parts Breakdown publications for the aircraft or piece of equipment. Occasionally, names may be shortened when they will not cause confusion.

3.1.6.1 Nomenclature appearing on placards and decals. The nomenclature used in the manual to identify controls and equipment shall contain the identical wording that appears on the applicable placards and decals. The only exception to this rule shall be when the decaled nomenclature has been definitely established as unsatisfactory by the procuring activity. In such cases, the approved nomenclature shall be used throughout the manual. However, the decaled nomenclature shall appear at least once in the manual in parentheses and immediately following the first reference to that item.

3.1.6.2 Nomenclature for controls and control positions. All controls shall be identified by titles that are descriptive of their configuration (e.g., “fuel selector handle” not “fuel selector control”; “flap lever” not “flap control”). Extremely common items (e.g., the throttle) are exceptions to this rule. Whenever reference is made to a specific decaled control position, it shall be shown using the same capitalization as on the decal. Quotes shall not be used.

3.1.6.3 Aircraft performance parameters terminology. Terminology shall be maintained in agreement with MIL-STD-3013, “DoD Standard Practice, Glossary of Definitions, Ground Rules, and Mission Profiles to Define Air Vehicle Performance Capability” definitions and usages.

3.1.7 Preferred usage. Preferred usage in NATOPS publications is provided in Appendix A. See also Chapter 6, “Compounding Rules,” in the GPO Style Manual.

3.1.8 General writing style. To achieve uniformity among publications under NAVAIR control, text in a new or revised publication shall be written and composed using the U.S. Government Printing Office Style Manual, 2000; The American Heritage Dictionary of the English Language; Webster’s Third International Dictionary; NTRP 1-02, Navy Supplement to the DoD Dictionary of Military and Associated Terms; and JOINT PUB 1-02, Department of Defense Dictionary of Military and Associated Terms.

Note: Material quoted from other publications shall be cited verbatim and without GPO standardization applied.

a. For a change to an existing publication, all Change pages shall be checked for spelling, punctuation, and content clarity. For capitalization, compounding, numerals, and abbreviations in new material, choose the usage that currently exists in the publication to ensure consistency throughout the publication.

3.1.8.1 Grammatical person and mood. The second person imperative mood shall be used for all operational procedures (e.g., “Check tip tank fuel level.”). The third person indicative

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mood shall be used for description and discussion (e.g., “When the No. 1 inverter fails, the caution light illuminates.”). Pronouns shall be used only to avoid confusion.

3.1.8.2 Use of “shall,” “should,” “may,” “need not,” and “will”. The words “shall,” “should,” “may,” “need not,” and “will” are extremely meaningful in NATOPS language, and their usage is very specific. When writing text for a NATOPS publication or a NATOPS change recommendation, the following definitions shall be followed:

a. “Shall” denotes action that is required.

b. “Should” denotes action that is recommended.

c. “May” and “need not” denote action that is optional.

d. “Will” denotes futurity only, and does not indicate any degree of requirement for application of a procedure or an action.

e. “Can” and “must” are not defined by NATOPS standards and are to be avoided when “shall,” “should,” or “may” can be used.

3.1.8.3 Development of text. The text shall contain only essential information of interest to the flightcrew. The text shall be developed in a factual, specific, concise, and clearly worded manner to ensure ready understanding and shall not resort to theoretical discussion. Use descriptive and unique paragraph headings and illustration titles, avoiding such words as general, chart, performance, description, and operation, unless accompanied by an identifying term. Superfluous words and phrases shall be avoided. Emphasis symbols such as bold capital letters, quotation marks, and underlining shall not be used. Use of italics or bold lower-case letters shall be used sparingly when phraseology reinforcement is considered necessary.

3.1.8.4 Capitalization. Capitalization shall generally be in accordance with the GPO Style Manual. For capitalization of certain words, refer to the preferred usage list in Appendix A.

a. When an acronym or abbreviation that is normally in lowercase is the first word of a sentence, the first letter only shall be capitalized; (e.g., “Dc power is available from two different buses.”).

b. The following rules shall be used to determine capitalization after 1-em dashes in listings:

(1) If an independent clause or complete sentence follows the 1-em dash, capitalize the first word, e.g., “Ballistics — Aircrews must understand weapon ballistics so they can employ the weapon system.”

(2) If a sentence fragment follows the 1-em dash, do not capitalize the first word.

Line one — negative.

Line two — positive.

c. In a primary paragraph heading, all uppercase letters shall be used for all principal words, including those in parentheses.

7.2 EXTERIOR PREFLIGHT (SEE FIGURE 7-1)

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d. In secondary and subordinate headings, use initial capitals for all principal words, including those in parentheses.

9.8.1 Tiedown Procedures (Shipboard)

e. All uppercase letters may be used for emphasis of a word or phrase. For consistency, both all uppercase and quotation marks shall not be used for emphasis in the same chapter.

f. For capitalization of sideheads and figure titles, apply rules 3.46 through 3.54 of the GPO Style Manual.

3.1.8.5 Spelling. Refer to the preferred usage list in Appendix A.

3.1.8.6 Compound words. Refer to the preferred usage list in Appendix A.

3.1.8.7 Punctuation. Punctuation shall generally be in accordance with the GPO Style Manual.

a. Apostrophe and possessive case.

(1) Do not use an apostrophe after a word that is more descriptive than possessive (not indicating personal possession) unless the plural does not end in s.

technicians guide children's hospital

(2) Decide if the modifier is descriptive or possessive by determining if its use is general or specific. For example, the phrase pilot procedures when used to connote procedures followed by any and all pilots would be considered descriptive. Conversely, in this example, "Proper separation shall be maintained between both wingmen and the *lead aircraft's lights*," the relationship expressed between modifier and noun is specific and limited and requires an apostrophe.

(3) Use an apostrophe and a lowercase "s" after an acronym only to show possession.

AIR-4.0P's workload commanding officer's order

(4) Use an "of phrase" to denote the possessive of inanimate objects:

The inspection of the aircraft was accomplished.

b. Colon. Use a colon to introduce a list.

The briefing guide shall include the following items:

a. Flight planning

b. Operational data

c. Mission specifics

c. Comma.

(1) Use commas in numbers of four or more digits, including metric units of measurement.

4,230 50,491 1,250,000

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Note: Radio frequencies (bands), meters pertaining to radios, and built-up fractions do not use a comma.

(2) Decimals, serial numbers, and the numerical portions of Universal Transverse Mercator (UTM) and Georef grid positions do not contain commas.

(3) Do not use a comma before a correlative conjunction (e.g., “either … or,” “not only … but also”).

Release the switch light as well as the pilot instrument light after the warning detection has been checked.

Hook safety pins shall be installed not only after landing but also while the No. 1 hydraulic system is powered.

(4) Use a comma after a long introductory phrase or clause.

When the helicopter is in a hover, the engine exhaust system acts as a power source.

With the control valve in the normally open position, foreign particles from the air will be removed.

(5) Use a comma to set off an introductory verbal phrase.

Participle: Designating minimum currency, the certificate shall be placed in the qualification jacket.

Infinitive: To verify system operation, depress the button and hold in.

Gerund: Since closing of the valve shuts off the fuel flow to the engine, avoid this action.

(6) Use commas with conjunctive adverbs (e.g., however, consequently, therefore) and sentence modifiers.

Approaches can be made to a hover; however, no-approach hovers are preferable.

Shore-based procedures, on the other hand, are discussed in Chapter 8.

(7) Use a comma to separate contrasted elements.

Turnouts after takeoff shall be accomplished using radius of turn, not power changes.

The heat can be operated with the sliding windows open, but only up to a 10-minute limit.

(8) Use commas to set off a nonrestrictive modifier.

The switch, which has three positions, controls the flow of air to the cabin.

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(9) Do not use commas to set off a restrictive modifier.

The circuit breaker that is next to the AIR COND switch is powered by the main ac bus.

(10) Use a comma after a subordinate clause that precedes its principal clause.

If visual contact is made, execute a normal autorotative landing.

(11) Do not use a comma when the subordinate clause follows the principal clause.

Execute a normal autorotative landing if visual contact is made.

(12) Do not use a comma between two compound verbs, subjects, complements, or predicates.

At 2 to 3 feet, the pilot applies the collective and adjusts forward airspeed.

(13) Do not use a comma before a coordinating conjunction (e.g., and, but, or) joining two dependent clauses.

The printer stated that the job was completed and that it was done properly.

(14) Do not use a comma after *such as* and *like*.

For restricted visibility takeoff, use any available object such as rocks for reference.

d. Hyphen. The hyphen is more frequently used for compounding words than as punctuation. Consult "Compound Words" in the GPO Style Manual.

(1) Use "to" for separating a range or sequence of figures or letters, rather than a hyphen or dash. "To" is preferred over "through."

5 to 10 knots (not 5 – 10 knots)

(2) There shall be no more than two successive end-of-line hyphenations dividing words in text. Where a hyphenated compound word appears at the end of a line, the division shall occur only at the existing hyphen (for example, "self-inflicted," not "self-in–flicted").

e. Parentheses.

(1) When employing both English and metric units of measurement, use the metric measurement parenthetically. Complete a series of English measurements before giving the metric measurements.

10 feet (3 m) 5 to 10 feet (1.5 to 3 m)

(2) Use "e.g." and "i.e." parenthetically

e.g. (exempli gratia = for the sake of example) is used to provide a simple, clear example that defines the meaning.

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If an emergency condition (e.g., fuel shortage) exists, a cell penetration can be made.

i.e. (id est = that is) is used to provide a further explanation or development of the intended meaning.

Restricted areas (i.e., any area that requires use of a magnetically coded badge) will be secured at 5:00 p.m.

(3) Do not annotate tolerances or limits with parentheses.

410 ± 50 rpm (*not* 410 (±50) rpm)

20° ± 1°F [*not* 20° (±1)]

(4) Numeration of equations in text is designated parenthetically against the right margin. Use the chapter number and number in sequence after the decimal.

$$a + b + c = x \quad (2.1)$$

f. Quotation marks.

(1) The period and the comma fall within the quotation marks.

Under the heading, “replenishment at sea,” will be found types of ships.

(2) The semicolon and colon fall outside the quotation marks.

The first column is “weapons”; the second column is “missiles.”

(3) The comma is used before quoted material.

The pilot states, “That is your tanker.”

(4) Set in quotation marks the following types of titles: articles in magazines, journals, and newspapers; chapter titles in books; and reports.

Chapter 1, “Search and Rescue,” is considered the most important chapter for the students to study.

(5) Quotation marks may be used for emphasis of a word or phrase. For consistency, do not use both quotation marks and all uppercase in the same chapter for this purpose.

g. Semicolon.

(1) Use a semicolon between clauses joined by conjunctive adverbs (e.g., however, moreover, nevertheless, therefore, then, for example, consequently).

Microorganisms are difficult to detect; therefore, if a biological attack is suspected, assume all exposed surfaces are contaminated.

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(2) Use a semicolon to separate lengthy statements/listed elements following a colon or when listed elements contain commas.

> When the levers are at STOP, the ignition is off and fuel flow to the engine is shut off; at CRANK, fuel flow remains shut off, the ignition system is energized, and the hydraulic start valve opens, admitting pressurized hydraulic fluid to the engine starting motor; at START, the fuel control provides starting fuel.

(3) Use a semicolon between clauses of a compound sentence that are not connected by a conjunction.

> The tank cell drain hose has a manual drain valve; the pump discharge hose has a directional flow check valve.

h. Italics. Limit the use of italics to occasions described below. Explain in the Preface any other use of italics. If the use of italics is confined to one or two pages, the explanation shall be provided in a footnote.

(1) Use italics for words that require special emphasis.

(2) Use italics for a block of text that is regulatory in nature.

(3) Use italics for mathematical letter symbols in text.

3.1.8.8 Abbreviations and acronyms. When editing text for a new or revised publication, a list of abbreviations and acronyms shall be prepared for use in the front matter of the publication. The first occurrence of an acronym shall be spelled out in the text.

a. Abbreviations and acronyms shall be avoided in text unless they are established as customary or unless repetition within text justifies the usage. Refer to NTRP 1-02, Navy Supplement to the DoD Dictionary of Military and Associated Terms, for established acronyms.

b. Abbreviations shall not be used excessively. Meanings of abbreviations shall be self-evident. When the use of an abbreviation or acronym could cause confusion, the word(s) shall be spelled out (e.g., the use of the abbreviation “in” for “inches” could be confused with the preposition “in”). A list of frequently used abbreviations appears in Appendix B.

(1) When only the first letter of each word is used to make up the acronym, use all uppercase for the acronym.

> MEU (Marine expeditionary unit)
>
> AOA (Airport operations area)

(2) In common-noun combinations made up of more than the first letter of lowercased words, use lowercase.

> loran (for long-range navigation)
>
> sonar (for sound navigation ranging)

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(3) Use a lowercase s without an apostrophe only to show plural.

DDGs F-14s

Note: Do not use a lowercase s with an abbreviation of a unit of measurement; the abbreviation includes the plural (e.g., lb, not lbs).

(4) Abbreviate terms used with numbers that are frequently encountered and readily understood in shortened form, and all metric units of measurement. Exceptions to this rule appear in 3.1.8.8.b.(5) following the note below. Examples of abbreviated terms used with numbers are the following:

100 m	20 mm	10 nm	7.5 km

Note: An abbreviation following a numeral shall not be separated from the figure at the end of a line in text. For example, if a line of text ends with the phrase "100 m," do not separate the "100" and the "m" at the end of the line; "drop" the "100" to the next line to keep the numeral and abbreviation together.

(5) Spell out the following words in text; but abbreviate them on figures and tables, if necessary.

Feet	Yards	Inches
Pounds	Knots	Miles
Years	Days	Hours
Minutes		

(6) Use the following as a guide when referring to ordnance:

Mine Mk 67 Mod 0, *but* Mk 67 mine

(7) Abbreviate words on figures and in tables, if necessary, to save space.

helo	ft	sub	lb

(8) Unless required by the GPO Style Manual, omit periods in abbreviations (e.g., use "fpm" instead of "f.p.m.").

(9) Avoid the use of stand-alone acronyms in chapter titles, paragraph headings, and figure titles (e.g., paragraph 17.1 SAR). Acronyms shall be used in lower-level headings if they have been identified at a higher level (e.g., "CHAPTER 17 Search and Rescue" followed by paragraph "17.1 OVERWATER SAR," or paragraph "17.1 ACOUSTIC DATA PROCESSOR" followed by paragraph "17.1.1 ADP Menu)."

(10) The existence of an acronym does not mean that it will be used in all textual instances. Repeated references justify the use of acronyms.

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c. The frequent use of an acronym in one portion of a chapter does not mean that the acronym will appear in a later part of that chapter or in a subsequent chapter. Solo reappearances of an obscure acronym many pages after its last use require the reader to refer to the List of Acronyms. For example, if paragraph 2.1.1.7 on page 2-4 discusses the augmenter fan temperature control (AFTC), this sentence on page 2-77, "Engine operation is automatically regulated by the AFTC," would be improved by spelling out the acronym. In most instances, once an acronym is first identified, it shall continue to be used until there is a sizable page break between mentions.

d. In some instances, even if an acronym has been identified, the spelled out form is the best choice for clarity and stylistic consistency. For example, this sentence, "The following paragraphs describe the hazards to personnel, the hazards of fuel ignition, and the hazards of electromagnetic radiation to ordnance," would be awkward if "HERO" were substituted for hazards to electromagnetic radiation to ordnance.

e. Acronyms shall not be used as stand-alone, first-level entries in indexes (e.g., SAR). The use of acronyms as subentries is based on the likelihood of their being understood by the reader. The use of ac, dc, UHF, etc., is acceptable at subentry levels.

3.1.8.9 Numerals. Numerals shall generally be used in accordance with the GPO Style Manual.

3.1.8.10 Signs and symbols. Refer to the GPO Style Manual for established symbols. For logic diagrams graphic symbols from IEEE 91 shall be used. For electrical and electronic parts graphic symbols from IEEE 315/315A shall be used.

a. Symbols with figures in text. Set the symbol without spaces against the number. Repeat the symbol with each number, when the numbers are in a series. Applications include: degrees (°), dollars ($), Greek letters such as mu (μ), plus or minus (±), plus (+), minus (–), multiplication (×), division (÷), equals (=), greater than (>), and less than (<). The following shall apply:

(1) For magnification, use X (capital) and a 1-en space before the number (e.g., X 4).

(2) For proportion and ratio, use a colon and set without spaces (e.g., 1:25,000).

(3) For percentage, spell out the word "percent"; do not use the symbol.

b. Mathematical and scientific expressions, formulas, and equations.

(1) Treat an equation as a sentence; the signs are substitutes for words. Punctuate as necessary.

(2) Set a simple expression or short equation (no more than one fractional component) within text.

(3) Do not break an expression or equation between two lines within text. Either expand the line and begin the expression or equation on the next line; or stop the line, center the expression or equation on the next line by itself, and continue the text on the next line.

(4) If a paragraph contains more than two equations and one is centered on its own line, center all the equations on their own lines. Numerated equations with reference numbers in the right margin shall be set on their own lines.

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(5) A fractional expression using the virgule, such as $\frac{p_2}{p_1}$ or $2g\frac{p_2}{v_1}$, can be edited and set as 2g (p_2/v_1), using parentheses and the virgule. If comprehension is better served, leave an expression in its built-up form and center it on its own line.

(6) A short equation that contains one fractional expression can sometimes be edited in the same way, provided no ambiguity results. When an equation with a fractional expression is set on its own line, it will be set in its built-up form.

(7) Align two or more equations in sequence on the = sign and center the longest equation.

(8) Words used to connect successive equations (e.g., “therefore” and “similarly”) shall be set flush on the left margin on their own line.

(9) If terms used in an equation are defined (e.g., A = height, B = width), set each term on the left margin on its own line; indent the longest term on the left-hand side of the = sign 1 pica; and align remaining terms on their = signs.

(10) Center a complex equation (one with two or more fractional components or a square root, an integral, or a summation sign) in space between the text.

(11) When a built-up fraction is used in one part of a complex equation, build up all fractions in the equation. Carefully align matter to the left and right of the = sign on the = sign.

(12) Fraction lines and square root signs shall be the same width as matter enclosed; braces, brackets, parentheses, square root signs, and integral signs shall be the same height as matter enclosed.

c. Mathematical terms for equations. Math signs (e.g., +, –, ×, ÷, =) are set close to (one space) numbers and symbols. Numbers and some symbols (e.g., sin, cos, and tan) are set in the text font. Letter symbols are set in italics in text or in the font style of the math signs in equations. Do not allow an expression, such as A + B, to expand through justification. Inferior and superior numbers are set without spaces against the number or letter symbol and in smaller type size than the text (e.g., $2^{(n+1)}$ and A_2P_1).

d. Chemical symbols. Set the letter symbols for the elements in the text font. Set numbers and letters in compounds without spaces.

3.1.8.11 System description. Any system description shall briefly state system purpose, identify major system components, and state the contribution of each component toward fulfilling the system purpose. The system description shall include a description of the major components. The identification of a component and its contribution to system purpose shall be combined within one discussion. Emphasis shall be placed on brevity.

a. System purpose. A statement of system purpose shall be specific but not an inflated version of the system name. It shall build upon the implications of the system name as in “the engine start system uses externally supplied air and electrical power to impart turbine rotation, to establish fuel flow, to energize start ignition, and to transition automatically to a self-sustaining condition.” A restatement of the obvious is not satisfactory, as in “the engine start system provides the power and controls to start the engine.”

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b. Major components. The requirement to identify only major components is intended to eliminate parts such as solenoids, relays, controls, and indicators. The components identified in a system shall be of approximately the same magnitude and mutually exclusive such as “oil cooler, scavenge pump,” and not “butterfly valve, carburetor.” Identification shall use official terminology and shall avoid adjectives (e.g., “120/208 volt, three-phase, 400 Hertz, ac generator”).

c. Component contribution to system purpose. All components have a purpose or an output. This output shall be transformed or refined by another component prior to finding its ultimate use regarding the system purpose. Thus, the outputs of the major components shall be identified or related to each other or to system purpose. These outputs and their relationships to each other convey an understanding of “how the system works.” An example follows: “Oil system — The pressure pump (major component) establishes a flow of oil from the supply tank (major component) to the frictional surfaces of the engine and reduction gearing.”

d. Component description. The major components of a system shall be described in terms of static characteristics and operating characteristics.

(1) Static characteristics include information concerning the physical attributes of a component that are normally important during preflight and postflight activities. These activities include briefing and flight planning as well as the preflight and postflight inspections conducted by the crew.

(2) Operating characteristics describe the dynamic attributes of a component during its normal operation. This information expands and details the outputs of components and their interrelationships that were introduced briefly in the beginning of the system description. The reason for providing this information is to tell how a component or series of components operate so that appropriate procedures can be applied effectively.

e. Depth of detail. Extent of coverage of any system shall be determined by the following:

(1) Criticality of system. Systems contributing to safety of flight are more critical than systems that are primarily for crew convenience. The more critical the system is, the more detailed its coverage.

(2) Complexity of system. A system is considered complex and warrants detailed description when one or more of the following characteristics are present: multiple components, multiple modes of operation, associated factors, and effect on other systems.

(3) Uniqueness of system. Two types of uniqueness shall warrant detailed description of the system: design innovation and limited occurrence.

f. Readability of systems descriptions. Writing style is most important in readability. Grouping and sequence are also important. Complementary artwork consisting of control figures and schematic designs shall be used. The following guidelines shall be emphasized:

(1) Directness. Begin new subjects with a positive statement expressed in the active voice.

(2) Wordiness. Avoid repetition, extraneous words, and unnecessary engineering detail.

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(3) Inferences. Avoid using inferences so that the reader does not have to infer the writer's intent regarding important points of the discussion.

(4) Lengthy sentences. Strive for conciseness.

(5) Parallel structure. Use parallel, grammatical structure for sentence elements that are logically parallel.

(6) Continuity. Maintain a logical continuity of thought, especially in describing time-oriented events.

(7) Balance. The amount of coverage for a subject shall be based on the subject's criticality, complexity, and uniqueness and shall not be influenced by writer interest or availability of reference material.

g. Grouping. Cover the description of a system in its entirety under a major heading. Group related systems so that they appear successively. Strict adherence to these rules is difficult because of the high degree of system interaction within an aircraft. The following rules include assistance in determining appropriate grouping for system description materials.

(1) Grouping by function. Descriptions of systems or hardware that are functionally related shall be grouped together. The functional grouping shall be meaningful in terms of pilot and crewmember activities. Mechanical relationships shall not be interpreted as functional.

(2) Grouping by flow. Descriptions of hardware devices that process energy, fluid, or gas shall be grouped under a common heading or series of related headings. The scope of the common heading or series of related headings is defined by tracing the process from the source to consumption and return to the source. Exceptions to this rule occur when functional differences are present within the flow, as in the fuel system and the hydraulic system.

(a) In the case of the fuel system, devices for the fueling, transfer, dumping, and filtering of fuel are functionally related to "fuel management" activities. The descriptions of the devices that control the metering and consumption of fuel are functionally related to "engine operation" activities. An accepted convention is to describe the former devices under the heading of "engine fuel supply system." The latter devices shall be described under the heading of "engine fuel control system" and within the description of the engine.

(b) Similarly, hydraulic systems typically include normal and emergency power for utility systems (landing gear, wing fold, bomb bay) and flight control systems. Each powered system often has separate reservoirs, pumps, and distribution networks, and, in terms of crewmember activities, can be operated independently. Each such system warrants a separate "source-to-use and return-to-source" description and appears as a separate write-up under the common heading of "Hydraulic Systems." The rules for grouping by function and flow have been applied to typical aircraft systems within the specification.

(3) Grouping of emergency-use items. The description of emergency-type hardware that forms a part of a system shall be included within the description of its system. Examples are emergency fuel pump within "Fuel Supply System," and ram air turbine within "Electrical System" (or "Hydraulic System"). Emergency equipment that is a self-contained system shall not be grouped.

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h. Sequence. Descriptive materials shall be presented in a sequence that is meaningful to the user. Sequence refers to materials within a major system. Three rules for establishing sequence are:

(1) Usage. Where possible, the sequence of equipment descriptions shall be the same as the operational usage of the equipment.

(2) Criticality. The sequence of descriptive materials shall be based on relative importance. The principle applies to components within a system.

(3) Flow. Many systems, particularly those that process fluids, possess a natural sequence of events. Adherence to this sequence in presenting descriptive materials will improve meaning.

3.1.9 Conversion formulas. Conventional usage of English or metric system units shall be observed. However, when necessary to convert the measurements from one system to the other, the equivalent units of the other system shall be used. This is especially true when both the English and metric units of a measurement are provided side by side. English system and metric system equivalent units of measure are presented on figure 7. When accuracy is an issue, use the smaller equivalent units (e.g., convert feet to centimeters instead of to meters). The factors for converting from one system to the other are provided on figure 8.

3.1.10 References. Duplicating material (including artwork) in two or more portions of a publication shall be avoided except where essential for clarity. Use references instead.

3.1.10.1 Same publication references. The preferred method of referencing material in another location in the same publication is by paragraph number(s) only (e.g., “see 4.4.5.3”). Vague references such as “Section,” “Part,” or “found elsewhere in the manual” shall not be used. Whenever possible, cite the most specific paragraph reference(s).

3.1.10.2 Illustrations. Illustrations shall be referred to by figure number only. “Spot art” shall not be referenced.

3.1.10.3 Illustration index numbers (callouts). Refer to index numbers in illustrations first, followed by the figure number (e.g., “34, figure 2-6”). However, when multiple references in a paragraph refer to the same figure, only the first reference need indicate the figure number.

3.1.10.4 Foldouts. Refer to foldouts in text by page number only (e.g., FO-1, FO-2).

3.1.10.5 Other Naval publications. Other naval publications shall be cited by publication number and title when first referenced. If the referenced publication is classified, place a “U” after the title to indicate that only the use of the publication number and title together is unclassified. Any reference to the same publication thereafter may be by publication number only. Avoid references to specific paragraphs. If a reference other than the publication number is necessary use the chapter number.

3.1.10.6 Technical publications other than Naval publications. Technical publications other than Naval publications shall be referred to by publication number and title when first referenced. Use publication number only for subsequent references. Do not refer to publications that are temporary in nature, including drafts or those with limited distribution.

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3.1.10.7 Switch positions and panel markings. Switch positions and panel markings shall be referred to as they are marked on the equipment. When unusual decal markings are encountered, define the approved nomenclature. Use uppercase letters unless they appear otherwise on the display.

3.1.10.8 Steps and substeps. Steps and substeps shall be referred to by both paragraph number and step or substep number (e.g., “Refer to paragraph 4.3.9, step 4” or “Refer to paragraph 10.3.8, step 4c”).

3.1.11 Steps in emergency procedures. The following conventions shall be observed when preparing emergency procedures.

3.1.11.1 Numbered steps versus narrative. All definite procedures shall be composed of numbered steps. If the crewmember must analyze the emergency to determine subsequent actions, additional information needed to help determine the correct action shall be presented in narrative form. The numbered steps shall contain all of the actions necessary to enable the best chance of avoiding injury to the aircrew and minimizing damage to the aircraft.

3.1.11.2 Critical steps in emergency procedures. In an emergency procedure, the critical steps are those that shall be performed immediately, without reference to a written checklist, to prevent injury or loss of the aircrew and/or minimize damage to the aircraft. Critical steps are as designated by placement of an asterisk at the beginning of their step numbers. The critical steps in an emergency procedure are consecutive steps, beginning with step 1 of the procedure. Steps that require actions that may be deferred until there is time to consult a checklist, are considered to be non-critical.

3.1.11.2.1 Determining critical steps. The need for a critical step shall be determined using the following criteria:

a. It is an action that is essential to avoid injury/incapacitation of the aircrew and/or damage to the aircraft.

b. It is to be acted upon immediately, without reference to the printed checklist.

3.1.11.2.2 Preparing critical steps. Each critical step shall conform to the following criteria:

a. It shall contain only an item that must be performed immediately to alleviate the immediate emergency sufficiently to allow time for the flight crew to refer to the printed checklist.

b. It shall precede all non-critical steps in the procedure.

c. It shall be easy to understand and learn.

d. It shall be written as briefly as possible.

e. Terms contained in the preface and glossary shall be used to the maximum extent possible.

f. Insofar as possible, the wording used regarding the action to be performed, shall be the same as in other similar emergency procedures and in emergency procedures for other aircraft with like procedures.

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3.1.12 Warnings, cautions, and notes. Warnings, cautions, and notes identify situations that require emphasis. Important text may be separated from the surrounding text and labeled as a warning, caution, or note for emphasis. A warning, caution, or note shall be read as applying to the paragraph, sentence, step, or item that immediately precedes it. Warnings, cautions, and notes may also be placed within figures. See figure 9 to determine when important text should be separated from the surrounding text and whether to label the important text as a warning, a caution, or a note.

a. A warning identifies an operating procedure, practice, or condition that may result in injury or death if not carefully observed or followed.

b. A caution identifies an operating procedure, practice, or condition that may result in damage to equipment if not carefully observed or followed.

c. A note identifies an operating procedure, practice, or condition that is essential to emphasize.

3.1.12.1 Wording warnings, cautions, and notes. The following procedures shall be observed when constructing warnings, cautions, and notes:

a. Warnings and cautions shall be worded factually in a cause and effect format (i.e., if "this" happens, then "that" adverse event can or will occur.)

b. Warnings, cautions, and notes shall not contain procedural steps; however, they may contain listings. Required actions to be taken during execution of a procedure shall be clearly numbered as steps in the procedure and not hidden within the subordinated warnings, cautions, or notes.

c. Directive statements such as "Do…" or "Do not…" should be avoided, since such wording when not followed causes the aircrewman to be in violation of NATOPS and casts a warning or caution as a procedural step.

d. Notes may be directive in nature when addressing contingency concerns that may occur during the execution of a procedure (e.g., "Note – Make smooth PCL adjustments while…").

e. Potential hazards shall be specifically identified to avoid having the aircrewman guess the consequential effects.

f. Warnings and cautions shall be kept as brief and to-the-point as possible. When additional explanatory text is necessary, the complete situation may be discussed in one or more larger paragraphs, followed by the warning or caution that captures the essence of the situation. The explanatory information for a warning or a caution, when required with a procedure, shall be placed within the background discussion that precedes the steps portion of the procedure. Only the warnings, cautions, and notes without the explanatory text shall appear within the steps portion of the procedure.

3.1.12.2 Order in which warnings, cautions and notes appear.

a. Warnings, cautions, and notes shall follow the paragraph/sentence/step/item to which they apply.

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b. When several warnings, cautions, and notes appear in sequence, they shall appear in order of their precedence: A warning may be followed by a caution or note; neither a caution nor a note shall immediately precede a warning; and a note shall not immediately precede a caution.

c. A warning, caution, or note that applies to the entire paragraph shall be placed between the paragraph title and the paragraphs, steps, or items rather than placed following a paragraph, sentence, step, or item that is narrower than its application.

d. All warnings, cautions, and notes that follow a paragraph/sentence/step/item shall be read as part of that paragraph/sentence/step/item. This is important in situations such as compound emergencies, where the procedures from more than one emergency must be combined and appropriately sequenced during execution.

3.1.13 Footnotes. Footnotes may be used to convey additional information that is not properly part of the text. A footnote to the text shall be placed at the bottom of the page containing the reference to the text. The Government Printing Office Style Manual shall be used as a guide for footnote usage and format.

3.2 Format requirements for NATOPS products.

3.2.1 Warnings, cautions and notes.

a. Warnings, cautions, and notes shall be 16 picas wide, centered in the column, and headed by the appropriate “WARNING,” “CAUTION,” or “Note” label as shown on figure 4.

b. Warnings deal with the subject of human safety and shall not be split between columns or pages. Cautions deal with hazards to equipment, and may be split between columns, but may not be split between pages. Notes may be split between columns and/or pages as necessary. When a note carries over to another page, carry over at least two lines of text.

c. When a warning, caution, or note is immediately followed by another respective warning, caution, or note on the same page, the warning/caution/note label shall appear prior only to the first warning/caution/note in the series. An 11-point bullet shall be placed to the left of the beginning of each warnings/caution/note of that series as illustrated on figure 9.

d. When bulleted warnings, cautions, or notes break from page to page, the warning, caution, or note symbol shall be repeated above the first warning, caution, or note on the next page.

e. When only one bulleted warning, caution, or note remains on either the bottom right column of the old page, or the upper left column of the new page, remove the bullet and repeat the warning, caution, or note label.

3.2.2 Footnotes. Footnotes shall be identified by number, not symbols such as asterisks. Two or more footnote references that appear together in text are separated by a space, not a comma (e.g., 1 2). If the footnote(s) applies to only one column, it shall appear at the bottom of the appropriate column. If the footnote(s) applies to both columns, it shall be placed at the bottom of the left column. Footnotes for 11-point text shall be set in 8-point type of the same font. Locate a 5-em (50 points) dash flush left above the footnote.

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3.2.3 Specifications and standards for digitally produced artwork. Line drawings, which include schematics, wiring diagrams, block diagrams, and other illustrations, such as aircraft drawings, “spot art,” circuit breaker panels, cockpit drawings, and flight patterns are classified as one of the following types:

a. Monochrome. Characterized by the use of only two distinct colors (such as black and white) — one color for the background, the other color for displaying the image/text.

b. Grayscale. Characterized by the use of black and varying tones of gray.

c. Full color. Characterized by use of color, gray tones, and black and white. Tones of individual colors are determined by the electronic color model, CMYK, RGB or PANTONE Matching System® (PMS) Color.

3.2.3.1 Style and technique for artwork. Style and technique shall be of a quality that will produce artwork that clearly and economically portrays the needed information. Use illustrative material to describe an item or idea more effectively through graphic presentation, to clarify text, to present phases or sequences difficult to understand by use of text alone, to call attention to details, and to furnish graphic identification of display and controls. Use the minimum number essential for such purposes. Do not use masthead or fill-in illustrations and other art solely for decorative purposes. All illustrations shall be functional and used to clarify text.

3.2.3.2 Conversion of legacy drawings, illustrations, and schematics to digital. Existing (legacy) hard-copy paper drawings, illustrations, and schematics that do not require updates shall be scanned and provided electronically as tagged image file format (TIFF) files or JPEG files. Computer-scanned, bit-mapped graphics may also be employed when line art and computer graphics are not available.

3.2.3.2.1 Scanning resolution requirements. Drawings and illustrations that consist of black lines and black text only (monochrome) shall be submitted electronically as TIFF or JPEG files at a resolution of 600 dpi (dots per inch). Those drawings and illustrations that also incorporate shades of grey shall be submitted electronically as TIFF or JPEG files at a resolution of 400 dpi. Drawings and illustrations that contain color shall be submitted electronically as TIFF or JPEG files at a resolution of 300 dpi/lpi (dots per inch/lines per inch). Computer-scanned, bit-mapped graphics may be used in reproducible copy, but the procuring activity shall be provided the original artwork from which the originals were prepared if available.

3.2.3.3 Vector art requirements. Vectorized (computer-generated, editable) line art shall be used whenever new artwork is prepared or existing artwork requires updating. Simple text callout updates to complicated drawings may not warrant vectorizing. Requirements shall be at the discretion of the procuring agency, and will be decided on a case-by-case basis. Line art shall meet the following requirements:

a. Prepare drawings in the same size that they will occupy on the printed page.

b. Detailed drawings shall be prepared to fit either full-page image width or column image width and shall not exceed page image height.

c. Major line weight for illustrations shall be 0.017 inch; minor line weight shall be 0.013.

d. When a line end option is available, line ends shall be rounded.

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3.2.3.3.1 Type size used within illustrations, charts, tables, and graphs. Type size used within illustrations, charts, tables, and graphs shall be no smaller than 8 points (6 points if all uppercase). Arial fonts (or equivalent; e.g., Helvetica) shall be used for column headers and for index numbers and legends. Approval of the procuring activity is required to use type smaller than specified. Use all uppercase letters for label callouts and descriptions.

3.2.3.4 Use of color in artwork. Color art, when approved by the procuring activity, shall meet the following requirements. Color shall be used in artwork only where necessary to clarify functional operations. The number of colors shall be kept to a minimum by the use of tint, cross-hatching, and dots. To maintain consistency within figures, all vectorized graphics, illustrations, and schematics that require color for clarity shall be produced using the spot colors of red (PMS 485), green (PMS 354), yellow (PMS Process Yellow), and blue (PMS Process Cyan) as appropriate. The use of black shall be PMS Process Black. When color is used, a legend containing an exact duplicate of the color or pattern shall be included in the illustration. The use of color shall be consistent.

3.2.3.5 Photographs. Photographs shall be high quality and have good contrast, and shall be screened at 300 dpi/lpi. The use of photographs shall be determined by their purpose and suitability in the publication. Editable line art is preferred over photographs, since details can be indicated and changes can be incorporated. For web-based publications, the lowest acceptable resolution possible shall be used for photographs in order to limit file size.

3.2.3.6 File naming. Electronic artwork (figures) files shall be named to facilitate identification for importing into NATOPS electronic documents and for output. The file naming convention is at the discretion of the provider, but shall, as a minimum, bear a unique control that includes publication identification. A combination of alpha and/or numeric characters may be used to maximize identification. A separate electronic document that cross-references the figure control numbers with the publication figure titles shall be submitted by the provider.

3.2.3.7 Layout of art in NATOPS publications. Avoid duplication of illustrations within a publication. Place illustrations as near as possible to related text except where this would require unnecessary duplication of illustrations in two or more locations in a publication. Locate full-page art on the left-hand page opposite related text or on the page following the related text. Use figure references within the text. Multiple-sheet illustrations are acceptable. When used, rotate turn-page illustrations so the top is at the left-hand margin, reading to the right-hand margin; place the figure number and title beneath the figure at the bottom of the page in the same place as other full-page illustrations.

3.2.3.7.1 Ruled boxes for figures. The current NATOPS templates contain ruled boxes for figures. Therefore, it is unnecessary to “box” figures/illustrations. If not using the current NATOPS templates (available from AIR-4.0P), figure box line weight shall be of 0.01-inch thickness in the final page image. “Spot art” shall not be boxed. Place classification and figure numbers and titles outside the boxed area (see figure 3).

3.2.3.7.2 Text for figures. Nomenclature, index numbers, and legends (or keys) shall be used, when necessary, for identification of significant features. Text shall be placed on the background area only, and shall not obscure any portion of the illustration.

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a. Nomenclature. Use justified left margins for nomenclature of more than one line. All lines of copy shall parallel the horizontal edges of the illustration. Place at least 2 extra points of leading between entries in a block of text within a figure. Do not use extra leading within an entry.

b. Notes to figures (including tables). Notes to figures (including tables) shall be numbered separately from textual footnotes within the publication. Notes shall be numbered consecutively and placed below the figure and above the title (within the figure frame). Place 6 points of space following the word, "NOTES," and separate notes by 6 points of space. The word, "NOTES:" followed by a colon as indicated shall be placed at the left margin of the figure followed by the explanatory information in Arabic numeral sequence under "NOTES." For example:

NOTES:

1. Time requirement considered met, upon completion of FRS training.

2. Qualification is normally required unless carrier availability precludes timely fulfillment.

c. Figure callout numbers. Figure callout numbers for each separate illustration shall always start with Arabic numeral 1 and continue consecutively. Exception: all multiple sheet illustrations shall be considered one figure. Sequence of numbering shall be from top to bottom or clockwise starting from the upper left corner. Capital alphabetical suffixes shall be added to any new callout numbers inserted between existing callout items when a Change to an illustration is issued (for example, 17A and 17B). Suffixed callout numbers need not be removed for a Revision unless the illustration is changed. In cases where figures contain multiple sheets, numbering shall be as indicated in the following example:

Figure 1-2. Hydraulic Lift Equipment (Sheet 1 of 3)

Figure 1-2. Hydraulic Lift Equipment (Sheet 2)

Figure 1-2. Hydraulic Lift Equipment (Sheet 3)

d. Legends (or keys). When callout numbers are used, a legend comprising the numerical listing and their identification shall be included on, or adjacent to, the illustration. If a callout is deleted from an illustration, the word "Deleted" shall be placed after the appropriate number in the legend. This statement is relevant to a Change only, and shall not be followed for a Revision.

e. Leader lines. Leader lines shall touch the object to which they apply. Lines shall be uniform, short, and as straight as possible; however, doglegs are permitted. Lines shall not cross or come in contact with other callout lines, and shall not hide essential details. A line shall be highlighted for clarity. Arrowheads shall be used. Arrowheads used throughout the publication shall be uniform.

3.2.3.7.3 Unacceptable artwork. The following practices and techniques are unacceptable as substitutes for original artwork:

a. Continuous tone screened film negatives or screened prints.

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b. Line illustrations containing weak or broken lines.

c. Illustrations with soft-edged, illegible, or broken type.

3.2.3.7.4 Schematic design. In designing most schematics, it will be necessary to compromise between completeness, simplicity, and making the diagram self-explanatory in order to facilitate reading and understanding. To prevent over-complexity because of automatic features, cover these characteristics by text in the diagram. Lists of large numbers of items shall be in alphabetical order. The major systems described above shall be arranged in alphabetical order. Refer to IEEE 315/315A for preparation.

a. Schematic diagrams. Schematic diagrams shall be used when necessary to show flow, such as the hydraulic, fuel, and electrical systems. The function of a schematic diagram is to illustrate the operation of the system in as straightforward a manner as possible. To accomplish this, present the components of the schematic diagrams in the following order of importance:

b. Flow of the system. The system flow shall receive primary importance by having a minimum of turns in the lines, and by starting at one of the outer edges of the page and continuing as straight as possible until the other side of the page is reached. Flow includes tanks or reservoirs that are considered the starting point of most schematic diagrams. The equipment need not assume the same relative position as it does in the aircraft. It is more important that the diagram be arranged so that the flow of the system can be traced with minimum effort. Avoid or eliminate crossovers whenever possible. Return lines need not be shown in entirety unless necessary to understand the system. To avoid resemblance to electrical wiring diagrams, all electrical flow lines on electrical schematics shall be wide bands as opposed to thin lines, except for electrical actuation lines that will be shown as thin black lines. Perspective shall not be used unless it improves the diagram and the system can be easily traced.

(1) In the interest of standardization, the following coding shall be used:

Solid line	Electrical actuation
Dashed line	Mechanical actuation
Black	Main portion of system
Black diagonal stripes	Emergency

(2) The following coding is recommended for use in those cases where an additional coding is warranted because of complexity of the system:

Blue	Secondary power, cold air
Green	Supply
Green diagonal stripes	Return
Blue diagonal stripes	Static fluid
White	Vent

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(3) The color coding, once assigned, shall be consistent throughout the manual. All symbology shall be listed in a legend on the schematic for easy interpretation of the schematic.

c. Controls and indicators used by flightcrew. These items shall be second in importance only to the flow of the system. The controls and indicators shall be presented in accordance with the appropriate military or commercial standards. Controls and indicators shall be set off slightly to the side of fluid flow lines. On electrical diagrams, controls shall be placed directly on flow lines. Valves shall be presented schematically in the shape of a "T," "L," etc.

d. Flow control devices. These include all flow control devices within the system, such as check valves, fuel pumps, fuses, relays, and restrictors. These flow control devices shall be presented in a simple, stylized version to show function of the device. Solenoid valves shall be indicated as such and shall include a note indicating whether the valve is spring-loaded to the open or closed position.

3.2.3.7.5 Graphs. The three types of acceptable graphs are illustrated on figure 10. When multiple plots appear in a graph, they shall be separated by a space in the grid structure. This space shall not exceed 1 inch. Shift graphs conserve space and are frequently used for correction functions on charts. However, their use shall be kept to a minimum. When used, the baseline shall be labeled "BASELINE" and the shift lines labeled "GUIDELINES."

a. Sample graphs. The graphs in this specification are samples to be used as guides in preparation of the manuals and use of better presentations is encouraged. However, all graphs shall be approved by the procuring activity to ensure that their presentation include the desired information (see 6.2).

b. Nomographs. Nomographs shall not be used.

c. Back-to-back scales. For certain functions such as simple value conversion, the use of back-to-back scales is recommended.

d. Grid construction. Grids shall appear on graphical- and profile-type charts. Grid structure is included to facilitate the tracing process, which is different from the scale reading process. The following rules shall apply to preparing grid structure:

(1) Grid interval. The distance between adjacent grid lines shall be less than 0.3 inch and no less than 0.1 inch. The larger grid intervals (0.2 to 0.3 inch) are preferred.

(2) Grid structure. Grid lines shall be composed of major and minor grid lines. There shall be four minor grid lines between each major grid lines. Major grid lines shall be 0.017 inch thick; minor grid lines shall be 0.013 inch thick.

(3) Contrast. Grid structure shall appear less bold than the scales, parametric lines, labels, and numbers.

e. Scale construction. To include optimum legibility and consistency, the following rules shall apply to scale construction:

(1) Axes selection. Graph axes shall be selected in terms of spatial orientation. Thus, altitude shall be located on the vertical axis (ordinate), and distance or range on horizontal axis

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(abscissa). Additionally, values shall increase in the upward, left-to-right, and clockwise directions, and shall decrease in the downward, right-to-left, and counterclockwise directions.

(2) Scale breakdown. Recommended scale breakdown patterns are depicted on figure 10. Scale breakdown selection shall take into account aircraft instrumentation. Minimum scale breakdown shall be established in accordance with actual instrument scales. Scale interval shall not be less than 0.1 inch. The number of graduation marks between numbered points shall not exceed nine. The height of major, intermediate, and minor graduation marks shall be approximately 0.22, 0.16, and 0.09 inch, respectively. The stroke width of major, intermediate, and minor graduation marks shall be approximately 0.0175, 0.015, and 0.0125 inch, respectively.

f. Parametric lines. The maximum thickness of parametric lines shall be approximately 0.015 inch. The minimum thickness shall be approximately 0.005 inch and shall be used only when adjacent curves are close and require increased separation distance. Distance between adjacent parametric lines shall not be less than 0.1 inch. No specific rule can be applied to the number of parametric lines that appear. Chart designers shall first determine the degree of accuracy required and then the number of parametric lines necessary to gain accuracy without clutter.

g. How-to-use description. A simple pictorial guide, as shown near the upper right corner of figure 11, example lines, or an explanation of how the chart is to be used shall appear on or with the first sheet(s) for each type of chart or in a paragraph explaining its use. In simple cases example lines, also shown on figure 11, may be sufficient.

h. Example lines. When a pictorial guide is not included to explain the use of a chart-type, a set of example lines shall be placed on the chart to depict its normal use. The word “EXAMPLE” shall be placed along the side of the grid scale where initial entry into the grid would normally occur. The end of the example line shall be extended beyond the grid and originate at the word “EXAMPLE.” Progression arrows shall be placed on each segment of the example lines to indicate the logical progression through the chart.

(1) Example lines shall be placed on additional charts as necessary to explain sample problems. Significant points along the example lines may be labeled alphabetically to explain how the chart-type is to be used or for reference in sample problems. However, the example lines and labeling shall not be allowed to interfere with daily use of the chart. Pictorial guides or additional figures shall be used to avoid cluttering charts essential for everyday use.

i. Limitations and marginal performance. Limitations and marginal performance areas shall be identified on graphs. The grid structure shall be deleted for regions that exceed limitations of the particular aircraft. Areas in which aircraft performance is marginal shall be identified and, where applicable, annotated with a warning note or reference to an additional chart that will include more detailed information concerning the marginal performance.

3.2.4 Source material guidelines for NATOPS products. Technical content of source data shall be delivered in accordance with the basic contract. Format of source data is at the discretion of the supplier; however, all illustrations shall meet industry standards for resolution and clarity. Electronic files are required, and editable files shall be provided if available. Computer graphic files shall be submitted on CD-ROM. Text shall be provided electronically in editable format; no

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proprietary software shall be used to generate the files. Text and illustrations shall be clearly annotated for location in each respective document. All material prepared under this chapter shall be government property, including original artwork and computer graphics. Additional guidelines are provided below.

3.2.4.1 Guidelines for delivery of source material using XML technologies. Documents or documentation generated using XML technologies shall be tagged in conformance with the currently approved version of the NATOPS document type definition (DTD). XML-tagged documents and documentation shall include intra-document link tagging (links to elements such as figures, paragraph references, part references, etc.). All tables shall be tagged according to the NATOPS DTD, which incorporates the Organization for the Advancement of Structured Information Standards (OASIS) Exchange table model.

3.2.5 Final product delivery guidelines. All final NATOPS products are delivered to the following organizations: NAVAIR-4.0P; the respective Prime Contractor who holds the contracting vehicle; the Contracting Officer Representative (COR) at the Naval Air Warfare Center, Aircraft Division (NAWCAD), Code 3.2.6; and the Logistics Element Manager (LEM) at the Naval Air Technical Data and Engineering Service Command (NATEC).

3.2.5.1 Delivery format requirements for final NATOPS products. All final NATOPS products shall be delivered on CD-ROM in portable document format (PDF). All PDFs shall meet the current standard for CD-ROM data formats. A separate PDF file on a separate CD-ROM disk shall be provided for each final NATOPS product. The PDF files shall be accompanied by the appropriate "reader" to successfully open and print the attached files. The PDF files shall employ the same fonts contained in the printed manual, and shall display figures at an appropriate resolution for on-screen viewing, as well as printing. Final deliverables shall be in accordance with the basic Contract Delivery Order or Task Order and may include hard-copy requirements, as well as electronic requirements.

3.2.5.1.1 Hyperlinked PDF files. Final deliverables required to be hyperlinked shall "open" on-screen in a one-column NATOPS format with bold headers. These PDF files shall include "links" to the table of contents (TOC), list of illustrations (LOI), glossary, list of abbreviations and acronyms, and index. Paragraph and figure references in text shall also be hyperlinked. Hyperlinked files shall be prepared to successfully upload to the NATEC website. NATEC will review all hyperlinked files for subsequent upload to their website at www.natec.navy.mil. Most unclassified NATOPS publications are posted and available for viewing on the NATEC website.

a. PDF files for NATOPS checklists. For those platforms deemed "paperless," PDF files for NATOPS checklists shall include links from the "bleed tabs" on the front cover to each respective section of the checklist document. Links shall also be provided from the emergency procedures index to individual emergency procedures and from the normal procedures index to individual normal procedures. For those platforms using paper manuals only, no hyperlinking is required for NATOPS checklists.

3.2.5.1.2 Print-ready PDF files and print run sheets. Each print-ready PDF file shall be accompanied by a comprehensive print run sheet. The print run sheet shall indicate emergency border pages for the NFM, with the notation "TRB" ("top, right, bottom") for right-hand pages, and "TLB" ("top, left, bottom") for left-hand pages. All other "bleeds," such as tab bleeds shall

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also be indicated on the print run sheet. The emergency side and bottom tab cuts for PCLs and other tabbed documents shall also be indicated on the print run sheets. Pages that require lamination shall be indicated on the respective print run sheet. All true folios, including those representing “blank” pages, shall also be identified on the print run sheet. Sample print run sheets for the NFM and checklists are provided as figure 12. As required for hyperlinked PDF files, print-PDF files shall represent the printed manual exactly. Foldouts shall be included in the PDF file, and shall be presented in the same size as in the printed manual.

a. CD-ROM Labels. CD-ROM labels shall contain the following data:

Respective NAVAIR number of publication

Specific NATOPS product description (e.g., NATOPS Flight Manual, Functional Check-flight Checklist)

Platform or Community (e.g., E-2C Hawkeye 2000; NATOPS Aircraft Refueling)

Classification of NATOPS product/publication

Description of NATOPS product (e.g., “print-ready” PDF; hyperlinked PDF)

Contract Number, Delivery Order Number, and Task Order Number

Appropriate CDRL number

Publication date (See 6.7d.)

Copy-freeze date (See 6.7b.)

Disk number (e.g., Disk 1 of 1 (if entire NATOPS product fits on one disk))

Contractor Agency that prepared the data

3.2.5.1.3 Web-compliant hypertext markup language (HTML) output. NATOPS products produced for web-compliant output shall be browser-independent, and shall not employ scripting as part of the output file. The HTML output shall conform to the HTML 3.2 (http://www.w3.org/TR/REC-html32) specification, and shall be syntactically correct and well formed. The HTML output shall be independent of style sheets for appearance. Dynamic HTML shall not be used.

a. HTML output table of contents. The HTML output shall include a type of table of contents (TOC) navigation. This navigation may be implemented using bookmarked frames as shown on figure 13, which shall remain visible during any browsing of the document/documentation content. The TOC shall reflect the same level of detail as the printed manual TOC, with the addition of figures, and shall not use a separate list of illustrations (LOI).

b. Embedded illustrations. Embedded illustrations in the HTML content shall appear in-line, in a size and format appropriate for web viewing. Specific size requirements shall be defined on a case-by-case basis in the basic contract. Content shall be displayed in one-column format, with fonts and font sizes that match the printed output.

3.2.5.1.4 Printing requirements.

a. NATOPS publication bar codes and stock numbers. Upon receipt of the NATOPS publication or Change package, the NATEC Logistic Element Manager (LEM) prepares a Print

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Order Sheet (POS). The LEM then passes the POS to NATEC Distribution, who requests a stock number from NAVICP Philadelphia via the Naval Logistics Library (NLL) website. The LEM then prepares the printing and mailing labels for the publication. Upon receipt of the stock number from NAVICP, NATEC Distribution passes the assigned stock number and the mailing labels for printing and distribution to the LEM. The LEM, in turn, sends the stock number and mailing labels along with the NATOPS publication or Change package to the contract printer. Upon receipt of the printing package, the printer generates a bar code, and places both the bar code and the stock number in the lower left corner of the NATOPS publication or change title/cover page or on the CD-ROM label in accordance with NAVSUPINST 5600.26-series before printing the publications for distribution.

b. NATOPS publication distribution. Upon receipt of the POS for the NATOPS publication or Change package from the NATEC LEM, NATEC Distribution prepares a set of mailing labels for distribution and mailing. A draft set of mailing labels is generated from the Automatic Distribution Requirements List (ADRL) accounts submitted by each unit in accordance with NAVAIR 00-25-100. The LEM “scrubs” the draft labels, adjusting the requested publication quantities to ensure that the appropriated quantities of paper and/or CD-ROM copies are printed for each unit. NATEC Distribution then prints the set of scrubbed mailing labels, which are forwarded by the LEM, along with the publication to be printed and its assigned stock number, to the contract printer. The printer then prepares, packages, and mails the publications directly to each unit on distribution.

c. Printing specifications for NATOPS manuals.

(1) Trim. Page trim size is 8-½ inches x 11 inches.

(2) Holes. Drill three round holes. The top and bottom holes are 5/16 inch in diameter and 8-1/2 inches center to center. The middle hole is 5/16 inch in diameter and 4-1/4 inches center to center. Center of all holes shall be ½ inch from bind edge (standard three-hole drill).

(3) Text stock. Print flight manuals on JCP Code A60, Offset Book, white, basis size 25 inches x 30 inches, basis weight 50 stock.

(4) Title and last page stock. Print unclassified cover page text on JCP Code L20, vellum finish cover, medium blue, basis size 20 inches x 26 inches, basis weight 65 stock. For Confidential NATOPS publications use vellum, Canary Yellow 111-5, 50-lb. For Secret NATOPS publications use vellum, Vermilion Red 111-22, 50-lb.

(5) Protective Tenite covers. Front and back covers shall be clear Tenite, high-impact polyethylene plastic, smooth finish, 0.035 gauge, hidden hinge with three-hole drill to match that of document.

(6) Binding. Bind text, printed cover, and Tenite cover with two metal (non-rust) screwposts of suitable capacity.

d. Printing specifications for NATOPS Pocket Checklists.

(1) Trim. Page trim sizes are one of the following as appropriate:

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Cover/Title page
5-¾ inch x 9 inch no tabs

Side tabbed pages
5-½ inch x 8 ¾ inch

Tabs appear on the sides and bottoms of pages as necessary. Die cut on the outside or bottom edge to form index tabs ½ inch in depth by varying lengths. The number of positions will vary depending upon the leaves.

(2) Holes. Spiral bound at the top.

(3) Text stock. All pocket checklist pages shall be printed on 250-lb (for tabs) and 130-lb Yupo paper.

Paper color shall be as follows:

Emergency procedures section title page and part – tinted yellow Bingo part – tinted blue

Front matter and normal/special and performance data parts – white

(4) Covers. Covers shall be clear high-impact linear polyethylene 0.035 gauge. Cover material shall:

(a) Withstand temperature changes of +180 °F to –50 °F without significant change of characteristics of rigidity or shape.

(b) Withstand impact of 1-pound round metal weight dropped to the material from a height of 10 feet at a temperature of –20 °F.

(c) Be unaffected by all common chemical solvents (e.g., gasoline, oil, cleaning fluids) at normal temperature.

(d) Be semi-rigid and retain its flat condition under normal use.

(e) Withstand a minimum of 50,000 180° flexings without significant change in tear strength.

(f) Face and back of the sheet shall be non-glossy.

(5) Binding. Pocket checklists shall be spiral bound at the top.

e. Printing specifications for NATOPS Card Checklists.

(1) Trim sizes. Trim sizes shall be one of the following:

Size A — 8-½ x 11 inches; intended to fit a standard-size clipboard.

Size B — 5 x 8 inches; intended to fit the aviator's kneeboard.

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(2) Text stock. Print card checklists on JCP Code A60, Offset Book, Basis Size 25 x 38 inches. Basis Weight 90 pound index, white.

(3) Lamination. Pages shall be laminated with clear acetate for durability when specifically requested and approved by the NATOPS Products Administrator. Both sides of the each selected page shall be laminated with 1.5-mil acetate or a suitable equivalent. Normally, laminated pages will be limited to heavy use pages such as normal procedures.

f. Printing specifications for Functional Checkflight Checklists (FCFCLs).

(1) Trim. Trim size is 5 x 8 inches.

(2) Text stock. The FCFCL shall be printed head to head on one side only on 32-pound white ledger paper (JCP J10) or equivalent.

(3) Binding. The FCFCL shall be stitched with one staple at the upper left corner parallel to the top edge as shown on figure 14.

g. Printing specifications for Passenger Information Cards.

(1) Trim. Trim size is 8-½ inches x 11 inches.

(2) Text stock. The passenger information card shall be printed head to head on white index paper, 250 pounds per 1,000 sheets (basis 25-½ by 30-½ inches) or equivalent.

(3) Lamination. The passenger information card shall be laminated on both sides with 1.5-mil acetate or equivalent.

h. Printing specifications for Servicing Checklists (SCLs).

(1) Trim size. Trim sizes are identical to those for the pocket checklist (see 3.2.5.1.4d.(1)).

(2) Text stock. All servicing checklist pages shall be printed head to head on 250-lb (for tabs) and 130-lb white Yupo paper.

(3) Covers. The front and back covers shall be translucent, high-impact linear polyethylene plastic (slightly frosted plastic), 35-gauge (0.035 inch) thick.

(4) Holes. Drill four round holes, 3/16 inch from edge of sheet to edge of holes, 5/16 inch diameter on left binding side, 1-5/8 inch, 3-1/8 inch, 5-1/8 inch, 6-5/8 inch center to center from bottom to center of holes.

(5) Binding. Servicing checklists shall be bound with four flexible wire (“flex rings”) rings of appropriate size.

i. Printing specifications for Takeoff and Landing Data (TOLD) Cards.

(1) Trim size. TOLD cards are normally 5-½ by 7 inches.

(2) Text stock. All sheets for the TOLD cards shall be printed head to toe on white index paper, 220 pounds per 1,000 sheets (basis 30-½ inches) or equivalent.

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(3) Lamination. The TOLD cards shall be laminated on both sides with 0.005-mil acetate or a suitable equivalent.

(4) Holes. Two 5/16-inch holes shall be drilled through the card deck, centered 5/8 inches from the top edge, with 2-¾ inches between hole centers.

(5) Binding. TOLD cards shall be bound using two flexible wire rings ("flex rings") of appropriate size.

j. Printing specifications for Ditching and Bailout Placards.

(1) Text stock. Ditching and bailout placards shall be printed on 5-mil white vinyl paper. Red emergency borders shall be reflective.

(2) Adhesive placards. The placards shall be printed on 3M Scotchcal™ film or equivalent with an adhesive back.

3.3 Requirements for NATOPS flight manuals.

3.3.1 Technical content requirements. The NATOPS Flight Manual shall contain information on all aircraft systems, performance data, and operating procedures required for safe and effective operations. The formal manual shall consist of all parts as listed in 3.3.1.1. Parts shall be marked with identifying tabs as described in 3.3.1.2.10. Material shall be included in the order and manner specified herein. Headings and subjects that are not applicable shall not appear.

3.3.1.1 Arrangement of publication. Publication arrangement for NATOPS Flight Manuals shall consist of the following:

Front Matter
Part I — The Aircraft
Part II — Indoctrination
Part III — Normal Procedures
Part IV — Flight Characteristics
Part V — Emergency Procedures
Part VI — All-Weather Operations
Part VII — Communications
Part VIII — Mission Systems
Part IX — Flightcrew Coordination
Part X — NATOPS Evaluation
Part XI — Performance Data
Back Matter

3.3.1.2 Front matter. The front matter shall consist of the following:

Front Cover
Title Page
Letter of Promulgation
Interim Change Summary (always Reverse Blank)
Summary of Applicable Technical Directives (always Reverse Blank)
Record of Changes (always Reverse Blank)

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List of Effective Pages (LEP)
Table of Contents
List of Illustrations
Bibliography (if applicable)
Glossary (if applicable)
List of Abbreviations and Acronyms (if applicable)
Preface
Three-View Illustration
Part Pages

3.3.1.2.1 Front cover. The front clear tenite cover shall remain free of printing so that the color and any printing on the front page can be viewed through it.

3.3.1.2.2 Title page. The title page (see figure 15) shall:

a. Be the first right-hand page (the reverse shall be blank)

b. Carry the NAVAIR publication number

c. Contain the classification and title

d. Carry the supersedure notice (when applicable)

e. Carry the appropriate classification and warning notice(s) (see 3.1.2.5)

f. Allow sufficient space (minimum of 1 inch) for the stock number and bar coding, which will be placed on the title page by the printer as noted in 3.2.5.1.4a.

g. Carry the applicable limited distribution statement contained in DoD Directive 5230.24. In most cases Distribution Statement C will be the distribution statement that appears on the title page of the publication:

> DISTRIBUTION STATEMENT C — Distribution authorized to U.S. Government agencies and their contractors to protect publications required for official use or for administrative or operational purposes only, determined on (date of current revision). Other requests for this document shall be referred to Commander, Naval Air Systems Command (PMA−____), 47123 Buse Rd, Bldg 2272, Patuxent River MD 20670-1547.
>
> Note: If the publication is revised, the distribution statement shall be confirmed by the procuring authority as still current, and the date within the distribution statement shall be changed to match the revision date on the title page of the publication.

h. If the publication is classified, classification, classification authority, and downgrading/declassification information shall be added (see 3.1.2).

i. Destruction notice. For unclassified publications, this statement shall appear on the title page:

> DESTRUCTION NOTICE — For unclassified, limited documents, destroy by any method that will prevent disclosure of contents or reconstruction of the document.

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j. For classified publications, this statement shall appear on the title page:

> DESTRUCTION NOTICE — For classified documents, follow the procedures in DoD 5220.22-M, National Industrial Security Program Operating Manual, Section II-19 or SECNAVINST 5510.36.

k. When NFM supplements exist, place a reference to the other related publications on the title pages of the NFM and any supplements: "THIS PUBLICATION IS INCOMPLETE WITHOUT NAVAIR 01-XXXX-XXX."

l. Issuing authority statement. The following statement shall be placed on the title page below the Destruction Notice:

> "ISSUED BY AUTHORITY OF THE CHIEF OF NAVAL OPERATIONS AND UNDER THE DIRECTION OF THE COMMANDER, NAVAL AIR SYSTEMS COMMAND."

m. Publication date. Revision and Change when applicable (See 6.7d.)

3.3.1.2.3 Letter of promulgation. Signature authority for the respective Letter of Promulgation (LOP) is delegated by CNO to COMNAVAIRSYSCOM. Within COMNAVAIRSYSCOM, AIR-4.0 has been designated as release authority for the LOP. The NATOPS Manual production organization shall request text and signature authorization for the respective LOP from AIR-4.0P. The LOP shall be the second right-hand page; the reverse shall be blank. The LOP shall carry the respective NAVAIR publication number. The LOP body text shall be justified. The LOP shall include the basic/revision issue date, centered between the COMNAVAIRSYSCOM line and the LETTER OF PROMULGATION line on the right margin. A sample LOP is presented on figure 16.

a. Supersedure. When a publication supersedes an existing publication, including a portion of another publication, such notice shall be included in the LOP provided by the procuring activity. Show the supersedure notice on the title page.

3.3.1.2.4 Interim change summary. The record of changes is used to document the incorporation of interim changes into the manual. The interim change summary shall be located on the third right-hand page. Figure 17 provides an example of an interim change summary page.

3.3.1.2.5 Summary of applicable technical directives. The summary of applicable technical directives shall contain the list of approved technical directives that are not fully incorporated within the aircraft, along with the visual cues that allow the aircrew to recognize incorporation of the technical directive in the aircraft. Technical directives that have been incorporated in all aircraft shall no longer appear here. Figure 18 provides an example of a technical directives summary page.

3.3.1.2.6 Record of changes. The record of changes is used to document the incorporation of changes and errata into the manual. Figure 19 provides an example of a record of changes.

3.3.1.2.7 List of effective pages. A list of effective pages (LEP) as shown on figure 20, shall follow the Record of Changes page in a new or revised publication and in a Change. The

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LEP provides accountability for a list of all numbered pages, including blank pages, and the page status of the pages. At the top of the LEP, dates of the revision and of each change issued to that revision shall be listed. Set the LEP in 11-point Times New Roman or equivalent.

3.3.1.2.8 Table of contents. The table of contents (TOC) shall identify each part, chapter, primary sidehead, secondary sidehead, and appendix within the publication. For all publications, show a page number for each primary and secondary heading. The TOC shall begin on a right-hand page and shall be updated as necessary with each Revision or Change (see figure 21). For TOC entries, omit all parenthetical figure references or cross-references in the paragraph headings.

3.3.1.2.9 List of illustrations. The list of illustrations (LOI) shall identify each figure within the publication by page number (see figure 22). The titles of the illustrations shall be updated as necessary with each Revision or Change. For LOI entries, delete all parenthetical figure references or cross-references in figure titles.

a. Location of LOI. The LOI shall follow the last page of the TOC and shall begin on a right-hand page. Use the same type sizes, styles, capitalization, and spacing as in the TOC. Set figure titles in 11-point Times New Roman or equivalent in uppercase letters. Include the titles of foldouts (if any) and their page numbers as the last entries in LOI.

3.3.1.2.10 Bibliography. A bibliography shall be included in publications that rely on extracts of data from other sources. Most NATOPS publications do not require a bibliography. When used, the bibliography shall identify which portions of the publications have been extracted, the source of the extracted data, and the effective date or edition of the source material. The bibliography shall begin on a right-hand page.

3.3.1.2.11 Glossary. Include in a glossary only those highly technical or unique terms pertinent to understanding the specific publication. Set the word(s) to be defined in lowercase (unless initial uppercase letter is required) in 11-point Arial bold or equivalent. Set glossary items flush left to the margin. Set the definitions in 11-point Times New Roman or equivalent. Indent carryover lines 1 pica. The glossary shall begin on a right-hand page.

3.3.1.2.12 List of abbreviations and acronyms. When editing text for a new or revised publication, a list of abbreviations and acronyms shall be prepared for use in the front matter of the publication. The first occurrence of an acronym shall be spelled out in the text. The list of abbreviations and acronyms shall contain all shortened and condensed terms found throughout the publication. See Appendix B for a list of frequently used abbreviations.

3.3.1.2.13 Preface. (See figure 4). The preface shall begin on a right-hand page. Include in the preface:

a. A brief statement of the publication's purpose, its intended audience, and its relationship to other publications, if applicable.

b. Instructions for submitting recommended changes to the publication.

c. A Change recommendation form as appropriate (see figure 23). Do not assign a figure number and figure title to the change recommendation form.

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d. Scope of the publication.

e. Arrangement of publication, if appropriate.

f. Ordering information.

g. Definitions of warnings, cautions, and notes, if appropriate.

h. Definitions of “shall,” “should,” “may,” “need not,” and “will,” if appropriate.

i. Identification of new or changed material.

3.3.1.3 Part I — The Aircraft.

3.3.1.3.1 Chapter 1 — Aircraft and Engine. Chapter 1 shall describe the aircraft, its arrangement, and its engine(s). A brief description of noteworthy features of the aircraft shall be presented. These shall include:

a. Aircraft type, class, model, and manufacturer

b. Engine type, thrust or horsepower, and manufacturer

c. Speed ranges for typical configurations

d. Missions

e. Typical takeoff gross weight

f. Aircraft arrangement. The exterior and interior arrangement, including illustrations displaying exterior, interior, cockpit layout, and aircraft dimensions shall be discussed (see figure 24).

(1) Overall dimensions of fixed-wing aircraft shall include:

(a) Maximum overall length.

(b) Width with wings folded and extended, incidence at aerodynamic chord, sweepback of 1/4-chord line, dihedral, aspect ratio, and main landing gear tread.

(c) Maximum height and static height with wings extended, folded, and during folding.

(2) Rotary-wing aircraft shall include:

(a) Maximum length with rotary-wing blades extended and folded, and main landing gear to tail wheel distance.

(b) Width with rotary-wing blades extended and folded, minimum rotary-wing ground clearance and ground clearance when folding, minimum rotary rudder ground clearance, and main landing gear tread.

(c) Maximum height, static height, and minimum height.

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3.3.1.3.2 Chapter 2 — Systems. Each system or item of equipment shall be described in accordance with 3.1.8.11. Minimum cooling requirements for heat-sensitive equipment shall be addressed. Include Navy or military designations for systems and equipment such as engines, ejection seat systems, avionics, and other equipment subject to significant modifications and upgrades. Systems or equipment for a particular aircraft that do not fit under a specific heading shall be included at the end of the general group with which they are most closely associated.

a. Powerplant systems.

(1) Engines. Include a general discussion of the development of thrust and horsepower. Discuss reversed/directed thrust systems with controls, indicators, and in-flight operation, and variable intake and variable exhaust area with sensors, controls, and indicators used, malfunction symptoms, and resulting engine operating characteristics.

(2) Engine fuel control system. Discuss metering systems, computing systems, and power management systems. Outline the functions of the system in a concise manner. Include modes of operation, system limitations, controls and indicators, and indications of system malfunction.

(3) Carburetor system. Include a complete description of all carburetor controls and indicators, including carburetor heat control, air temperature indicator, throttle control, manifold pressure indicator, and mixture control. Optimum range of carburetor air temperature, symptoms of carburetor icing, and preventive and corrective actions shall also be discussed.

(4) Supercharger system. Describe the type of supercharger, the supercharger control, and use of supercharger in cabin pressurization. Discuss the operation of the supercharger covering altitude and power requirements for blower shifting and the effect of automatic shift on engine operation.

(5) Manifold pressure regulation system. Discuss the purpose of the system. Outline consequences of system malfunction, including throttle movement precautions, limitation of available power, and back-up system operation.

(6) Water injection system. Discuss quantity, duration of supply, use, and control.

(7) Alternate air system. Describe function of the system including source of protected air, controls and indicators, and method of testing system operation.

(8) Start system. Describe components, controls and indicators CSD and external power required. Discuss indications of start system malfunction, and system capabilities and limitations.

(9) Ignition system. Include controls and methods of ground checking.

(10) Torque sensing system. Describe function of system and components, and indications of system malfunction.

(11) Overspeed protection system. Describe function of system and components, and indications of and response to system malfunctions.

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(12) Afterburner system. Describe system functions, controls, and indicators. Include a brief discussion of safety precautions, effect of system on operating characteristics, and indications of afterburner malfunction.

(13) Engine oil system. Discuss oil supply, oil pressure, oil cooling, oil-to-oil/oil-to-fuel heat exchange, oil dilution, and chip detector systems.

(a) Oil supply system. Describe oil quantity indicator, emergency oil shutoff control, and oil-quantity-low warning indicator. Where appropriate, discuss in-flight replenishment of oil supply, oil consumption for extended flights, and emergency shutoff control use. Limitations of the oil supply system under aerobatic flight and maximum time indications shall also be mentioned when oil supply is a factor.

(b) Oil pressure system. Identify the types of oil pressure and scavenging pumps and their source of power; describe the oil pressure indicator and normal indicator readings; discuss oil pressure as the governing factor for the selection of low-cruise engine rpm. Also include abnormal oil pressure and the need for cross-checks, remedial action to extend flight, and expected time to engine seizure.

(c) Oil cooling system. Describe the oil cooler bypass control, oil cooler shutter control, oil cooler door/flap control, oil temperature gauge and normal oil temperature range, and oil cooler door/flap position indicator. Discuss warmup sequence, ground and in-flight operation, and normal and bypass oil cooler operation.

(d) Oil-to-oil heat exchange system. Discuss the purpose of the system. Emphasize the effect of improper engine oil temperature management on the second oil system and the equipment that it lubricates. Describe action to relieve high oil temperature condition. Identify systems that should be shut down as a precaution when oil temperature is high.

(e) Oil-to-fuel heat exchange system. Discuss the purpose of the system. Emphasize the effect of high oil temperature on fuel system and engine operation, the limitations imposed on maximum permissible engine rpm, and means of detecting heat exchanger malfunction from engine performance.

(f) Oil dilution system. Describe the oil dilution switch, and note the point where fuel is injected into the oil system. Show in tabular form the relationship of ambient temperature, percentage of dilution required, and time from actuation of control to achieve percentage oil dilution required, for each type of oil specified. Describe engine warmup techniques required after oil has been diluted to ensure proper lubrication during takeoff.

(g) Chip detector system. Discuss the purpose of the chip detector system including the location of the detector plugs. Describe the chip detector warning light with press-to-test feature, and its source of power. Indicate inability of magnetic plug to attract aluminum or other nonferrous metal chips. Describe the action recommended when warning light illuminates, such as cross-checking of other engine instruments and feathering when possible to guard against additional engine damage.

(14) Engine controls and instruments. All engine controls and instruments that indicate engine condition and operation, such as tachometer, manifold pressure gauge, torque meter,

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engine analyzer, engine oil pressure, engine oil temperature, engine fuel pressure, and fuel flowmeter shall be discussed.

(15) Engine overheat system. Describe indicators, sensors, and function of the system.

(16) Engine fire detection system. Describe this system, including its components, system checks, and proper operation.

(17) Engine fire extinguishing system. Describe this system, including its controls, indicators, and operational procedures.

b. Propeller systems.

(1) Transmission/reduction gear. Describe the function of each gearbox, transmission oil systems, oil coolers, source of cooling air and accessory drive systems. Discuss capabilities and limitations.

(2) Propeller rpm control system. Describe the propeller pitch control, source of power, propeller governor, tachometer, and rpm range limit indicators. Indicate the normal propeller governing range. Discuss the relationship of oil pressure and rpm, and minimum oil pressure as the determining factor in selecting low rpm for cruise. Discuss oil temperature management and its effect on governor operation.

(3) Propeller synchronization system. Describe the propeller synchronization and synchrophaser system. Include synchronizer control, power source, master engine synchronization selector, manual synchronization override, and synchronization reset switch. Discuss automatic synchronization limits, when reset should be actuated, and implications of securing (feathering) engine selected as “master.”

(4) Propeller reverse pitch control system. Describe the reversing control, control lock, power source for reversing propeller, auxiliary pump and propeller-in-reverse indicator. Emphasize throttle manipulation required to achieve and control reverse pitch power. Discuss safety features incorporated to prevent use of system while airborne, the setup of air conditioning system during operation of reverse pitch, implication of system malfunction on landing roll distance, and possibility of asymmetric reverse thrust and its control. Describe propeller unreversing cycle if different from the opposite of the reversing cycle.

(5) Automatic/manual feathering system. Describe system purpose, method of sensing power loss, and the throttle or power lever position in relation to system operation for automatic feathering. Also describe auto-feather arming control and system-armed light. Discuss time delay as a safety feature and the automatic deactivation of the automatic system when one propeller has been feathered, and the caution involved. Mention test switch and test indicators. For the manual system, briefly describe the feathering pump or motor and its source of power. Describe the feathering control, feathering indicators, and warning indicators. Discuss the effect of feathering on other systems operating from each respective engine, such as air-conditioning, pressurization, and ac generator.

(6) Propeller blade fold system. Describe the propeller blade fold system and its operation.

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c. Rotor systems.

(1) Rotary-wing system. Briefly describe the characteristics of the rotors, rotary-wing head, wing blades, operation of antiflap restrainers, droop stops with respect to rotary-wing speed, and function of dampers. The rotary-wing system includes rotor mechanism, automatic blade fold system, and rotor brake system.

(2) Tail rotor system. Describe the tail rotor assembly, hub, and blades. Indicate the source of tail rotor drive power and the type of power transmission used. Discuss pilot directional controls including pedal feel and position as affected by the negative force gradient system or inoperative auxiliary servo. Include any interaction between tail rotor movement and movement of the collective pitch control.

(3) Transmission system. Include transmission oil system and accessory drive system. Describe the transmission oil system covering the supply tank (capacity and fill point), quantity indicator, oil pressure pump, scavenge pump, and oil cooler. Discuss the purpose of the accessory drive system. List the accessories that can be operated without turning the rotary wing. Describe accessory drive/rotary-wing control, accessory drive warning indicator, and test features. Indicate limitations for system operation such as engine speed, throttle/power lever position, and rotor brake control position and interlock features.

(4) Bearing monitor system (BMS). Discuss the purpose, system components, and operation of the bearing monitor system, including sensor types and locations, and bearing and system fault indications.

d. Aircraft fuel system.

(1) Fuel supply system. Fuel grades, specifications, and capacities shall be covered. Recommended and emergency fuels shall also be covered. Discuss drop tank release controls, fuel tanks, tank venting, boost pumps, gravity feed, jettisoning, controls, and indicators. Include procedures for fuel system management and fuel heater. Discuss single-point ground refueling.

(2) Fuel pressure system. Discuss engine-driven and electrical pressure and boost pumps and their interrelationship. Describe boost pump control, fuel pressure indicator and normal range of fuel pressure, low-fuel-pressure warning indicator, and fuel flowmeter.

(3) Air refueling system. Discuss air-refueling system in detail and include all aircraft-peculiar refueling procedures and techniques. Include tanker and receiver systems. Refer to NATOPS Air-to-Air Refueling Manual (NAVAIR 00-80T-110) for standard air refueling procedures.

(4) Aircraft fuel system schematic. Schematic shall include tanks and up to the engine-driven fuel pump.

e. Assist takeoff (ATO) system. Indicate conditions under which the system is to be used. Briefly describe the system covering nomenclature and number of units or mounts. Describe arming switches, system armed indicator light, electrical power source, firing switch, and safety interlocks. Indicate capability of replacing units from within the aircraft. Discuss jettisoning of ATO units, precautions to be observed, and sequence of jettisoning.

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f. Auxiliary power unit (APU). Discuss the APU system, compressor assembly, power turbine assembly, accessory assembly, oil system, fuel system, starting system, special control, fire detection and warning system, fire extinguisher system, controls, indicators, and operational procedures.

g. Electrical power supply system.

(1) Electrical power distribution system. Discuss ac power supply and system, dc power supply and system, auxiliary power supply and system, emergency power supply and system, electrical power distribution system, circuit breakers, electrical bus shedding, and fault clearing techniques. Include schematic diagrams.

(2) Air turbine motor/ram air turbine. Discuss air turbine motor/ram air turbine components, cooling, speed controls, controls and indicators, and operation.

(3) Lighting systems. Discuss exterior and interior lighting. Include diagram to show areas illuminated by exterior lighting.

h. Hydraulic power supply system. Identify the aircraft hydraulic systems and controls and indicators. The primary and secondary systems are generally those that serve aircraft flight systems. The auxiliary system serves only non-flight related systems (e.g., cargo doors, radomes).

(1) Primary system (PC-1, flight booster, etc.). Describe major components and identify aircraft systems served by primary system. Outline degradation or loss of aircraft systems for primary system malfunction.

(2) Secondary system (PC-2, combined, utility, etc.). Describe major components and identify aircraft systems served by secondary system. Outline degradation or loss of aircraft systems for secondary system malfunction.

(3) Auxiliary system. Describe major components and identify aircraft systems served by auxiliary system.

(4) Backup system. Describe the major components, controls, and indicators. Indicate the capabilities and limitations of the system, and include system tests.

(5) Hydraulic power distribution. Summarize the flight controls and aircraft systems served by each hydraulic system.

(6) Hydraulic power supply system schematic. Include all flight controls, hydraulically operated aircraft systems, major system components, and controls and indicators.

i. Flight controls. Describe effectiveness and possible unusual reactions encountered in operations and use of flight controls. Consider all types listed below; state their function, power source, capabilities, and limitations.

(1) Pilot's/copilot's (cockpit) controls

(2) Ailerons/flaperons

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(3) Horizontal stabilizer

(4) Rudder

(5) Flaps, slats, and boundary layer control

(6) Collective pitch

(7) Cyclic controls

(8) Trim controls

(9) Speed brakes/spoilers

j. Automatic flight controls. Describe major components and capabilities of automatic flight control systems, automatic stabilization equipment, approach power compensation systems, and similar equipment.

k. Pneumatic power supply system. Describe major components, aircraft systems served by pneumatic power system, indications of system malfunction, and backup modes of operation for pneumatically powered systems.

l. Landing gear systems. The landing gear systems listed below, including their controls and indicators, limitations, and emergency extension and retraction provisions shall be discussed.

(1) Attachments (skis, floats, etc.)

(2) Ground steering system

(3) Wheel brake system. Include normal and auxiliary braking and anti-skid system.

(4) Catapult system

(5) Arresting hook system

(6) Drag chute. Include safety release features.

m. Wing fold or stowage system. Describe the wing fold or stowage system including power requirements and safety precautions.

n. Pylon fold system. Describe the pylon fold system.

o. Flight instruments.

(1) Angle-of-attack system. Describe function of the system, interaction among angle-of-attack indicator, approach indexer, approach lights, and head-up display.

(2) Pitot-static system. Cover location of probes and static ports, alternate static source, instruments served by the system, and pitot heat.

(3) Vertical gyro system. Include power sources.

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(4) Horizontal situation systems

(5) Compass system

(6) Acceleration indication systems

(7) Radar altimeter

(8) Miscellaneous flight instruments. Include all controls and indicators not specifically described under another heading.

p. Warning, caution, and advisory lights/system. This discussion shall include a summary table that is organized by flight station. For each light or advisory provide name, location, meaning when illuminated/present, and action to be taken.

q. Aircraft fire detection and extinguishing systems. Describe function, components, test procedures, and operating procedures for the aircraft fire detection and built-in fire extinguishing systems other than those described elsewhere in the NFM.

r. Ingress/egress systems.

(1) Canopy. Discuss all canopy controls, both external and internal.

(2) Doors/hatches. Discuss operation and operating procedures.

(3) Ladders. Discuss operation and operating procedures.

(4) Ejection seats. Cover in detail ejection seat (module) and controls, emphasizing how they are affected by other systems, such as canopy. Discuss applicable hardware and list attached survival equipment.

(5) Parachutes. Discuss operation and operating procedures, including location when not in use.

s. Environmental control systems.

(1) Cockpit (cabin air-conditioning and pressurization systems). Discuss air source, air-conditioning units, distribution, temperature regulation, airflow regulation, controls and indicators, and normal and alternate operation. Discuss the source of pressurized air and method of controlling it. Cover any effect of the pressurization system on any other systems or vice versa.

(2) Avionic equipment cooling system. Discuss any system or method included for cooling avionic or electronic equipment.

(3) Heating and ventilating. Discuss heating and ventilating systems as specified for air-conditioning and pressurization systems, as applicable.

(4) Oxygen system. Discuss crew and passenger oxygen systems, in that order. Include the effects of temperature on pressure reading, and an oxygen duration chart showing man-hours

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of oxygen available for various combinations of oxygen pressure versus altitude. Show duration for normal and for 100 percent use of oxygen. Establish location of all portable oxygen bottles and all recharger points in the aircraft. Include items such as the types of regulators, types of masks, and procedures for checking and operating the oxygen system.

t. Defogging system. Include controls and indicators and system operation.

u. Windshield washing/rain removal system. Include controls and system operation.

v. Anti-icing/de-icing systems.

(1) Windshield/canopy. Include controls, indicators, and system operation.

(2) Engine. Include engine pressure ratio (EPR) probes, and engine intake. Cover controls and indicators.

(3) Propeller/rotor. Describe function, controls, and indicators. Outline system capabilities and limitations.

(4) Wings and empennage. Describe function, controls, and indicators. Outline system capabilities and limitations.

(5) Ice detection system. Describe function, controls, and indicators. Specify if the system automatically actuates the on-board ice protection system or if pilot interaction is required.

(6) Other. Describe function, controls, and indicators of any other anti-ice/de-ice system installed in the aircraft, such as ice protection for probes, radomes, etc. Outline the system capabilities and limitations.

w. Personnel equipment.

(1) Pilot's and copilot's seats (if not ejection seats), crew seats, and passenger seats. Describe seats, adjustments, and positions.

(2) Seat belts/harnesses. Discuss completely the automatic safety belt controls. Include appropriate illustrations of belt and parachute controls to show proper attachment of parachute harness and safety belt.

x. Rescue equipment. Describe function and location of all rescue equipment, rescue hoists, life rafts, etc., including capabilities and limitations.

y. Cargo provisions.

(1) Cargo deck. Discuss access and space included for cargo loading, tie-down points, and protective devices. Reference the applicable cargo-handling manual.

(2) Cargo sling. Describe cargo sling and tiedown provisions; include precautions to observe, capabilities, and limitations.

(3) Winches/hoists. Describe and include lifting capacity, hoisting speeds, and emergency jettison and braking provisions.

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(4) Aerial delivery equipment. Include references to applicable cargo-handling manual. Describe aircraft-unique equipment.

z. Emergency equipment.

(1) Survival equipment. Identify name, purpose, and location. Include first-aid kits.

(2) Portable fire extinguishing systems and access points. Discuss operation and procedures; identify location of hand-held fire extinguishers. Discuss location and use of all ground firefighting provisions, such as punch panels and other fire access points.

(3) Flotation systems. Discuss purpose of the system including stability on water with rotor stopped. Describe the emergency flotation units or bags and the emergency air supply. Cover the pressure gauge, emergency flotation control panel including off/armed switch, system armed indicator, activation switch or button, source of system power, and test switches and indicators. Indicate normal inflation time and if the bags should be inflated for other than water landing, such as, as energy absorbers for rough terrain. Discuss asymmetric inflation and if bags can or should be deflated to provide symmetrical flotation.

(4) Emergency signaling equipment. Discuss names and purposes of all emergency signaling equipment.

aa. Miscellaneous equipment. Include description and location of all equipment listed below, and any additional equipment of a general nature.

(1) Armor plate

(2) Flak curtains

(3) Radiation shields

(4) Stowage/storage compartments

(5) Relief tubes

(6) Rear view mirrors

(7) Equipment destruct systems

(8) Tool kits

(9) Checklists mounted to airframe

ab. Loose gear. Include instructions for securing all loose gear carried inside the aircraft.

3.3.1.3.3 Chapter 3 — Servicing and Handling. Chapter 3 includes all information required for servicing and handling the aircraft at a naval installation unfamiliar with the aircraft, or at an Air Force, Army, NATO (reference the Cross-Servicing Manual when available), or commercial installation.

a. Servicing data. Servicing data include information on fuels, oils, hydraulic fluid, lubricants, and other servicing items. Servicing data shall be presented in tabular form, and shall include:

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(1) Item — Fuel, oil, oxygen, electrical starting unit, windshield wash fluid, etc. Alternate use items shall be indicated in order of preference.

(2) Navy designation— Include the Navy or military designation of the item; (e.g., MIL-L-7807, NC-5, or JP-5).

(3) NATO equivalent

(4) Commercial equivalent

(5) Enroute supplement code — This is the code for the item as indicated in the Enroute Supplement, Flight Information Publication, such as "A+," "ADI," or "W."

(6) Remarks — Include whether an item is an alternate or emergency replacement for the standard item.

b. Fueling procedures. Describe fueling procedures including safety precautions.

(1) Pressure fueling. Discuss the requirement for external power during pressure fueling, and the aircraft attitude required to allow maximum amount of fuel to enter tanks. Describe the pressure fueling procedure, and include an illustration of pressure fueling receptacles and fueling panel switch setup during refueling.

(2) Pressure fueling top-off method. Describe procedures for pressure fueling when only a small quantity of fuel is required. Emphasize the limit below which this method is to be used and the requirement for strict adherence to this procedure to prevent tank damage or rupture.

(3) Pressure fueling alternate method. Where appropriate, describe the pressure fueling procedure when external electrical power is not available.

(4) Gravity fueling. Describe the gravity fueling procedure. Include an illustration of the gravity fueling setup. Emphasize position of fueling nozzle and location of grounding jack. Discuss nozzle length or diameter restrictions to prevent tank or filler port damage. Discuss precautions that must be observed in routing filler hose to prevent damage to wing slots, vortex generators, and wing fences. If the gravity fueling procedure differs from one integral tank to another, or for external tanks, include separate procedures and illustrations.

(5) Fuel control/fuel selector. Describe the procedure for setting the fuel selector adjustment for alternate grades of fuel. Include illustrations suitable for use by maintenance personnel totally unfamiliar with the aircraft. Insert warning notice indicating that the adjustment shall be noted on the aircraft yellow sheet upon return to home base to preclude operation with an improper fuel control fuel selector setting.

(6) Hot refueling. Describe switch settings, safety precautions, crew manning level, and ground crew requirements. Include emergency shutdown procedures.

c. Oil system servicing. Describe the procedure for checking oil system level and for filling. Identify the maximum oil consumption figure (the rate of consumption that, when reached or

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exceeded, requires maintenance action). For helicopters, discuss transmission gearbox oil system servicing procedures. Include a table listing grade of oil versus ambient temperature.

d. Hydraulic system servicing. Describe servicing procedures for each hydraulic system if separately serviced. If utility hydraulic system quantity indication varies with aircraft configuration, such as wings folded versus spread, indicate the normal servicing level for each configuration. Include an illustration to depict features such as the portable hydraulic test stand and fill and bleed lines.

e. Pneumatic system servicing. Describe servicing procedures. Indicate location of servicing points, system preloads, and gases to be used. Specify safety precautions.

f. Oxygen servicing. Describe the servicing procedure for the oxygen system to include the specific tasks required, precautions to be observed, and the use of portable servicing units. Include an illustration of the system being serviced. For liquid oxygen systems, show the danger areas. In tabular and chart form, describe the process of determining how much oxygen is required for various aircraft missions (for example, a chart showing oxygen duration by altitude and the conversion of liters to pounds). Where appropriate, include a discussion of emergency (bailout) and portable oxygen system servicing.

g. Battery servicing. Indicate the location of and the steps for gaining access to the aircraft battery. Describe filling procedure, normal servicing level, minimum specific gravity per cell, and minimum acceptable battery voltage (no load).

h. Anti-icing/de-icing system servicing (fluid systems only). Indicate the location of filler point and normal servicing level.

i. Windshield wash system servicing. Treat this in the same manner as the anti-icing system.

j. Water injection system servicing. Indicate the location of filler point and normal servicing level. Include the percentage of water and alcohol/methanol for those systems requiring a particular mix.

k. Fire extinguishing system servicing. State that there are no provisions for servicing the fire extinguishing systems while installed in the aircraft, and that if servicing is needed, the fire extinguishing systems shall be removed from the aircraft and serviced in accordance with existing maintenance instructions.

l. Auxiliary power unit (APU) servicing. Describe the fueling and oil system servicing procedures for the APU.

m. Constant speed drive (CSD) servicing. Describe the procedure for gaining access to the CSD. Discuss inspection, filling procedure and normal servicing level. Discuss the consequences of using other than approved fluid.

n. Assisted takeoff (ATO) servicing. Describe the installation procedure for JATO (jet assisted takeoff) bottles. Indicate the checks that the pilot can perform to ensure that the system is in a ready condition.

o. Drag chute system servicing. Discuss the procedure for removing, repacking, and installing the drag chute assembly. Emphasize items that the pilot can check visually to ensure

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proper attachment and deployment. Include illustrations that will permit the system to be serviced by a parachute rigger who has never seen the system.

p. Aircraft jacking points. Describe location and type of jack fitting to be used for each jacking point. Specify capacity required of each jack.

q. Tire servicing. Specify minimum acceptability requirements for tire, tread, cuts, and abnormal wear. Include inflation pressures and tire sizes and types.

r. External power requirements. Reference MIL-HDBK-274 for aircraft electrical grounding requirements.

(1) Electrical. Describe the external ac and dc electrical power requirements. Indicate voltages, current, phases, and the location of the electrical power receptacle. Indicate whether service power requirements are identical to starting unit requirements.

(2) Starting unit requirements. Indicate the starting unit requirements for both air and electrical power and combinations thereof. State minimum and maximum air pressure flow. Identify voltages, current rating, and phases for electrical power. List approved starting units. Indicate modification or change to unacceptable units to make them acceptable under emergency conditions.

s. Danger areas.

(1) Engine danger areas. Include an illustration showing the hazard areas that exist during ground operation with engines at idle power and military power. Show both induction and exhaust hazard areas. Show the length and width of the exhaust pattern indicating both temperature and velocity relative to the distance from the exhaust nozzle or tailpipe and the distance from either the exhaust centerline or aircraft longitudinal axis. Indicate ear protection zones, type of ear protection required in zones, and maximum exposure time, as applicable.

(2) Other aircraft danger areas, including those for rf, radar, and laser equipment emissions.

t. Hazardous equipment and devices. List the types and locations of hazardous equipment and devices installed in the aircraft. Provide information on the nature of the hazard, injurious effects, and precautions to be taken. Reference may be made to NAVAIR 00-80R-14-1 or other applicable NAVAIR publication for source of this information when the risk of death or physical injury to aircrew and aircraft servicing and handling personnel during flight operations is remote.

u. Turning radii and ground clearance. Include an illustration showing turning radii, ground clearances, and overall aircraft height in different configurations (see figure 25).

v. Towing aircraft. Discuss special towing procedures, including towbar, tractor, maximum velocity, and safety requirements.

w. Securing aircraft. Address hazards and requirement to ground aircraft in accordance with MIL-HDBK-274. Describe the tiedown fittings, jury struts, and methods of securing the aircraft. Include illustrations of tiedown arrangements for normal and heavy weather tiedown.

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3.3.1.3.4 Chapter 4 — Operating Limitations. Chapter 4 shall present a summary of the operating limitations for the aircraft. Operating limitations manage the integrity of the aircraft by ensuring that essential strength, stability, and control limits are not exceeded. Coverage shall include both aerodynamic and system limitations and restrictions. Aerodynamic limitations are associated with flight of the aircraft and include maneuvering limits, weight and balance considerations, minimum and maximum speeds, and accelerations in various aerodynamic configurations. System operating limitations concern the normal and abnormal ranges of system variables and include factors such as pressures, temperatures, voltage loading, and revolutions per minute.

a. Identify the aircraft (airframe and engine) configuration for which the limitations are presented.

b. Provide the aircraft's operating limitations and restrictions to include the following:

(1) Aircraft and system limits.

(a) Provide power plant and auxiliary devices, including engine operating limits, transmission and gearbox limits, APU limits, starter limits, and an altitude versus airspeed chart of the airstart envelope.

(b) Provide wind limitations associated with starting engines. For rotocraft, also provide wind limits for engagement and disengagement of rotors, including gust limitations. Also include rotor brake speed limit for applying rotor brake while on the ground.

(c) Present ejection seat limitations.

(2) Instrument markings. Illustrate or list each instrument that indicates an operating limit, along with the restriction. For each instrument that contains an engine power instrument, include a conspicuous notation of the fuel grade on which the limits are based.

(3) Altitude. Provide absolute ceiling and service ceiling for the aircraft, minimum height for safe landing after engine failure, and for single-engine aircraft, include a chart for no-thrust glide distance for both clean and dirty configurations.

(4) Airspeed.

(a) Provide an altitude-versus-airspeed diagram.

(b) Include sideslip limitations. For rotorcraft, present allowable aircraft sideslip versus airspeed envelope. Where no sideslip gauge is present, present diagrams relating allowable sideslip to the turn and slip indicator.

(c) Include system limitations landing gear, flaps, tires, canopy, etc.

(d) Provide fuel dumping limits and restrictions, including maximum airspeed and maximum descent rate for fuel dump. Maximum nacelle angle for fuel dump, if appropriate. Identify any prohibited maneuvers or conditions associated with fuel dump operations (e.g., ramp upper door open, gear down, cabin door open, etc.)

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(5) Angle of attack. Provide AOA limits for approaches to stalls, maneuvering flight, landing gear and flap extension, and landing approaches.

(6) Acceleration.

(a) Include maximum acceleration with tip tanks and maximum bank at high gross weights.

(b) Present maximum permissible accelerations under various flight conditions at specific gross weights and fuel load distributions.

(c) Include a V-N diagram.

(7) Maneuvers.

(a) List prohibited maneuvers and maneuvering flight restrictions, including those for aerobatic flight.

(b) Include restrictions on control movements, bank angle limits, and limits for slipping and skidding during asymmetric power conditions and landing approaches.

(c) For rotocraft, include restrictions for practice autorotations, defensive combat maneuvering (DCM), nacelle movements or angles, descent rates, gravity fueling with engines running, boost off flight operations, etc.

(8) Ground operations.

(a) Provide limits associated with wheel brake application, taxi turn speeds and slope landing and takeoff angles.

(b) For rotocraft, provide blade fold, wing fold, tail pylon fold limitations. List conditions under which fold operations are authorized. Include ambient temperature and wind speed envelopes when appropriate.

(9) Weight and balance.

(a) Provide the maximum takeoff, inflight, and landing weight and allowable center-of-gravity limits for the aircraft. Define maximum flight weight, which may be greater than the maximum takeoff and landing weights because of external sling load capacity. Include gross weight limitations for field and carrier catapult and non-catapult launches; touch-and-go and field carrier landing practice landings; and arresting hook and barricade engagements. Provide weight and center-of-gravity limitations for aircraft in which weight and balance is a problem. Include a tabulation for normal, emergency and overload takeoff gross weights, and for zero-fuel weight for applicable aircraft. Include a tabulation of normal and emergency weight distributions. Include a reference to the individual aircraft's weight and balance handbook (prepared in accordance with NAVAIR 01-1B-40 and NAVAIR 01-1B-50) for detailed weight and balance information.

(b) Provide a figure showing the allowable gross weight and center-of-gravity envelope for the aircraft. Label limits on the figure that are mission specific (e.g., aft c.g. extensions for deployment/retrieval of minesweeping gear, fastrope operations, air drops, etc.)

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(c) Include cargo ramp loading during ground loading and while in flight. Include weight and airspeed limits for opening and closing the cargo ramp on ground and while airborne. Identify allowable floor loading capability and cg distributions for internal distribution of cargo. For rotorcraft, include what floor loading distributions are required to ensure continued control and weight-over-gear loading.

(d) List lateral weight asymmetry limits for internal and external cargo, stores, and loads.

(10) External carriage. If not contained in aircraft Naval Aviation Technical Information Product (NATIP) or legacy tactical manual publications for the aircraft, present carriage, release, launch, and jettison limitations for external stores, external cargos, and external mission stores, including:

(a) Bombs, rockets, and missiles.

(b) Tow banners and towing equipment.

(c) Mission Pods – Inflight refueling, cameras, jamming pods, etc.

(11) External cargo sling transport-related limits.

(a) Present single, dual, and tandem cargo hook limits and forward flight speed and angle-of-bank limits associated with external sling loads. Include limits associated with opening and closing cargo hook access doors, and maximum forward flight speeds with cargo hook doors opened without sling loads. Also include aircraft power and rotor speed operating limits associated with external sling loads.

(b) Identify day, night and weather restrictions on cargo sling operations.

(12) Aerial refueling. Address Hover-In-Flight Refueling (HIFR) and forward flight refueling including:

(a) A reference to ATP-56, Air-to-air refueling, and the NAVAIR 00-80T-110, NATOPS Air-to-Air Refueling Manual, for authorized tanker and receiver aircraft and equipment and procedures.

(b) Allowable airspeed range for refueling probe extension and retraction and for forward flight refueling with probe extended.

(c) A chart of the gross-weight-versus-altitude inflight refueling envelope.

(d) Modes of operation and circumstances in which aircraft refueling is authorized (or prohibited). Address conditions such as turbulence, system failures, and maneuvers that are authorized (or prohibited) during inflight refueling.

(e) For rotorcraft, air refueling pod rotor speed restrictions.

(f) External slight loads and/or stores authorized or prohibited during inflight refueling operations.

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3.3.1.4 Part II — Indoctrination. Part II normally comprises one chapter.

3.3.1.4.1 Chapter 5 — Indoctrination.

a. Ground training syllabus. Include a brief and general outline of the material that shall be included in the ground training syllabus. The syllabus outline shall be in tabular form. For each training entry, include crewmember applicability and the number of hours to be devoted to the training item. Include the following:

(1) Familiarization

(2) Safety and survival training

(3) Weapon system training

(4) Weapon delivery training

(5) Crew resource management training

b. Flight training syllabus. Outline the flight syllabus by identifying the topics to be treated during each training flight. Include the following:

(1) Takeoff

(2) Landings

(3) Navigation instruments

(4) Shipboard procedures

c. Personal flying equipment. List only aviation and survival equipment to be worn or carried by crewmembers on all flights that is being covered by the manual. Refer to the current OPNAVINST 3710.7 for all standard equipment.

d. Flightcrew designations. Define the designations and basic duties of each crewmember.

e. Flightcrew qualifications and currency requirements. List the initial qualification requirements for each crewmember for various flight phases and the requirements for maintaining current qualification.

3.3.1.4.2 Waivers. Include a statement stating that unit commanders are authorized to waive, in writing, minimum flight and training requirements in accordance with the current OPNAVINST 3710.7.

3.3.1.5 Part III — Normal Procedures.

3.3.1.5.1 Chapter 6 — Flight Preparation. Flight preparation specifies procedures to be followed prior to manning the aircraft.

a. Mission planning. Mission planning specifies procedures including aircraft-unique information on the following mission aspects unless a separate tactical manual is required. Information shall be in tabular or graphic form.

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(1) Mission capabilities

(2) Fuel (range and endurance capabilities)

(3) Navigation capabilities

(4) Tactics

(5) Communication capabilities

(6) Conventional weapons

b. Briefing/debriefing. Specify briefing/debriefing responsibilities for briefing officer and pilot in command. Reference applicable briefing/debriefing publications, if any. Specify that when crew compositions are non-standard (VIP, MIDS, or non-qualified personnel), additional consideration shall be given to Crew Resource Management. Specify the general areas that shall be covered during briefing/debriefing, including the following:

(1) Target or destination. Cover location, characteristics, mission purpose, alternates, and interaction with other participating units.

(2) Navigation and flight planning. Cover launch and recovery points, local operating procedures, and divert and emergency fields.

(3) Communication. Cover frequencies, controlling agencies, and identification procedures.

(4) Weapons or cargo. Cover special routing or loading, tactics, emergencies, and safety precautions.

(5) Weather. Cover local, en route, destination, alternate and divert field weather conditions, and winds aloft.

(6) Emergency procedures. Generally cover aborted missions to include communication, navigation, and search and rescue.

(7) Crew resource management. Address ORM, crew communications doctrine, CRM-related mission topics/threats, and crew responsibilities and procedural initiatives outlined in 3.3.1.11.1.

(8) Intelligence information. Identify friendly and enemy dispositions, targets of opportunity, reports and authentications, escape and evasion, and NOTAMs.

3.3.1.5.2 Chapter 7 — Shore-Based Procedures. Include general information and detailed procedures to formulate crewmember checklists. The crewmember checklist shall be in a challenge-response format for normal shore-based ground and flight procedures. Information and procedures shall include the following:

a. Line operations. Cover general information on operating from a flight line.

b. Preflight check. Specify in checklist format those checks that must be made in the cockpit and on the exterior of the aircraft prior to boarding, and the interior of the aircraft immediately after boarding; include diagrams as required.

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(1) Include procedures for preflight of external stores and pylons, internal stores, and related equipment when not provided in the tactical manual publications.

(2) Ejection seat check. Specify checks to be made prior to strapping in, the procedures for strapping in and for the removal and stowage of safety pins.

c. Before start. Include checks for preparing the aircraft and cockpit prior to starting engines.

d. Starting engines. Include checklists for prestart, start, and post-start engine procedures. Include indications of and procedures for abnormal starts (hot starts, wet starts, etc.).

e. Before-taxi checks.

(1) Include equipment to be turned on and system checks to be performed.

(2) Wing/rotor spread. Describe procedures, including flight and ground crew duties during spread, cycling procedures, and locked/unlocked indication.

(3) Control checks. Describe procedures for checking all flight controls for free and correct movement.

(4) Weapon systems. Include weapon systems checks when not included in the aircraft tactical manual publications.

f. Taxi.

(1) Include all checks to be accomplished before taxiing, such as hatches and doors, fire extinguishing system readiness, chocks, taxi clearance, crew readiness, safety lock pins, steering, and brakes.

(2) Taxiing. Include all information useful to the pilot while taxiing such as differential power, braking, precautions to help avoid ground accidents during day and night, engine operation, engine taxi power settings, flight instruments, and crosswinds.

g. Before takeoff checks.

(1) Engine run-up. Include complete instructions for checking items such as engine and propeller operation including power and ignition. Cover the means of ensuring that the propeller is not in reverse pitch. Describe the proper use of brakes during run-up.

(2) Aircraft arming/dearming procedure.

h. Takeoff. Include a discussion of the takeoff, covering the technique and procedures necessary to complete a normal takeoff. State that a normal takeoff is one in which the takeoff data matches that predicted by the aircraft performance data in Part XI. Also consider the following items in the discussion of takeoff:

(1) Use of controls to overcome engine torque if applicable.

(2) Force required to lift nose wheel or tail wheel.

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(3) Conditions that may affect takeoff such as runway surface covering, runway condition reading crosswind, runway length, and obstacle clearance.

(4) Takeoff configuration such as external stores, center of gravity (cg) location, gross weight, and flap position.

(5) Effect of trim changes that may be required after breaking ground.

(6) Recommended techniques such as the use of brakes, anti-skid, flaps, and trim and effect of deviations from the recommended techniques.

(7) Peculiarities and unfavorable tendencies of the particular type or model of aircraft.

(8) High-altitude takeoffs and runway requirements.

(9) Means of determining when a takeoff should be aborted.

(10) Any other items requiring consideration.

i. After take-off checks.

j. Transition to forward flight (helicopter/VTOL). Describe procedures for normal transition, including altitude, airspeed, crosswind limitations, and post-takeoff cleanup procedures.

k. Climb. Discuss the normal climb procedures and techniques that will be required to produce the results stated in the performance data climb charts. Include post-takeoff cleanup procedures for fixed-wing, conventional aircraft. Describe trim settings, and power settings for normal, cruise, and military type climb.

l. Cruise. Cover action that must be taken when the transition from climb to cruise is made. Include any particular matters that must be considered during cruise flight such as periodic crewmember checks, oxygen system checks, and any peculiarities during instrument flight. Refer to the applicable series of performance charts in Part XI for each aircraft cruise configuration for specific speeds, power settings, and fuel consumption.

m. Descent. Include the procedural steps and a discussion of the descent phase of operation. The checklist shall include all checks that must be made immediately before and during descent to land. Cover in detail the normal procedures that will be required to produce the results stated in the descent charts in the performance data. As applicable, include special instructions regarding various types of descent such as en route, teardrop, rapid (with spoiler), and rapid (clean) penetration. Add any special devices that may be included to facilitate descent.

n. Approach to landing. Describe all activities performed prior to traffic pattern entry and during traffic-pattern flight, including procedures for the following:

(1) Instrument approaches. Include text descriptions and checklists for instrument approach patterns with a discussion of aircraft configuration, and techniques for all instrument approaches within aircraft capability. Include complete coverage on any special precautions or restrictions peculiar to the aircraft. Refer to applicable instrument flight procedures manual.

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(2) Automatic approach. Cover the preparation and procedure to be followed during automatic approach and special precautions and techniques. Discuss unsatisfactory approach indications and procedures for discontinuing such an approach.

o. Transition to hover (helicopter). Indicate all airspeed, altitude, power, and crosswind requirements for transition from forward flight to hover. Reference the applicable performance charts in Part XI.

p. Transition to vertical flight (VTOL). Indicate all airspeed, power, and crosswind requirements for transition from forward to vertical flight. Reference the applicable performance charts in Part XI.

q. Landing (fixed-wing). Discuss problems that may be encountered during the landing phase. Cover the normal landing technique that will be required to produce the results stated in the landing charts of the performance data in Part XI. Discuss techniques such as use of brakes, nose wheel on or off the runway, reverse thrust, transition from aerodynamic to mechanical directional control, and drag chute deployment. Include procedures for each type of landing that can be made. Include procedures for touch-and-go, over-weight, crosswind, minimum run, and adverse weather landings.

r. Waveoff or go-around. Discuss any special procedures required for waveoffs from a normal landing approach. Include go-arounds with engine(s) failed for multi-engine aircraft.

s. Landing (helicopter). Cover special landing problems such as mountaintop, confined areas, snow and dust, run-on landing, landing from hover, and crosswind landing.

t. After landing. Include all checks and operations to be accomplished after turnoff from runway and before the parking area is reached.

u. Before shutdown. List all actions necessary to accomplish after the parking area is reached.

v. Wing/rotor fold. List procedures for unlocking and folding wings or rotors.

w. Engine shutdown. Include a checklist covering the proper procedures for engine shutdown, including all precautions to be observed in accomplishing this procedure.

x. Postflight procedures. List all actions necessary to accomplish the postflight check or inspection.

3.3.1.5.3 Chapter 8 — Ship-Based Procedures. Include general information and detailed procedures to formulate crewmember checklists that differ from shore-based procedures and are unique to the aircraft. Include the following:

a. Flight deck/hangar deck procedures. Include flap settings, turn-up limitations, recommended trim control settings for various loading configurations, and control manipulation. For helicopters, emphasize moving of aircraft, and blade folding and spreading.

b. Preflight procedures. Describe any actions that are unique to ship-based operations.

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c. Post-start procedures. List any special turn-on, checkout, or alignment procedures that are unique to ship-based operation.

d. Taxi. Briefly describe normal ship-based taxi procedures.

e. Launch procedures. Include control settings, gross weight limits, and ranges of launching airspeed for STOL/VTOL launches.

f. Catapult launch. Include flap settings, trim, control settings, ranges of launching airspeeds, power setting for tensioning catapult, and acceleration characteristics after launch.

g. Recovery. Include procedures for helicopter/VTOL recovery.

h. Landing patterns. Include day, night, VFR, and IFR procedures.

i. Approach. Cover positions required during the approach (abeam, ninety, and final), optimum angle-of-attack and airspeeds, and aircraft-unique procedures for fresnel lens, angle-of-attack index, and APCS approaches.

j. Wave-off/touch-and-go/bolter procedures. Emphasize wave-off/clearing turn procedures.

k. Arrested landing/recovery. Specify special procedures for arresting hook malfunction.

l. Exit from landing area. Describe procedures for clearing landing area, including procedures for arresting hook, steering, or brake malfunction.

m. Carrier controlled approach. Describe aircraft-unique procedures for carrier controlled or ACIS approaches, including manual override of ACLS system.

n. Aviation facility ships. Specify any procedures for helicopters or VTOL aircraft operating from non-aviation ships that differ from standard carrier-based procedures.

o. Field carrier landing practice. Describe procedures that differ from standard ship-based procedures.

3.3.1.5.4 Chapter 9 — Special Procedures. Include general information and checklists for special flight evolutions not involved in the basic mission profile of the aircraft and for abnormal, but non-emergency procedures. Refer to applicable manuals for standard procedures.

a. Formation flying procedures

b. Air-to-air refueling

c. Aerobatics

d. Parachute drops

e. Cargo drops

f. Vertical replenishment

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g. Helicopter in-flight refueling

h. Aircraft ferry procedures

i. Three-engine takeoff procedures (multi-engine aircraft)

j. Search and rescue procedures

k. Fastrope operations

l. Autorotation

m. Windmilling starts

3.3.1.5.5 Chapter 10 — Functional Checkflight Procedures. Include general information and procedures for functional checkflights, as follows:

a. General information about the conduct of functional checkflights.

b. A designation procedure for functional checkflight check pilots and aircrewmen.

c. The minimum qualifications required of each aircrew member to be eligible for designation.

d. The conditions requiring functional checkflights, with each assigned an individual profile designator.

(1) The list of conditions requiring functional checkflights shall be compiled from those listed in COMNAVAIRFORINST 4790.2, additional conditions as cited in the NATOPS flight manual, and other identified functional/operational conditions. Assign a different uppercase letter (e.g., A, B, etc.) as an individual profile designator for each listed condition. When identical checks are required for two (or more) different conditions, the conditions and the profiles for both conditions may be combined into one individual profile.

(2) Individual checkflight check items shall be numbered consecutively as performed within each flight profile segment. Group the check items into flight profile segments under the following headings:

PREFLIGHT
BEFORE START
ENGINE START
BEFORE TAXI
ROTOR ENGAGEMENT & ENGINE RUN-UP (HELICOPTERS)
TAXI
PRETAKEOFF
TAKEOFF CHECKS AND PROCEDURES
AFTER TAKEOFF
HOVER
TRANSITION TO FORWARD FLIGHT (HELICOPTER & VTOL)
CLIMB

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CRUISE
DESCENT
APPROACH
TRANSITION TO LANDING (HELO)
TRANSITION TO VERTICAL FLIGHT (VTOL)
LANDING PROCEDURES AND CHECKLIST
AFTER LANDING
BEFORE SHUTDOWN
ENGINE SHUTDOWN
POST FLIGHT INSPECTION

(3) Segments not applicable shall be omitted. Additional segments shall be added where required.

(4) Each checked item shall be annotated with the individual profile designator(s) of the condition(s) for which the check is required; and, when applicable, the position of the crewmember(s) required to perform the check. When more than one crewmember is involved in acting on a check item, the crewmember(s) responsible for the action shall be identified by crew position. The checklist shall provide pilot data and crewmember data (where required). If all involved crewmembers are located in the cockpit, a single integrated checklist shall be provided. When required, an appropriate code (e.g., FE – flight engineer, T – TACCO) indicating the crewmembers responsible for performing each check shall be used. Where crewmembers involved are not all located in the cockpit, either a single integrated checklist or a sectionalized checklist shall be provided, dependent upon the number and location of crewmembers involved and the complexity and scope of checks to be performed. In sectionalized checklists, the pilot's section shall contain the complete number-sequenced items, and the other sections shall contain only those items required by the applicable crewmember with enough additional information to permit intelligent integration into the check sequence. The profile diagram and profile descriptions shall be included only in the Pilot's section of the sectionalized functional checkflight checklist. In sectionalized checklists, numbered checks shall be numbered identically to the master list in the pilot's section.

3.3.1.6 Part IV — Flight Characteristics and Control Procedures. Describe all significant flight characteristics and control procedures that are unique to the aircraft.

3.3.1.6.1 Chapter 11 — Flight Characteristics (Fixed-Wing Aircraft).

a. General characteristics. Include a description of how the aircraft is controlled.

(1) Flight controls. Include a complete discussion of the effectiveness and unusual reactions that may be encountered in the operation and use of flight controls. Cover all types of controls such as ailerons, elevators, rudders, trim tabs, speed brakes, slats, and directional thrust valves. State when and how the controls are used to achieve maximum benefits and what precautions must be observed. Cover the capabilities and limitations of power boosted systems and when power boost is inoperative.

(2) Automatic flight control system (AFCS) and approach power compensator system (APCS). Describe the sensing, coupling, and control features. Cover each mode of operation and emergency disconnect procedures.

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b. Climb characteristics. Describe aircraft characteristics at various climb configurations, including indications of approaching stalls.

c. Level flight characteristics. Discuss the characteristics of the aircraft through the entire speed range. Include characteristics in level flight in the transonic and supersonic regimes and with external stores.

d. Maneuvering flight. Describe the aircraft characteristics in accelerating and constant speed maneuvers. Include stick forces, emphasizing conditions that result in control reversal. Include recovery from unusual attitudes if procedures are unique to the aircraft.

e. Stall characteristics. Describe the aircraft characteristics in various conditions and include procedures for practice stalls. Discuss power-off and power-on stall characteristics of the aircraft in takeoff, landing, and clean configurations. Include stall characteristics for the approach configuration if sufficiently different from landing. Include a definition of power-on as used in the discussion. Include information on stall warning, emphasizing stall recovery procedures and complete instructions regarding the method of accomplishing practice stalls. Apply the above specifications to normal stalls, stall penetration, accelerated stalls, and post-stall gyrations.

f. Spin characteristics. Describe characteristics of the aircraft in spin conditions and procedures for spin recovery. Include an explanation of how spin characteristics of each type of spin differ from adverse-yaw induced spirals. Discussion shall cover the following spin conditions:

(1) Erect spins and recovery

(2) Inverted spins and recovery

(3) Spin recovery on instruments

g. Compressor stall. Describe characteristics of aircraft in compressor stall, indications of approaching stall, and any unique characteristics associated with compressor stalls in the aircraft.

h. Degraded mode operations. Address handling qualities in degraded modes such as during engine failure, engine shutdown in flight, and AFCS failure.

3.3.1.6.2 Chapter 11 — Flight Characteristics (Helicopters).

a. General characteristics. Describe how the helicopter is controlled as follows:

(1) Flight controls. Include a complete discussion of the effectiveness and unusual reactions that may be encountered in the operation and use of flight controls. Cover all types of flight controls such as rudders, trim system, cyclic stick, collective pitch, and stabilator controller.

(2) ASE/AFCS stabilization systems. Describe the relationship with sensing systems, doppler, radar altimeter, gyro systems, and system test features. Cover control features, each mode of operation, and emergency disconnect procedures.

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b. Ground handling characteristics. Describe the nature and cause of, and prevention, avoidance and recovery techniques for ground resonance and dynamic rollover. Discuss proper and improper use of the rotor brake.

c. Hover and low speed flight characteristics. Describe the characteristics of the aircraft in the hover and low speed flight regimes, including sideways and rearwards flight. Characterize for both hover in ground effect (HIGE) and hover out of ground effect (HOGE).

d. Forward flight characteristics. Describe characteristics of the aircraft in forward flight. Include a description of climbs, descents, and turns over the entire range of airspeeds. Describe the effects of external stores.

e. Blade stall characteristics. Emphasize corrective action in the event of severe blade stall and provide an incipient blade stall chart. Include complete instructions for accomplishing practice stalls.

f. Maneuvering flight. Describe the aircraft characteristics and control forces during accelerating and constant speed maneuvers in high-g and low-g flight. Address recovery from unusual attitudes and mast bumping.

g. Loss of tail rotor effectiveness. Describe conditions and characteristics for loss of tail rotor effectiveness (LTE). Describe corrective action. Describe critical azimuth data as it relates to LTE.

h. Vortex ring state. Describe power settling and vortex ring state characteristics, and avoidance and recovery procedures.

i. Vibrations. Describe characteristic vibrations encountered during normal flight conditions and unusual vibrations that are indicative of a particular problem.

j. Autorotation and degraded mode operations. Address handling qualities in degraded modes such as during engine failure, engine shutdown in flight, autorotation and AFCS failure. Describe aircraft characteristics and maneuvering limitations for autorotation over the entire altitude and airspeed range of the autorotation envelope.

k. Shipboard flight characteristics. Describe handling qualities as they relate to shipboard operations including, but not limited to, launch, recovery, deck recovery systems (i.e., clear deck, free deck, haul down with RAST) and ship airwake effects.

l. Turbulence. Describe severe turbulence flight characteristics and corrective actions.

m. Hoist/rappel operations. Describe aircraft characteristics important to rescue hoist and FASTROPE operations. Include special AFCS functions, such as crewman hover controller characteristics.

n. External and internal load operations. Describe any peculiar aircraft characteristics that may be encountered when flying with external loads, including during attachment, pickup, transit, and drop. For internal loads, discuss flight characteristics and corrective action if a load snags on the aircraft ramp placing the cg outside of normal aircraft limits.

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3.3.1.6.3 Chapter 11 — Flight Characteristics (Tiltrotor Aircraft).

a. General characteristics. Identify and define the flight mode configurations for the aircraft (e.g., VTOL, CONV, APLN), and include a description of how the aircraft is controlled in each flight mode.

(1) Flight controls. Include a complete discussion of the effectiveness and unusual reactions that may be encountered in the operation and use of flight controls. Cover all types of controls such as ailerons, elevators, rudders, speed brakes, slats, cyclic, and collective. State when and how the controls are used to achieve maximum benefits and what precautions must be observed. Cover the capabilities and limitations of power boosted systems and when power boost is inoperative.

(2) AFCS (Automatic Flight Control System) stabilization system. Describe the sensing, coupling, and control features. Cover flight characteristics peculiar to each mode of operation and emergency disconnect.

b. Ground handling characteristics. Describe the nature and cause of, and prevention, avoidance and recovery techniques for ground resonance and dynamic rollover. Discuss proper and improper use of the rotor brake.

c. Takeoff. Describe takeoff characteristics and handling qualities during vertical and short takeoff.

d. VTOL mode flight characteristics. Describe flight characteristics of the aircraft throughout the altitude and airspeed envelope for VTOL mode, including for climbs and descents, and for sideward and rearward flight. Characterize for both hover in ground effect (HIGE) and hover out of ground effect (HOGE). Discuss directional control limits, including critical azimuth considerations.

e. CONV mode flight characteristics. Describe the aircraft's flight control and handling characteristics while in conversion mode. Include characteristics encountered throughout the range of altitudes, airspeeds, and nacelle angles while in CONV mode flight and during transitions to and from VTOL and APLN modes.

f. APLN mode flight characteristics. Include discussions of level flight, climb, and descent characteristics throughout the altitude and airspeed envelope for the aircraft.

g. Wing stall characteristics. Describe the aircraft characteristics in various stall conditions in the APLN and CONV flight modes, and include procedures for practice stalls. Discuss power-off and power-on stall characteristics of the aircraft configurations. Include a definition of power-on as used in the discussion. Include information on stall warning, emphasizing stall recovery procedures and complete instructions regarding the method of accomplishing practice stalls. Apply the above specifications to normal stalls, stall penetration, accelerated stalls, and post-stall gyrations.

h. Blade stall characteristics. Emphasize corrective action in the event of severe blade stall in the VTOL and CONV flight modes. Include complete instructions for accomplishing practice stalls.

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i. Maneuvering flight. Describe the aircraft characteristics and control forces during accelerating and constant speed maneuvers in high-g and low-g flight for all three flight modes (VTOL, CONV, and APLN). Address recovery from unusual attitudes.

j. Settling during low speed flight. Describe vortex ring state characteristics, avoidance, and recovery procedures.

k. Vibrations. Describe characteristic vibrations encountered during normal flight conditions and unusual vibrations that are indicative of a particular problem.

l. Landing characteristics. Discuss handling qualities during approach, vertical landing, and roll-on landing evolutions.

m. Autorotation and degraded mode operations. Address handling qualities in degraded modes such as during engine failure, engine shutdown in flight, autorotation, and AFCS failure. Describe aircraft characteristics and maneuvering limitations for autorotation over the entire altitude and speed range of the autorotation envelope.

n. Shipboard and close proximity airwake effects. Describe handling qualities as they relate to shipboard operations including, but not limited to, launch, recovery, STO, and ship airwake effects. Describe unique handling qualities as they relate to the aerodynamic interaction between aircraft during shipboard operations and when operating in close proximity to other rotorcraft.

o. Turbulence. Describe severe turbulence flight characteristics and corrective actions.

p. Hoist/rappel operations. Describe aircraft characteristics important to rescue hoist and FASTROPE operations. Include special AFCS functions, such as crewman hover controller characteristics.

q. External and internal load operations. Describe any peculiar aircraft characteristics that may be encountered when flying with external loads, including during attachment, pickup, transit, and drop. For internal loads, discuss flight characteristics and corrective action if a load snags on the aircraft ramp, placing the cg outside of normal aircraft limits.

3.3.1.7 Part V — Emergency Procedures. Describe all emergency procedures, including the use of emergency features of primary systems as well as use of backup systems. All pages in this part shall contain emergency page borders.

3.3.1.7.1 Chapter 12 — Ground Emergencies. Include general information and challenge-response checklists for ground emergencies. Describe and illustrate as necessary emergency entrance and exit procedures. Include procedures to be followed in the event of engine starting malfunctions, emergency engine shutdown, and engine fire on the ground.

3.3.1.7.2 Chapter 13 — Takeoff Emergencies. Include general information and challenge-response checklists for takeoff emergencies. Describe abort procedures as necessary. Include procedures to be followed in the event of engine failure or fire with takeoff aborted, engine failure with takeoff continued, fire warning with takeoff continued, and tire failure during takeoff roll.

3.3.1.7.3 Chapter 14 — In-Flight Emergencies. Notes/cautions/warnings call particular attention to aspects of the emergency that may not be readily apparent. Cover general information and checklists for in-flight emergencies to include the following:

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a. Engine failure and air restart. Include critical and non-critical steps. Include procedures to be followed in case of afterburner overheat for jet aircraft and recovery procedures for compressor stall.

b. In-flight fire. Include instructions and procedures for fuselage fire, electrical fire, and wing fire. Include smoke and fumes elimination, crew protection, and use and limitations of onboard extinguishing agents.

c. Emergency stores and equipment jettison. Cover the jettisoning of external stores and semipermanent or loose gear carried inside the aircraft.

d. Emergency fuel dumping.

e. Refueling hose jettison.

f. Oil system failure. Discuss all aspects of engine operation with partial or complete loss of oil and oil pressure.

g. Fuel system failure. Discuss symptoms of known possible causes of failure, such as fuel inlet pressure, pump, excessive fuel consumption, or likely combinations of these symptoms. Present procedures for meeting these emergencies.

h. Electrical system failure. Describe procedures for controlling aircraft and operating critical aircraft systems in case of electrical system failure. Include procedures for operating back-up electrical power sources.

i. Hydraulic system failure. Describe procedures for controlling aircraft in case of failure of one or more hydraulic systems.

j. Propeller/rotor failure.

k. Rotary rudder failure (helicopter).

l. Torque sensing system failure.

m. Transmission failure.

n. Aileron/flaperon failure.

o. In-flight refueling emergency. Include procedures for both tanker and receiver.

p. Hung ordnance. Procedures to be taken when ordnance has failed to leave aircraft after triggering shall be included.

3.3.1.7.4 Chapter 15 — Landing Emergencies. Include general information and checklists for landing emergencies, to include the following:

a. Landing with engine(s) inoperative. Include the changes in configuration and procedures and the recommended precautions required for a forced or engine(s)-out landing. In the event of complete loss of power, emphasize that ejection or abandonment of the aircraft may be prefera-

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ble to forced landing. Include procedures for single-engine and twin-engine aircraft landing without thrust, a discussion of missed approach with inoperative engine(s), and procedures for ejection or abandonment from aircraft. For helicopters, describe procedures for autorotation.

b. Flaps/slats failure. All procedures for leading edge and trailing edge flap/slat failures shall be included. Discuss any unusual conditions, such as chatter, failure to close, and asymmetrical (split flaps) condition, that may occur.

c. Landing gear failure. Procedures for emergency extension of the landing gear shall be included. Describe procedures for combinations such as main gear down nose gear up, one main gear up, nose wheel cocked, and all gear up.

d. Forced landing. Include procedures and warnings and cautions as applicable. Consider altitude, aircraft configurations, pattern speeds, and engine speed.

e. Field arresting gear.

f. Barricade arrestment.

g. Blown tire(s). A discussion of the procedures required in the event of tire failure during landing roll shall be included.

h. Nosewheel steering malfunctions.

i. Wheel brake system failure. Procedures for accomplishing a landing when brakes are inoperative shall be discussed. If failure of these systems affects any other areas, mention the affected area(s). Note that emergency brakes (if applicable to the aircraft) are limited to a certain number of applications.

j. Emergency exits. Include illustrations showing emergency exits and entrance points.

k. Ground emergency egress. Address exiting aircraft quickly, as may be required after a forced landing, running off runway, or landing with smoke and/or fire. Address procedures and hazards encountered during egress in mission flight gear (e.g., NVG, CBR, etc.) until safely clear of aircraft.

l. Ditching. Include complete instructions regarding the method and best configuration for ditching the aircraft. Cover the capabilities of the aircraft after ditching and the advantage of ditching versus bailout. Include night ditching, partial power ditching, power-off ditching, preparation for ditching, after ditching, and crew member duties. Address procedures and hazards.

3.3.1.7.5 Chapter 16 — Ejection/Bailout. General information shall include the ejection envelope and checklist for ejection/bailout. Describe the techniques, precautions, and warning signals for leaving the aircraft in flight. Include complete coverage for bailout from aircraft not equipped with ejection systems and a brief explanation of parachute characteristics such as deployment speeds. Address procedures and hazards encountered during ejection/bailout sequence in mission flight gear (e.g., CBR, NVG, etc.) until parachute is released and on ground or entering liferaft. Cover ejection seat procedures and bailout procedures in the event of ejection

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seat failure. Indicate the proper procedure for preparing the aircraft for bailout and the method of jettisoning cockpit enclosures and doors. Provide a pictorial sequence of operation for ejection/ bailout.

3.3.1.8 Part VI — All-Weather Operation.

3.3.1.8.1 Chapter 17 — Instrument Procedures. Include general information and checklists for procedures unique to instrument flight. Reference the applicable manual for standard instrument flight procedures.

a. Simulated instrument procedures. Include procedures for flight under simulated instrument conditions. Include procedures for deactivating these instrument systems and equipment. Describe safety precautions, chase plane requirements, instrument hood, radio checks, simulated instrument maneuvers, confidence maneuvers, and instrument patterns.

b. Instrument flight procedures. Include procedures for flight under actual IFR conditions to include the following:

(1) Instrument takeoff. Provide engine anti-icing, setting of directional indicators, activation of automatic tracking or computer equipment, and maneuvering limitations while cleaning up and transitioning to climb.

(2) Instrument climb. Describe the speed and attitude parameters for normal instrument climbout and best fuel settings.

(3) Instrument cruising flight. Cover best economy or maximum endurance power settings, configurations, and airspeeds, referring to Performance Data, Part XI, where necessary.

(4) Holding. Cover power settings, airspeeds, and configurations for optimum fuel conservation and maneuverability.

(5) Instrument descent. Provide procedures to employ for normal instrument descent, dirty and clean penetration procedures, and operating limitation to be observed.

(6) Instrument approaches. Provide instrument approach patterns, and discuss aircraft configuration, procedures, and techniques for all instrument approaches within aircraft capability. Include complete coverage of any special precautions or restrictions.

(7) GCA/CCA (ground controlled approach/carrier controlled approach). Address the preparation and procedure(s) to be followed during GCA/CCA approach, and cover special precautions and techniques. Provide unsatisfactory approach indications and the procedures for discontinuing such an approach.

(8) Formation-unique procedures. Cover section penetration and rendezvous.

(9) Carrier-unique procedures. Cover special holding patterns, loiter configurations, bingo fuel, etc.

(10) Aviation facility ship-unique procedures

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3.3.1.8.2 Chapter 18 — Extreme-weather operation. Include general information and checklists for operation under the following extreme-weather conditions:

a. Ice, rain, and snow. Provide the precautions and remedial actions appropriate for flight in ice, snow, rain, and hail, covering each phase of flight. Include preflight removal of ice and snow from aircraft surfaces and taxiing on ice or snow. For inflight, cover use of pitot heat, alternate air and defogging equipment, propeller and rotor de-icing provisions, and special empennage or radome de-icing equipment. Cover the freezing of moving parts, the resulting flight characteristics, and flight procedures to be followed under the circumstances. Include precautions to be observed in applying ice removal and defogging compounds and fluids.

b. Thunderstorms and turbulence. Briefly describe performance of the aircraft under moderate and severe turbulence conditions. Describe the counteracting procedures to be used, including preparation for unavoidable thunderstorm penetration. Cover penetration airspeed, use of autopilot, and control restrictions to be observed.

c. Cold weather. Describe the operating requirements unique to extreme low temperatures and cold environments by phase of flight. Address the use of carrier-type approach for wet, slush-covered, or frozen runways. Describe the use of ground-effect techniques to counteract adverse runway conditions.

d. Hot weather. Provide the operating requirements unique to extreme high temperatures and hot environments by phase of flight. Describe inspection for overinflation of tires in hot weather, and address the use of high temperature portions of the Performance Data Charts.

e. Desert operations. Describe the operating requirements unique to desert environments by phase of flight. Emphasize removal and installation of dust and sand-protective devices.

f. Arctic/Antarctic operations. Provide the operating requirements unique to arctic environments by phase of flight.

3.3.1.9 Part VII — Communications-Navigation Equipment and Procedures. This part shall consist of descriptions of the communication-navigation-identification systems, controls and indicators, and the procedures for operating them. Limitations and peculiar modes of operation shall be discussed. Provide sufficient information for the aircrew to operate effectively in normal and degraded modes of operation.

3.3.1.9.1 Chapter 19 — Communications Equipment.

a. Equipment descriptions. Describe the following systems and equipment:

(1) Radios: satellite, UHF, VHF, and HF

(2) Speech and data security system(s)

(3) Identification radar (IFF/SIF)

(4) Intercommunication system

(5) Tacan, SATNAV, marker beacon receiver, VOR/DME, RDF (radio direction finder)

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(6) VOR/ILS (VHF omni-range/instrument landing system)

(7) Global positioning system (GPS)

(8) Non-integral part of the weapon system (if applicable)

(a) Terrain avoidance system

(b) Weather radar

(c) Doppler radar

(d) Inertial navigation systems

(e) Other satellite-based systems

(f) Midair collision avoidance system

(g) Emergency locator transmitter (ELT)

b. Equipment description elements. Description of the systems and equipment shall include:

(1) Type of equipment

(2) A/N nomenclature or applicable Navy designation

(3) Function

(4) Horizontal or slant range of equipment in miles

(5) Location of controls used to operate equipment

(6) Identification of the crewmember(s) responsible for control of the equipment

(7) Notes or warnings important to operation or effects on other systems in the aircraft

c. Equipment operating procedures. Following each description, outline the controls and turn-on and shutdown of the equipment unless complexity or uniqueness warrants added detail.

3.3.1.9.2 Chapter 20 — Communications Procedures. Include illustrations and explanatory text, as necessary, for the operation of the communication-navigation-identification system.

a. Visual communications. Routine visual communications procedures are contained in the aircraft signals manual. Include illustrations and explanatory text, as necessary, for the following:

(1) In-flight visual communication unique to the aircraft

(2) Ground handling signals unique to the aircraft

3.3.1.10 Part VIII — Mission Systems. This part shall include system descriptions, controls and indicators, and operating procedures for mission systems and associated equipment, when

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not contained in aircraft NATIP or legacy tactical manual publications for the aircraft. Typical coverage includes a brief discussion of the function of the system, location of equipment, source of power, functional description and illustration of each control and display panel, normal turn-on procedures, modes of system operation, and shutdown procedures. Detailed information is required for those mission systems that are highly specialized and unique to the aircraft.

3.3.1.10.1 Chapter 21 — Armament Systems. Chapter 21 describes all hardware associated with weapons delivery other than avionics (see 3.3.1.10.2), including the following:

a. Gunnery equipment

b. Bombing equipment

c. Rocket and missile equipment

d. Torpedo and mine equipment

e. Chemical and tank equipment

3.3.1.10.2 Chapter 22 — Avionics. Include all mission system avionics hardware.

a. Fire control systems. Include computer function and operation, peripheral equipment, and inputs received from other avionic systems.

b. Radar systems. Include search, attack, and navigation systems. Include the following if part of a weapon system; if aircraft is not equipped with a weapon system, include the following in a communications-navigation equipment and procedures chapter.

(1) Terrain avoidance system

(2) Weather radar

(3) Doppler radar

c. Inertial navigation system. If the aircraft is not equipped with a weapon system, include the inertial navigation system in a communications-navigation equipment and procedures chapter.

d. Target detection systems. Include TV, infrared, laser, and other devices for detection of airborne and surface targets.

e. Antisubmarine warfare systems. Include all systems for detection of underwater targets.

f. Tactical situation display systems. Include scope presentations, heads-up displays, and any associated equipment used to present the attack or intercept situation.

g. Electronic countermeasure systems. Include electronic countermeasure and electronic intelligence systems.

h. Photographic equipment. Include photoreconnaissance and radar recorder systems.

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3.3.1.10.3 Chapter 23 — Special Missions Systems. Include the following systems not associated with the basic attack/intercept/reconnaissance mission.

a. In-flight refueling system (tanker)

b. Glider tow system

c. Tow target system

d. Target drone control system

e. Searchlight system

3.3.1.10.4 Chapter 24 — Software. Include computer software functions for computerized, airborne-programmable mission systems, addressing the following:

a. Data processing, including computer software description and the procedure for loading the computer/software.

b. Display system, including display hardware and off-line controls.

c. Operator controls, including keyboard, mouse trackball, buttons, etc; controls operation; and built-in-test(s).

d. Loading the program, including normal loading procedures, initialization, program loading faults and fault recovery, program shut-down procedures, and built-in test (BIT) procedures.

e. Display features, including display formats; display conventions; alerts, warnings, and errors; and operator data entry/cues.

f. Key functions, including system-specific key software functions not addressed above or in the Systems sections.

3.3.1.10.5 Chapter 25 — Degraded Modes of Operation. Include capabilities and limitations of mission systems and their components when individual systems or parts of systems are down.

3.3.1.10.6 Chapter 26 — Troubleshooting — Techniques and Procedures. Include charts, diagrams, and text for airborne troubleshooting of mission systems.

3.3.1.11 Part IX — Flightcrew Coordination.

3.3.1.11.1 Chapter 27 — Crew Resource Management. This part shall contain information regarding flightcrew coordination, responsibilities, duties, and procedures.

a. Address the goal of CRM and how its application/utilization can lead to improved mission effectiveness while reducing crew-preventable errors.

b. Address the importance of squadron/wing validation and attitudes toward CRM, to include continual CRM training to increase aggregate levels of knowledge and skills.

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c. Address the critical skills of CRM and provide in-depth type/model specific information on each.

d. Address operational risk management (ORM) as a sub-set of CRM. Include discussions of the following subjects introduced in OPNAVINST 3710.7:

(1) The process for hazard identification and making risk decisions.

(2) Using the different levels of ORM application (time-critical, deliberate, and in-depth).

(3) Principles used as guidelines for risk acceptance.

e. Address threat and error management and the prevalence of errors in compromising mission effectiveness. Include the following topics:

(1) Cockpit interruptions and distractions.

(2) Strategies for reduction of human errors.

(3) Specific internal and external threats that lead to CRM breakdowns.

(4) Common causes of errors and the importance of crew coordination as a means for managing hazards and reducing errors.

f. Address flight crew responsibilities, duties, and procedures as a means to defining roles/responsibilities for mitigating hazards and reducing crew preventable errors:

(1) List aircrew positions and related responsibilities pertaining to CRM.

(2) List/detail crewmember procedural initiatives for minimizing exposure to the effects of hazards and reducing errors, including:

(a) Communicating the presence of all significant hazards.

(b) Announcing when performing head-down functions and periods of uncovered responsibilities.

(c) Communicating the loss of personal situation awareness.

(d) Implementing dual-concurrence concepts when performing critical measures such as shutdown of an engine during flight.

(e) Using a standard procedure for transferring control of the aircraft to another crewmember.

(f) Monitoring the performance/non-performance of other aircrew members.

(g) Announcing spatial disorientation or sub-optimal physiological states.

(h) Maintaining a positive CRM environment.

(i) Any other means of reducing crew preventable errors.

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3.3.1.12 Part X — NATOPS Evaluation.

3.3.1.12.1 Chapter 28 — NATOPS Evaluation. This part shall explain the concept and implementation of the NATOPS evaluation program and define specific meanings of terms as they are used in the program.

a. Ground evaluation. Describe the requirements of ground evaluation: open-book examination, closed-book examination, oral examination, OFT/WST (operational flight trainer/ weapons system trainer) procedures examination, and question format and grading instructions for each of these types of NATOPS examinations.

b. Flight evaluation. Include a description of the flight evaluation phase of NATOPS evaluation, including the objective and format of flight evaluations. The description shall follow the chronology of flight beginning with the mission planning and briefing phase and ending with postflight procedures and debriefing. List communication and emergency procedures; include the procedure for determining the flight evaluation grade.

c. Final grade determination. Describe the method of determining the final grade.

d. Records and reports. Specify records and reports that will be used and include an illustration of each form. Show entries to be made in the Flight Log Book.

e. NATOPS evaluation question bank. Include a list of questions concerning standard normal and emergency procedures that form a basis for preparation of ground examinations. Questions are intended to be representative, not all inclusive, and shall be discussion, completion, true or false, multiple choice, or a combination of these.

3.3.1.13 Part XI — Performance Data. Performance charts required for each type of aircraft (See 3.3.1.13.4 through 3.3.1.13.7) are specified in the following:

3.3.1.13.1 Chapters 29 through 38 — Performance Data Arrangement. Flight manual performance data for all types of aircraft shall be arranged as follows:

Chapter 29 — Standard Data
Chapter 30 — Takeoff
Chapter 31 — Climb
Chapter 32 — Range
Chapter 33 — Endurance
Chapter 34 — In-Flight Refueling
Chapter 35 — Descent
Chapter 36 — Landing
Chapter 37 — Mission Planning
Chapter 38 — Emergency Operation

a. Each chapter shall contain the following:

(1) An introduction to the use of the charts in the chapter and any background information necessary for using the charts

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(2) Lists and descriptions of the specific types of charts contained in the chapter. For each type of chart, describe:

(a) Chart function. In general terms, describe the type of chart and the information it provides. Include any chart limitations.

(b) Method of use. Without providing specific numbers, describe how the type chart is used, including how data are projected throughout the chart to obtain various parameters.

(c) Example problem(s). Provide specific (given) information, and follow the example lines on the charts to obtain the numerical (find) solutions(s).

b. Place the performance charts by type following the sample problems.

c. Avoid duplication of information contained in other parts of the flight manual.

3.3.1.13.2 General performance chart data requirements.

a. Title block. Each chart depicting performance data shall contain positive and concise means of identification in a title block at the top of the chart. The following information shall be given in the title block:

A title descriptive of the data (e.g., time to climb – clean configuration)
Aircraft model number
Engine model number
Propeller model number
Configuration (e.g., takeoff flaps)
Power (e.g., two engines, maximum power)
Pressure altitude (if applicable)
Atmosphere (e.g., standard day, if applicable)
Data as of (e.g., date)
Data basis (estimated or flight test)
Fuel grade
Fuel density

b. Engine inoperative charts. Charts providing data with an engine inoperative shall be marked with an emergency border.

c. Notes in data charts. Notes providing information that is required in the interpretation of the charts shall be placed at the bottom of the charts and shall be numbered.

d. Example lines. For each chart that provides data as a function of several variables, insert example lines that pictorially explain the use of the chart.

e. Operating procedures. Information contained on performance charts shall be based on, and consistent with, the recommended operating procedures set forth in the flight manual. If special piloting techniques are required to achieve performance and these techniques are described in the manual, footnotes referring the reader to the appropriate description elsewhere in the manual shall be provided on the performance charts.

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f. Data basis.

(1) Source of data. The engine power, fuel flow, and aerodynamic drag used in preparing the charts shall be derived from flight tests whenever available; otherwise, the data presented in the performance data substantiation reports described in g. of this specification shall be used. Flight test data obtained in tests conducted at Navy testing facilities shall be used in preference to data obtained during contractor flight tests. The procuring agency shall grant approval for the contractor to use his flight test results as the basis for performance after a review of the contractors' methods of flight tests, instrumentation, and data reduction techniques (see 6.2).

(2) Propulsion data. If engine performance has been verified by flight tests, performance data calculated using engine manufacturer's thrust and fuel flow data, with appropriate corrections for installation losses, shall be labeled flight test data. If engine performance has not been verified by flight tests, engine manufacturer's fuel flows shall be increased 5 percent and performance charts shall be labeled estimated data. In all cases, the status computer deck, with appropriate corrections for installation losses, shall be used as the source of engine manufacturer's data.

(3) Drag data. Performance data calculated using flight verified drag shall be labeled flight test data. Drag shall change as much as 5 percent for configuration differences between the flight test aircraft and flight manual aircraft, using estimated drag increments to make the change.

(4) Propeller efficiency. Propeller efficiency data used in deriving flight test drag polars and propeller efficiency data used in calculating performance shall be consistent.

g. Performance data substantiating reports. The basic aerodynamic data report and the substantiating performance data report, required by SD-8706C, shall serve to substantiate flight manual performance data. These reports shall present the derivation of drag data for all configurations for which performance data is provided in the flight manual and shall also describe assumptions and techniques used to calculate flight manual performance data.

h. Airspeed. Unless specifically called for otherwise, all airspeeds shall be provided as CAS, true Mach number, or TAS, if applicable. IAS shall be given for minimum control speeds, stall speeds, and for takeoff and landing data. IAS shall be based on the primary airspeed system calibration.

i. Atmosphere. International Civil Aviation Organization (ICAO) standard atmosphere shall be used unless otherwise specified. Performance data provided as a function of temperature shall be indicated in terms of ambient air temperature and not in terms of deviation from standard atmosphere temperature, if at all possible.

j. Drag count. Performance data shall be presented in chart form with a chart input variable for drag count of external stores. A drag count of one corresponds to a drag coefficient of 0.0001. In cases where an input variable of drag count cannot be used, a complete set of charts shall be included for each alternate configuration when a variation in performance between alternate configurations exceeds 5 percent. These instances occur when charts are applicable to one drag count only, or where the use of drag count is not desirable because a small number of external stores configurations are being used.

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k. Power setting. For takeoff performance and climb performance with flaps extended, intermediate (military) thrust for turbojet and turbofan engines is synonymous with takeoff power for turboprop engines. For climb performance with flaps retracted, intermediate (military) thrust for turbojet and turbofan engines is synonymous with maximum continuous power for turboprop engines.

3.3.1.13.3 Aircraft performance definitions.

a. Minimum control speeds.

(1) Ground minimum control speed (V_{MCG}) shall be the minimum airspeed at which directional control can be maintained on the ground using aerodynamic control only, with the critical engine inoperative and propeller feathered on the inoperative engine (if applicable).

(2) Air minimum control speed (V_{MCA}) shall be the minimum airspeed at which directional control can be maintained in the air with the critical engine inoperative, not more than 5° bank away from the inoperative engine, and propeller feathered on the inoperative engine (if applicable).

b. Runway coefficients of friction.

(1) The rolling coefficient of friction used in the calculation of takeoff distance shall be determined by ground tests. When ground test data are not available, a rolling coefficient of friction of 0.025 shall be used for dry, hard surfaces.

(2) The braking coefficient of friction used in the calculation of stopping distance shall be determined by ground tests. When ground test data are not available, a braking coefficient of friction of 0.3 shall be used for dry, hard surfaces.

(3) Runway condition reading (RCR) corrections to the braking coefficient of friction used in the calculation of stopping distance shall be based on the authorized coefficient of friction for dry, hard surfaces at an RCR of 23, and a coefficient of friction of 0 at an RCR of 0, with a linear relationship in between.

c. Takeoff.

(1) Acceleration check distance or time is the distance or time from start of takeoff to the point at which a particular airspeed of interest is reached during the takeoff ground run.

(2) Lift-off speed (V_{LOF}) shall be the speed at which the main landing gear leaves the ground. For conventional aircraft, V_{LOF} shall be at least 10 percent greater than the speed represented by 90-percent maximum lift coefficient, power-on, including ground effect. If ground clearance limits the angle of attack at main gear lift-off, V_{LOF} shall be increased to coincide with at least the speed corresponding to maximum obtainable angle of attack as limited by ground clearance with shock absorbers in the static, no-lift position. (See 3.3.1.13.3c.(11) for lift-off speed criteria of fixed-wing V/STOL aircraft.)

(3) Takeoff ground run distance shall be the ground run in feet to V_{LOF}.

(4) Minimum go speed (V_1) shall be the minimum airspeed at which the aircraft can experience an engine failure and then continue to accelerate to V_{LOF} within the remaining

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runway length. The data are based on an engine failure occurring at the minimum go speed. Engine failure is followed by a 3-second decision period with the remaining engines operating at the initial thrust setting. In the case of an intermediate thrust takeoff, an additional time period shall be allowed for advancing the operating engine throttles to maximum thrust. The time period used shall be applicable to the aircraft configuration and shall be approved by the procuring activity. V_1 shall not be less than V_{MCG}.

(5) Rotation speed (V_{RO}) is the airspeed at which transition from ground run attitude to lift-off attitude is begun. V_{RO} shall not be less than 1.05 times V_{MCA}, nor shall it be less than V_1.

(6) The speed at the 50-foot obstacle height (V_2) for conventional aircraft shall be the highest of the following three speeds, determined with takeoff flaps, landing gear retracted: a) speed for maximum climb gradient out of ground effect with the critical engine inoperative, takeoff thrust on the operative engines, and propeller feathered on the inoperative engine (if applicable); b) 1.2 times power-off stall speed; or c) 1.1 times V_{MCA}. (See 3.3.1.13.3c.(11) for corresponding criteria of fixed-wing V/STOL aircraft.)

(7) Maximum abort speed ($V_{MAX\ ABORT}$) shall be the maximum airspeed at which to start an abort and stop the aircraft within the remaining runway length. The data are based on a 3-second decision period after reaching maximum abort speed with the engines operating at the initial thrust setting during this time. At the end of the 3-second decision period, a time period shall be allowed for wheelbrake application, and a time delay allowed for movement of engine throttles to the idle position and activation of deceleration devices (if applicable). The time periods to be used shall be applicable to the aircraft configuration and shall be approved by the procuring activity.

(8) ATO (assisted takeoff) ignition time or distance shall be the time or distance from start of takeoff at which rocket-assist devices are ignited. Unless otherwise directed by the procuring activity, ATO ignition shall be timed to produce the minimum takeoff ground run distance with all engines operating (see 6.2).

(9) Minimum afterburner blowout speed shall be the minimum airspeed at which the aircraft can experience an afterburner failure and then continue to accelerate to V_{LOF}, within the remaining runway length.

(10) Maximum braking speed shall be the highest speed from which the aircraft can be brought to a stop without exceeding the maximum design energy absorption capability of the brakes. The data are based on engines producing idle thrust and on the aircraft being brought to a stop with wheel brakes only.

(11) For fixed-wing V/STOL aircraft, short takeoff (STO) performance shall be based on engines delivering 100 percent of rated power output up to the point at which aircraft configuration is changed from the acceleration configuration to the lift-off configuration. From lift-off to attainment of a height of 50 feet, engines shall be assumed to deliver 95 percent of rated power output. At lift-off, the lift coefficient for single-engine aircraft shall not exceed 80 percent of maximum lift coefficient including thrust-induced lift effects. For multiengine aircraft, V_{LOF} shall be such that in the event of engine failure, 1.0g flight can be maintained at lift-off speed with one engine inoperative at a lift coefficient not to exceed 80 percent of

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maximum lift coefficient determined with one engine inoperative and including thrust-induced lift effects with no change in thrust vector angle. For multiengine aircraft, V_2 shall be the speed for maximum climb gradient with one engine inoperative, landing gear retracted, and flaps in the takeoff position. For single-engine aircraft, V_2 shall be the speed for maximum climb gradient with landing gear retracted and flaps in the takeoff position.

(12) For fixed-wing V/STOL aircraft, vertical takeoff (VTO) performance shall be based on engines delivering 95 percent of rated power output.

d. Cruise and endurance speeds.

(1) Maximum-range cruise speed for turbojet, turbofan, and turboprop-powered aircraft shall be the cruise speed that results in attainment of 100 percent of the maximum nautical miles per pound of fuel.

(2) Long-range cruise speed shall be highest of the two cruise speeds that results in attainment of 99 percent of the maximum nautical miles per pound of fuel.

(3) Best range cruise speed for helicopters shall be the cruise speed that results in attainment of 100 percent of the maximum nautical miles per pound of fuel.

(4) Maximum endurance speed shall be the speed that results in the lowest fuel flow rate. Maximum endurance speed shall not be less than V_{MCA}, nor shall it be less than airframe buffet speed.

e. Ceilings.

(1) Service ceiling shall be the pressure altitude at which the maximum rate of climb is 100 feet per minute at the stated power.

(2) Cruise ceiling shall be the pressure altitude at which the maximum rate of climb is 300 feet per minute at maximum continuous thrust or the equivalent power setting.

(3) Combat ceiling shall be the pressure altitude at which the maximum rate of climb is 500 feet per minute at the stated power (maximum or intermediate).

(4) Optimum cruise altitude (not to exceed cruise ceiling) shall be the pressure altitude at which cruise at maximum-range cruise speed, long-range cruise speed, or best range cruise speed results in the highest possible nautical miles per pound of fuel.

(5) Optimum endurance altitude (not to exceed cruise ceiling) shall be the pressure altitude at which cruise at maximum endurance speed results in the lowest possible fuel flow rate.

f. Wind corrections. All wind correction plots shall be for 100-percent accountability; the full force of the reported wind shall be assumed to act on the aircraft.

g. Landing.

(1) Land-based aircraft. Landing distances for conventional fixed-wing aircraft shall be based on the following speeds determined out of ground effect.

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(a) The speed at the 50-foot obstacle height shall not be less than 1.2 times power-off stall speed.

(b) Touchdown speed shall not be less than 1.1 times power-off stall speed.

(2) Carrier-based aircraft. Approach and touchdown speeds shall correspond to 1.05 times V_{PA}(min) using the definition of V_{PA}(min) indicated in the aircraft detail specification. Final approach and touchdown speeds shall be based on flight test results.

(3) Fixed-wing V/STOL aircraft.

(a) Short landing (SL) performance shall be based on engines delivering not more than 85 percent of rated power output from a height of 50 feet down to touchdown. Aerodynamic lift shall be based on no more than 60 percent of maximum lift coefficient at the 50-foot height point and on no more than 80 percent of maximum lift coefficient at touchdown. Aerodynamic lift as defined above includes thrust-induced lift forces.

(b) Vertical landing (VL) performance shall be based on engines delivering 95 percent of rated power output.

3.3.1.13.4 Fixed-wing turbojet and low bypass ratio turbofan aircraft performance data requirements.

a. Chapter 29 — Standard Data. (Fixed-wing turbojet and low bypass ratio turbofan aircraft) Chapter 29 includes standard data that are independent of the aircraft and aircraft performance data that are applicable to more than one flight regime.

(1) Explanatory text. The explanatory text shall include the following:

(a) Glossary of terms and abbreviations. Include terms that explain the conversion of airspeed from indicated to calibrated to equivalent to true.

(b) Weights. Include a gross weight table indicating the following weights:

Aircraft basic weight. Note that aircraft basic weight includes fixed equipment, oil, and unusable fuel.

Crew. Note that crew weight includes crew equipment and disposables such as oxygen.

Internal fuel.

External fuel.

Gross weight including internal fuel only but no payload.

Gross weight including internal and external fuel but no payload.

(2) Performance charts. The performance charts shall be arranged as follows:

(a) External store drag count and weight table. (See figure 26). The drag count and weight for each externally carried store and its associated suspension equipment shall be

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indicated. If stores are intended to be carried asymmetrically, producing appreciable rolling moment and trim drag, methods of accounting for these effects shall be provided.

(b) Standard unit conversion. (See figure 27).

(c) Fuel density. This shall be a plot of primary and alternate fuel specific weight versus temperature, covering a temperature range of –50 to +50 °C.

(d) Standard atmosphere table. (See figure 28).

(e) Temperature deviation from standard. (See figure 29).

(f) Compressibility correction to calibrated airspeed. (See figure 30).

(g) Airspeed Mach number conversion. (See figure 31).

(h) Outside air temperature compressibility correction. This shall be a plot of indicated outside air temperature versus true ambient air temperature with parameters of Mach number or true airspeed.

(i) Airspeed position error correction. (See figure 32). Charts covering flap deflection, landing gear extension, speedbrakes, and ground effect shall be provided. Data shall be shown for sufficient conditions to provide a ±2-knot accuracy.

(j) Altimeter position error correction. (See figure 33). Charts covering flap deflection, landing gear extension, speedbrakes and ground effect shall be provided. Data for subsonic aircraft shall be plotted versus indicated airspeed. For supersonic aircraft, data shall be plotted versus indicated Mach number also. Data for sufficient conditions to provide a ±100-foot accuracy shall be indicated.

(k) Mach meter position error correction. The correction shall be plotted versus indicated Mach number with parameters of altitude.

(l) Takeoff and landing crosswind. (See figure 34). Include minimum nosewheel lift-off (touchdown) speed as a function of runway crosswind component. At the minimum lift-off (touchdown) speed, the aircraft will be capable of ground steering with aerodynamic control only in a crosswind.

(m) V_{MCG} (ground minimum control speed). This shall be a plot of ambient air temperature versus ground minimum control speed with lines depicted on the plot for pressure altitudes of sea level, 4,000, 8,000, and 12,000 feet. Temperatures shall cover a range from –50 to +50 °C. Data shall be given for the engine power settings and the flap settings that are normally used for takeoff.

(n) V_{MCA} (air minimum control speed). This shall be a plot of ambient air temperature versus air minimum control speed with lines shown on the plot for pressure altitudes of sea level, 4,000, 8,000, and 12,000 feet. Temperatures shall cover a range from –50 to +50 °C. Data shall be provided for the engine power settings that are normally used for takeoff. The effect of flap setting shall be indicated if applicable.

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(o) Stall speed. This shall be a plot of gross weight versus power-off stall speed, based on the pilot airspeed system, for bank angles of 0°, 30°, 45°, and 60°. Separate plots shall be included for takeoff, cruise, combat, approach, and landing configurations. Data shall be labeled as being applicable to altitudes below 10,000 feet.

(p) Angle-of-attack. This shall be a plot of indicated airspeed versus indicated angle-of-attack units, based on the pilot airspeed system, with lines depicted on the plots for various gross weights. Separate plots at sea level shall be furnished for takeoff, cruise, approach, and landing configurations. For supersonic aircraft, data shall also be included at an altitude of 35,000 feet plotted versus indicated Mach number for cruise and combat configurations.

(q) Center-of-gravity versus gross weight. Forward and aft stability limits shall be indicated and effects of external store and other configuration changes shall be provided if applicable. A note referring to the individual aircraft's Weight and Balance Handbook, prepared in accordance with NAVAIR 01-1B-40 and NAVAIR 01-1B-50, for detailed weight and balance information, shall be included.

(r) Fuel flow. Graphical data shall be provided for maximum continuous power or the equivalent power setting with parameters of pressure altitude and ambient air temperature covering a temperature range of –50 to +50 °C, and airspeed or Mach number.

b. Chapter 30 — Takeoff Data (Fixed-Wing Turbojet and Low Bypass Ratio Turbofan Aircraft). Include separate charts for intermediate (military) thrust and for maximum (afterburner) thrust where applicable. Data shall cover a temperature range from –50 to +50 °C, an altitude range from sea level to 12,000 feet, and winds from –40 to +40 knots.

(1) Graphical illustration of multiengine aircraft takeoff. For multiengine aircraft, sketches similar to those on figure 35 depicting safe takeoff (maximum abort speed greater than minimum go speed) and unsafe takeoff (maximum abort speed less than minimum go speed) conditions shall be presented in the text. Explanatory text shall emphasize that takeoff is unsafe if maximum abort speed is less than minimum go speed, because if engine failure occurs at a speed higher than maximum abort speed, insufficient runway may remain to complete the takeoff.

(2) V_1 (minimum go speed). (See figure 36). Include parameters of runway length, ambient temperature, pressure altitude, wind velocity, and gross weight.

(3) $V_{MAX\ ABORT}$ (maximum abort speed). (See figure 37). Include parameters of runway length, ambient temperature, pressure altitude, runway condition reading, and gross weight.

(4) V_{LOF} (lift-off speed) and V_2 (speed at the 50-foot obstacle height). (See figure 38). Parameters shall include ambient temperature, pressure altitude, and gross weight.

(5) Takeoff distance. (See figure 39). Takeoff ground run to lift-off speed and total distance to clear a 50-foot obstacle, reaching a speed of V_2 at that point, shall be shown with all engines operating. Parameters shall include ambient temperature, pressure altitude, gross weight, and wind velocity.

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(6) ATO ignition time or distance. This shall be a plot of ATO ignition time or distance with parameters of ambient temperature, pressure altitude, runway length, gross weight, and wind velocity. Format shall be similar to the minimum go-speed chart.

(7) Takeoff gross weight limit. (See figure 40). For multiengine aircraft, the gross weight that results in a rate of climb of 200 feet per minute at lift-off speed with one engine inoperative shall be indicated. Parameters shall include ambient temperature, pressure altitude, wing flap position, and landing gear position.

(8) Velocity during takeoff ground run. (See figure 41). This shall be a plot entered with takeoff ground run and lift-off speed. Following a guideline to a lower speed will yield the ground roll distance at which this speed will be reached.

(9) Minimum afterburner blowout speed. Include parameters of runway length, ambient temperature, pressure altitude, wind velocity, and gross weight. Format shall be similar to the minimum go-speed chart.

(10) Maximum braking speed. This shall be a plot of maximum braking speed, Knots Indicated Air Speed (KIAS), for aborted takeoff from a dry, hard-surface runway. Include parameters of ambient temperature, pressure altitude, and gross weight.

c. Chapter 31 — Climb Data (Fixed-Wing Turbojet and Low Bypass Ratio Turbofan Aircraft).

(1) Climb performance. (See figure 42). Include a climb speed schedule table as a function of drag count with fuel allowances for takeoff and acceleration to climb speed provided as a note. Unless otherwise directed by the procuring activity, fuel allowance shall be fuel for 4 minutes intermediate thrust at sea level, static, standard day conditions. Include graphical data for time, fuel, and distance to climb with parameters for initial gross weight, pressure altitude, drag count, and temperature deviation from standard day. For fighter and attack aircraft with all engines operating, include charts for intermediate (military) thrust and for maximum (afterburner) thrust. For all other types of aircraft, with all engines operating, include charts for maximum continuous thrust and for intermediate (military) thrust. For all types of aircraft, with one engine inoperative, include charts for intermediate (military) thrust. Charts shall be provided for the clean configuration (flaps up, landing gear retracted) only.

(2) Instantaneous rate of climb. Parameters shall include air temperature, pressure altitude, gross weight, and drag count. The format shown on figure 44 shall be used. The altitude range shall extend from sea level to service ceiling. These charts shall be included for all conditions for which climb performance data are given (see 3.3.1.13.4c.(1)).

(3) Service ceiling. (See figure 43). Parameters shall include gross weight, drag count, and temperature deviation from standard day. These data shall be provided for maximum continuous thrust with all engines operating and with one engine inoperative.

(4) Combat ceiling. (See figure 43). For fighter and attack aircraft only, parameters shall include gross weight, drag count, and temperature deviation from standard day. Include charts for intermediate (military) thrust and for maximum (afterburner) thrust with all engines operating.

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(5) One engine inoperative climb performance. (See figure 44). Parameters shall include air temperature, pressure altitude, gross weight, and drag count. Climb speed shall be depicted on the gross weight lines. These data shall be provided for the takeoff and landing configurations (flaps down, landing gear extended) for intermediate (military) thrust.

d. Chapter 32 — Range Data (Fixed-Wing Turbojet and Low Bypass Ratio Turbofan Aircraft). At the option of the procuring activity, range charts of the types specified for fixed-wing turboprop and high bypass ratio turbofan aircraft in 3.3.1.13.5d. shall be substituted for or specified as additional charts for fixed-wing turbojet and low bypass ratio turbofan aircraft. The following types of charts shall be provided for all engines operating and one engine inoperative conditions (see 6.2).

(1) Optimum cruise altitude. This shall be a plot of gross weight versus optimum cruise altitude with parameters of drag count and temperature deviation from standard. Format shall be the same as used for figure 43.

(2) Cruise performance. (See figure 45). Design of charts is as follows:

(a) Phase I. Enter chart at aircraft average gross weight and proceed right to intersect pressure altitude line. Proceed up into the top block of data and intersect zero drag count line. From this point, move parallel to guidelines to desired Mach number and project right to read transfer scale. If it is desired to cruise at recommended Mach number for maximum range, move parallel to guidelines to intersect drag count line for appropriate drag count. At this intersection, read Mach number for maximum range and associated transfer scale reading.

(b) Phase II. Enter with Mach number, drag count, and transfer scale reading, and read reference number.

(c) Phase III. Enter with Mach number, reference number, and pressure altitude and read pounds of fuel per nautical mile at zero wind.

(d) Phase IV. Enter with Mach number, temperature, and pounds of fuel per nautical mile and read fuel flow at zero wind.

(3) Maximum range cruise at constant altitude. (See figure 46). These charts provide cruise Mach number, true airspeed, groundspeed, cruise time, pounds of fuel per nautical mile, fuel flow, and fuel required for maximum range cruise at constant altitude as a function of average gross weight, pressure altitude, drag count, ambient temperature, wind velocity, and distance to be traveled.

(4) Speed, time, and fuel to cruise. (See figure 47). Include parameters of Mach number, ambient temperature, windspeed, distance to be traveled, and fuel flow.

(5) Low-altitude cruise. (See figure 48). For fighter and attack aircraft only, these tables present total fuel flow values for various combinations of true airspeed and drag count at various selected low altitudes. Provide separate charts for several gross weights. Fuel flow values are tabulated for standard day; however, correction factors are provided for nonstandard temperatures.

(6) Range wind correction. (See figure 49). This chart includes a means of correcting specific or total range for existing wind effects.

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(7) Bingo chart. (See figure 50). These charts show time, fuel, and airspeed required to travel a given distance using a combination of climb, maximum-range cruise, and maximum-range descent. Fuel required values include a fuel allowance for reserve. Unless otherwise directed by the procuring activity, the reserve fuel allowance shall be 10 percent of maximum internal usable fuel. Provide data for cruise at optimum cruise altitude and for cruise at sea level (see 6.2).

e. Chapter 33 — Endurance Data (Fixed-Wing Turbojet and Low Bypass Ratio Turbofan Aircraft). At the option of the procuring activity, endurance charts of the types specified for fixed-wing turboprop and high bypass ratio turbofan aircraft in 3.3.1.13.5e.(1) shall be substituted for or specified as additional charts for fixed-wing turbojet and low bypass ratio turbofan aircraft. The following type of chart shall be provided for all engines operating and one-engine inoperative conditions (see 6.2).

(1) Maximum endurance. (See figure 51). Include parameters for average gross weight, bank angle, pressure altitude, drag count, and temperature deviation from standard.

f. Chapter 34 — In-flight refueling data (Fixed-Wing Turbojet and Low Bypass Ratio Turbofan Aircraft).

(1) Tanker aircraft. Data shall be included for combat aircraft that are equipped to carry a refueling system as an external store.

(a) Air refueling transfer time. (See figure 52). Parameters shall include gross pounds of fuel transferred, fuel flow rate, and fuel density.

(b) Fuel consumption rate during air refueling. (See figure 53). Parameters shall include Mach number or calibrated airspeed, gross weight, drag count, and pressure altitude.

(2) Receiver aircraft. Charts shall be the same as described in 3.3.1.13.4f.(1) except that they are applicable to receiver aircraft rather than tanker aircraft.

g. Chapter 35 — Descent Data (Fixed-Wing Turbojet and Low Bypass Ratio Turbofan Aircraft).

(1) Maximum range descent. (See figure 54). Include parameters for initial gross weight, pressure altitude, and drag count. Provide data for all engines operative and one-engine inoperative conditions.

(2) Normal descent with constant calibrated airspeed. The format of figure 49 shall be used. Provide data for all engines operative at flight idle thrust, speedbrakes retracted only condition. The airspeed to be used shall be applicable to the aircraft configuration and be approved by the procuring activity.

(3) Quick descent at limit airspeed. The format of figure 49 shall be used. Using a simplified speed schedule is optional. Provide data for all the engines operative at flight idle thrust, speedbrakes extended only condition.

h. Chapter 36 — Landing Data (Fixed-Wing Turbojet and Low Bypass Ratio Turbofan Aircraft). Data shall cover a temperature range from –50 to +50 °C, an altitude range from sea level to 12,000 feet, winds from –20 to +40 knots, and slopes from –2° to +2°.

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(1) Landing speeds. (See figure 55). For carrier-based aircraft, include initial stall warning speed and approach/touchdown speed as a function of gross weight for no flaps and normal landing flaps configurations. Recommended angle-of-attack setting shall appear on the charts. For land-based aircraft, provide initial stall warning speed, approach speed, and touch-down speed as a function of gross weight for no flaps and normal landing flaps configurations.

(2) Landing performance — ground roll. (See figure 56). Include parameters for temperature, altitude, gross weight, wind velocity, runway condition reading, and runway slope. Include landing speed versus gross weight as a sub-graph. Include data with hard braking only for no flap and normal landing flap configurations.

(3) Landing performance — total distance from 50-foot height. (See figure 57). Same conditions as 3.3.1.13.4h.(2).

i. Chapter 37 — Mission Planning (Fixed-Wing Turbojet and Low Bypass Ratio Turbofan Aircraft). Chapter 37 includes information regarding overall mission planning and presents special mission and tactical charts. Do not duplicate charts contained in tactical manuals in this chapter. External store loadings for charts shall be typical of the aircraft primary mission.

(1) Fuel transferred versus tanker mission radius. (See figure 58). For tanker aircraft only, include parameters for loiter time and refueling rate.

(2) Loiter time versus tanker mission radius. (See figure 59). For search and patrol aircraft only, include parameters for loiter altitude.

(3) Level flight acceleration. (See figure 60). For fighter and attack aircraft only, show time and distance to accelerate from cruise Mach number to combat Mach number. Fuel used can be obtained from initial and final readings of gross weights by following weight guidelines. Include separate charts for intermediate and maximum thrust. Include separate charts for individual altitudes covering the range of operating altitudes of the aircraft.

(4) Combat allowance. (See figure 61). For fighter and attack aircraft only, provide combat time as a function of initial gross weight, pressure altitude, drag count, and fuel for combat under conditions of straight, level, stabilized flight. Include separate charts for intermediate and maximum thrust.

(5) Turn rate versus airspeed. (See figure 62). For fighter and attack aircraft only, at a gross weight representative of combat weight, show turn rate that can be sustained in level flight with maximum thrust. Show instantaneous turn rate at maximum usable angle of attack, as defined by maximum lift coefficient under static stall conditions, and at four units below maximum angle of attack. Include parameters for altitude. Include separate charts for intermediate and maximum thrust.

(6) Turn radius versus airspeed. (See figure 63). For fighter and attack aircraft only, include data for the same conditions as turn rate.

(7) Altitude lost in pullout. (See figure 64). Parameters shall include altitude, speed, and dive angle at start of pullout. For fighter and attack aircraft, typically 4g and 6g charts are included for both low altitude and high altitude for intermediate thrust.

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(8) Level flight envelope. (See figure 65). For fighter and attack aircraft only, include an envelope of airspeed or Mach number versus altitude formed by the minimum and maximum straight, level flight operating airspeeds of the aircraft. Include parameters for gross weight and for various configurations, if applicable.

(9) Tanker speed envelope. (See figure 66). For tanker aircraft only, include an envelope of Mach number or calibrated airspeed formed by the upper and lower speed limits. Provide data for refueling drogue extended and retracted configurations.

(10) V-n envelope. (See figure 67). For fighter and attack aircraft only, at a gross weight representative of combat weight, include an envelope of symmetrical limit load factor versus calibrated airspeed and Mach number with lines drawn showing load factor at constant angle of attack. Include separate envelopes for individual altitudes covering the range of operating altitudes of the aircraft.

j. Chapter 38 — Emergency Operation (Fixed-Wing Turbojet and Low Bypass Ratio Turbofan Aircraft).

(1) Glide performance. (See figure 68). Provide time, distance, and speed for maximum-range glide descent to sea level with all engines inoperative. Parameters shall include gross weight, drag count, and initial altitude.

(2) Airstart envelope. (See figure 69). Include an envelope of airspeed or Mach number versus altitude formed by the minimum and maximum speeds at which the engine can be airstarted.

3.3.1.13.5 Fixed-wing turboprop and high bypass ratio turbofan aircraft performance data requirements.

a. Chapter 29 — Standard Data (Fixed-wing turboprop and high bypass ratio turbofan aircraft). Chapter 29 includes standard data that are independent of the aircraft and aircraft performance data that are applicable to more than one flight regime.

(1) Explanatory text. Same as 3.3.1.13.4a.(1).

(2) The performance charts shall be prepared and arranged the same as described in 3.3.1.13.4a.(2).

b. Chapter 30 — Takeoff Data (Fixed-wing turboprop and high bypass ratio turbofan aircraft). Include charts for intermediate (military), maximum (afterburner), and any reduced power setting normally used for takeoff, when applicable.

(1) Graphical illustration of multiengine aircraft takeoff. Same as 3.3.1.13.4b.(1).

(2) Static power check for takeoff. (See figure 70). For turboprop-powered aircraft only, include shaft horsepower or torque pressure available for takeoff with parameters for ambient temperature and pressure altitude.

(3) V_1 (minimum go speed). Same as 3.3.1.13.4b.(2).

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(4) $V_{MAX\ ABORT}$ (maximum abort speed). Same as 3.3.1.13.4b.(3).

(5) V_{LOF} (lift-off speed) and V_2 (speed at the 50-foot obstacle height). Same as 3.3.1.13.4b.(4).

(6) Takeoff distance. Same as 3.3.1.13.4b.(5).

(7) ATO ignition time or distance. Same as 3.3.1.13.4b.(6).

(8) Takeoff gross weight limit. For multiengine aircraft, same as 3.3.1.13.4b.(7). Propeller of inoperative engine is feathered.

(9) Velocity during takeoff ground run. Same as 3.3.1.13.4b.(8).

(10) Maximum braking speed. Same as 3.3.1.13.4b.(10).

(11) Climbout factor. (See figure 71). For turboprop-powered aircraft only, the climbout factor is a reference number used on climbout flightpath charts (see 3.3.1.13.5b.(12)), to simplify the determination of climbout performance. Parameters shall include ambient temperature, pressure altitude, and gross weight. For multiengine aircraft, include charts with all engines operating and with one engine inoperative.

(12) Climbout flightpath. (See figure 72). For turboprop-powered aircraft only, these shall be plots of vertical height above takeoff point versus horizontal distance from brake release with the parameter of climbout factor. All engine operating charts shall be constructed to a minimum of either 8,000-foot vertical height or 20-nautical mile horizontal distance. One-engine inoperative charts shall be constructed to a minimum of either 1,000-foot vertical height or 10-nautical mile horizontal distance. One-engine inoperative charts shall be based on engine failure occurring during the takeoff run at a speed such that, under conditions of no wind with a dry runway, the runway lengths required for continued acceleration to lift-off speed or braking to a stop are equal. Landing gear retraction shall be initiated 3 seconds after lift-off. The speed at the 50-foot obstacle height, V_2, shall not be exceeded during landing gear retraction. After gear retraction, the aircraft shall be accelerated to flap retraction speed, at which time flap retraction shall be initiated. After flaps are up, the aircraft shall be accelerated to the best climb speed. The final climb segment shall be performed with flaps up at best climb speed using maximum continuous power. Charts shall be based on maintaining takeoff power until the start of the final climb segment or when engine time limit is reached, whichever occurs first.

c. Chapter 31 — Climb Data (Fixed-wing turboprop and high bypass ratio turbofan aircraft). All data with one engine inoperative for turboprop-powered aircraft shall be provided with propeller feathered on the inoperative engine.

(1) Climb performance. Same as 3.3.1.13.4c.(1).

(2) Instantaneous rate of climb. Same as 3.3.1.13.4c.(2).

(3) Service ceiling. Same as 3.3.1.13.4c.(3).

(4) Combat ceiling. Same as 3.3.1.13.4c.(4).

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(5) One-engine inoperative climb performance. Same as 3.3.1.13.4c.(5).

d. Chapter 32 — Range Data (Fixed-wing turboprop and high bypass ratio turbofan aircraft). The data presentation format provided in 3.3.1.13.4d. for fixed-wing turbojet and low bypass ratio turbofan aircraft are not applicable to fixed-wing turboprop and high bypass ratio aircraft. Therefore the following format shall be used. The following types of charts shall be provided for all engines operating and one-engine inoperative conditions. Charts are required for alternate configurations when the variation in range exceeds 5 percent.

(1) Mission profile maximum range. (See figure 73). This chart includes a simplified method of flight planning at standard day conditions. The chart is based on a sequence of maximum power takeoff, intermediate power climb, and maximum range cruise (zero wind). A fuel allowance is included for engine start, taxi, takeoff, and acceleration to climb speed. Unless otherwise directed by the procuring activity, this allowance shall be fuel for 4 minutes at intermediate power, or the equivalent, at sea level, static, standard day conditions. No fuel allowance is included for descent, landing, or reserve. The mission profile chart can be used to determine directly the total fuel required to fly a given distance or the total distance available for a given fuel load at any altitude (see 6.2).

(2) Mission profile minimum time. Same as 3.3.1.13.4d.(1) except that cruise is at maximum continuous power or the equivalent.

(3) Maximum range summary. (See figure 74). This chart includes nautical miles per 1,000 pounds of fuel and calibrated airspeed for maximum range cruise at constant altitude at standard day conditions.

(4) Maximum range cruise. (See figure 75). This table presents torque per engine, fuel flow per engine, total fuel flow, and calibrated airspeed for maximum range cruise speed for various combinations of altitude and gross weight. Include separate tables for temperature deviations from standard of –20, 0, 20, and 40 °C.

(5) Minimum time cruise. Same as 3.3.1.13.4d.(4) except that cruise is at maximum continuous power or the equivalent.

(6) Nautical miles per 1,000 pounds of fuel. (See figure 76). This shall be a plot of nautical miles per 1,000 pounds of fuel versus Mach number, true airspeed, and calibrated airspeed at standard day conditions extending from maximum endurance speed to speed with intermediate power. Lines of constant total fuel flow shall be indicated on the plots. Separate plots shall be provided for altitudes separated by increments of 5,000 or 10,000 feet covering a range from sea level to approximately cruise ceiling for the lightest weight.

(7) Range wind correction. Same as 3.3.1.13.4d.(6).

(8) Bingo chart. Same as 3.3.1.13.4d.(7).

e. Chapter 33 — Endurance Data (Fixed-wing turboprop and high bypass ratio turbofan aircraft). The following types of charts shall be provided for all engines operating and one-engine inoperative conditions. Charts are required for alternate configurations when the variation in fuel flow exceeds 5 percent.

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(1) Maximum endurance profile. (See figure 77). This chart shows maximum endurance that is available for any fuel remaining quantity at any altitude and also what is available by climbing from sea level to optimum endurance altitude. A climb speed schedule is also provided. A fuel allowance is included for reserve. Unless otherwise directed by the procuring activity, the reserve fuel allowance shall be 10 percent of maximum internal usable fuel (see 6.2).

(2) Maximum endurance summary. (See figure 78). This chart includes total fuel flow rate and calibrated airspeed for maximum endurance speed.

f. Chapter 34 — In-Flight Refueling Data (Fixed-wing turboprop and high bypass ratio turbofan aircraft).

(1) Tanker aircraft. Data shall be included for combat aircraft that are equipped to carry a refueling system as an external store.

(a) Air refueling transfer time. Same as 3.3.1.13.4f.(1)(a).

(b) Fuel consumption rate during air refueling. Same as 3.3.1.13.4f.(1)(b).

(2) Receiver aircraft. Charts shall be the same as described in 3.3.1.13.5f.(1) except that they are applicable to receiver aircraft rather than tanker aircraft.

g. Chapter 35 — Descent Data (Fixed-wing turboprop and high bypass ratio turbofan aircraft).

(1) Maximum-range descent. Same as 3.3.1.13.4g.(1).

(2) Normal descent. Same as 3.3.1.13.4g.(2).

(3) Quick descent at limited airspeed. Same as 3.3.1.13.4g.(3).

h. Chapter 36 — Landing Data (Fixed-wing turboprop and high bypass ratio turbofan aircraft).

(1) Landing speeds. Same as 3.3.1.13.4h.(1).

(2) Landing performance ground roll. Same as 3.3.1.13.4h.(2).

(3) Landing performance total distance from 50-foot height. Same as 3.3.1.13.4h.(3).

i. Chapter 37 — Mission Planning (Fixed-wing turboprop and high bypass ratio turbofan aircraft). Chapter 37 includes information regarding overall mission planning and presents special mission and tactical charts. Do not duplicate charts contained in tactical manuals in this chapter.

(1) Fuel transferred versus radius. For tanker aircraft only. Same as 3.3.1.13.4i.(1).

(2) Loiter time versus radius. For search and patrol aircraft only. Same as 3.3.1.13.4i.(2).

(3) Level flight acceleration. For fighter and attack aircraft only. Same as 3.3.1.13.4i.(3).

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(4) Combat allowance. For fighter and attack aircraft only. Same as 3.3.1.13.4i.(4).

(5) Turn rate versus airspeed. For fighter and attack aircraft only. Same as 3.3.1.13.4i.(5).

(6) Turn radius versus airspeed. For fighter and attack aircraft only. Same as 3.3.1.13.4i.(6).

(7) Altitude lost in pullout. Same as paragraph 3.3.1.13.4i.(7).

(8) Level flight envelope. For fighter and attack aircraft only. Same as 3.3.1.13.4i.(8).

(9) Tanker speed envelope. For tanker aircraft only. Same as 3.3.1.13.4i.(9).

(10) V-n envelope. For fighter and attack aircraft only. Same as 3.3.1.13.4i.(10).

j. Chapter 38 — Emergency Operation (Fixed-wing turboprop and high bypass ratio turbofan aircraft).

(1) Glide performance. Same as 3.3.1.13.4j.(1).

(2) Airstart envelope. Same as 3.3.1.13.4j.(2).

3.3.1.13.6 Fixed-wing V/STOL aircraft performance data requirements.

a. Chapter 29 — Standard Data (Fixed-wing V/STOL aircraft). Includes standard data that are independent of the aircraft and aircraft performance data that are applicable to more than one flight regime.

(1) Explanatory text. Same as 3.3.1.13.4a.(1).

(2) Performance charts. The performance charts shall be prepared and arranged the same as described in 3.3.1.13.4a.(2).

b. Chapter 30 — Takeoff Data (Fixed-wing V/STOL aircraft). Include charts for the power settings generally used for V/STOL. Charts are required for the following types of takeoff: 1) Conventional takeoff data are required with all engines operative, and with engine failure considered to occur during the ground run; 2) short takeoff data are required with all engines operative, and with engine failure considered to occur only at lift-off speed; 3) vertical takeoff data are required only with all engines operative; and 4) rolling vertical takeoff data are required only with all engines operative.

(1) Static power check for takeoff. Same as 3.3.1.13.5b.(2).

(2) Conventional takeoff, graphical illustration of multiengine aircraft takeoff. Same as 3.3.1.13.4b.(1).

(3) Conventional takeoff, V_1 (minimum go speed). Same as 3.3.1.13.4b.(2).

(4) Conventional takeoff, $V_{MAX\ ABORT}$ (maximum abort speed). Same as 3.3.1.13.4b.(3).

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(5) Conventional takeoff, V_{LOF} (lift-off speed) and V_2 (speed at the 50-foot obstacle height). Same as 3.3.1.13.4b.(4).

(6) Conventional takeoff, takeoff distance. Same as 3.3.1.13.4b.(5).

(7) Conventional takeoff, gross weight limit. Same as 3.3.1.13.4b.(7). Data shall be provided for the thrust vector angle (nozzle angle) used for conventional takeoff.

(8) Conventional takeoff, velocity during takeoff ground run. Same as 3.3.1.13.4b.(8).

(9) Conventional takeoff, maximum braking speed. Same as 3.3.1.13.4b.(10).

(10) Short takeoff, nozzle rotation speed. (See figure 79). This chart provides the speed at which the thrust vector angle (nozzle angle) is changed from the ground run setting to the lift-off setting. Parameters shall include ambient temperature, pressure altitude, and gross weight.

(11) Short takeoff, nozzle angle. (See figure 80). This chart provides the thrust vector angle (nozzle angle) at lift-off. Parameters shall include ambient temperature, pressure altitude, and gross weight. A note shall indicate the ground run nozzle angle.

(12) Short takeoff, V_{LOF} (lift-off speed) and V_2 (speed at the 50-foot obstacle height). Same as 3.3.1.13.4b.(4).

(13) Short takeoff, takeoff distance. Same as 3.3.1.13.4b.(5).

(14) Short takeoff, gross weight limit. Same as 3.3.1.13.4b.(6). Data shall be provided for the thrust vector angle (nozzle angle or nacelle angle) used at lift-off and a note indicating this fact shall be placed at the bottom of the chart.

(15) Vertical takeoff, gross weight limit. (See figure 81). This chart shall include the vertical takeoff weight capability with parameters of ambient temperature, pressure altitude, and power setting.

(16) Rolling vertical takeoff distance. (See figure 82). This chart shall include parameters of ambient temperature, pressure altitude, gross weight, and wind velocity. The takeoff technique including thrust vector angle (nozzle angle) and power setting shall be described in notes.

c. Chapter 31 — Climb Data (Fixed-wing V/STOL aircraft).

(1) Climb performance. Same as 3.3.1.13.4c.(1).

(2) Instantaneous rate of climb. Same as 3.3.1.13.4c.(2).

(3) Service ceiling. Same as 3.3.1.13.4c.(3).

(4) Combat ceiling. Same as 3.3.1.13.4c.(4).

(5) One-engine inoperative climb performance. Same as 3.3.1.13.4c.(5). These data shall be provided for conventional takeoff and landing flight modes only. For aircraft with

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manual control of the thrust vector angle (nozzle angle), data shall be provided for the thrust vector angles used at lift-off and touchdown, respectively. Notes indicating this fact shall be placed at the bottom of the chart.

d. Chapter 32 — Range Data (Fixed-wing V/STOL aircraft). Charts are required for alternate configurations when the variation in range exceeds 5 percent. The charts shall be as described in 3.3.1.13.5d. and are provided for all engines operating and one-engine inoperative conditions.

e. Chapter 33 — Endurance Data (Fixed-wing V/STOL aircraft). Charts are required for alternate configurations when the variation in fuel flow exceeds 5 percent. The following types of charts shall be provided for all engines operating and one-engine inoperative conditions:

(1) Maximum endurance profile. Same as 3.3.1.13.5e.(1).

(2) Maximum endurance summary. Same as 3.3.1.13.5e.(2).

(3) Hover fuel flow. This shall be a plot of total fuel flow in hover with parameters of ambient temperature, pressure altitude, and gross weight.

f. Chapter 34 — In-Flight Refueling Data (Fixed-wing V/STOL aircraft).

(1) Tanker aircraft. Data shall be included for combat aircraft that are equipped to carry a refueling system as an external store.

(a) Air refueling transfer time. Same as 3.3.1.13.4f.(1)(a).

(b) Fuel consumption rate during air refueling. Same as 3.3.1.13.4f.(1)(b).

(2) Receiver aircraft. Charts shall be the same as described in 3.3.1.13.4f.(1) except that they are applicable to receiver aircraft rather than tanker aircraft.

g. Chapter 35 — Descent Data (Fixed-wing V/STOL aircraft).

(1) Maximum-range descent. Same as 3.3.1.13.4g.(1).

(2) Normal descent. Same as 3.3.1.13.4g.(2).

(3) Quick descent at limited airspeed. Same as 3.3.1.13.4g.(3).

h. Chapter 36 — Landing Data (Fixed-wing V/STOL aircraft). Data shall be provided with braking action supplied by means of wheel brakes only and by means of wheel brakes plus thrust vector braking (nozzle angle), where applicable. Charts are required for the following types of landing: 1) conventional landing, 2) short landing, and 3) vertical landing.

(1) Conventional landing, landing speeds. Same as 3.3.1.13.4h.(1).

(2) Conventional landing, landing performance ground roll. Same as 3.3.1.13.4h.(2).

(3) Conventional landing, landing performance total distance from 50-foot height. Same as 3.3.1.13.4h.(3).

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(4) Short landing, landing distance. Include ground roll and total distance from a height of 50 feet with parameters of ambient temperature, pressure altitude, gross weight, and wind velocity. A format similar to that depicted on figure 56 shall be used. The landing technique including thrust vector angle (nozzle angle), power settings, and angle of attack or airspeed shall be described in footnotes or in separate charts if required. For multiengine aircraft, charts shall be included for all engines operative and for one-engine inoperative conditions.

(5) Vertical landing, gross weight limit. This chart includes the vertical landing weight capability with parameters of ambient temperature, pressure altitude, and power settings. A format similar to that depicted on figure 82 shall be used. For multiengine aircraft, charts shall be included for all engines operative and for one-engine inoperative conditions.

i. Chapter 37 — Mission Planning (Fixed-wing V/STOL aircraft). Chapter 37 includes information regarding overall mission planning and presents special mission and tactical charts. Do not duplicate charts contained in tactical manuals in this chapter.

(1) Fuel transferred versus radius. For tanker aircraft only. Same as 3.3.1.13.4i.(1).

(2) Loiter time versus radius. For search and patrol aircraft only. Same as 3.3.1.13.4i.(2).

(3) Level flight acceleration. For fighter and attack aircraft only. Same as 3.3.1.13.4i.(3).

(4) Combat allowance. For fighter and attack aircraft only. Same as 3.3.1.13.4i.(4).

(5) Turn rate versus airspeed. For fighter and attack aircraft only. Same as 3.3.1.13.4i.(5).

(6) Turn radius versus airspeed. For fighter and attack aircraft only. Same as 3.3.1.13.4i.(6).

(7) Altitude lost in pullout. Same as 3.3.1.13.4i.(7).

(8) Level flight envelope. For fighter and attack aircraft only. Same as 3.3.1.13.4i.(8).

(9) Tanker speed envelope. For tanker aircraft only. Same as 3.3.1.13.4i.(9).

(10) V-n envelope. For fighter and attack aircraft only. Same as 3.3.1.13.4i.(10).

j. Chapter 38 — Emergency Operation (Fixed-wing V/STOL aircraft).

(1) Glide performance. Same as 3.3.1.13.4j.(1).

(2) Airstart envelope. Same as 3.3.1.13.4j.(2).

3.3.1.13.7 Helicopter Performance Data Requirements.

a. Chapter 29 — Standard Data (Helicopters). Chapter 29 contains data that are independent of the aircraft and aircraft performance data that are applicable to more than one flight regime.

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(1) Explanatory text. Same as 3.3.1.13.4a.(1).

(2) Performance charts. Unless otherwise specified in the contract or specification, the performance charts shall be prepared and arranged as follows. (see 6.2).

(a) External store drag count and weight table. Same as 3.3.1.13.4a.(2)(a).

(b) Standard unit conversion. Same as 3.3.1.13.4a.(2)(b).

(c) Fuel density. Same as 3.3.1.13.4a.(2)(c).

(d) Standard atmosphere table. Same as 3.3.1.13.4a.(2)(d).

(e) Temperature deviation from standard. Same as 3.3.1.13.4a.(2)(e).

(f) Center-of-gravity versus gross weight. Same as 3.3.1.13.4a.(2)(q).

(g) Airspeed calibration. (See figure 83).

(h) Altitude calibration. (See figure 84).

(i) Density altitude/temperature conversion. (See figure 85).

(j) Shaft horsepower versus torque. Parameters shall include indicated torque (percent), rotor speed (percent), and shaft horsepower (see figure 86).

(k) Fuel flow versus torque. Chart shall be a plot of fuel flow per engine versus torque at a specific power turbine rpm (N_{pt}) (see figure 87).

(l) Power available. (See figure 88). This chart includes engine torque with parameters of pressure altitude, inlet air temperature, turbine speed, and ambient temperature. Charts are required for maximum, intermediate, and maximum continuous power ratings.

b. Chapter 30 — Takeoff Data (Helicopters).

(1) Hover gross weight limits. Provide maximum gross weight allowable for hover at military power when entered with pressure altitude, ambient air temperature, and wind velocity for in and out of ground effect conditions. Altitude range shall be from sea level to 20,000 feet and outside temperature range from –60 to +60 °C. Wind velocity shall be from 0 to 30 knots (see figure 89).

(2) Torque required to hover. Include charts for in and out of ground effect hovering. Charts shall be plots of gross weight versus torque (percent) with parameters of pressure altitude and correction plots of rotor rpm and outside air temperature. The rotor rpm correction is applicable only if the helicopter has an operational band of rotor rpm. This chart shall be placed in the manual on the right-hand page opposite the chart for maximum gross weight to hover (see figure 90).

c. Chapter 31 — Climb Data (Helicopters).

(1) Climb performance. This chart shall include necessary climb performance for all appropriate power settings. Parameters shall include gross weight, pressure altitude, climb time, distance, fuel, and a climb speed schedule (see figure 91).

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(2) Service ceiling. Parameters shall include gross weight, temperature, pressure altitude, and indicated airspeed (see figure 92).

d. Chapter 32 — Range Data (Helicopters).

(1) Best range. Parameters shall include gross weight, pressure altitude, unit range, IAS, CAS, fuel flow, and approximate torque at standard temperature and 100 percent rotor rpm (N_r) (see figure 93).

(2) Range at maximum continuous power. Parameters shall include gross weight, pressure altitude, unit range, IAS, CAS, fuel flow, and approximate torque at standard temperature and 100 percent rotor rpm (see figure 94).

(3) Time and range versus fuel. Parameters shall include ranges of fuel quantity, fuel flow, time, TAS, and range in nautical miles (see figure 95).

e. Chapter 33 — Endurance Data (Helicopters).

(1) Maximum endurance. Parameters shall include gross weight, pressure altitude, IAS, CAS, fuel flow, time, and approximate torque at standard temperature and 100 percent rotor rpm (see figure 96).

(2) Hovering endurance. Include charts for out-of-ground-effect hovering. Include parameters of gross weight, pressure altitude, ambient temperature, and fuel flow (see figure 97).

f. Chapter 34 — Emergency Operation (Helicopters).

(1) Single-engine range. Same as 3.3.1.13.7d.(1) except for single-engine operation (see figure 98).

(2) Single-engine endurance. Same as 3.3.1.13.7e.(1) except for single-engine operation (see figure 99).

(3) Single-engine service ceiling. Parameters shall include pressure altitude, temperature, rotor speed, and gross weight (see figure 100).

(4) Ability to maintain flight on one engine. Parameters shall include altitude, IAS, temperature, and gross weight (in increments of 2,000 pounds throughout operating range). Charts shall be provided for –20, 0, 20, and 40 °C (see figure 101).

(5) Minimum airspeed for flight with one engine. Parameters shall include gross weight, temperature, and CAS based on 100 percent rotor RPM, military power, sea level, and out-of-ground effect conditions (see figure 102).

g. Chapter 35 — Special Charts (Helicopters). Radius of turn at constant airspeed. This chart shall plot turn radius (feet) versus TAS based on 100 percent rotor rpm and bank angle. Plot standard (3° per second) and double standard (6° per second) turns (see figure 103).

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3.3.1.14 Back matter. Back matter of all publications comprises the following:

Appendices
Foldout pages
Indexes
Last page
Back cover

3.3.1.14.1 Appendices. Material that supplements one or more chapters shall be placed in appendixes located after the last chapter.

3.3.1.14.2 Foldout pages (if appropriate). A publication shall be arranged so that foldout pages immediately precede the index, regardless of their relationship to the text. Foldouts are always right-hand pages.

3.3.1.14.3 Indexes. An alphabetical index, (see figure 104), shall be provided as the last portion of the publication, except for foldout pages. Reference subjects by page numbers. Figures shall not be indexed. Update the index as necessary for each Revision or Change. Alphabetize indexes word by word rather than letter by letter. Include in the index a reference to headings by using the keyword system. Multiple words or phrases are considered keywords. For example, in the heading "MOUNTAIN AND ROUGH TERRAIN FLYING," there will be entries at both mountain and rough terrain. As a minimum requirement, all primary and secondary headings shall be indexed.

a. Acceptable indexing. If the same subject falls in different sections of a chapter or in different chapters, ensure that all references are included. When the same subject accumulates five separate entries, subentries are required. An example is:

Rotor system 2-9
  Blade stall 11-7
  Droop 11-10
  Inspection 7-10
  Limitations 4-6
  Main dish vortex 11-9

b. Cross-referencing. If a series of entries would be duplicated under another first-level entry, cross-reference from the inverted heading to the normal heading after five entries. Using the word see, cross-reference from one heading (or subheading) to an alternative heading, under which all the relevant references to an item in the text are collected. Cross- referencing requires no page or other references; (e.g., System, direct current. (*See* Direct current system.))

c. Unacceptable indexing. If a keyword is identified once at a higher level, it is not entered again if it appears at a lower level in the same chapter. Thus the entry:

Weapons delivery . . . . . Chapter 2

eliminates the need for:

Weapons delivery:
  Ballistics 2-46
  Techniques 2-1

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Similarly, a series of entries could become an “alphabetized table of contents.” The example below illustrates:

Exterior light system 1-110
Anticollision lights 1-110
Controllable searchlight 1-111
Formation lights 1-112
Landing/hover lights 1-113
Navigation lights 1-114

d. Singular and plural entries. Separate entries are not made for singular and plural forms of the same noun. For example, the correct entry for Approach, Approaches is Approach(es).

e. Multiple entries. Multiple entries from the same page that have small differences shall be combined into one entry. Thus the entry:

Climb:
Military power charts 20-1

replaces the series:

Climb:
Military power 5,000 feet chart 20-1
Military power 10,000 feet chart 20-2
Military power 20,000 feet chart 20-3

f. Cross-references in indexes. Use a cross-reference to avoid using acronyms or abbreviations as keyword entries. An example is: SAR. (See Search and rescue).

g. Capitalization in indexes. Use an initial uppercase letter on the first word of entries of all levels.

h. Punctuation in indexes. Entries with multiple page numbers in page sequence; separate the first and last page references by a 1-en dash.

3.3.1.14.4 Last Page. The last page of the manual (see figure 105) shall:

a. Be the last left-hand page (the obverse page shall be blank).

b. When the publication is a classified one:

(1) Carry the classification on the publication centered at the top and bottom of the page.

(2) Carry the NAVAIR number of the publication centered in the middle of the page.

3.3.1.14.5 Back cover. The back clear tenite cover shall remain free of printing so that the color and any printing on the back page can be viewed through it.

3.3.2 Format requirements for NATOPS Flight Manuals. NATOPS Flight Manuals follow a strict format that standardizes ground and flight operational procedures, training requirements,

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and other technical operational information. This standardized format shall be closely adhered to when developing, producing, and updating NATOPS Flight Manuals.

3.3.2.1 Page size and layout. Page size for the NATOPS Flight Manual shall be 8-½ x 11 inches (51 x 66 picas). Page format shall be one-column. The right- and left-hand margins will be 4-½ picas each. Top margin shall be 5-½ picas; bottom margin shall be 5-½ picas. (See figure 106).

3.3.2.2 Marginal copy (including corner markings). Marginal copy shall consist of the following:

a. Publication NAVAIR number. The publication number and volume (if applicable) assigned by the procuring activity shall be placed at the top of each page, 2-½ picas above the top of the text matter and aligned flush right with the right margin of the text for right-hand pages and flush left with the left margin for left-hand pages. When the publication is classified, also place the NAVAIR number on the last page of the publication as shown on figure 106.

b. Page number. The page number shall be centered on each page, 2-½ picas below the bottom of the text. Page numbers shall consist of a chapter number and consecutive page numbers that are separated by a hyphen, with odd numbered pages right and even numbered left.

c. Page status. The status of the page (e.g., ORIGINAL, CHANGE 1, CHANGE 2 with IC 29 and IC 32, INTERIM CHANGE 27) shall be placed in the lower corner on the same base line as the page number, 2-½ picas below the bottom of the text; align flush right with the right margin of text for right-hand pages and flush left with the left margin for left-hand pages. Uppercase letters shall be used.

d. Classification markings. Where applicable, marginal copy shall also include security classification markings and the status of the reverse or obverse side, if blank. When the publication is classified, place the security classification of the manual at the top and bottom of the last page as shown on figure 106.

3.3.2.3 Text. All text shall be set in justified, 11-point Times New Roman or equivalent with 12-point interline spacing. Elements of text, such as procedural steps, shall be separated by a 1-pica space. Unnumbered paragraphs shall not be indented and shall be separated from the preceding paragraph by 2 picas of space.

3.3.2.3.1 Runover text. Runover lines of text shall be brought back to the left margin of the page regardless of the level of heading to which they apply. The only exceptions to this are: items of a listing or sublisting; warnings, cautions, and notes; bibliographies; glossaries; list of abbreviations and acronyms; and index.

3.3.2.3.2 Table of contents. Within the TOC, parts and chapters shall be set in all upper case using 11-point Arial bold or equivalent. Set paragraph headings in 11-point Times New Roman (primary in all uppercase; secondary in initial uppercase). Use 12-point interline spacing. Use a 1-pica space to separate preceding and succeeding material from parts and chapters and between the last secondary heading and the next primary heading. There shall be a 1-pica space above all primary headings. Include the heading INDEX and the page number Index-1 as the last entry in the table of contents.

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3.3.2.3.3 Parts. A NATOPS Flight Manual (NFM) is divided into groups of chapters called parts (see figure 107). Parts shall be identified with consecutive Roman numerals and a descriptive title. Separate them by divider pages that contain the part number and title centered on the right-hand face. Number divider pages with consecutive italic Arabic numerals following the page sequence of front matter; set in 11-point Times New Roman italic. For rapid reference, divider pages shall be tabbed with a bleed on the outside edge. The part page and the first four right-hand pages of each part shall be tabbed. Use the part page to list the chapter numbers and titles within the part.

3.3.2.3.4 Chapters. The NFM consists of chapters titled and numbered consecutively using Arabic numerals (see figure 108). The numbers, letters, and titles for chapters shall be centered at the top of the first page of text. Always begin chapters on right-hand pages. Set chapter numbers and letters in 14-point Arial bold and chapter titles in 24-point Arial bold. The space between the bottom of the top corner marking and the top of the chapter number is 5 picas. The chapter number and letter line is set in all uppercase letters; the chapter title line is set in initial capitals (main words only).

3.3.2.3.5 Paragraph headings and numbering. A decimal system shall be used to distinguish paragraphs down through the quaternary level. Divide text into paragraphs using sideheads. Use primary sideheads to divide text within chapters into main portions. There shall be at least one primary sidehead in each chapter. Identify all sideheads by appropriate sequential numbers and decimals. For example, 3.10 is the tenth primary sidehead in Chapter 3. Use secondary, tertiary, and quaternary sideheads to provide additional breakdown of material. For example, 9.5.2 is the second secondary sidehead under the fifth primary sidehead in Chapter 9. Figure 2 illustrates the method of numbering paragraphs.

a. Sideheads beyond the fifth level of indenture (e.g., 1.2.3.4.5) shall not be used. If necessary to further subdivide text beyond the fifth level, examine the material to see if editing would improve or clarify the arrangement. Breakdowns beyond the fifth level shall be alphabetized paragraphs such as “a” or “c” with descriptive titles. Treat an alphabetized paragraph with a title the same as a secondary sidehead with regard to type size and style and classification.

b. All sideheads shall be set flush left in 11-point Arial bold or equivalent with 12-point interline spacing. All sideheads (primary, secondary, tertiary, and quaternary) shall be standalone (not run in with text). Primary sideheads shall be set in all uppercase letters, unjustified, and separated by 10 points of space from preceding material (if followed by a sidehead) and 6 points of space from succeeding material. Secondary and subordinate sideheads shall be set in initial capitals, flush left.

c. Runover lines of all sideheads shall be aligned with the first word of the preceding line.

3.3.2.3.6 Listings and procedural steps. Listings and procedural steps may occur under any level of heading. Items in a list shall be numbered sequentially beginning with Arabic numeral 1. If it is necessary to use a sublisting, identify items in the following order: 1., a., (1), (a), 1), a). To separate the items of the list or procedure from regular body text, indent the Arabic numeral 1 pica and align all runover text with the first word of the first line. If necessary to use a sublisting or substeps, indent the lettered items of the sublisting or substeps 2 picas and align all runover text with the first word of the first line. Use a 1-pica space between listings and steps, sublisting and substeps, and to separate the first and last listings or steps from text.

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a. Always use a period at the end of each entry in a listing.

b. Do not supply headings for listings.

c. Bullets shall not be used for listings and sublistings.

d. Whether a list comprises complete or incomplete sentences, place a period after each step in the list.

e. List procedural steps numerically. In case of complicated procedures, divide into substeps listed alphanumerically. Procedural steps shall be in text form. Emergency procedures and commonly used procedures shall be listed in challenge-response checklist form. The former shall be presented in the third person indicative mood (e.g., “When the inverter fails, the caution light illuminates”); the latter shall be presented in the second person imperative mood (e.g., “Brakes — OFF.”).

3.3.2.3.7 Indexes. Letters of the alphabet shall be centered in 11-point Arial bold or equivalent and set off from the preceding and succeeding index entries by a 1-pica vertical space; set index entries in 11-point Times New Roman or equivalent with 12-point interline spacing. Place first-order entries flush-left with the margin of the column; indent second-order entries 1 pica and third-order entries, if necessary, 2 picas. Indent runover lines 2 picas to the right of deepest entry indent. Use an initial capital on the first word of all order entries.

3.3.2.3.8 Layout and readability. Page layouts shall be designed to conserve space without reducing usability or clarity of material. Avoid blank pages and spaces. Choose leading for best readability and conservation of space. Obvious expansion or reduction of leading or spacing to fill a page or balance columns is discouraged. Do not double space text. Slight variations are permitted to avoid layout practices that would result in:

a. A stand-alone heading or the first line of a paragraph falling at the bottom of a page or column or the last line of a paragraph beginning a new page or column. Paragraphs and listings that are two or three lines in length cannot be divided between columns or pages and shall be set as a unit.

b. Improperly split warnings, cautions, and notes (see 3.2.1).

c. Buried text. Unless the continuity is unmistakably clear, a small block of text shall not be set between or beneath illustrations to fill up space on the page. At least 10 lines of text in each column shall appear beneath an illustration.

3.3.2.3.9 Text requirements of figures, tables, and graphics. Reference data presented in tabular pages, charts, or graphs shall be designed for clarity and understanding. Material that takes up a relatively small space and is not otherwise referenced shall be set within the text without identification by title (e.g., a “spot” table — a condensed table with no title). Consider other tabular pages, charts, and graphs as illustrations; assign them figure numbers and reference them only by figure number within the text. Place each illustration as close as possible to the text where the primary discussion appears. Examine small, related artwork (“spot art”) to see if it can be combined into one illustration.

a. Figure numbers and titles. Assign figure numbers and titles to illustrations; place them beneath the illustration. Put figure numbers and titles in initial uppercase letters in the same size

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and style of type used for text. Center them approximately 1 pica below the figure. Use more than one line if necessary. Align the second and subsequent lines under the classification for classified manuals or with the first word of the figure title for unclassified manuals. If a figure is more than one page, place "(Sheet _____ of _____)" after the title. When text is set below a figure, allow at least 2 picas of open space below the last line of the figure number and title and above the first line of text (see figure 109).

3.3.2.3.10 Thumb indexing. Thumb or edge indexing tabs (bleeds) shall appear on the outer edge of each part page and the first four right-hand pages of each part (see figure 107). These tabs are aligned with the appropriate front cover tabs. Parts, appendixes, foldout pages, and the alphabetical index shall have edge indexing tabs. Edge indexing shall be superimposed over emergency page markings.

3.3.2.4 Figures.

a. Figures shall be inserted as close as possible following the primary textual reference.

b. When the textual reference and full-page width/not-full-page height figure appear on the same page, insert the figure at the top margin, not at the bottom margin of the page.

c. When a multisheeted figure includes one or several sheets that are not full page, treat these sheets as full-page art and do not "bury" text on them.

3.3.2.4.1 Foldouts. Foldout pages shall be used only when necessary to aid in locating information while reading text or when the illustration requires more than a full page width. Candidate illustrations for foldouts include cockpit arrangements and electrical systems. Foldout pages shall fall at the end of the respective NATOPS publication regardless of relationship to the applicable text. Foldouts shall follow the last page of the index. Foldouts shall always be presented as right-hand pages. The figure and foldout page number shall be visible when the page is folded.

a. Foldout size. Foldouts shall be prepared in the same size that they will appear in the respective NATOPS publication. The maximum image height is 9-½ inches; the maximum length is 46 inches, including a 9-inch apron (blank space to the left of art). Leave 3/8-inch margin at the unbound edge and a 3/8-inch margin outside the apron fold to ensure that the illustration can be seen when preceding pages cover the apron. The size of the apron is 8-½ inches.

3.3.2.5 Numbering.

a. Title page. Number the title page Arabic numeral 1 (11-point Times New Roman italic or equivalent).

b. Letter of promulgation. Number the LOP Arabic numeral 3.

c. Balance of the front matter. After the LOP, number the remainder of the publication front matter consecutively with Arabic numerals (11-point Times New Roman italic or equivalent). Refer to 3.3.1.2 for sequence of front matter.

d. Part pages. Number part pages by consecutive italic Arabic numerals (11-point Times New Roman italic or equivalent). Part pages are considered extensions of the front matter. If the

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last page of the front matter is 14, the first part divider page (Part I) will be numbered 15 (Reverse Blank).

e. Blank pages. Blank pages are normally left-hand pages, identified by (Reverse Blank) immediately after the preceding page number [e.g., 1 (Reverse Blank)]. If necessary to leave a right-hand page blank, mark the left-hand page immediately following "(Obverse Blank)" [e.g., 16 (Obverse Blank)]. The printing of "intentionally blank" pages is not permitted.

f. Page and illustration numbering.

(1) Chapters and appendices. Within chapters, pages and illustrations shall be numbered consecutively using Arabic numerals set in 11-point Arial bold or equivalent for page numbers, and 11-point Times New Roman or equivalent for figure numbers. All shall be assigned two-part numbers. For example, page 2-6 is the sixth page in Chapter 2, and figure 3-1 is the first figure in Chapter 3. Appendixes have titles and are identified using consecutive uppercase letters (see figure 110). Center the numbers, letters, and titles for chapters and appendixes at the top of the first page of text. Always begin chapters and appendixes on right-hand pages. Set chapter/appendix numbers and letters in 14-point Arial bold and chapter/appendix titles in 24-point Arial bold. The space between the bottom of the top corner marking and the top of the chapter/appendix number is 5 picas. The chapter/appendix number and letter line is set in all capitals; the chapter/appendix title line is set in initial capitals (main words only). Appendixes are considered chapters for page and figure numbering. Replace chapter numbers with letter designations. For example, page A-5 is the fifth page in Appendix A.

(2) Alphabetical indices. Number the pages of alphabetical indexes consecutively in 11-point Times New Roman italic, but replace the chapter number with the word "Index." For example, *Index-3* is the third page of the index.

(3) Foldout pages. Foldouts shall not carry figure numbers; the figure title shall be centered below the illustration. Foldouts shall be page numbered sequentially and carry (Reverse Blank). For example, the first three foldouts would be numbered FO-1 (Reverse Blank), FO-2 (Reverse Blank), and FO-3 (Reverse Blank). Identify multisheeted figures parenthetically by sheet number in the figure title. For example, Electrical System (Sheet 1 of 2) would be FO-1, and Electrical System (Sheet 2 of 2) would be FO-2. Do not assign a separate part or chapter designation to the group of foldouts (see 3.3.1.14.2).

(4) List of effective pages. The LEP pages shall be numbered as part of the front matter (see figure 20 and 3.3.1.2.7). Do not assign a figure number to the LEP.

3.4 Requirements for General Series NATOPS Manuals.

3.4.1 Technical content requirements. The General Series NATOPS Manuals present operations-oriented aviation-related subjects that usually encompass more than one type/model aircraft. They address subjects that are peripheral to those contained in the aircraft NATOPS Flight Manuals, and are not subject to the very detailed and standardized content requirements that are presented for NFMs. Although there are very few standardized contents requirements for the General Series NATOPS Manuals, they shall be developed with their purpose in mind and with emphasis on simplicity and understanding. The volume of information contained therein shall be presented in a direct manner with brevity in mind. Inclusion of excessive and overly detailed information shall be avoided.

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3.4.1.1 Front matter. The front matter required for the General Series NATOPS Manuals shall be the same as for the NFMs, except for the following:

a. The Summary of Applicable Technical Directives is neither required nor relevant.

b. The aircraft three-view illustration is not relevant.

c. Part pages are not required.

3.4.1.2 Main text.

a. The main body of text shall be organized into Chapters. Division of the main body into both parts and chapters, similar to NATOPS flight manuals, is not necessary but remains optional.

b. The contents of the manual shall be organized in a logical and useful manner. Consider the ease with which all subjects can be located within the table-of-contents structure.

c. The general writing style shall be the same as followed in the NATOPS flight manuals.

d. When discussing operations or procedures, the organizational structure shall be defined and discussed, addressing the more senior personnel and/or those with the greater authority first. Provide lists of responsibilities, duties, and functional relationships for each of the significant entities in the organization.

e. When describing systems and equipment, the same standards as provided for the NATOPS flight manuals shall be followed. (See 3.1.8.11).

f. Following a description of systems and equipment, equipment limitations, normal and abnormal operating procedures, and emergency procedures shall be addressed in that order, as appropriate.

g. Whenever practicable, emergency procedures shall be collected into one chapter or section of the manual so that cross-hatched emergency pages appear together in one location.

h. Training requirements and standards to be met for operator qualifications shall be presented, along with the length of time for which the qualification is valid and the procedures for renewing the qualification. A NATOPS question bank may be prepared for use during periodic review and requalification and attached as an appendix.

3.4.1.2.1 Chapter 1 — Introduction. Chapter 1 is the only chapter for which there are specific content requirements.

a. Chapter 1 shall be titled “Introduction.”

b. The first paragraph of the chapter shall address the purpose of the manual, as well as the subject of the publication and the users for whom it is intended.

c. The scope of the publication also shall be addressed. This shall be in the form of a brief overview of the manual.

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d. When parts are used, their contents shall be noted in this chapter.

e. The members of the NATOPS Advisory Group for the publication shall be identified.

f. Those commands selected to act as general approval and/or waiver authorities for qualifications, requirements, and procedures shall be identified. Such designations may be placed in the text of other chapters when considered more appropriate.

3.4.1.3 Back matter. Back matter shall be prepared and organized in the same manner as for the NATOPS flight manuals.

3.4.1.3.1 Appendices. Liberal use of appendices may be made in the general series NATOPS manuals. Appendices shall be relevant to the manual and shall be lettered alphabetically. Each appendix shall be titled, and its purpose and connection with the main body chapters and text shall be identified in the first paragraph of the appendix. Reference(s) to each appendix and its relevance to the main body subject matter shall also appear somewhere within the main body of the document.

3.4.2 Format requirements for General Series NATOPS Manuals. The format requirements for general series NATOPS manuals are the same as those for the equivalent sections of NATOPS flight manuals.

3.5 Requirements for NATOPS checklists.

3.5.1 Technical content requirements for NATOPS checklists. All information contained in checklists shall be extracted from the NATOPS flight manual or another equivalent source document and presented in condensed form. When the source document is other than the aircraft's NATOPS flight manual, the source document shall be identified in those checklists where the information appears. For example, launch and recovery wind envelopes are published in NAVAIR 01-80T-122. The NFM and checklist procedures shall agree in the following respects:

a. When changes are made to either the NFM or checklist publication, the same additions, revisions, or deletions should appear in both publications. Approval for inclusion or deletion of procedures/artwork in or from either or both publications shall be as obtained through either the NATOPS review conference or interim change process.

b. Both shall contain the identical number of steps. However, step subsets may not be necessary in the abbreviated checklist format.

c. The same capitalization shall be used in both publications (see figure 111).

3.5.1.1 Pocket Checklist (PCL). NATOPS pocket checklists are used by aircrew personnel as a ready–reference for normal procedures, emergency procedures, special procedures, and reference data during ground and flight operations in aircraft in which the aircraft NATOPS flight manual cannot be carried and readily used in the cockpit.

a. Arrangement of Pocket Checklist (PCL). The typical content and arrangement of data appearing in the pocket checklist is listed below. Variations in content and layout are permitted

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with the approval of the NATOPS Products Administrator when requested by the NATOPS Model Manager.

(1) Typical PCL data order:

Front protective Tenite cover
Title page (See figure 112)
List of effective pages (LEP) ("A" page) (See figure 113)
Interim Change Summary Page ("B" page) (See figure 114)
Normal/special procedures and reference data section –
Part 1: Normal procedures –
Flight briefing guide(s)
Normal procedures
Part 2: Special procedures –
Air refueling procedures
Special equipment preflight/alignment/operating procedures
Weapons procedures
In–flight troubleshooting and maintenance procedures
Part 3: Reference data/Limitations
Aircraft/equipment operating limitations and prohibited maneuvers (includes aircraft weight and balance limitations)
Performance charts and data (to include):
Aircraft systems data
Weapons data
Servicing checklist data
Climb, cruise and descent data
Takeoff and landing data
Emergency procedures section –
Part 1: Emergency procedure title page (See figure 112)
Emergency procedure index (See figure 115)
Emergency procedures
Special alerts and warnings, cautions, and advisory indications
Part 2: Bingo charts
Back page
Back protective Tenite cover

(2) Divider tabs. With the exception of the front matter, each section and part shall begin with a table of contents or an index of its contents. Typically, PCLs use tabs along the left–hand side of the section title pages to identify the locations of the emergency procedures and the normal/special procedures and reference data parts indexes and tabs along the bottoms of the pages to identify the special alert, warning, caution, advisory indicators, and the bingo portions of the PCL (see figure 116). Divider tab dimensions for top-fold PCLs are listed on figure 117.

(3) Divider tab labels. A tab shall be provided for access to each part in the PCL. At a minimum, tabs shall be provided to each of the parts identified in 3.5.1.1a(1), above. Subject to the approval of the NATOPS Products Administrator, when requested by the NATOPS Model Manager, standard parts may be divided and tabbed as separate parts, or additional parts may be added to address information unique to the aircraft. In the emergency procedures part, emergency procedures groupings shall be by aircraft system or phase of flight and will vary by aircraft

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model as determined by the assigned NATOPS Model Manager. Emergency procedures group titles shall be the same as the titles used for the title page side tabs or side locator bars. In the normal/special procedures and reference data section the normal procedures part may, subject to NATOPS Products Administrator approval, be tabbed separately as “flight briefing guide” and “normal procedures”, or the reference data part may be separated and tabbed as ”limitations” and “performance data”. Other part titles may be used when the substituted title more accurately describes the content, providing any such substitutions are first approved by the NATOPS Products Administrator.

b. Content requirements for the PCL. The PCL shall conform to the following requirements:

(1) Front protective Tenite cover. The PCL shall have a clear Tenite protective cover as described in 3.2.5.14d(4).

(2) Front matter section. The front matter section of the PCL contains the front title page, LEP and interim change summary. This section has no table of contents or index of its contents.

(3) Front title page. (See figure 112). The front title page shall contain the NAVAIR publication number; title; aircraft model; supersedure, distribution, and destruction notices; authorization notice, the basic or revision date, and the change number and change date (if applicable). It shall also contain the tab identification markings for the front (forward) facing portions of the PCL. An abbreviated distribution statement may be used if there are space restrictions.

(4) List of effective pages (“A” page). (See figure 113). The “A” page shall back up to the title page. It shall incorporate standard “A” page information.

(5) Interim change summary page (“B” page). (See figure 114). The “B” page shall be the same size as the title page.

(6) Table of contents or index for normal/special procedure and reference data section. The table of contents or index shall be the first tabbed page of each section. It shall be a complete table of contents or alphabetical index. Groupings will vary by aircraft model as determined by the Model Manager. Grouping titles shall be the same as the titles used for the title page side locator tabs (see figure 115).

(7) Normal/special procedures and reference data section. The normal procedures, special procedures, and reference data parts shall be separated from each other with tabbed pages that contain the text “Norm”, “Spec”, or “Ref” on each tab of the section. The first tab of each section visible from the front of the unopened document also shall be marked with the number that corresponds to the tab number on the cover page. Page numbering shall begin with page 1 on the first page of the normal procedures part index and run sequentially through the end of the normal/special procedures and reference data section.

(8) Normal procedures. Part 1 of the normal/special procedures and reference data section shall begin with a table of contents or index and include, but not be limited to, the following checks and procedures (see figure 120):

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Preflight inspections
Before start
Engine start
Before taxi
Taxi
Before takeoff checks
Takeoff
After takeoff
Climb
Cruise
Descent
Approach
Landing procedures and checklist
Waveoff
After landing
Before shutdown
Engine shutdown
Postflight

Further information for crewmen shall include equipment turnon/turnoff procedures, equipment alignment information, mission procedures, and other procedures appropriate to the crew position(s) for which the checklist is designed. When this type of information becomes too voluminous to be contained in one PCL, it shall be placed into a separate operator's or crewman's PCL. This decision shall be approved by the NATOPS Products Administrator.

(9) Special procedures. Part 2 of the normal/special procedures and reference data section shall begin with a table of contents or index and include information on special mission procedures applicable to the aircraft. Information shall include, but is not limited to:

(a) Air–to–air refueling.

(b) Special equipment preflight/alignment/operation (unless contained in the aircraft's tactical manual pocket guide(s)).

(c) Practice weapon delivery patterns (unless contained in the aircraft's tactical manuals pocket guide(s)).

(10) Reference data. Part 3 of the normal/special procedures and reference data section shall begin with a table of contents or index and include, but not be limited to, the following:

(a) Limitations. This portion shall include, but is not limited to, the following:

Aircraft/system operating limitations.
Weight and balance limitations.
Inflight refueling limitations.
Prohibited maneuvers.

(b) Performance data. This portion shall include the following information. Figure 119 contains typical performance data pages.

General information on aircraft, engine, gross weight, maneuvering, and system/ equipment limitations; and aircraft prohibited maneuvers.

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Weight factors, including stores
Weapons data (when no aircraft tactical manual pocket guide exists).
Servicing information (when no NATOPS servicing checklist exists) such as:
Fuels
Hydraulic fluid
Engine oils
Constant speed drive lubricants
Ground power units (including connecting procedures)
Tire pressures
Oxygen servicing
Pneumatic servicing
Tiedown points
Towing points
Jacking points
Aircraft climb, cruise, and descent data.
Takeoff and landing data (when no NATOPS TOLD cards exist)

(11) Table of Contents or index of emergency procedures. The emergency procedures table of contents or index shall begin on the first page that follows the emergency procedure section title page.

(12) Emergency procedures. Part 1 of the emergency procedures section shall begin with a complete table of contents or alphabetical index (see figure 115) and contain the procedures for all emergencies that are likely to be encountered. These procedures shall be in complete conformance with the procedures in the parent publication as specified by 3.5.1. The emergency procedures shall be presented in checklist form and abbreviated from the amplified checklist or procedures in the NATOPS flight manual (see figure 118). The abbreviation is to be accomplished by omitting explanatory material and reducing each check item to the minimum information necessary to describe the required action. For example, the step "Reduce airspeed to 130 KIAS for best glide" can be abbreviated "Airspeed – 130 KIAS glide." Explanatory information such as warnings, cautions, and notes shall be excluded from the PCL unless inclusion is considered essential. Authorization to include warnings, cautions, and notes shall be requested from the NATOPS Products Administrator.

(13) Bingo Data. Part 2 of the emergency procedures section shall begin with a table of contents or index and contain bingo tables for various probable normal and emergency aircraft divert configurations. These tables shall be separated from the emergency procedures by a tab titled "Bingo".

(14) Back page. The back title page shall be the front page of the emergency procedures section, and shall contain NAVAIR publication number, the basic or revision date and the change and change date (if applicable) as shown on figure 112 (Sheet 2) as well as the emergency procedure section title and the tab identification markings for the back (rearward) facing portion of the PCL.

(15) Back protective Tenite cover. See 3.5.1.1b.(1).

3.5.1.2 Card Checklist (CCL). Card checklists are two-sided laminated cards that normally contain aircraft normal operating procedures and common emergency procedures. Card check-

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lists are intended for use in aircraft in which a NATOPS flight manual can be carried and readily used in the cockpit.

a. Technical content and arrangement of CCL. The card checklist shall be arranged in a manner similar to that depicted on figure 121. The top of the card checklist shall indicate the publication (NAVAIR) number, followed by the aircraft designations covered, "NATOPS CHECKLIST," distribution statement, and supersedure notice. Separate headings shall identify "NORMAL PROCEDURES" and "EMERGENCY PROCEDURES." Normal and emergency procedures shall not appear on the same page. Emergency card checklists shall have black diagonal border markings bleeding off all four edges after trimming. Individual checklists shall be separated and identified by the checklist title that describes the evolution for which it is to be used (e.g., "TAKEOFF," "LANDING," etc.). Individual checklists shall not be continued from one side of the card to the other. Interim Changes that have been incorporated shall be indicated in the bottom right-hand corner of the first page.

b. The card checklist shall contain the normal aircraft and emergency checklists as agreed to at the formal NATOPS review conference. The Model Manager shall provide clarification in cases when the conference report is incomplete or unclear. The individual check items are derived by abbreviating the amplified checklists appearing in the NATOPS flight manual. This abbreviation is accomplished by omitting all explanatory material, including warnings, cautions, notes, and substeps, and retaining only the minimum challenge-response/action check item. For example, "Reduce airspeed to 130 KIAS for best glide" can be abbreviated "Airspeed — 130 KIAS glide." Individual checklists shall contain the same number of steps and appear in the same sequence as the checklists in the parent NATOPS flight manual. Card checklists shall not contain illustrations or classified information.

3.5.1.3 Servicing Checklist (SCL). The servicing checklist is a PCL-type publication for use by aircrew as a reference for servicing the aircraft in preparation for flight at remote locations. The servicing checklist shall contain information condensed from the NATOPS flight manual and NAVAIR maintenance publications to include:

a. Servicing equipment

b. Consumables

c. Electrical grounding

d. Cockpit entry

e. Fueling

f. Oil servicing

g. Hydraulic servicing

h. Oxygen servicing

i. Battery charging

j. External power application and removal

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k. External air application and removal

l. Aircraft jacking

m. Tire servicing

n. Danger areas

o. Turning radii and ground clearance

p. Towing and spotting

q. Tiedown and securing

r. Hazmat icons (where applicable to emphasize the hazards associated with toxic materials)

Note: In cases where the information is drawn from a NAVAIR maintenance publication, reference shall be made to that publication.

3.5.1.3.1 Arrangement of SCL. The SCL shall consist of the following pages:

a. Front cover. The SCL shall have a clear Tenite protective cover as described in 3.2.5.1.4d(4) for the pocket checklist.

b. Title page. The title page shall be the same size as the cover and contain the NAVAIR publication number; title; aircraft model; supersedure, distribution, and destruction notices; authorization notice; revision date; and change date and change number, if applicable, (see figure 122).

c. List of effective pages ("A" page). The "A" page shall back up the title page and be arranged as shown on figure 113. It shall incorporate standard "A" page information.

d. Interim change summary page ("B" page). The "B" page shall be arranged as shown on figure 114. It shall incorporate standard "B" page information.

e. Table of contents. Page 1 of the servicing checklist shall be a table of contents.

f. Servicing data pages. Each servicing data page shall contain a summary of all equipment required to perform any servicing. Both nomenclature and designation shall be included.

3.5.1.4 Functional Checkflight Checklist (FCFCL). The FCFCL is designed to provide aircrew members with the procedures to be followed when performing a post-maintenance checkflight, and provide maintenance personnel with an archivable record of the results of the checks performed. The FCFCL is an abbreviated checklist containing the individual procedures to be followed to verify proper operation of equipment for the condition(s) referred to in 3.3.1.5.5.

3.5.1.4.1 Technical content of FCFCL.

a. Front matter pages. (See figures 123 and 124.) The front cover/title page shall be unnumbered. The NAVAIR publication number shall be assigned by the procuring agency. The

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"A" page contains the List of Effective Pages and the Interim Change Summary. The page letter "A" shall be centered at the bottom of the page. The publication initial issue or revision date shall be placed in the lower right corner of the title page. The Change number and date shall be placed below this date, when applicable. The security classification shall be centered on the top and bottom of the title and "A" page. The supersedure, distribution, and destruction notices shall be centered below the aircraft model number.

b. Technical content pages and numbering. The pages of the FCFCL shall be numbered in Arabic numerals, beginning with the number "1." Technical content pages include the following information:

(1) List of conditions requiring a checkflight.

(2) Individual checkflight profiles. The individual checkflight profiles with their flight profile designators shall be included as provided in the NATOPS flight manual.

c. Illustrations. Use of illustrations shall be held to a minimum. Where illustrations are required, they shall be extracted and reduced from the parent NATOPS manual(s), if feasible, and located on the same page as the section of the step (or substep) that requires its use. Where collocating step and illustration on the same page is not possible, the illustration shall be on the following page and the statement "See the illustration depicting ____ on following page" added to the step.

d. Nomenclature. Nomenclature shall be consistent with that used in the applicable NATOPS flight manual.

e. Designations. Text in FCFCLs shall be limited to that pertaining to the model(s) of aircraft designated on title page. To facilitate later incorporation of additional series (type, model, BuNo effectivity) reference to type or model designation shall be held to a minimum consistent with clarity. These references shall be expressed in definite terms, such as type or model, BuNo applicability, or similar means.

f. Warnings, cautions, and notes. Warnings, cautions, and notes from the parent NATOPS manual(s) shall be omitted unless their inclusion is considered essential and is specifically authorized by the NATOPS Product Administrator.

3.5.1.4.2 Arrangement of the FCFCL. The FCFCL shall be arranged as follows:

a. Title Page (no Tenite cover). The title page is a self-cover. It shall contain the applicable information shown on figure 123.

b. Page "A". Page "A" (combined list of effective pages and interim change summary) shall back up to the title page and shall conform to the layout shown on figure 124.

c. Page 1. Page 1 within the FCFCL shall include the aircraft serial number, pilot's signature block, date, aircraft model, line drawing of all flight profiles, and a brief description of the individual flight profiles with their profile designators as depicted on figure 125. A diagram of the general checkflight profile and the list of the individually numbered check items in abbreviated form shall follow.

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d. General checkflight profile diagram. A diagram of the individual checkflight flight profile segments from pretakeoff to postlanding shall be depicted in a presentation of the combined flight profiles from pretakeoff through postlanding and shall provide the most efficient sequence for safely completing all of the checkflight item requirements. The depiction shall include the segment profile designators and altitudes at which applicable checks shall be accomplished. Each check item applicable to more than one profile shall have the applicable profile designators attached to it (see figure 125).

e. Numbered checks (individual checkflight check items). The individual checkflight items shall be presented in the same order as they are presented in the NATOPS flight manual. The information presented with each numbered check shall be organized as follows:

(1) Checkflight check items. (See figure 126). Each numbered item listed in the FCFCL shall be numbered identically with the corresponding item in the applicable NATOPS flight manual. The individual check items are derived by abbreviating the amplified functional checkflight procedures that appear in the NATOPS flight manual. This abbreviation is normally accomplished by omitting all explanatory material, including warnings, cautions, and notes, and retaining only the basic check item. For example, a 50-item procedure in the NATOPS flight manual shall have 50 identically numbered abbreviated items in the FCFCL. Subitems may not appear in the checkflight checklist.

(2) Profile column. All typical pages shall contain a profile column on the left image area with the profile designator letter(s) preceding all applicable numbered checks. The appropriate profile designator letter(s) shall precede each numbered check item throughout the FCFCL on the same line as the step number.

(3) Recording performance. A square or line following each check item shall be provided for pilot notations. A 1-pica square as provided where a simple notation of satisfactory (check mark) or unsatisfactory (X) performance is all that is required. Where a specific value or pressure is to be recorded, a line at least 5/8 inch long shall be provided for each data element to be recorded. In instances where multiple parameters are to be recorded, the use of a chart to organize the data required is recommended.

Note: When more than one crewmember is involved in acting on a check item, the crewmember(s) responsible for the action shall be identified by crew position identically as presented in the NATOPS flight manual.

f. Back cover. No special back cover will be provided. The last page containing the checks shall be used as the back cover.

3.5.2 Format requirements for NATOPS checklists.

3.5.2.1 Pocket Checklist (PCL).

a. Tab markings. All emergency sections have tabs with the tab number and major heading, as it appears on the locator bar on the cover, printed on the tab. It may be necessary to abbreviate the major heading. Do not use a font smaller than 7-point Arial bold (or equivalent) or run the heading beyond three lines. See figure 116. Ten emergency tabs and four normal procedures tabs are shown on the figures referenced above. It is possible to have less than ten or four depending on the particular aircraft. Actual page sizes are listed on figure 117.

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b. Warning/caution/advisories/bingo charts. Warning/caution/advisory and bingo tables appear at the end of the emergency section if they are to be included in the PCL. Each section has a bottom tab with the titles “Warning”, “Caution”, “Advisory” and “Bingo” or other appropriate title. The actual type of caution/warning/advisory or bingo chart is denoted on the side tabs.

c. Illustrations. The preparation and assembly of illustrations shall be in accordance with MIL-STD-38784. Illustrations shall normally be line drawings. Cartoons are not permitted. The use of color shall be approved by the NATOPS Products Administrator. Multisheeted figures do not use continued bars, but sheet numbers (e.g., Sheet 1 of 2, Sheet 2).

d. Type sizes, typefaces, and leading (spacing). Specified type sizes and faces may be varied to improve definition or clarity. Point sizes and leading may be adjusted to meet page requirements. Such variations shall have prior approval of the NATOPS Products Administrator.

(1) Emergency procedures. The leading between steps and sub-steps in the emergency section will depend on the size of the section. Ideally, a full line space between steps and substeps is maintained if this will not inordinately increase the size of the section. In large emergency sections, a half line of space is preferable. When granted by the NATOPS Products Administrator, emergency section text in large checklists may appear on both right-hand and left-hand pages. The type sizes and faces for presentation of emergency procedures shall be as follows:

(a) Primary heading (all caps) — 14-point Arial bold reverse (or equivalent).

(b) Secondary heading (all caps) — 14-point Arial bold (or equivalent). The first second-level reversed heading of an emergency tab is always run into the first-level heading.

(c) Tertiary heading (all caps) — 11-point Arial bold (or equivalent).

(d) Text — 10-point Arial bold (or equivalent).

(e) Procedural steps — 10-point Arial bold (or equivalent). Procedural steps shall be separated from the procedural responses and/or actions by dot leaders.

(f) Procedural responses and/or actions (all caps) — 10-point Arial bold (or equivalent). The response part of the step shall appear in initial capitals except where placarded/decaled control positions are used. Nomenclature shall be consistent with that used in the applicable NATOPS flight manual. Procedural responses and/or actions shall be justified flush right.

(g) Figure titles (all caps) — 10-point Arial bold (or equivalent). Figure titles shall be centered below the figure.

(h) NAVAIR number, page number, and change number (if applicable) — 9-point Arial bold (or equivalent).

(2) Performance data. Type used in charts, notes, etc., shall be of sufficient size to achieve desired definition. Performance data titles should be 10-pt Arial bold (all caps) or equivalent and centered beneath the data.

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(3) Other sections. Type for normal/special procedures and reference data shall be as follows:

(a) Primary heading (all uppercase) — 11-point Sans bold (or equivalent).

(b) Secondary headings (all uppercase) — 11-point Sans bold (or equivalent).

(c) Tertiary headings (initial uppercase) — 11-point Sans bold (or equivalent).

(d) Quaternary headings (all uppercase) — 10-point Times New Roman (or equivalent).

(e) Text — 10-point Times New Roman (or equivalent).

(f) Procedural steps — 10-point Times New Roman (or equivalent). Procedural steps shall be separated from the procedural responses and/or actions by dot leaders.

(g) Procedural responses and/or actions (all caps) — 10-point Times New Roman (or equivalent). The response part of the step shall appear in uppercase letters except where placarded/decaled control positions are used. Nomenclature shall be consistent with that used in the applicable NATOPS flight manual. Procedural responses and/or actions shall be justified flush right.

(h) Figure titles (all caps) — 10-point Arial bold (or equivalent). Figure titles shall be centered below the figure.

(i) NAVAIR number, page number, and change number (if applicable) — 9-point Arial bold (or equivalent).

3.5.2.2 Card Checklist (CCL).

a. Type sizes, typefaces, and leading (spacing). Specified type sizes and faces may be varied to improve definition or clarity. Point sizes and leading may be adjusted to meet page requirements. Such variations shall have prior approval of the NATOPS Products Administrator.

(1) Checklist title (all uppercase): Size B — 16-point Arial bold (or equivalent); size A — 24-point Arial bold (or equivalent)

(2) Individual checklist titles (all uppercase): Size B — 10-point Arial bold (caps) (or equivalent) (shall be boxed for emphasis); size A — 12-point Arial bold (or equivalent)

(3) Check items (all uppercase): Size B — 8-point Arial bold (or equivalent); size A — 10-point Arial bold (or equivalent)

(4) NAVAIR publication number: Size B — 8-point Arial bold (or equivalent); size A — 10-point Arial bold (or equivalent)

(5) Publication date (all uppercase): Size B — 12-point Sans bold (or equivalent); size A — 12-point Sans bold (or equivalent)

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3.5.2.3 Servicing Checklist (SCL).

a. Type sizes, typefaces, and leading (spacing). Specified type sizes and faces may be varied to improve definition or clarity. Point sizes and leading may be adjusted to meet page requirements. Such variations shall have prior approval of the NATOPS Products Administrator.

(1) Titles and primary side heads (all uppercase) — 11-point Arial bold (or equivalent)

(2) Subordinate side heads (all uppercase) — 11-point Arial bold (or equivalent)

(3) All line items (upper and lower case) — 10-point Times New Roman (or equivalent)

3.5.2.4 Functional Checkflight Checklist (FCFCL).

a. Image area. The image area (see figure 14) of an FCFCL page shall be 4-½ by 7 inches, excluding marginal copy. The image area shall be positioned as follows: approximately 3/8 inch from the left border, 1/8 inch from the right border, 3/8 inch from the bottom edge, and 5/8 inch from the top edge.

b. Marginal copy. The marginal copy (see figure 14) shall consist of the publication number, page number, and when applicable, the security classification. The publication number shall be centered approximately 1/4 inch above the image area. The page number shall be centered approximately 1/4 inch below the image area. When required, the security classification shall be centered above the publication number and the page number.

c. Type size. FCFCL title page type size shall be in accordance with figure 14. Figure titles shall be centered beneath the figures in 10-point Arial or equivalent. All other text and marginal copy shall be 10- or 11-point type. Arial boldface type or equivalent shall be used in the preparation of checklist pages (see figure 14).

d. Page layout. Whenever possible, pages shall be composed to avoid crowding and continuing related substeps for numbered steps from one page to another. When unrelated substeps under a numbered step are continued on a subsequent page, the segment heading shall be repeated followed by “(CONT).”

e. Indentation and spacing of checkflight items. The first line of each check shall be indented 1 pica for single-digit numbered checks and ½ pica for double-digit numbered checks. The first line of each subordinate check shall be indented 2 picas. The first line of each sub-subordinate check shall be indented 3 picas. The second and subsequent lines of main, subordinate, and sub-subordinate checks shall be aligned with the first word or number of text shown in the line above.

3.6 Requirements for additional NATOPS products.

3.6.1 Technical content requirements for additional NATOPS products.

3.6.1.1 Ditching and Bailout Placards (DBPs). Ditching and bailout placards are designed to be installed at each seat location for the aircrew member or passengers assigned to that station. The placards shall contain ditching procedures, bailout procedures, and the fire bill and depict

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the location of the parachute, liferaft, assigned emergency equipment, and the exit routes for use during ditching and bailout and situations (see figure 127). The ditching and bailout placards for each crewstation shall identify the exact same procedures as found for that crewstation on the ditching and bailout procedure pages in Part V of the flight manual. The ditching and bailout placards shall contain responsibilities to be performed by that position in the event of ditching and bailout. Depiction of the crewstation location within the aircraft, the location of the assigned parachute, and the routes to be taken in the event of ditching or bailout shall be included. The locations of emergency equipment for which that crewstation has responsibility also shall be depicted.

a. Cover page. A cover page shall be prepared for each set of placards (see figure 128). Content requirements for the cover page shall include the NAVAIR publication numbers, the aircraft and title, the list of effective pages, supersedure notice, a handling and distribution statement, a destruction notice, the date of the basic Change or Revision, and the Change number and date (if applicable). The supersedure notice shall be deleted when preparing the first Change.

3.6.1.2 Takeoff and Landing Data (TOLD) Cards. Takeoff and landing data cards are prepared as a deck of data cards for quick reference by flight crew personnel during takeoff and/or landing. The cards provide precalculated aircraft rotation and takeoff airspeeds, approach and landing airspeeds, takeoff and landing runway distances, and critical field lengths for various aircraft gross weights and configurations, and ambient conditions extracted and condensed from the performance charts in the aircraft NATOPS flight manual for ready reference.

a. Combined front cover and title page. The deck of TOLD cards shall contain a title page (which also serves as a cover page), and an A-page. The front cover shall contain the applicable information contained on figure 129. Content requirements for the cover page shall include the NAVAIR publication number, the aircraft and title, distribution statement, destruction notice, the list of effective pages, supersedure notice, the publication notice, the date of the basic Change or Revision, Change number and date (if applicable), and stock number with barcode. The supersedure notice shall be deleted when preparing the first Change.

b. "A" page (combined list of effective pages and interim change summary). The "A" page shall conform to the layout shown on figure 130 and shall reflect current data applicable to the TOLD card deck being delivered.

c. Takeoff and landing data pages. The data cards that follow the title page and combined List of Effective Pages and Interim Change Summary shall be printed on both sides and numbered sequentially with Arabic numerals, centered at the bottom of each page. The first page following the combined List of Effective Pages and Interim Change Summary page (i.e., page 1) shall contain the table of contents for the TOLD card package. Page 2 and subsequent pages shall include:

(1) V_1, V_r, and V_2 Speeds charts at standard flap/slat settings and outside air temperatures.

(2) RCR correction factors at standard flap/slat settings.

(3) Maximum recommended takeoff weight — single engine at standard flap settings.

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(4) Critical field length at standard flap settings.

(5) Minimum runway lengths at standard flap settings and outside air temperatures.

(6) A wind component table.

(7) Individual takeoff and landing data cards for aircraft gross weights at 1,000-pound increments with airspeeds and runway lengths required as extracted from the aircraft performance charts, as depicted on figure 131.

(8) A stall speed table for standard flap settings.

d. Back cover. The last data card in the TOLD card package shall serve as the back cover of the deck.

3.6.1.3 Passenger Information Card (PIC). Passenger information cards are single-sheet laminated cards printed on each side with passenger emergency data and are stored in the pocket on the backs of the passenger seats. The PIC shall contain following information extracted from the NATOPS flight manual, which may be simplified and condensed where necessary:

a. Location and use of emergency equipment, including fire extinguishers.

b. Procedures for seats and seatbelts, including brace position information.

c. Procedure for donning supplemental oxygen masks.

d. Procedures for donning lifejackets.

e. Emergency door and window locations and exit procedures, including procedures for opening emergency exit doors and removal of emergency exit windows.

f. Aircraft emergency slide and life raft locations and deployment procedures.

3.6.1.3.1 Arrangement of information. The passenger information card shall be arranged in a manner similar to that shown on figure 132. Information shall be simplified as much as possible to prevent ambiguity, so that a passenger will readily understand the information under emergency conditions. Use of illustrations is strongly encouraged to aid in passenger comprehension.

Note: Passenger information cards shall not contain classified information.

3.6.2 Format requirements for additional NATOPS products.

3.6.2.1 Ditching and Bailout Placards (DBPs).

a. Size. Ditching and bailout placards are normally sized so that two crew station diagrams fit on one side of a standard NFM page. A reasonable attempt to standardize the size of the entire set of ditching placards shall be made. When necessary, ditching placard sizes may be altered to contain additional text or to fit into a restricted space on the aircraft bulkhead. Size shall not exceed 6 inches by 12 inches.

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b. Layout. The top of the placard shall show the publication (NAVAIR) number. Under the NAVAIR number, the outline of the aircraft with the station to which the placard pertains shall be depicted. The aircraft orientation on the placard shall be such that its nose on the placard points in the general direction of the aircraft's nose when affixed at the crewstation in the aircraft. The basic or latest Revision or Change date shall be depicted in the lower right-hand corner. A red 3/16-inch reflective border shall be used to outline the perimeter of each placard. A 1-pica space shall be maintained between the inside of the red border and all text margins.

c. Type sizes, typefaces, and leading (spacing). Specified type sizes and faces may be varied to improve definition and clarity. Point size and leading may be adjusted to meet page requirements. Such variations shall have prior approval of the NATOPS Products Administrator.

(1) Placard title (all uppercase) — 14-point Arial bold (or equivalent)

(2) Ditching and bailout headers (all uppercase) — 14-point Arial bold (or equivalent)

(3) Procedural text (all uppercase) — 11-point Arial bold (or equivalent)

(4) NAVAIR publication number (all uppercase) — 8-point Arial bold (or equivalent)

(5) Publication date (all uppercase) — 8-point Arial bold (or equivalent)

3.6.2.2 Takeoff and Landing Data (TOLD) Cards.

a. Printed area. The size of the print area on each card is 5 inches by 5-7/8 inches, excluding the page number.

b. Type sizes, typefaces, and leading (spaces). Specified type sizes and faces may be varied to improve definition or clarity. Point sizes and leading may be adjusted to meet page requirements. Such variations shall have prior approval of the NATOPS Products Administrator.

(1) TOLD card title — 20-point Arial bold (or equivalent)

(2) Platform title — 30-point Arial bold (or equivalent)

(3) NAVAIR publication number — 16-point Arial bold (or equivalent)

(4) Procedural text (all uppercase) — 9-point Arial bold (or equivalent)

(5) Supersedure, Distribution Statement C, and destruction notice — 8-point Arial bold (or equivalent)

(6) Publication date(s) — 16-point uppercase Arial bold (or equivalent)

(7) Header text — 10-point uppercase Arial bold (or equivalent)

(8) Page number — 12-point Arial bold (or equivalent)

(9) Issuing authority statement — 8-point uppercase Arial bold (or equivalent)

(10) Figure titles — 12-point Arial bold (or equivalent)

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(11) LEP and interim change summary page — 8-point uppercase Arial bold (or equivalent)

(12) Column headers — 8-point uppercase Arial bold (or equivalent)

(13) Interim change summary box — 10-point uppercase Arial bold (or equivalent)

3.6.2.3 Passenger Information Card (PIC).

a. Type sizes, typefaces, and leading (spaces). Type sizes and faces may be varied to improve definition or clarity. Point sizes and leading may be adjusted to meet page requirements. If the information does not fit on two pages using 9-point type with 11-point leading, a smaller font or leading may be used; however, such variations shall have the prior approval of the NATOPS Products Administrator.

(1) NAVAIR publication number — 8-point uppercase Arial bold (or equivalent)

(2) PIC title — 11-point uppercase Arial bold (or equivalent)

(3) Platform title — 11-point uppercase Arial bold (or equivalent)

(4) Page number — 8-point uppercase Arial bold (or equivalent)

(5) Publication date — 9-point uppercase Arial bold (or equivalent)

(6) Procedural text (all uppercase) — 9-point Arial bold (or equivalent)

3.7 Submission and acceptance requirements for NATOPS information, publications and products.

a. Submission of information for incorporation into NATOPS publications and products. Acceptance by the procuring activity does not constitute final approval for incorporation of the information into NATOPS publications and products. The approval process for incorporation of information into the NATOPS documents is contained in the OPNAVINST 3710.7 series instruction. Since it is not the final NATOPS approving authority, the procuring activity shall ensure that copies of NATOPS documents are forwarded to (1) the NAVAIR NATOPS Products Administrator (AIR 4.0P) and (2) the Logistics Element Manager (LEM) at the Naval Air Technical Data and Engineering Service Command (NATEC) for review and assistance in obtaining the necessary NATOPS approvals. Timelines shall be coordinated in advance with the NAVAIR NATOPS Products Administrator (AIR 4.0P). Submission shall be made in sufficient time for deficiencies to be identified and recommended corrections to be returned to the procuring activity for incorporation prior to final acceptance of the material by the government. Following acceptance by the government, final copies of the material shall be forwarded to the Prime Contractor that holds the contracting vehicle and the Contracting Officer's Representative (COR) at the Naval Air Warfare Center Aircraft Division (NAWCAD), Code 3.2.6, as well as AIR 4.0P and the NATEC LEM for final approvals and incorporation into or production and distribution as NATOPS publications or products.

b. Acceptance of information for incorporation into NATOPS publications and products. Upon receiving information which has not yet been approved by a NATOPS review conference

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for incorporation into a NATOPS publication, the procuring activity shall, concurrent with its own review, submit the information to AIR 4.0P through the NATOPS Airworthiness Issue Resolution System (AIRS) website at https://airworthiness.navair.navy.mil as an AIRS item. In order to ensure the acceptability of the information for the NATOPS publications, the procuring activity shall obtain comments and approval for incorporation into the NATOPS publication as a NATOPS interim change from AIR 4.0P prior to accepting the procured information for the government.

c. Inspection of NATOPS publications and products during editorial production. Inspection of NATOPS publications and products during production. Inspection of NATOPS publication revisions and changes to ensure that the requirements of this specification are met should be conducted through in-process reviews and a final review as detailed in the current OPNAVINST 3710.7-series instruction. The number and location of inspections shall be as required in the solicitation or procurement contract, with additional inspections authorized if necessary to ensure proper incorporation of the information. In-process inspections of newly created source data, review conference reports, and interim change replacement pages may be waived by the procuring activity when considered unnecessary.

d. Inspection of NATOPS publications and products after delivery. Except for copies of draft publications and products that are to be replicated locally, master copies of each paper and electronic NATOPS publication or product shall be forwarded by the procuring activity to AIR 4.0P and to the NATOPS LEM at NATEC for inspection, posting, replication, and distribution. The NATOPS LEM will inspect the paper NATOPS products to ensure they meet publishing standards for printing and distribution, and will coordinate the inspection of electronic NATOPS products to ensure that they conform with information technology standards for CD-ROM and web compatibility. This includes inspecting the first-copy prior to mass replication, and passing any editorial deficiencies to the procuring activity for correction. The procuring activity shall delay acceptance of the NATOPS publications and products for the government until after notification by the NATEC LEM that the delivered products are acceptable.

4. VERIFICATION

4.1 Verification. Verification of the NATOPS publications and products produced under this specification is normally accomplished through in-process and final review. Verification for acceptance by the government is the responsibility of the procuring activity and shall be specified in the contract or solicitation (see 6.2).

4.1.1 NATOPS Model Manager responsibility. The NATOPS Model Manager shall have responsibility for ensuring that all NATOPS changes/updates/modifications resulting from a NATOPS review conference have been correctly and accurately incorporated into the appropriate NATOPS publications and products. The Model Manager shall also have responsibility for developing Preliminary NATOPS publications and products, with the requisite engineering and test and evaluation support from NAVAIRSYSCOM. Additionally, the Model Manager shall have responsibility for a draft and final review of NATOPS publications and products prior to submission to NATEC for subsequent mass production (printing and distribution of hard-copies and electronic copies and uploading to the NATEC website).

4.1.2 COMNAVAIRSYSCOM responsibility. COMNAVAIRSYSCOM (specifically AIR-4.0P) shall have responsibility for all engineering data incorporated into NATOPS publica-

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tions and products (for all phases of NATOPS development, including draft, partial, preliminary, and promulgated). AIR-4.0P shall also have responsibility for acceptance of final NATOPS publications and products prior to delivery to NATEC.

4.1.3 NATEC responsibility. The NATEC logistics element manager (LEM) shall have responsibility for ensuring that all NATOPS publications and products and the requisite electronic files and documentation submitted for mass production meet the required specifications for printing, web-based accessibility, and data standards. The NATEC LEM shall have responsibility for acquisition of mass-produced, "finished" products for distribution. Prior to mass-production of NATOPS publications and products, the GPO-approved contract printer shall generate a "blue-line review" copy of each publication or product. This is a *final* pre-press check of exactly what the printer will print in hard-copy. The NATEC LEM shall have primary responsibility for the "blue-line review."

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. NATOPS publications and products prepared in accordance with the requirements of this specification are intended for use by the flight crews operating the aircraft and ground personnel supporting aircraft operations.

6.2 Acquisition requirements. Acquisition documents should specify the following information.

a. Title, NAVAIR number, and date of each publication or product to be prepared.

b. Aircraft models for which the NATOPS publications and products are to be prepared (see 1.2).

c. Specific issues of the individual documents cited in Section 2 for use with this acquisition (see 2.2.1, 2.2.2 and 2.3).

d. Preparation of a change or revision.

e. Requirement for copyright release (see 3.1.3).

f. Approval of graphs (see 3.2.3.7.5a.).

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g. Flight test data requirements (see 3.3.1.13.2f.(1)).

h. ATO ignition time (see 3.3.1.13.3c.(8)).

i. Type of range charts (see 3.3.1.13.4d.).

j. Bingo chart reserve fuel allowance (see 3.3.1.13.4d.(7)).

k. Type of endurance charts (see 3.3.1.13.4e.).

l. Mission profile – maximum range fuel allowance (see 3.3.1.13.5d.(1)).

m. Standard data charts for helicopters (see 3.3.1.13.7a.).

n. Verification requirements (see 4.1 and OPNAVINST 3710.7-series).

o. Packaging requirements (see 5.1).

6.3 Technical Manuals. The requirement for technical manuals should be considered when this specification is applied to a contract. If technical manuals are required, specifications and standards that have been authorized and assigned an Acquisition Management System Control (AMSC) Number must be listed on a separate Control Data contract. The technical manuals must be acquired under separate contract line items in the contract.

6.4 Guidance documents. The documents cited in this section are provided for guidance and information only.

6.4.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks comprise a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

6.5 Tailoring guidance. To ensure proper application of this detail specification, invitation for bids, requests for proposals, and contractual statements of work should tailor the requirements in sections 3 and 4 of this detail specification to exclude any unnecessary requirements. For example, if the statement of work requires a revision to pocket checklist publication, then paragraphs in this detail specification related to other types of publications may be excluded provided their requirements are not referenced by the pocket checklist requirements.

6.6 Submission and acceptance of NATOPS information, publications and products. (See 3.7).

6.7 Definitions.

a. Change. A change is a modification of information in an existing publication.

b. Copy-freeze date. The copy-freeze date is the date after which no further changes to the contents of a publication will be accepted. Copy-freeze dates are established to avoid undue delays and increased costs in preparing updates to publications. Procuring activities establish a copy-freeze date for each NATOPS publication that will be placed into production. All completed, approved changes submitted to the editors by the copy-freeze date are incorporated into

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the resultant change or revision to the applicable publication. Unless otherwise directed by the procuring activity, changes received after the copy-freeze date are not incorporated, but are held by the NATOPS Model Manager for incorporation into the next change or revision of the publication. If required, refer to OPNAVINST 3710.7-series for additional information about assigning the copy-freeze date for a NATOPS publication.

c. Preliminary issue. A preliminary flight manual is normally intended for interim use to make the information available for test, verification, or training purposes.

d. Publication date. The publication date is set by the procuring activity and always falls on the first or fifteenth of the month. If the copy-freeze date falls on the first or fifteenth day of the month, the publication date will normally be the same as the copy-freeze date. If not, the publication date will normally be on the one that immediately follows the copy freeze date.

e. Reprint. A reprint is a second or subsequent printing of a publication Change or Revision.

f. Revision. A Revision is a second or subsequent edition of a publication that supersedes the preceding edition.

g. Supplement. A supplement is a document that complements or augments information in a NATOPS flight manual.

6.8 Changes from previous issues. Because of the extensive reorganization and large number of changes incorporated in this revision, marginal notations (*) have not been used in this revision to identify information that has been changed from that in the previous issue. However, major departures in this MIL-DTL from the information contained in the superseded NATOPS specifications are as follows:

a. Combines NATOPS flight manual requirements and NATOPS checklist requirements into one specification document.

b. Adds requirements for NATOPS manuals and deletes those for NATOPS scroll checklists.

c. Creates shell NATOPS flight manuals for use with selected commercial derivative aircraft Original Equipment Manufacturer's (OEM) manuals and checklists and avoids the added overhead of producing separate parallel NATOPS publications for those aircraft.

d. Replaces camera-ready copy and photographic-based requirements with direct-image copy, computer graphics, and CD-ROM requirements to reflect the use of computers in preparing and publishing NATOPS paper and electronic publications and products.

e. Replaces obsolete text styles and fonts with those currently available in computer publishing programs.

f. Adds use of hypertext links between tables of contents, indexes, definitions, text strings, numbered paragraphs, figures, non-text objects, and other related information in electronic NATOPS publications and products.

g. Reflects the use of websites and e-mail for accessing referenced specifications and data, for reporting errors in NATOPS products, and submitting change recommendations to NATOPS publications and products.

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6.9 Subject term (keyword) listing:

Arrangement
Artwork
Card checklist
Ditching and bailout placards
Flight manual supplements
Functional checkflight checklist
NATOPS checklist
NATOPS flight manual
Partial flight manual
Pilot pocket checklist
Preliminary flight manual
Servicing checklist

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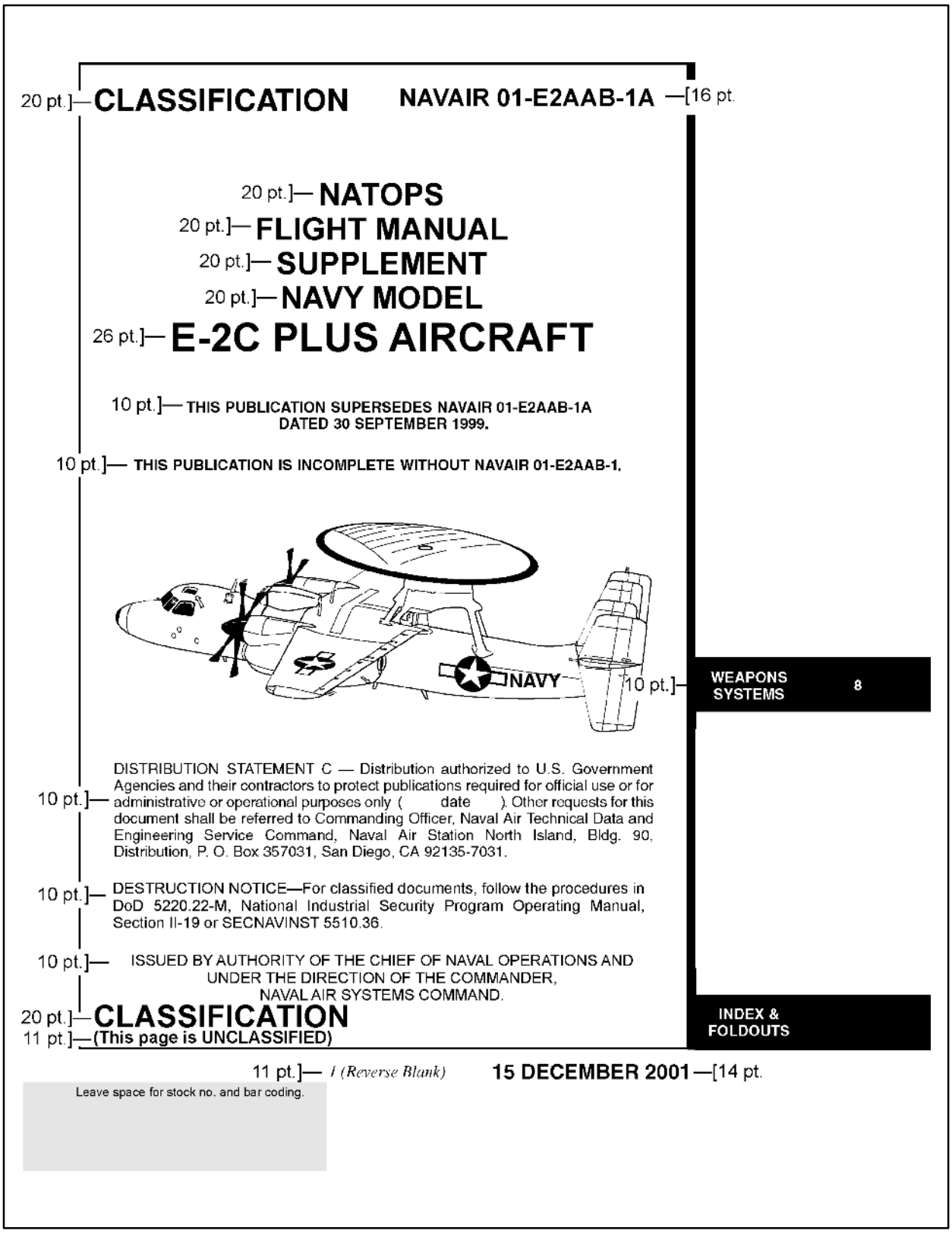


FIGURE 1. Sample classification markings on NATOPS flight manual title page.

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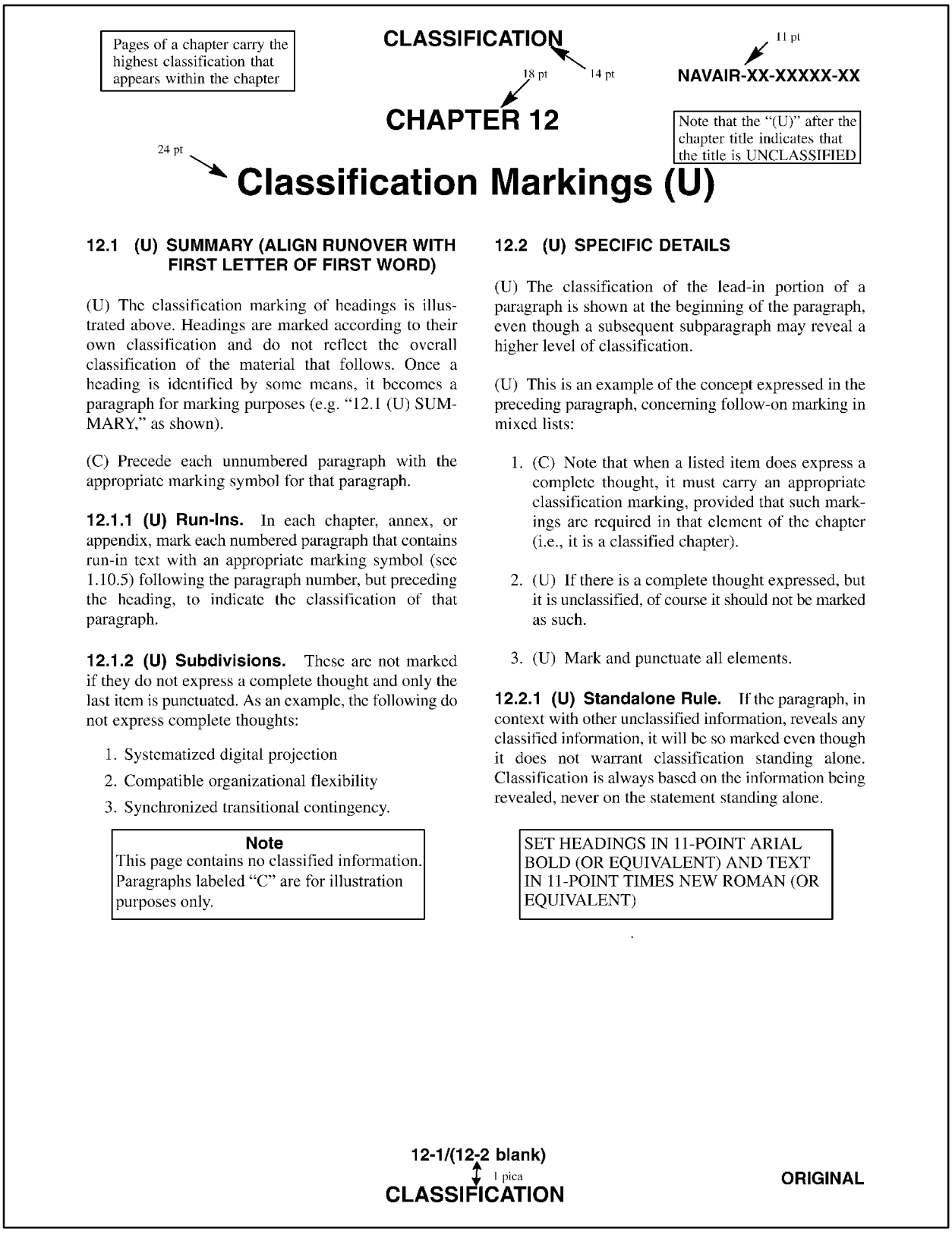
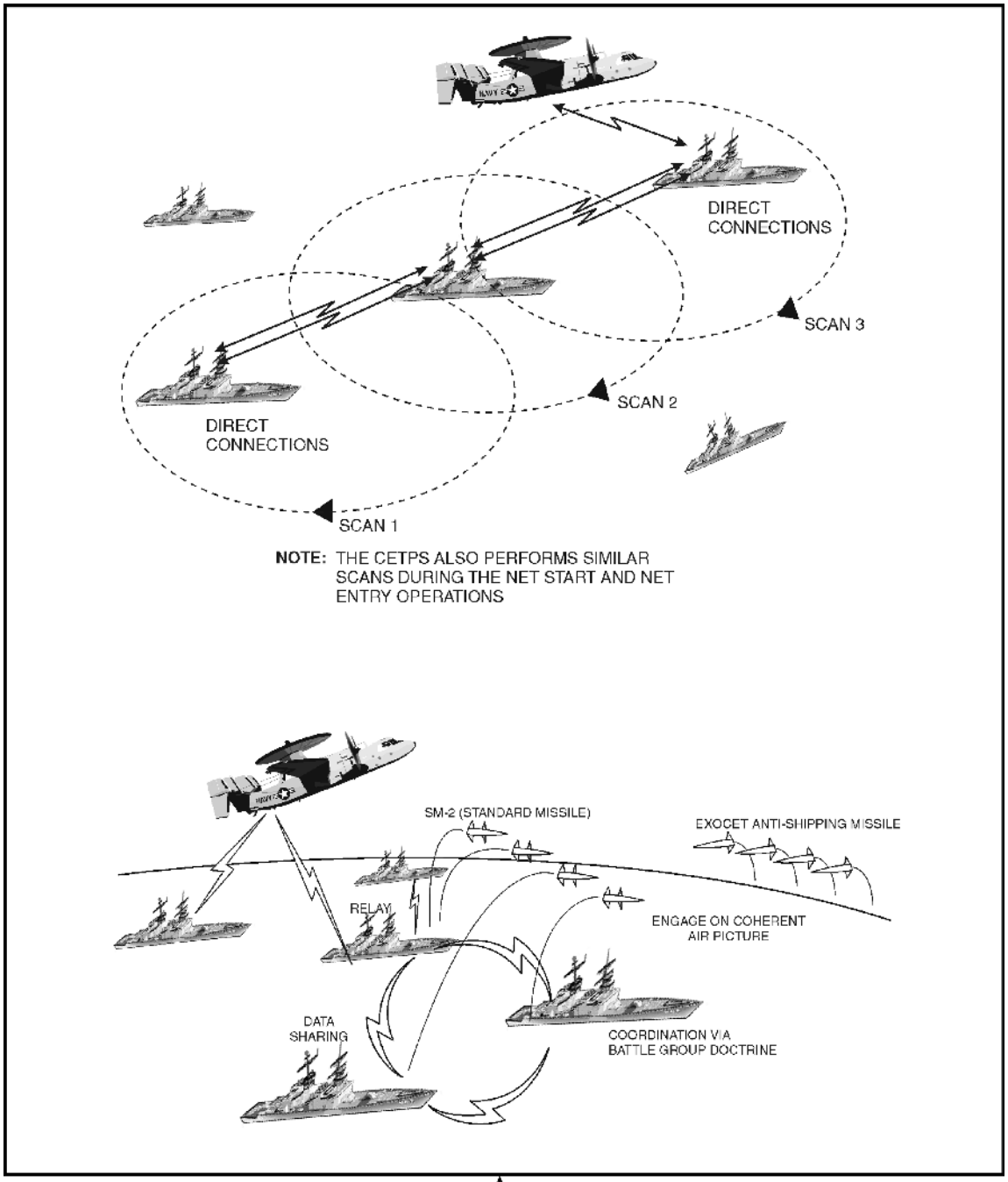


FIGURE 2. Sample of paragraph classification and numbering.

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CLASSIFICATION

PUBLICATION NO.



10 pt. Arial] CLASSIFICATION (1 pica above and 1 pica below)

Figure 7-2. (U) CETPS Communications

7-8

ORIGINAL

CLASSIFICATION

FIGURE 3. Sample figure classification markings.

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PREFACE

SCOPE

The NATOPS Flight Manual is issued by the authority of the Chief of Naval Operations and under the direction of Commander, Naval Air Systems Command in conjunction with the Naval Air Training and Operating Procedures Standardization(NATOPS) Program. This manual contains information on all aircraft systems, performance data, and operating procedures required for safe and effective operations. However, it is not a substitute for sound judgment. Compound emergencies, available facilities, adverse weather or terrain, or considerations affecting the lives and property of others may require modification of the procedures contained herein. Read this manual from cover to cover. It is your responsibility to have a complete knowledge of its contents.

APPLICABLE PUBLICATIONS

The following applicable publications complement this manual:

A1-H53ME-NFM-500 (Pilot's Pocket Checklist)

A1-H53ME-NFM-700 (Functional Checkflight Checklist)

NTTP 3-22.5-MH53E TPG (Vol 1 and 2) A1-H53BE-CLG-000 (Cargo Loading Manual)

FMFRP 5-3; NWP 42-1 VOL I, II, and III (Basic Operation/Equipment and Single-/Dual-point Hook Procedures)

NAVAIR 00-80T-110 (NATOPS Air Refueling Manual)

HOW TO GET COPIES

Additional copies of this manual and changes thereto may be procured through the local supply system from NAVICP Philadelphia via DAAS in accordance with NAVSUP P-409 (MILSTRIP/MILSTRAP), or a requisition can be submitted to Naval Supply Systems Command via the Naval Logistics Library (NLL) website, www.nll.navsup.navy.mil. This publication is also available to view and download from the NATEC website, www.natec.navy.mil.

Automatic Distribution (with Updates)

This publication and changes to it are automatically sent to activities that are established on the Automatic Distribution Requirements List (ADRL) maintained by Naval Air Technical Data and Engineering Service Command, in San Diego, CA. If there is continuing need for this publication, each activity's Central Technical Publication Librarian must send a revised ADRL report on floppy disk to Naval Air Technical Data and Engineering Service Command. If an activity does not have a library, send a letter to the Commanding Officer, Naval Air Technical Data and Engineering Service Command, Naval Aviation Depot North Island, Bldg. 90, Code 3.3A, P.O. Box 357031, San Diego, CA 92135-7031, requesting assignments of a distribution account number (if necessary) and automatic mailing of future issues of the publications needed.

Note

The ADRL floppy disk can be used only to place an activity on the mailing list for automatic distribution of future issues of the publication. It cannot be used to make one-time orders of publications from current stock.

Once established on automatic distribution for this or any other NAVAIR technical publication, an activity must submit an ADRL report on floppy disk at least once every 12 months to update or confirm their automatic distribution requirements.

FIGURE 4. Sample NATOPS flight manual preface including change bar and warning, caution, and note symbols.

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Note

Activities not submitting an ADRL report on floppy disk for more than 12 months may be dropped from distribution of all NAVAIR technical publications.

UPDATING THE MANUAL

To ensure that the manual contains the latest procedures and information, NATOPS review conferences are held in accordance with OPNAVINST 3710.7 series.

CHANGE RECOMMENDATIONS

Recommended changes to this manual or other NATOPS publications may be submitted by anyone in accordance with OPNAVINST 3710.7 series.

Routine change recommendations are submitted directly to the Model Manager on OPNAV 3710/6 (4-90) shown herein. The address of the Model Manager of this helicopter is:

Commanding Officer
Department of the Navy
Airborne Mine Countermeasures Weapon
Systems Training School (AWSTS)
483 A Street
Norfolk, Virginia 23511-4289
Attn: Model Manager MH-53E
Telephone:
Commercial (757) 444-1615
Autovon 564-1615

Change recommendations of an URGENT nature (safety of flight, etc.), should be submitted directly to the NATOPS Advisory Group Member in the chain of command by priority message.

YOUR RESPONSIBILITY

NATOPS Flight Manuals are kept current through an active manual change program. Any corrections, additions, or constructive suggestions for improvement of its content should be submitted by routine or urgent change recommendation, as appropriate, at once.

NATOPS FLIGHT MANUAL INTERIM CHANGES

Flight Manual Interim Changes are changes or corrections to the NATOPS Flight Manuals promulgated by CNO or NAVAIRSYSCOM. Interim changes are issued either as printed pages, or as a naval message. The Interim Change Summary page is provided as a record of all interim changes. Upon receipt of a change or revision, the custodian of the manual should check the updated Interim Change Summary to ascertain that all outstanding interim changes have been either incorporated or canceled; those not incorporated shall be recorded as outstanding in the section provided.

CHANGE SYMBOLS

Revised text is indicated by a black vertical line in either margin of the page, like the one printed next to this paragraph. The change symbol shows where there has been a change. The change might be material added or information restated. A change symbol in the margin by the chapter number and title indicates a new or completely revised chapter.

FIGURE 4. Sample NATOPS flight manual preface including change bar and warning, caution, and note symbols. — Continued

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WARNINGS, CAUTIONS, AND NOTES

The following definitions apply to WARNINGs, CAUTIONs, and Notes found throughout the manual.

WARNING

An operating procedure, practice, or condition, etc., that may result in injury or death, if not carefully observed or followed.

CAUTION

An operating procedure, practice, or condition, etc., that may result in damage to equipment, if not carefully observed or followed.

Note

An operating procedure, practice, or condition, etc., that is essential to emphasize.

WORDING

The concept of word usage and intended meaning adhered to in preparing this Manual is as follows:

1. "Shall" has been used only when application of a procedure is mandatory.
2. "Should" has been used only when application of a procedure is recommended.
3. "May" and "need not" have been used only when application of a procedure is optional.
4. "Will" has been used only to indicate futurity, never to indicate any degree of requirement for application of a procedure.

FIGURE 4. Sample NATOPS flight manual preface including change bar and warning, caution, and note symbols. — Continued

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ERRATUM ERRATUM

ERRATUM
16 JUNE 2002

TO

E-2C
NATOPS FLIGHT MANUAL
NAVAIR 01-E2AAB-1
01 FEBRUARY 1998
CHANGE 2 — 15 JUNE 2002

This Erratum package includes this Erratum Notice and attached pages as follows:
a. Erratum 16 June 2002 replacement title page.
b. Erratum 16 June 2002 replacement pages 31-107/31-108.

Error(s) addressed in this erratum:
a. Incorrect barcode appears on existing title page.
b. Red overlay omitted in Figure 31-7 on page 31-107.

Enter this erratum into NAVAIR 01-E2AAB-1 as follows:
1. Replace existing title page with enclosed title page.
2. Replace existing pages 31-107/31-108 with enclosed page 31-107/31-108.
3. Annotate incorporation of this erratum on the Record of Changes page.
4. Destroy removed pages by any method that will prevent disclosure of the contents or reconstruction of the deleted pages and discard this erratum sheet.

DESTRUCTION NOTICE — For unclassified, limited documents, destroy by any method that will prevent disclosure of contents or reconstruction of the document.

ISSUED BY AUTHORITY OF THE CHIEF OF NAVAL OPERATIONS AND UNDER THE DIRECTION OF THE COMMANDER NAVAL AIR SYSTEMS COMMAND

Page 1 of 1

FIGURE 5. Sample erratum.

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CHAPTER 12

Emergency Procedures

12.1 MEMORY ITEMS

Checklist items indicated with an * shall be completed without reference to the checklist.

12.2 INTRODUCTION

The terms IMMEDIATELY, POSSIBLE, and PRACTICAL, as used in this manual, refer to the degree of urgency with which a landing must be made.

1. Land Immediately — Self-explanatory.
2. Land as Soon as Possible — Means land at the first site at which a safe landing can be made.
3. Land as Soon as Practical — Means extended flight is not recommended. The landing site and duration of flight is at the discretion of the pilot-in-command.

Any helicopter experiencing trouble in flight shall immediately notify the flight leader by radio or by visual signals, as the situation dictates. If the nature of the emergency warrants an immediate return to base, a radio call shall be made to enable the base to prepare for an emergency landing. In any case, the following information shall be transmitted:

*1. Side number of the helicopter.

*2. Position.

*3. Difficulty.

*4. Intentions.

*5. Fuel state.

If the helicopter with the emergency does not have radio communications, all possible information is relayed visually to the wingman who makes the necessary radio transmission.

Weight Reduction — Any emergency situation discussed in this section assumes that jettisoning of external cargo is an acceptable step in any procedures.

Note

The term PICKLE means: Jettison external loads, AMCM equipment, or fuel, as applicable.

12.3 PRECAUTIONARY LANDINGS

A precautionary landing is described as a landing when further flight is possible but inadvisable. Such landings will be governed by the following:

FIGURE 6. Emergency borders.

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Equivalent Units of Measure

Measure	English System	Metric System
Diameter	inches feet or yards miles (statute)	millimeters meters kilometers
Dimensions	all in inches feet and inches all in feet	millimeters centimeters meters
Distance	inches feet or yards miles (statute)	millimeters meters kilometers
Liquid Measure	gallons	liters
Pressure	pounds/square inch	kiloPascals
Volume	cubic feet, gallons, or barrels	cubic meter
Weight	ounces pounds tons(1)	grams kilograms tonnes or kilograms

NOTES:

1. A ton in the English system is a short ton (2,000 lb). Use kg (2.2046 lb) unless metric tons are specified. A metric ton or tonne (t) is 2,204.6 lb (or 1,000 kg). Avoid using MT (megaton) as the abbreviation for metric ton.

FIGURE 7. English and metric units of measurement.

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Factors for Converting Between English System and Metric System Units

	UNITS KNOWN:	UNITS TO FIND:	MULTIPLY BY:
LENGTH/DISTANCE	inches	millimeters	25.4 mm/in
	feet	centimeters	30.48 cm/ft
	feet	meters	0.3048 m/ft
	yards	meters	0.9144 m/yd
	statute miles	kilometers	1.6093 km/mi
	nautical miles	kilometers	1.852 km/nm
	millimeters	inches	0.0394 in/mm
	centimeters	inches	0.3937 in/cm
	meters	feet	0.36453 ft/m
	meters	yards	1.0936 yd/m
	kilometers	miles	0.621 37 mi/km
WEIGHT	ounces	grams	28.35 g/oz
	pounds	kilograms	0.4536 kg/lb
	short tons (2,000 lb)	metric tons (t)	0.907 t/T
	long tons (2,240 lb)	metric tons (t)	1.016 t/T
	grams	ounces	0.007626 oz/gm
	kilograms	pounds	2.2046 lb/kg
	metric tons (t)	short tons (2,000 lb)	1.1023 short ton/t
	metric tons (t)	long tons (2,240 lb)	0.9842 long ton/t
PRESSURE	pounds per square inch	kiloPascals	6.894757 kPa/psi
VOLUME	cubic inches	cubic centimeters	16.39 cm^3/in^3
	cubic feet	cubic meters	0.0283 m^3/ft^3
	cubic yards	cubic meters	0.76446 m^3/yd^3
	barrels	cubic meters	0.15899 m^3/bbl
	cubic centimeters	cubic inches	0.061 in^3/cm^3
	cubic meters	cubic feet	35.3146 ft^3/m^3
	cubic meters	cubic yards	1.3079 yd^3/m^3
LIQUID VOLUME	pints	liters	0.47315 L/pt
	quarts	liters	0.94625 L/qt
	gallons	liters	3.785 L/gal
	liters	pints	2.11344 pt/L
	liters	quarts	1.05672 qt/L
	liters	gallons	0.26418 gal/L
TEMPERATURE	degrees Fahrenheit	degrees Celsius	5/9 (after subtracting 32)

Note: 1 barrel = 42 gallons

FIGURE 8. Factors for converting between English units and metric units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
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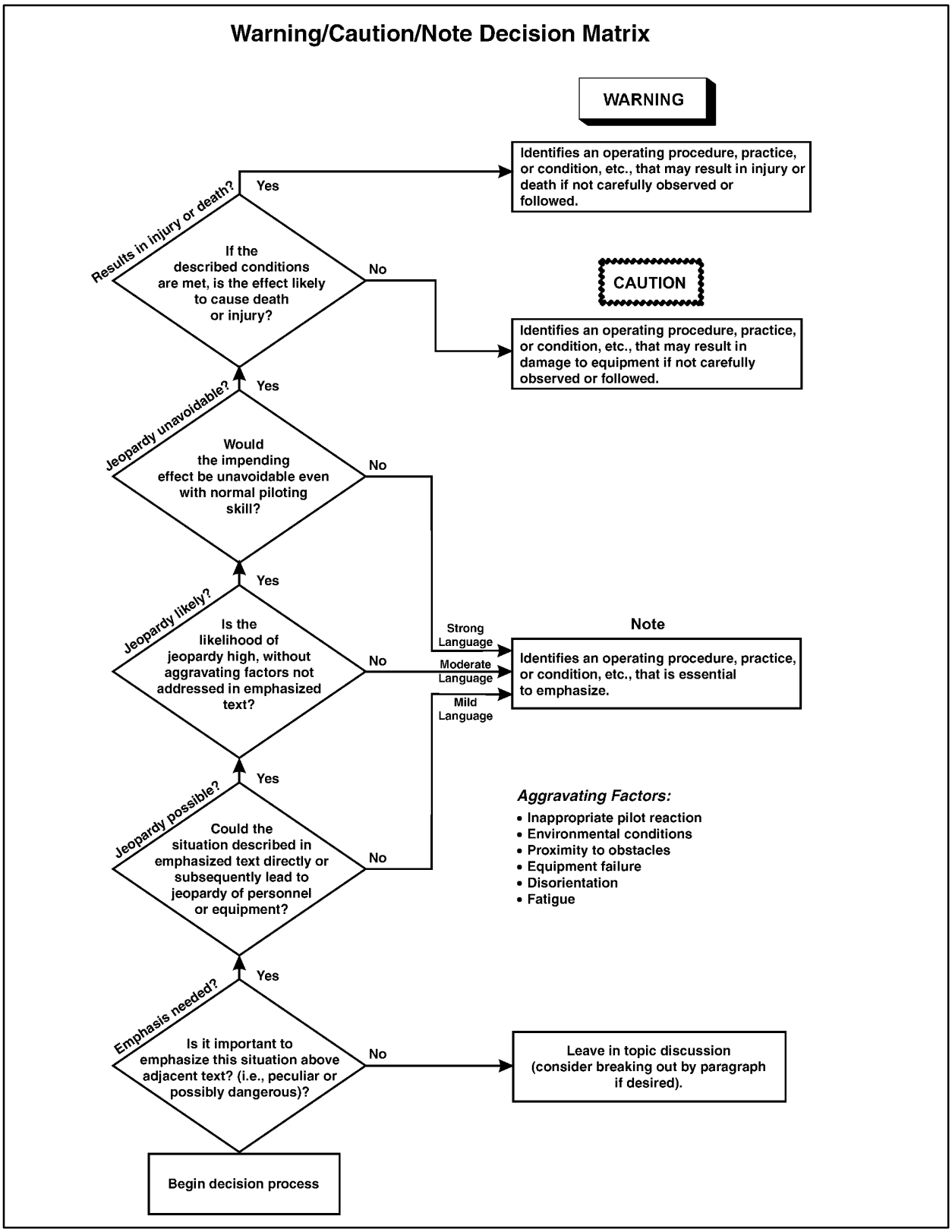
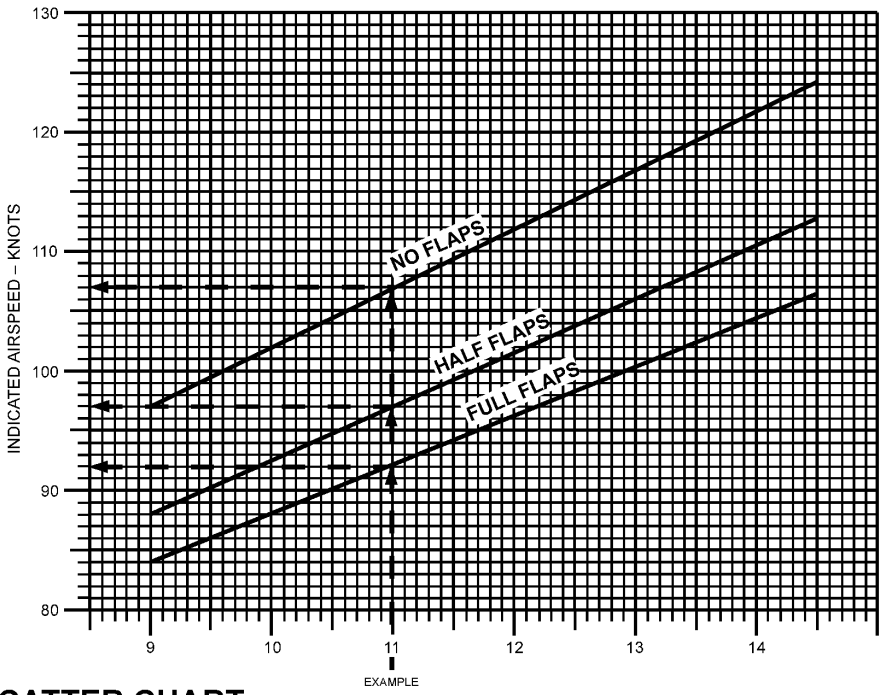


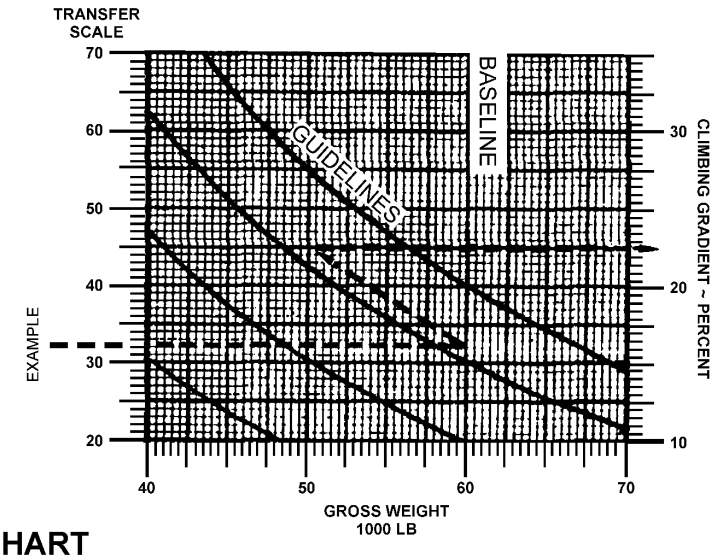
FIGURE 9. Warning/caution/note decision matrix.

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SCATTER CHART

This chart type is sometimes called a “scatter” chart. In this chart, one input variable enters into the determination of more than one output. The example line extending from the initial input axis intersects more than one function line or family of lines without changing direction. Each time a line is intersected, the example line “scatters” to an output axis and another output is determined.

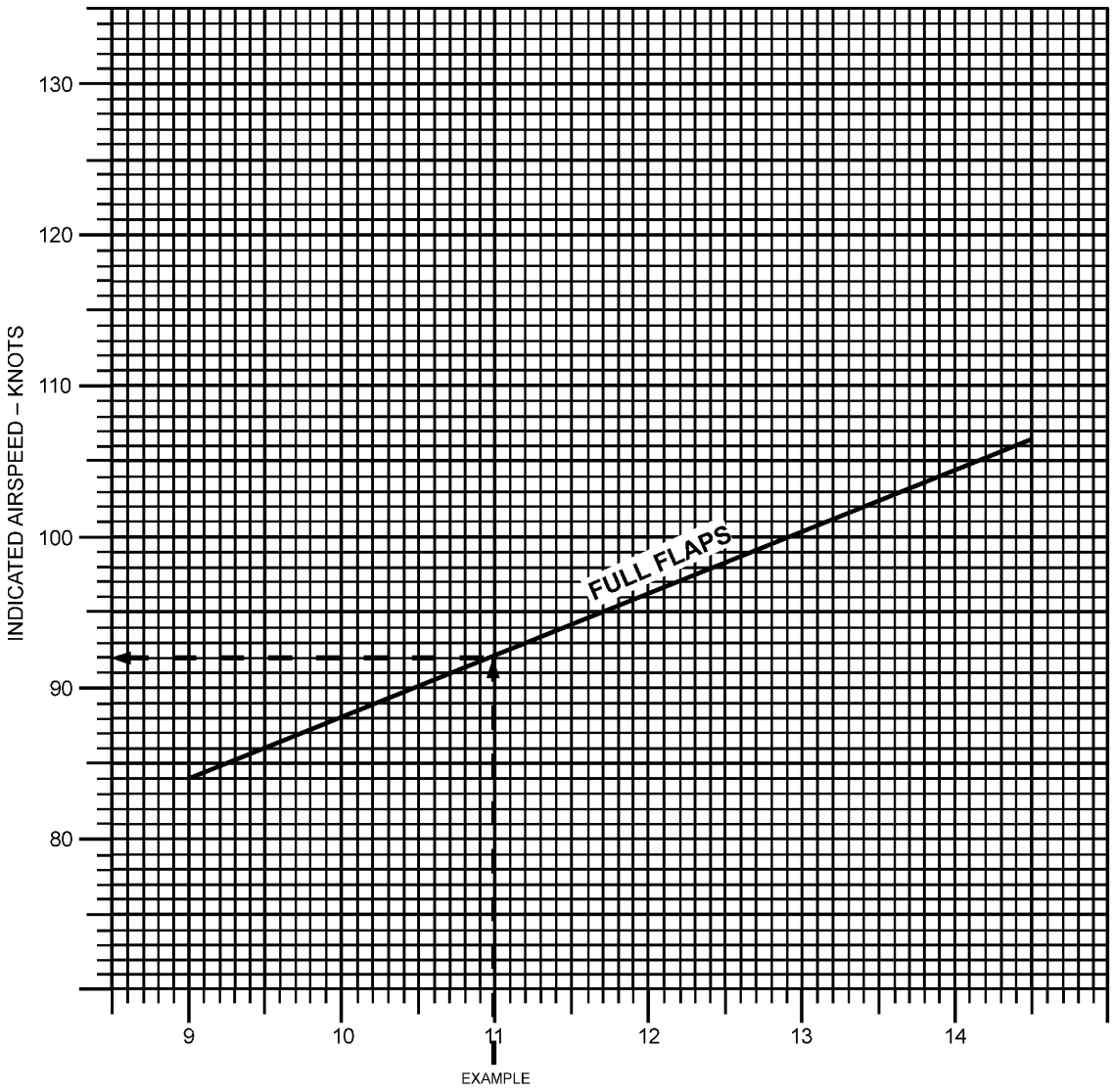


SHIFT CHART

This chart type is sometimes called a “shift” chart. The “shift” chart differs from other chart types because (1) the example line lies on a curve or proportionately between two curves for some portion of its travel, and (2) by the presense of a set of curves which are not labeled as representing any variable. The unlabeled curves are called guidelines. In order to locate the correct interval point, the user must trace the initial example line to the baseline and then parallel the guideline to the second input variable. The example line is then traced in the same direction as the initial example line to the output axis to determine the output value.

FIGURE 10. Sample graphs and scale breakdown.

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BOUNCE CHART

This chart type has sometimes been called a “bouncer” chart. In the simplest three-variable form, as shown above, there are two scaled axes and a set of parametric lines representing the interaction between two input variables in relation to an output variable. The user traces (or imagines) an example line extending perpendicularly from an input axis to a parametric line. Each line is labeled with a value of the second input. The initial example line is extended to the line representing the second input, or proportionately between two of the labeled lines. At the point of intersection, the example line pivots 90° and proceeds to intersect the output axis or (when the number of input variables exceeds two) a second input curve.

FIGURE 10. Sample graphs and scale breakdown. — Continued

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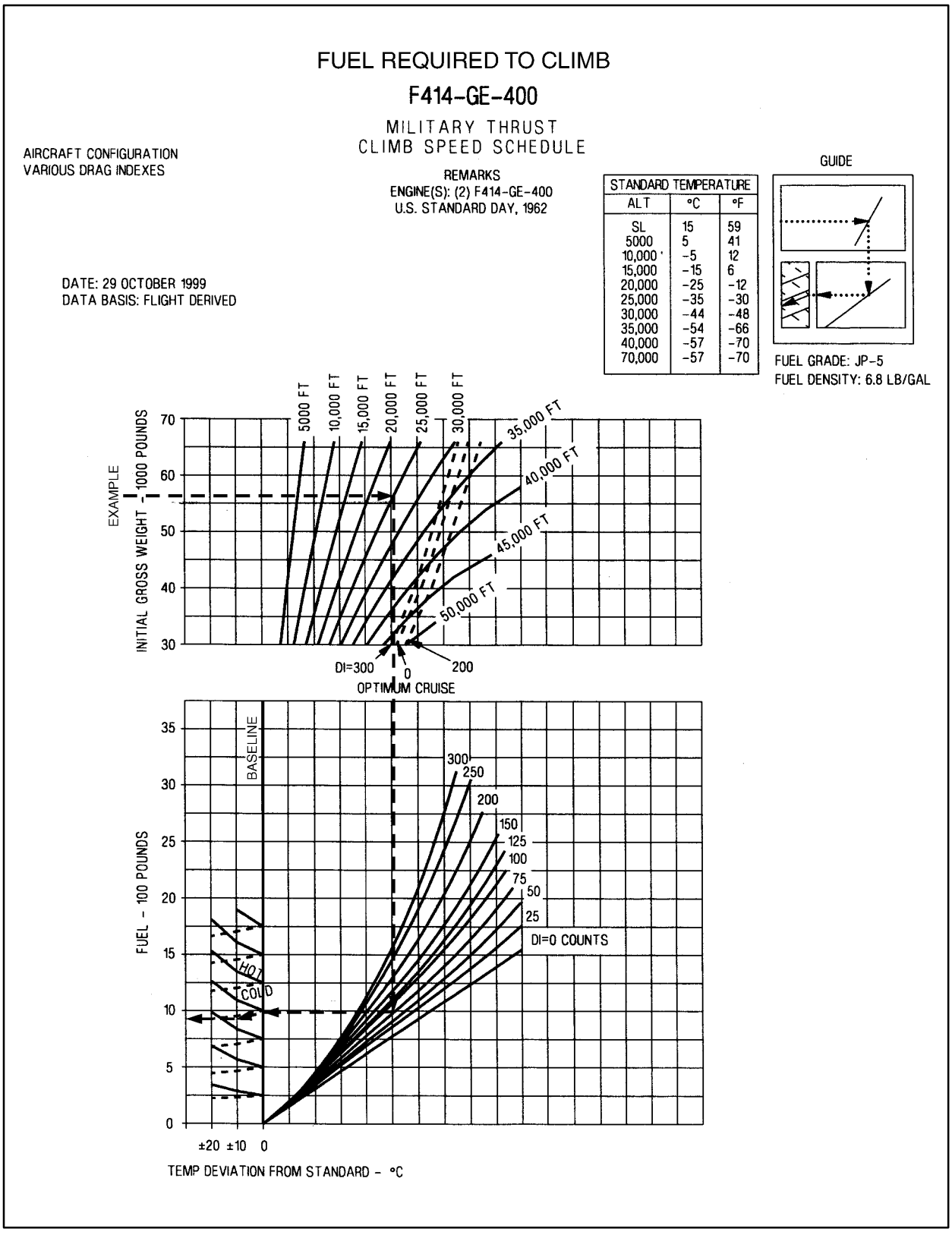


FIGURE 11. Sample performance chart with pictorial guide and example lines.

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Publication Running Sheet
NAVAIR NUMBER HERE
MANUAL TITLE HERE

Job Number: _____
Sheet No.: 1 of ?

	True Folio	Printed Folio	True Folio	Printed Folio
BLEED TABS	1	*1 (cover)*	2	BLANK
	3	*3*	4	BLANK
	5	*5*	6	*6*
TAB	7	*7*	8	BLANK
TAB	9	1-1	10	1-2
TAB	11	1-3	12	1-4
TAB	13	1-5	14	1-6
TAB	15	1-7	16	1-8
	17	1-9	18	BLANK
	19	2-1	20	2-2
	21	2-3	22	BLANK
	23	3-1 (CMYK)	24	3-2 (CMYK)
	25	3-3	26	BLANK
	27	*9*	28	BLANK
	29	4-1	30	4-2
	31	*11*	32	BLANK
	33	5-1	34	5-2
	35	5-3	36	5-4
	37	6-1	38	BLANK
	39	*13*	40	BLANK
	41	7-1	42	7-2
	43	7-3	44	7-4
	45	*Index-1*	46	*Index-2*
	47	*Index-3*	48	BLANK
	49	LEP (11X17)	50	BLANK

FIGURE 12. Sample print run sheet.

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	No. & Security Classification NAVAIR 01-E2AAF-1B (U)	REPRODUCTION ASSEMBLY SHEET E-2C POCKET CHECKLIST				Type of Job:
	1 September 2006					Date of Change: 1 September 2006
						Note: All pages will be printed on White, Blue or Yellow 130# or 250# Yupo/Hop-Syn paper
			Paper Specification			
	# Page Number	Emergency Bleed Edge	Yupo	Weight	Size Length X Width (inches)	Comments / Special Requirements
F	115 110		white	130	8 x 5	NP left tab 9 pg
B	116 111					
B	316 E191	BLANK				
F	315 E190		blue	130	8.25 x 5	Bottom Tab D Tab 9
B	314 E189					
F	313 E188		blue	130	8.25 x 5	Bottom Tab D Tab 8
B	312 E187					
F	311 E186		blue	130	8.25 x 5	Bottom Tab D Tab 7
B	310 E185					
F	309 E184		blue	130	8.25 x 5	Bottom Tab D Tab 6
B	308 E183					
F	307 E182		blue	130	8.25 x 5	Bottom Tab D Tab 5
B	306 E181					
F	305 E180		blue	130	8.25 x 5	Bottom Tab D Tab 4
B	304 E179					
F	303 E178		blue	130	8.25 x 5	Bottom Tab D Tab 3
B	302 E177					
F	301 E176		blue	130	8.25 x 5	Bottom Tab D Tab 2
B	300 E175					
F	299 E174		blue	130	8.25 x 5	Bottom Tab D Tab 1
B	298 E173	BLANK				
F	297 E172		yellow	250	8.25 x 5	EP Tab 9 – Cont B
B	296 E171					
F	295 E170		yellow	250	8 x 5	EP Tab 9 – Step 1
B	294 E169					
F	293 E168		yellow	250	8.25 x 5	EP Tab 9 – Cont A
B	292 E167					
F	291 E166		yellow	250	8 x 5	EP Tab 9 – Step 1
B	290 E165					
F	289 E164		yellow	250	8 x 5	EP Tab 8 – Step 1
B	288 E163					
F	287 E162		yellow	250	7.5 x 5	EP Tab 8 – Step 2
B	286 E161					
F	285 E160		yellow	250	8 x 5	EP Tab 7 – Step 1
B	284 E159					
F	283 E158		yellow	250	8.25 x 5	EP Tab 7 – Cont B
B	282 E157					
F	281 E156		yellow	250	8 x 5	EP Tab 7 – Step 1
B	280 E155					
F	279 E154		yellow	250	7.5 x 5	EP Tab 7 – Step 2
B	278 E153					
F	277 E152		yellow	250	7 x 5	EP Tab 7 – Step 3
B	276 E151					
F	275 E150		yellow	250	8.25 x 5	EP Tab 7 – Cont A
B	274 E149					
F	273 E148		yellow	250	8 x 5	EP Tab 7 – Step 1
B	272 E147					
F	271 E146		yellow	250	7.5 x 5	EP Tab 7 – Step 2
B	270 E145					
F	269 E144		yellow	250	7 x 5	EP Tab 7 – Step 3
B	268 E143					
F	267 E142		yellow	250	8.25 x 5	EP Tab 7 – Cont B
B	266 E141					
F	265 E140		yellow	250	8 x 5	EP Tab 7 – Step 1
B	264 E139					
F	263 E138		yellow	250	7.5 x 5	EP Tab 7 – Step 2
B	262 E137					
F	261 E136		yellow	250	7 x 5	EP Tab 7 – Step 3

10/25/2006 *3 of 6*

FIGURE 12. Sample print run sheet. — Continued

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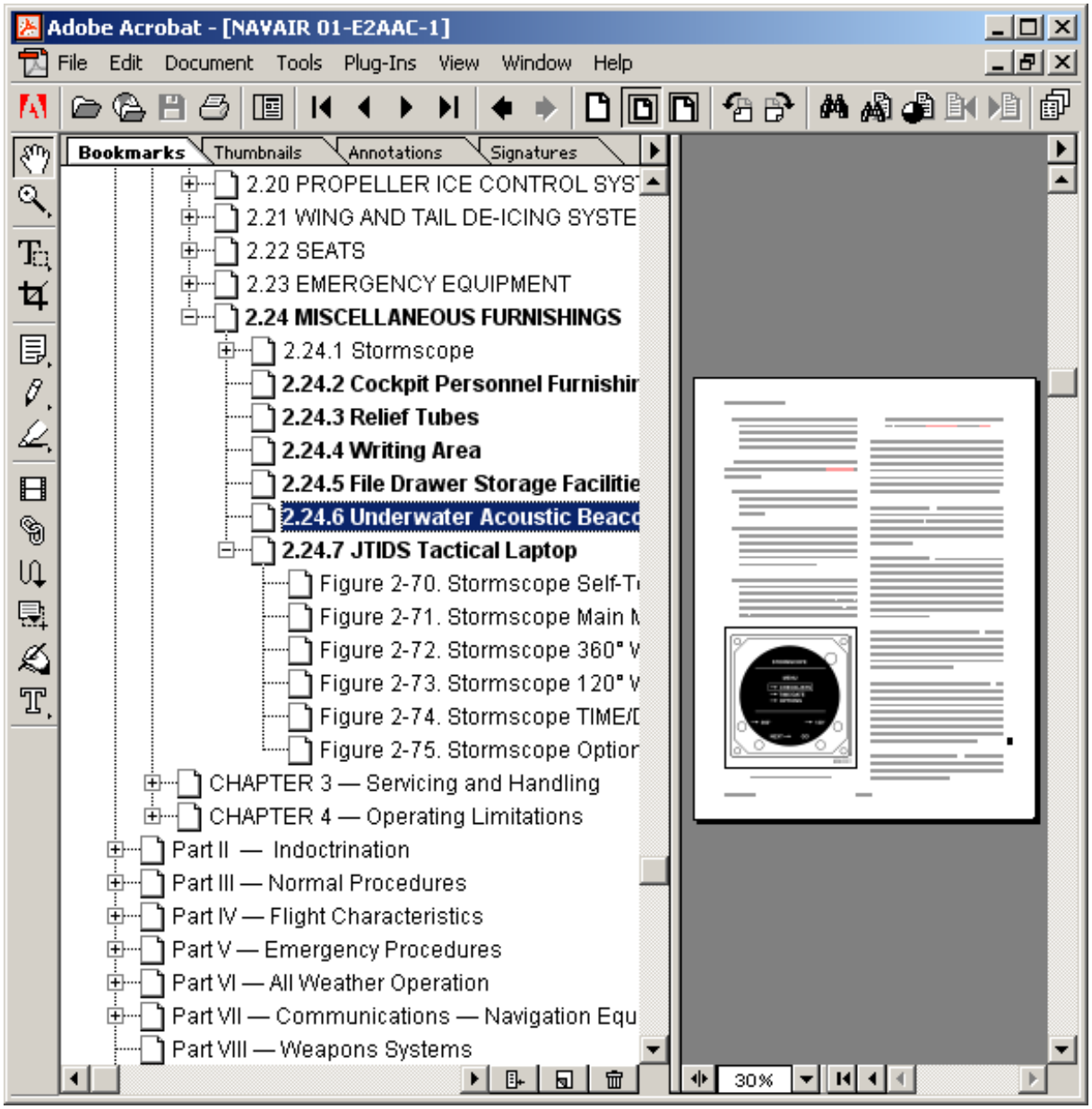


FIGURE 13. Example of HTML output table of contents.

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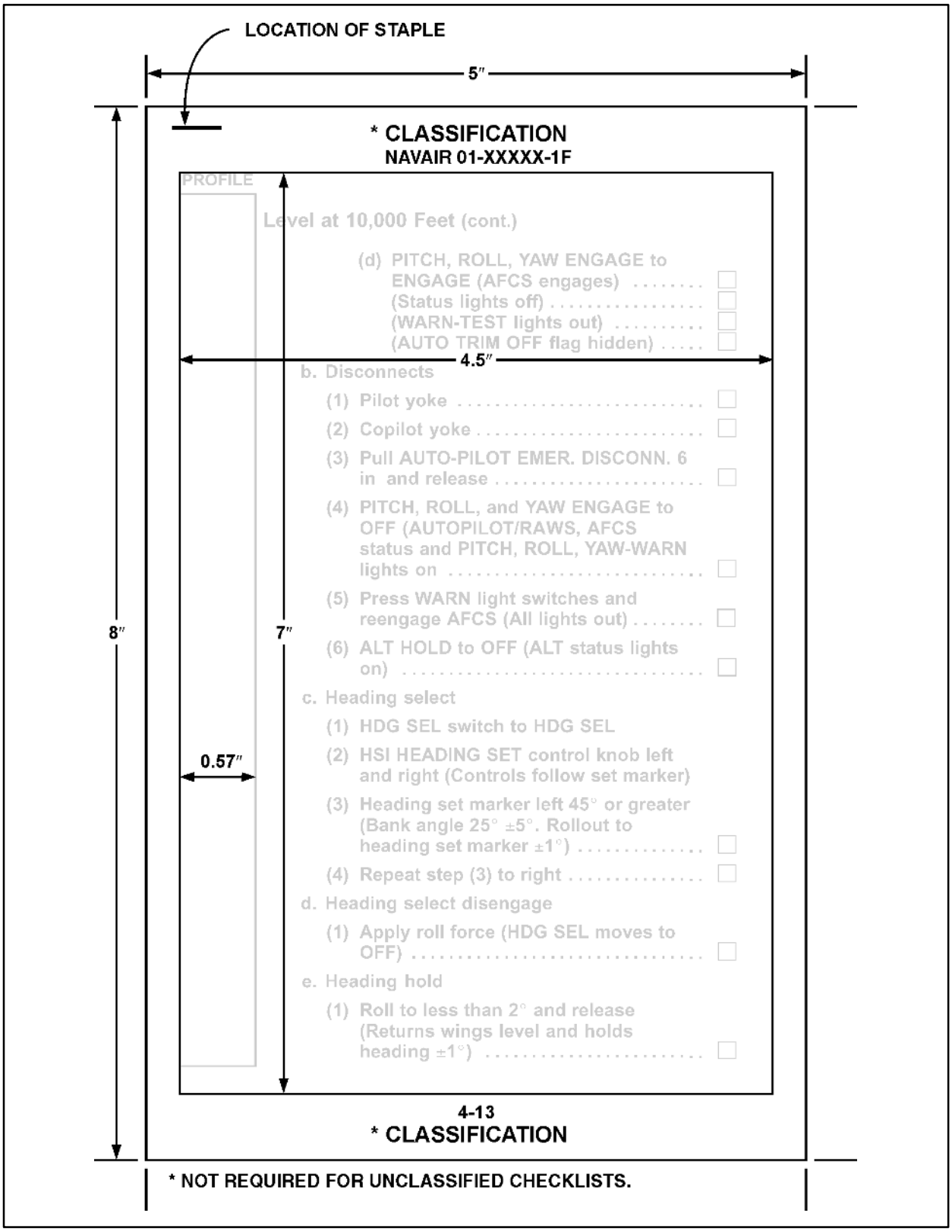


FIGURE 14. Functional checkflight checklist format.

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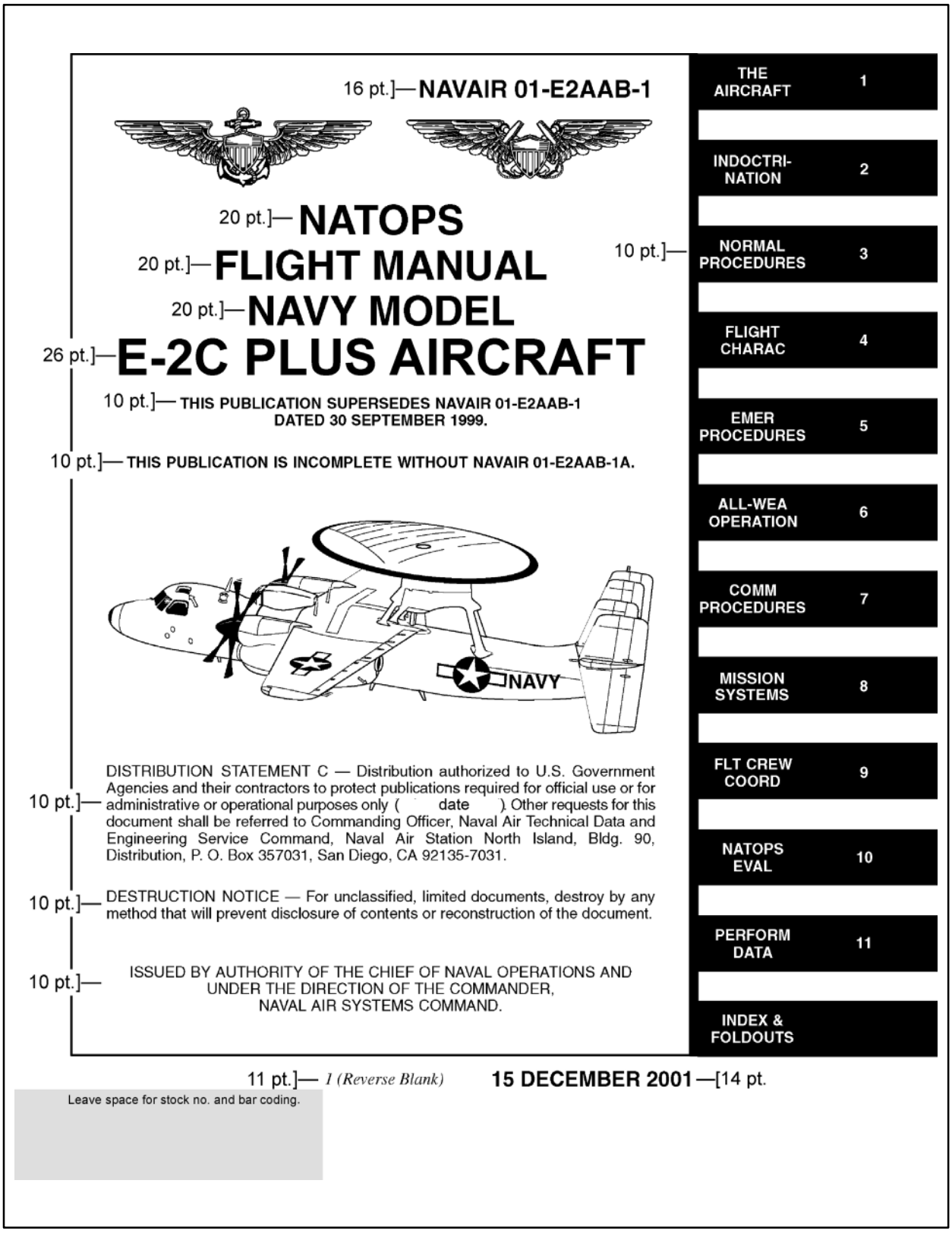


FIGURE 15. Sample title page for NATOPS flight manual (unclassified).

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NAVAIR 01-E2AAD-1

DEPARTMENT OF THE NAVY
NAVAL AIR SYSTEMS COMMAND
RADM WILLIAM A. MOFFETT BUILDING
47123 BUSE ROAD, BLDG 2272
PATUXENT RIVER, MARYLAND 20670-1547

IN REPLY REFER TO

01 October 2003

LETTER OF PROMULGATION

1. The Naval Air Training and Operating Procedures Standardization (NATOPS) Program is a positive approach toward improving combat readiness and achieving a substantial reduction in the aircraft mishap rate. Standardization, based on professional knowledge and experience, provides the basis for development of an efficient and sound operational procedure. The standardization program is not planned to stifle individual initiative, but rather to aid the commanding officer in increasing the unit's combat potential without reducing command prestige or responsibility.

2. This manual standardizes ground and flight procedures but does not include tactical doctrine. Compliance with the stipulated manual requirements and procedures is mandatory except as authorized herein. In order to remain effective, NATOPS must be dynamic and stimulate rather than suppress individual thinking. Since aviation is a continuing, progressive profession, it is both desirable and necessary that new ideas and new techniques be expeditiously evaluated and incorporated if proven to be sound. To this end, commanding officers of aviation units are authorized to modify procedures contained herein, in accordance with the waiver provisions established by OPNAVINST 3710.7, for the purpose of assessing new ideas prior to initiating recommendations for permanent changes. This manual is prepared and kept current by the users in order to achieve maximum readiness and safety in the most efficient and economical manner. Should conflict exist between the training and operating procedures found in this manual and those found in other publications, this manual will govern.

3. Checklists and other pertinent extracts from this publication necessary to normal operations and training should be made and carried for use in naval aircraft.

T.L. HEELY
Rear Admiral, United States Navy
By Direction of
Commander, Naval Air Systems Command

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ORIGINAL

FIGURE 16. Sample letter of promulgation page.

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INTERIM CHANGE SUMMARY

The following Interim Changes have been canceled or previously incorporated into this manual.

INTERIM CHANGE NUMBER(S)	REMARKS/PURPOSE

The following Interim Changes have been canceled or incorporated into this Change/Revision.

INTERIM CHANGE NUMBER(S)	REMARKS/PURPOSE

Interim Changes Outstanding — To be maintained by the custodian of this manual.

INTERIM CHANGE NUMBER(S)	ORIGINATOR/DATE (or DATE/TIME GROUP)	PAGES AFFECTED	REMARKS/PURPOSE

FIGURE 17. Sample NATOPS flight manual interim change summary page.

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Summary of Applicable Technical Directives

Information relating to the following technical directives has been incorporated in this manual.

CHANGE NUMBER	DESCRIPTION	DATE INC. IN MANUAL	VISUAL IDENTIFICATION
AFC-299	Install TACAN AN/ARN-118(v)	1 May 1980	AN/ARN-118 TACAN Control Panel
AVC-2229 (AFC-290)	Link-11 Data Network	1 May 1980	E-2 Scan Control Link
AFC-293 (ECP-032R2)	Installation of 10 KVA Emergency Generator	1 June 1981	Essential AC/DC Bus No. 2, Electrical System
ECP-295/295R1	Fixed Fairing and Serviceability and Maintainability of Rotodome Drive System, Incorporation of	1 June 1981	Center Engine Control Console
(RAMEC P-58-79)	Aircraft Crew Seat Arm Rest Deletion, Incorporation of	1 June 1981	Crew Seats
AFC-309 (ECP-299)	AFCS Safety Power Interlock Relay	1 June 1981	—
AFC-312 (ECP-303)	Oil Cooler Control Unit, Removal of	1 June 1981	—
AFC-324 (ECP-301)	Installation of Secure Voice KY-75	1 October 1983	RCU-2B and RCU-3B KY-75 Control Panels
AFC-310 (ECP-166R4)	Installation of Attitude Director Indicator and Rate Gyro	1 October 1983	Pilot and Copilot Instrument Panels
AFC-330 (ECP-331)	Installation of AN/ARC-182 VHF/UHF Radio Sets, UHF Guard Receiver and Provisions for Mobile Sea Range Transponder	1 June 1985	ICS Select and KY Mode Select Control Panel
(RAMEC P-11-85) (QEC-4)	Propeller Oil Cooling Pump, Removal of	15 August 1988	Left and Right Engines
AYC-937	Constant Speed Rotor Reliability Improvement	1 January 1991	—
AFC-332	Structural Modification of Tail Vertical Control Surfaces	15 March 1994	—
AYC-1000	Improved Power Lever Lock	15 March 1994	Base of Power Lever Quadrant
AFC-350 AFC-383	Removal of Exposure Suit Vent Blower System	15 March 1994	Right Side of Crew Seats
AFC-366	IFF Mode IV Code Hold Modification	15 March 1994	—
AFC-394	Fire Extinguisher Access MPDP	15 March 1994	Installation of Access Panel on MPDP
AFC-402	Illumination of MAX RUDDER Light and Master Caution Light When Pitch Feel Airspeed Is Less than 130 to 150 Knots with Less Than 20° Rudder Available	15 March 1994	—
AFC-404	AN/USC-43(V)3 ANDVT	15 September 1997	Terminal Located at RO Station

FIGURE 18. Sample NATOPS flight manual summary of applicable technical directives page.

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RECORD OF CHANGES

Record entry and page count verification for each printed change and erratum:

Change No. and Date of Change	Date of Entry	Page Count Verified by (Signature)

9/(10 blank)

ORIGINAL

FIGURE 19. Sample NATOPS flight manual record of changes page.

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LIST OF EFFECTIVE PAGES

Effective Pages	Page Numbers
Change 2	1 (Reverse Blank)
Original	3 (Reverse Blank)
Original	5 (Reverse Blank)
Original	7, 8
Original	9 (Reverse Blank)
Original	11 thru 63 (Reverse Blank)
Original	65 (Reverse Blank)
Original	1-1 thru 1-3 (Reverse Blank)
Original	2-1 thru 2-123 (Reverse Blank)
Original	3-1 thru 3-40
Change 2	4-1 thru 4-10
Original	67 (Reverse Blank)
Change 1	5-1 thru 5-4
Original	69 (Reverse Blank)
Original	6-1 thru 6-2
Original	7-1 thru 7-54
Original	8-1 thru 8-12
Original	9-1 thru 9-10
Original	10-1 thru 10-7 (Reverse Blank)
Original	11-1 thru 11-39 (Reverse Blank)
Original	71 (Reverse Blank)
Original	12-1 thru 12-14
Original	73 (Reverse Blank)
Original	13-1 thru 13-2
Original	14-1 thru 14-2
Original	15-1 thru 15-43 (Reverse Blank)
Original	16-1 thru 16-17 (Reverse Blank)
Original	17-1 thru 17-6
Original	18-1 thru 18-5 (Reverse Blank)
Original	75 (Reverse Blank)
Original	19-1 thru 19-4
Original	20-1 thru 20-10
Original	77 (Reverse Blank)
Original	21-1 thru 21-51 (Reverse Blank)
Original	22-1 thru 22-108
Original	79 (Reverse Blank)

Effective Pages	Page Numbers
Original	81 (Reverse Blank)
Original	29-1 thru 29-5 (Reverse Blank)
Original	83 (Reverse Blank)
Original	30-1 thru 30-23 (Reverse Blank)
Original	85 (Reverse Blank)
Original	31-1 thru 31-20
Original	32-1 thru 32-32
Original	33-1 thru 33-10
Original	34-1 thru 34-29 (Reverse Blank)
Original	35-1 thru 35-3 (Reverse Blank)
Original	36-1 (Reverse Blank)
Original	37-1 thru 37-7 (Reverse Blank)
Original	38-1 thru 38-21 (Reverse Blank)
Original	39-1 thru 39-6
Original	40-1 thru 40-26
Original	Index-1 thru Index-26
Original	FO-1 (Reverse Blank)
Original	FO-2 (Reverse Blank)
Original	FO-3 (Reverse Blank)
Original	FO-4 (Reverse Blank)
Original	FO-5 (Reverse Blank)
Original	FO-6 (Reverse Blank)
Original	FO-7 (Reverse Blank)
Original	FO-8 (Reverse Blank)
Original	FO-9 (Reverse Blank)
Original	FO-10 (Reverse Blank)
Original	FO-11 (Reverse Blank)
Original	FO-12 (Reverse Blank)
Original	FO-13 (Reverse Blank)
Original	FO-14 (Reverse Blank)
Original	FO-15 (Reverse Blank)
Original	FO-16 (Reverse Blank)
Original	FO-17 (Reverse Blank)
Change 2	LEP-1 (Reverse Blank)

*Use UNCLASSIFIED in headers and footers
if publication is classified.

CHANGE 2

UNCLASSIFIED*

FIGURE 20. Sample NATOPS flight manual list of effective pages.

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24 pt. Helvetica]— **NATOPS FLIGHT MANUAL**

24 pt. Helvetica]— **E-2C PLUS NP2000**

18 pt. Helvetica]— **Contents**

Page No. —[11 pt. Times.

12 pt. Helvetica]— **PART I — THE AIRCRAFT**

12 pt. Helvetica]— **CHAPTER 1 — AIRCRAFT AND ENGINE**

11 pt. Times]— 1.1 AIRCRAFT 1-1
1.1.1 Crew 1-1
1.1.2 Aircraft Weights 1-1
1.1.3 Aircraft Dimensions 1-1

1.2 ENGINE 1-3

CHAPTER 2 — SYSTEMS

2.1 POWER PLANT SYSTEMS 2-1
2.1.1 Engine 2-1
2.1.2 Engine Fuel Control System 2-2
2.1.3 Digital Electronic Control (DEC) System 2-6
2.1.4 Engine Controls 2-9
2.1.5 Engine Start and Ignition System 2-10
2.1.6 Engine Monitoring System 2-11
2.1.7 Engine Instruments 2-13
2.1.8 Engine Oil System 2-16
2.1.9 Engine Cooling 2-17

2.2 PROPELLER SYSTEM 2-17
2.2.1 Major Propeller Subassemblies 2-17
2.2.2 Propeller Operation 2-21
2.2.3 Propeller Feathering 2-22
2.2.4 Propeller Safety Features 2-23

FIGURE 21. Sample NATOPS flight manual table of contents page.

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LIST OF ILLUSTRATIONS

Page No.

CHAPTER 1 — AIRCRAFT AND ENGINE

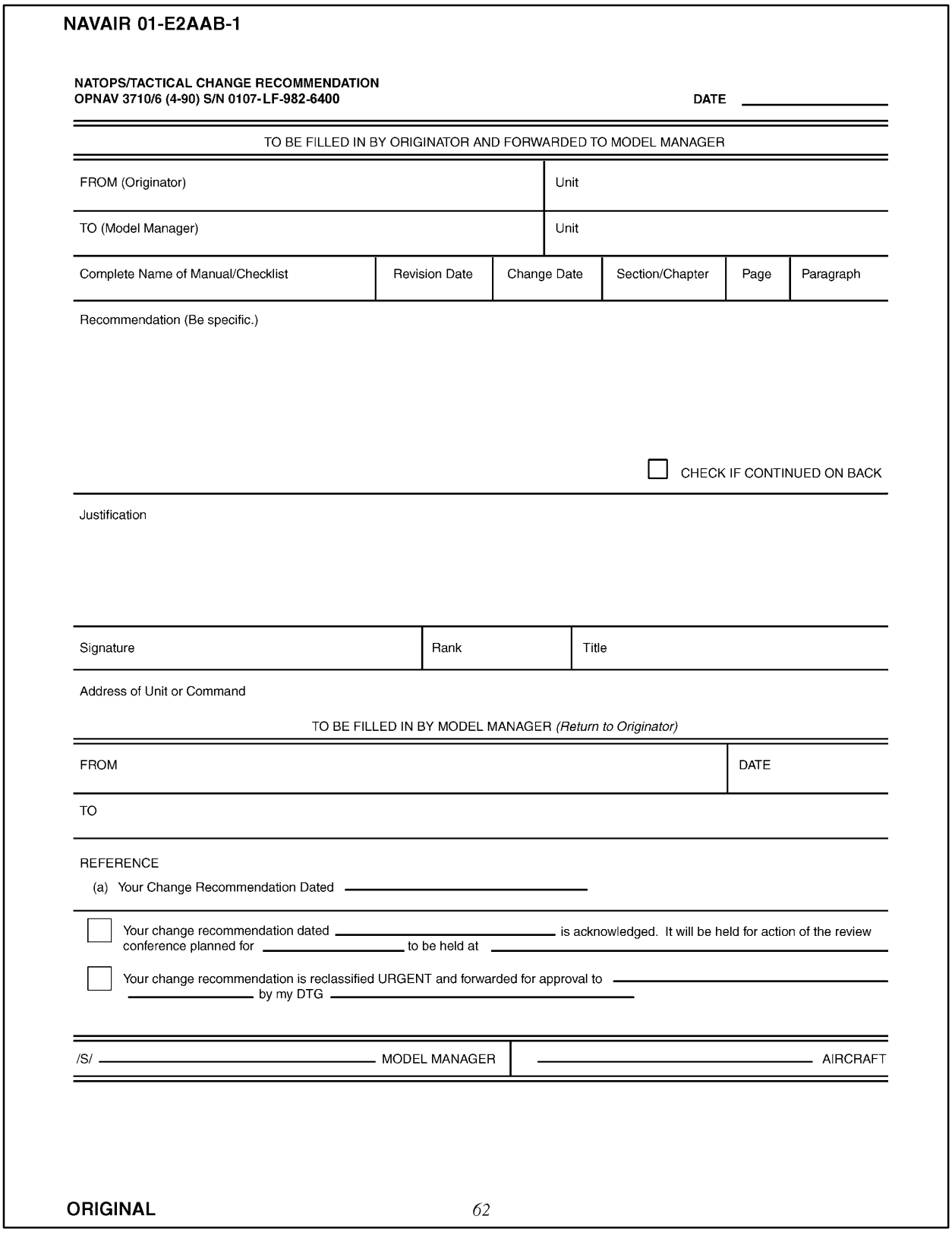
Figure 1-1. Aircraft Dimensions 1-2

CHAPTER 2 — SYSTEMS

Figure 2-1. T56-A-427 Engine — Typical 2-1
Figure 2-2. Engine Fuel Control System 2-3
Figure 2-3. Power Plant Controls and Indicators 2-4
Figure 2-4. DEC Fault Matrix 2-7
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FIGURE 22. Sample NATOPS flight manual list of illustrations page.

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NAVAIR 01-E2AAB-1

NATOPS/TACTICAL CHANGE RECOMMENDATION
OPNAV 3710/6 (4-90) S/N 0107-LF-982-6400 **DATE** _____

TO BE FILLED IN BY ORIGINATOR AND FORWARDED TO MODEL MANAGER

FROM (Originator)	Unit
TO (Model Manager)	Unit

Complete Name of Manual/Checklist	Revision Date	Change Date	Section/Chapter	Page	Paragraph

Recommendation (Be specific.)

☐ CHECK IF CONTINUED ON BACK

Justification

Signature	Rank	Title

Address of Unit or Command

TO BE FILLED IN BY MODEL MANAGER *(Return to Originator)*

FROM	DATE

TO

REFERENCE

(a) Your Change Recommendation Dated _____

☐ Your change recommendation dated _____ is acknowledged. It will be held for action of the review conference planned for _____ to be held at _____

☐ Your change recommendation is reclassified URGENT and forwarded for approval to _____ _____ by my DTG _____

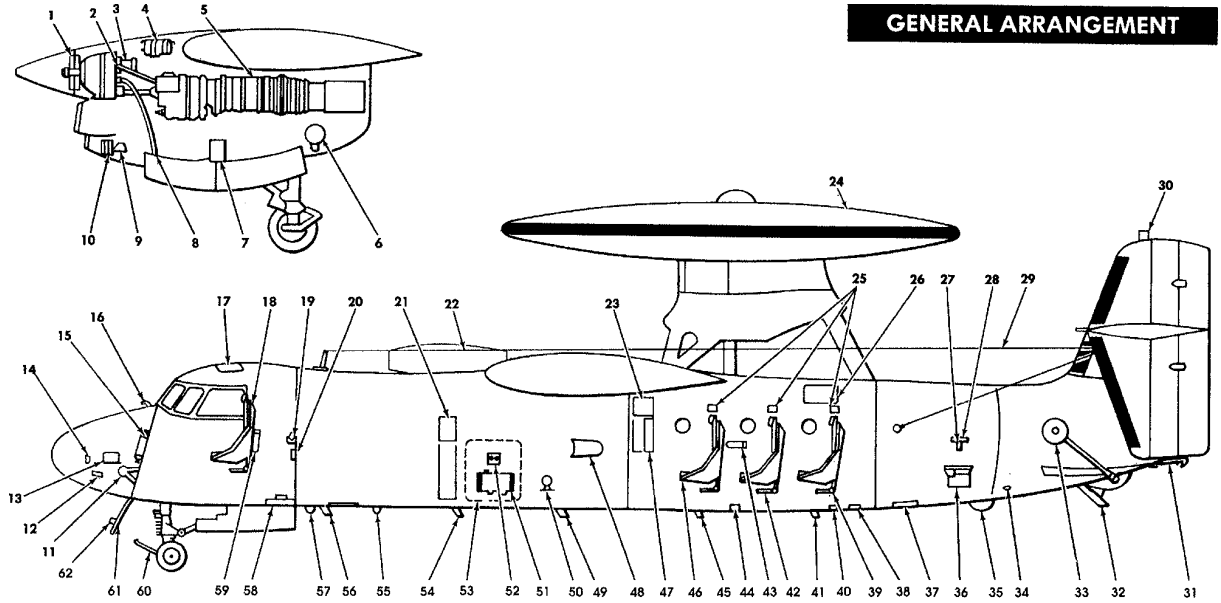
/S/ _____ MODEL MANAGER	_____ AIRCRAFT

ORIGINAL

62

FIGURE 23. Sample NATOPS change recommendation form.

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1. PROPELLER CONTROL
2. OIL BOOST PUMP
3. AC GENERATOR
4. OIL TANK
5. ENGINE
6. FIRE EXTINGUISHER CONTAINER
7. PRESSURE FUELING PANEL (RIGHT NACELLE)
8. PNEUMATIC AIR CONNECTOR
9. PRESSURE OILING PANEL
10. OIL COOLER
11. EMERGENCY LANDING GEAR AIR BOTTLE
12. APPROACH LIGHTS
13. LIQUID OXYGEN CONVERTER
14. TAXI LIGHTS (2)
15. EMERGENCY BRAKE ACCUMULATOR
16. PITOT HEADS (2)
17. COCKPIT DITCHING HATCHES (2)
18. SEATS (2)
19. CANTEENS (2)
20. PARACHUTES (2)
21. AVIONICS JUNCTION BOX (LEFT SIDE)
22. VAPOR CYCLE SCOOP
23. RADAR JUNCTION BOX (RIGHT SIDE)
24. ROTODOME
25. OXYGEN MASKS (3)
26. DITCHING HATCH (RIGHT SIDE)
27. EMERGENCY ARRESTING HOOK HANDLE
28. RUDDER STOP RELEASE HANDLE
29. HIGH-FREQUENCY FIXED WIRE ANTENNA
30. ANTI-COLLISION STROBE LIGHT
31. ARRESTING HOOK
32. TAIL SKID AND TAIL SKID FAIRING
33. TRAILING WIRE ANTENNA
34. ATTITUDE LIGHT (AFT)
35. TAIL BUMPER
36. CHEMICAL TOILET
37. UHF 4 COMMUNICATION ANTENNA
38. TACAN ANTENNA
39. ACO SEAT
40. RADAR ALTIMETER ANTENNA
41. V/UHF 1 COMMUNICATION ANTENNA
42. CICO SEAT
43. CIC COMPARTMENT FIRE EXTINGUISHER
44. RADAR ALTIMETER ANTENNA AS-1858/APN-171(V)
45. V/UHF 6 COMMUNICATION ANTENNA
46. RO SEAT
47. MAIN POWER DISTRIBUTION BOX (RIGHT SIDE)
48. AIR CYCLE SCOOP (RIGHT SIDE)
49. UHF 3 COMMUNICATION ANTENNA
50. FLIGHT HYDRAULIC SYSTEM RESERVOIR (RIGHT SIDE)
51. COMBINED HYDRAULIC SYSTEM RESERVOIR (RIGHT SIDE)
52. CABIN PRESSURE REGULATOR (RIGHT SIDE)
53. MAIN ENTRANCE DOOR AND BAILOUT HATCH (LEFT SIDE)
54. V/UHF 2 COMMUNICATION ANTENNA
55. TRANSPONDER ANTENNA
56. UHF 5 COMMUNICATION/DATA LINK ANTENNA
57. ANTI-COLLISION STROBE LIGHT
58. UHF/ADF ANTENNA (AS-909/ARA-48)
59. COCKPIT FIRE EXTINGUISHER
60. NOSE TOW LINK
61. NOSE WHEEL FAIRING
62. UHF GUARD RCVR ANT

FIGURE 24. Example of aircraft arrangement illustration.

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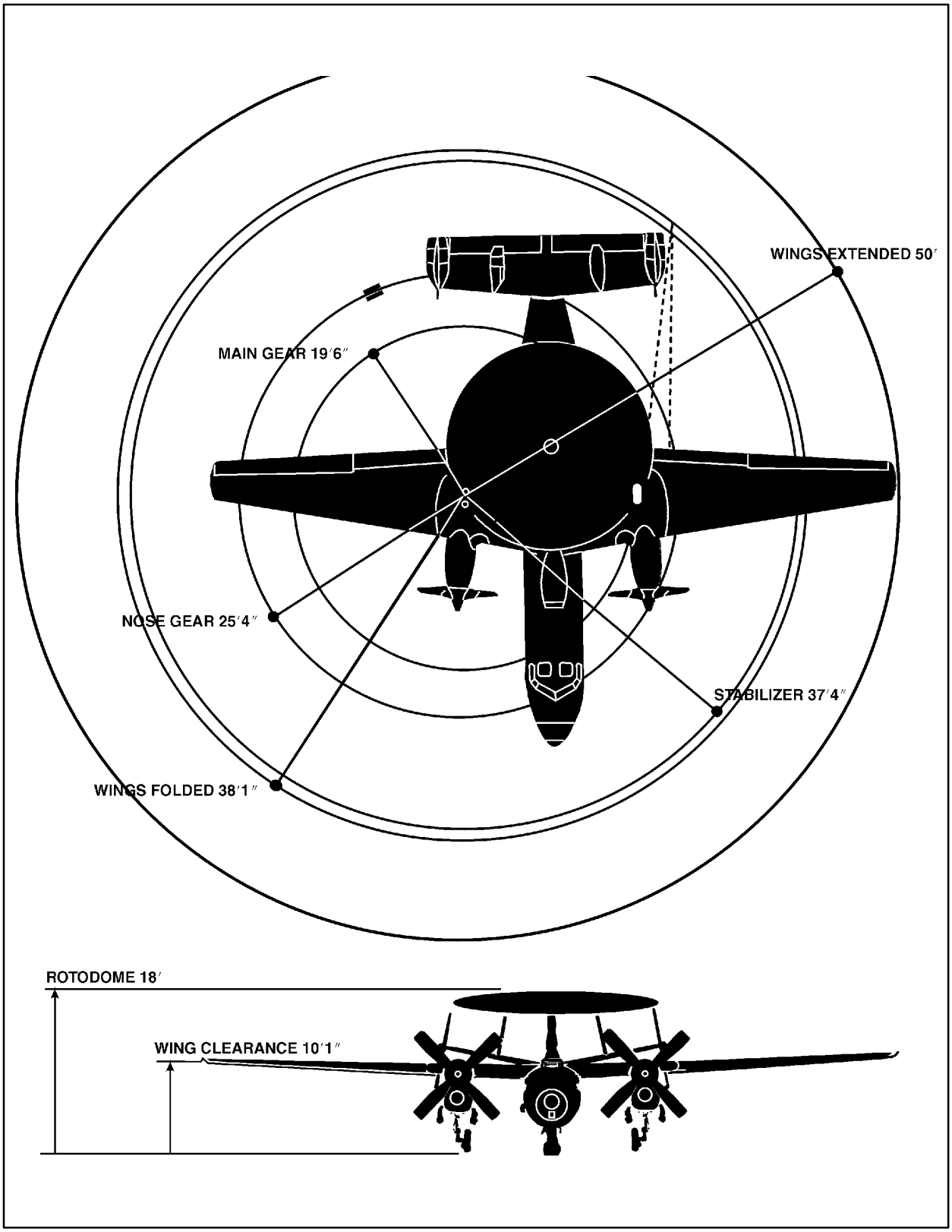


FIGURE 25. Example of turning radii and ground clearance.

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EXTERNAL STORES DRAG COUNTS AND WEIGHTS
STORES COMPUTATIONS

NOTES

1. Basic store drag count includes drag for pylons, racks and launchers.
2. Where two distances are given, the smaller is the distance to: a store mounted on the bottom station of a MER or TER.
3. Weights include all mount/store weight except pylon and parent rack.
4. Rolling moments include all mount/store weights except pylon and parent rack.

(1) STORE	(2) MOUNT/NO. OF STORES	(3) BASIC DRAG COUNT (Note 1) MN =0.6	MN =0.7	MN =0.8	MN =0.9	(4) DISTANCE FROM PYLON CENTER TO EDGE OF STORE IN INCHES (Note 2)	(5) WEIGHT IN LBS (Note 3)	(6) ROLLING MOMENT FT LBS (Note 4) Sta 1 or 8 Sta 2 or 7 Sta 3 or 6	REMARKS
FUSELAGE									
PYLON, LAUNCHER									
Pylon							13		
LAU-7/A		2.5	2.5	2.5	2.5		87		
STORES									
AIM86	LAU-7/A/1	7.5	7.5	7.5	11.5		242		
AIM-9D/G	LAU-7/A/1	7.5	7.5	7.5	11.5		276		
WING									
PYLONS, RACKS AND LAUNCHERS									
Pylon 1 or 8		5.0	5.5	9.0	13.0		204		
Pylon 2 or 7		5.0	5.5	9.0	13.0		201		
Pylon 3 or 6		5.0	5.5	9.0	13.0		173		
Parent Rack (PR)							100		
TER		12.0	14.0	23.0	37.5		105		
MER		23.0	24.0	29.0	40.0		223		
LAU-7/A		7.5	8.5	11.5	18.5		87		
AERO 5 (A-5)		7.0	7.0	10.5	16.5		91		
BOMBS									
Mk81	MER/6	44.5	46.0	51.5	71.5	12.75	2,029	23,110	
Snakeye								16,435	
								10,348	

FIGURE 26. Example of external store drag count and weight table.

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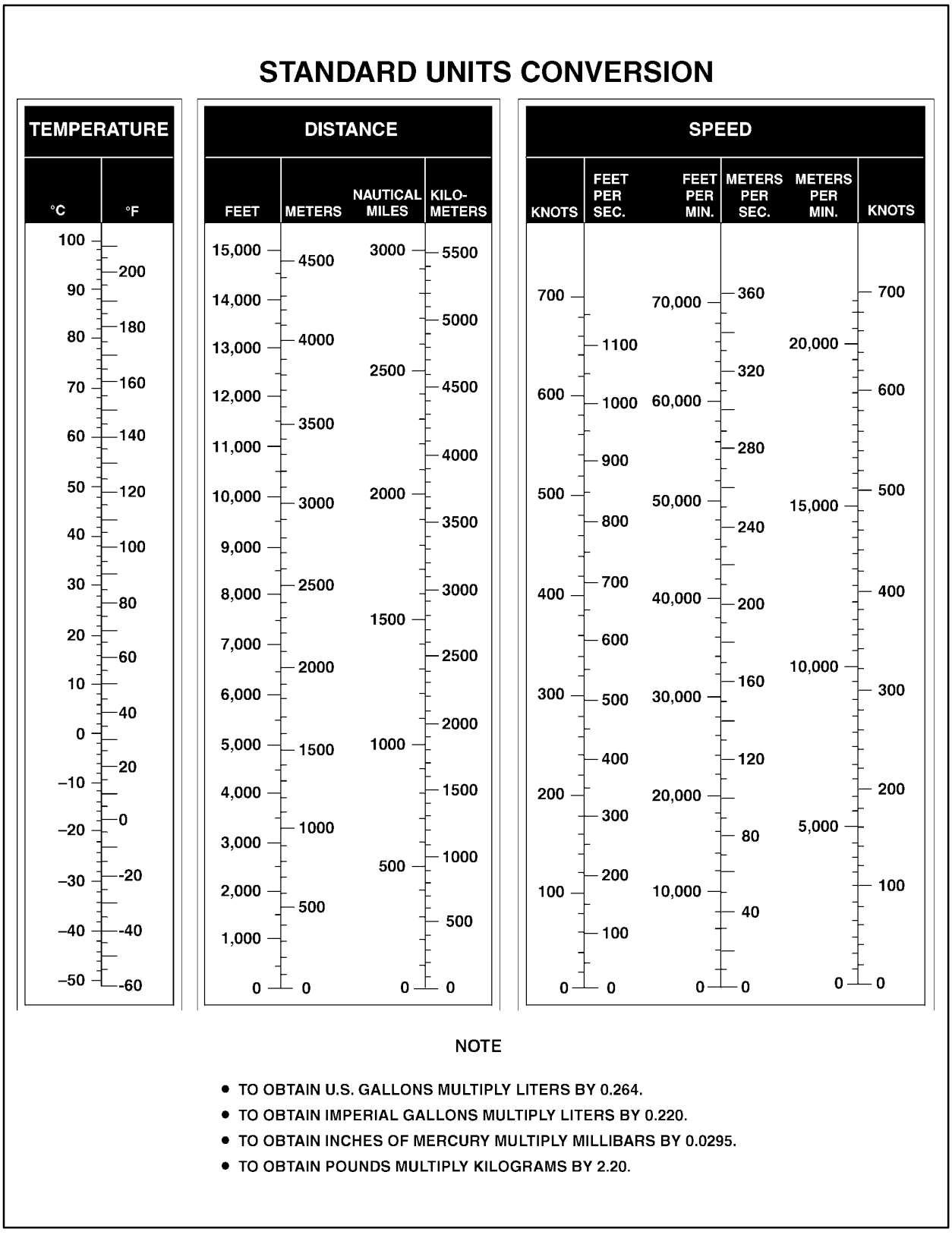


FIGURE 27. Example of standard units conversion chart.

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STANDARD ATMOSPHERE TABLE

STD. SEA LEVEL AIR
T = 15 °C
P = 29.921 IN of Hg.

NOTE: THE INTERNATIONAL, THE ICAO, AND THE U.S. STANDARD ATMOSPHERE ARE ALIKE FROM –16,405 TO +65,620 FEET (–5 TO +20 KILOMETERS) ALTITUDE.

W = .07648 LB/CU. FT. ρo = .0023769 SLUGS/CU. FT.
1″ OF Hg. = 70.727 LB/SQ. FT. = 0.4912 LB/SQ. IN.
A_o = 1117 FT./SEC. (661.7 KNOTS)
σ = DENSITY RATIO

U.S. STANDARD ATMOSPHERE, NASA, 1962 (ISA)

ALTITUDE FEET	DENSITY RATIO ρ/ρ_o	$\frac{1}{\sqrt{\sigma}}$	TEMPERATURE		SPEED OF SOUND		PRESSURE	
			DEG. C	DEG. F	RATIO A/A_o	KNOTS	IN. OF Hg	RATIO ρ/ρ_o
0	1.0000	1.000	15.0	59.0	1.000	661.7	29.92	1.0000
1000	0.9711	1.015	13.0	55.4	0.997	659.5	28.86	0.9644
2000	0.9428	1.030	11.0	51.9	0.993	657.2	27.82	0.9298
3000	0.9151	1.045	9.0	48.3	0.990	654.9	26.82	0.8962
4000	0.8881	1.061	7.1	44.7	0.986	652.6	25.84	0.8637
5000	0.8617	1.077	5.1	41.2	0.983	650.3	24.90	0.8320
6000	0.8359	1.094	3.1	37.6	0.979	647.9	23.98	0.8014
7000	0.8107	1.111	1.1	34.0	0.976	645.6	23.09	0.7716
8000	0.7860	1.128	–0.8	30.5	0.972	643.3	22.23	0.7428
9000	0.7620	1.146	–2.8	26.9	0.969	640.9	21.39	0.7148
10000	0.7385	1.164	–4.8	23.3	0.965	638.6	20.58	0.6877
11000	0.7156	1.182	–6.8	19.8	0.961	636.2	19.79	0.6614
12000	0.6932	1.201	–8.8	16.2	0.958	633.9	19.03	0.6360
13000	0.6713	1.221	–10.8	12.6	0.954	631.5	18.29	0.6113
14000	0.6500	1.240	–12.7	9.1	0.951	629.1	17.58	0.5875
15000	0.6292	1.261	–14.7	5.5	0.947	626.7	16.89	0.5643
16000	0.6090	1.282	–16.7	2.0	0.943	624.3	16.22	0.5420
17000	0.5892	1.303	–18.7	–1.6	0.940	621.9	15.57	0.5203
18000	0.5699	1.325	–20.7	–5.2	0.936	619.4	14.94	0.4994
19000	0.5511	1.347	–22.6	–8.8	0.933	617.0	14.34	0.4791
20000	0.5328	1.370	–24.6	–12.3	0.929	614.6	13.75	0.4595
21000	0.5150	1.394	–26.6	–15.9	0.925	612.1	13.18	0.4406
22000	0.4976	1.418	–28.6	–19.5	0.921	609.6	12.64	0.4223
23000	0.4807	1.442	–30.6	–23.0	0.918	607.2	12.11	0.4046
24000	0.4642	1.468	–32.6	–26.6	0.914	604.7	11.60	0.3876
25000	0.4481	1.494	–34.5	–30.1	0.910	602.2	11.11	0.3711
26000	0.4325	1.521	–36.5	–33.7	0.906	599.7	10.63	0.3552
27000	0.4173	1.548	–38.5	–37.3	0.903	597.2	10.17	0.3398
28000	0.4025	1.576	–40.5	–40.9	0.899	594.6	9.725	0.3250
29000	0.3881	1.605	–42.5	–44.4	0.895	592.1	9.297	0.3107
30000	0.3741	1.635	–44.4	–48.0	0.891	589.6	8.885	0.2970
31000	0.3605	1.665	–46.4	–51.6	0.887	587.0	8.488	0.2837
32000	0.3473	1.697	–48.4	–55.1	0.883	584.4	8.106	0.2709
33000	0.3345	1.729	–50.4	–58.7	0.879	581.9	7.737	0.2586
34000	0.3220	1.762	–52.4	–62.2	0.876	579.3	7.382	0.2467
35000	0.3099	1.796	–54.3	–65.8	0.872	576.7	7.041	0.2353
36000	0.2981	1.832	–56.2	–69.4	0.868	574.0	6.712	0.2243
37000	0.2844	1.875	–56.5	–69.7	0.867	573.8	6.397	0.2138
38000	0.2710	1.921	–56.5	–69.7	0.867	573.8	6.097	0.2038
39000	0.2583	1.968	–56.5	–69.7	0.867	573.8	5.811	0.1942
40000	0.2462	2.016	–56.5	–69.7	0.867	573.8	5.538	0.1851
41000	0.2346	2.065	–56.5	–69.7	0.867	573.8	5.278	0.1764
42000	0.2236	2.115	–56.5	–69.7	0.867	573.8	5.030	0.1681
43000	0.2131	2.166	–56.5	–69.7	0.867	573.8	4.794	0.1602
44000	0.2031	2.219	–56.5	–69.7	0.867	573.8	4.569	0.1527
45000	0.1936	2.273	–56.5	–69.7	0.867	573.8	4.355	0.1455
46000	0.1845	2.328	–56.5	–69.7	0.867	573.8	4.151	0.1387
47000	0.1758	2.385	–56.5	–69.7	0.867	573.8	3.965	0.1322
48000	0.1676	2.443	–56.5	–69.7	0.867	573.8	3.770	0.1260
49000	0.1597	2.502	–56.5	–69.7	0.867	573.8	3.593	0.1201
50000	0.1522	2.563	–56.5	–69.7	0.867	573.8	3.425	0.1145
51000	0.1451	2.625	–56.5	–69.7	0.867	573.8	3.264	0.1091
52000	0.1383	2.689	–56.5	–69.7	0.867	573.8	3.111	0.1040
53000	0.1318	2.755	–56.5	–69.7	0.867	573.8	2.965	0.0991
54000	0.1256	2.822	–56.5	–69.7	0.867	573.8	2.826	0.0944
55000	0.1197	2.890	–56.5	–69.7	0.867	573.8	2.693	0.0900

FIGURE 28. Example of standard atmosphere table.

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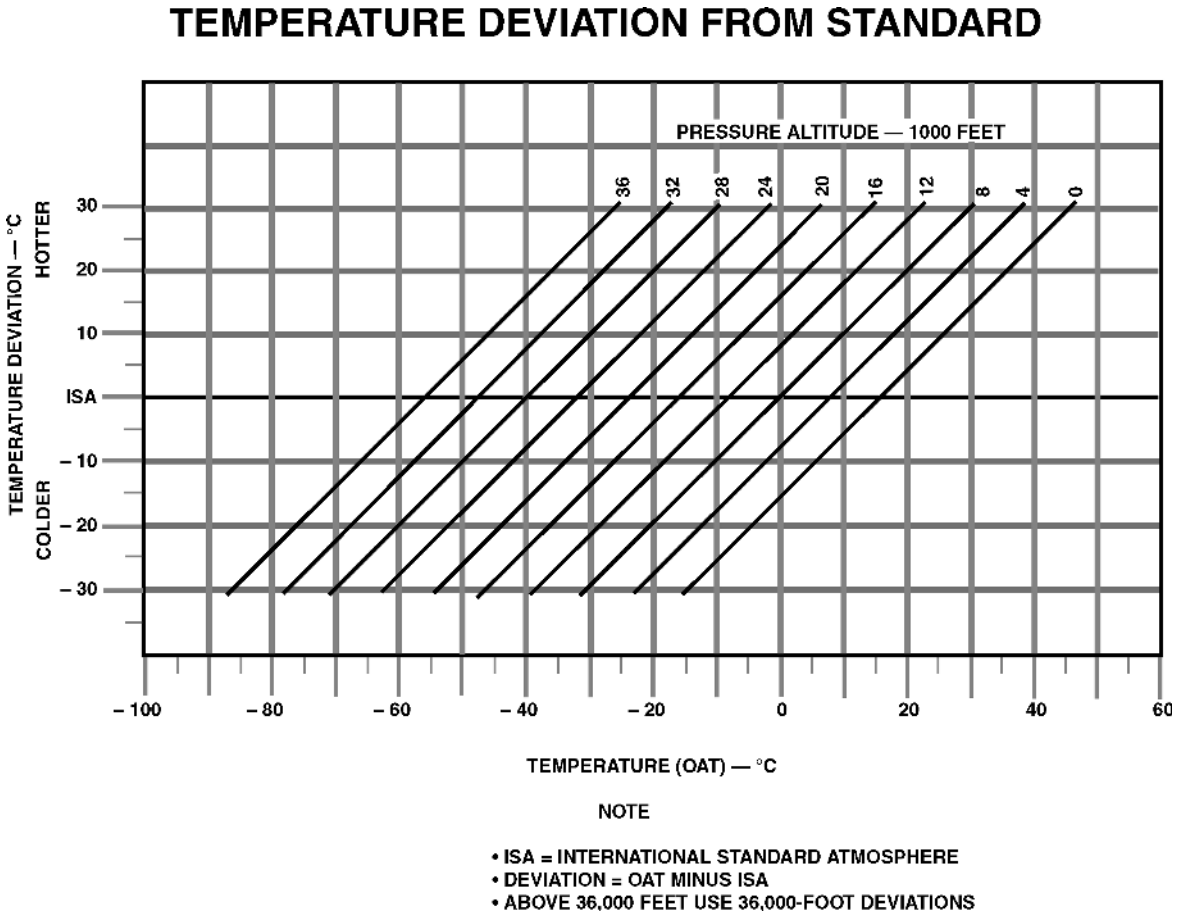


FIGURE 29. Example of temperature deviation from standard chart.

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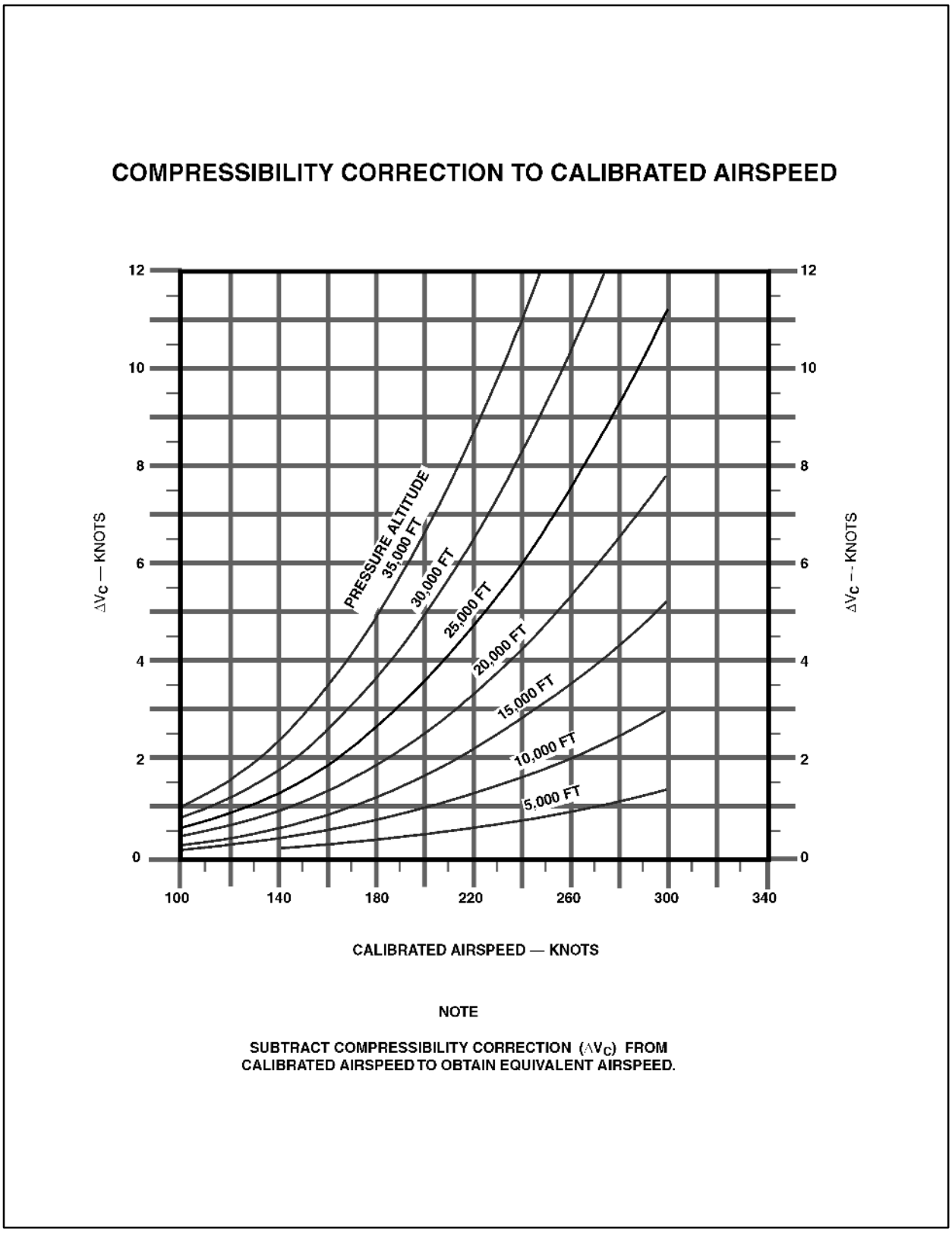


FIGURE 30. Example of compressibility correction to calibrated airspeed chart.

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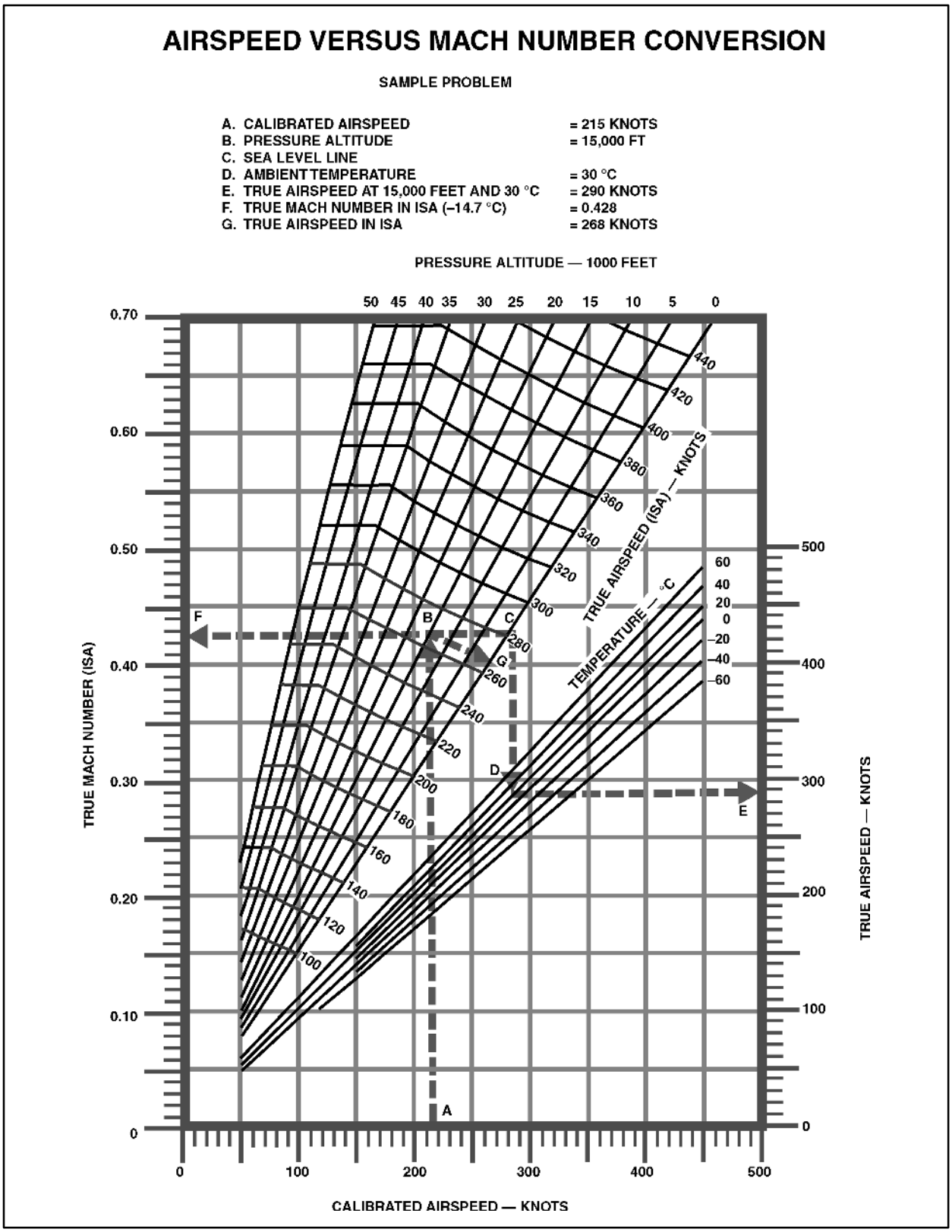


FIGURE 31. Example of airspeed Mach number conversion chart.

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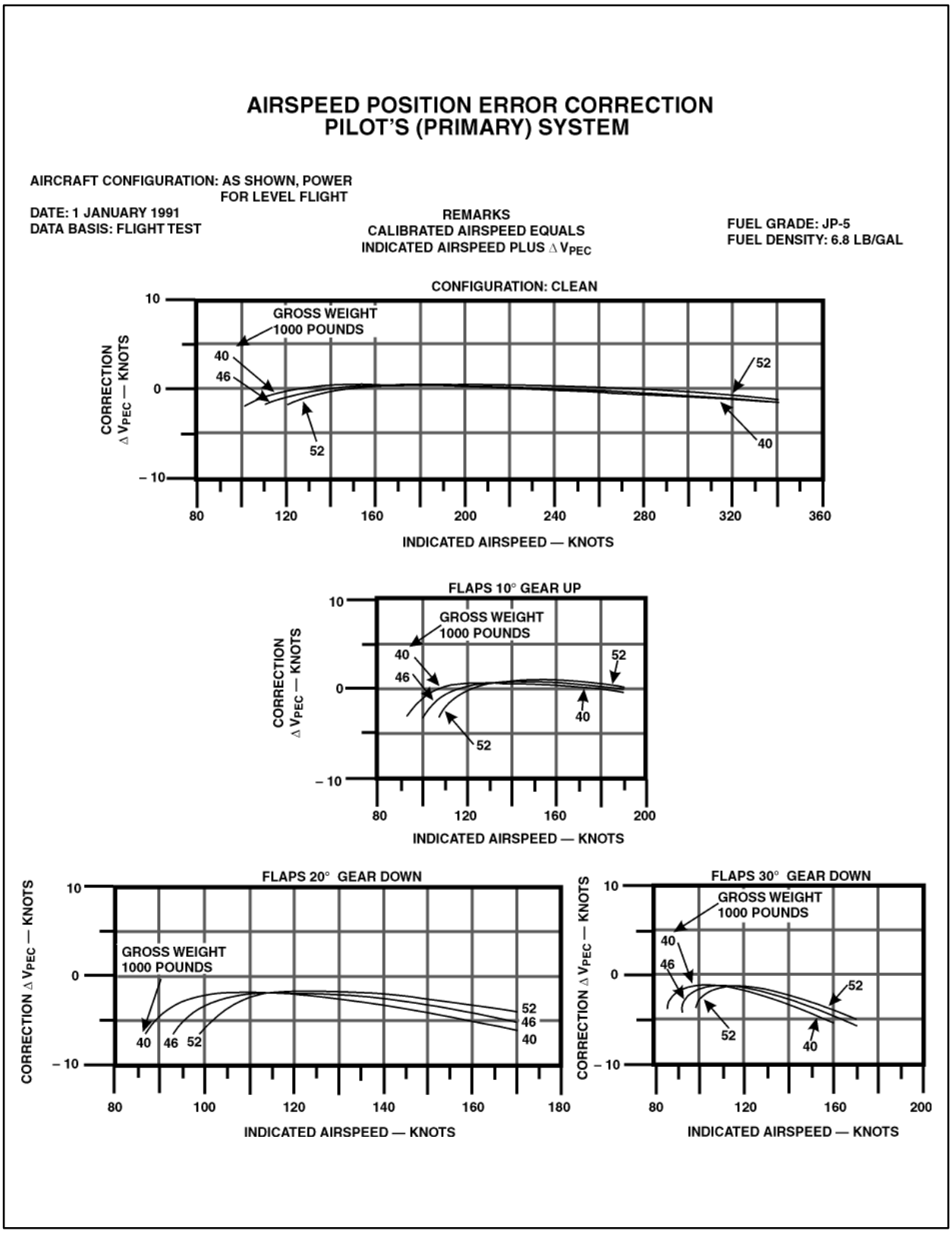


FIGURE 32. Example of airspeed position error correction chart.

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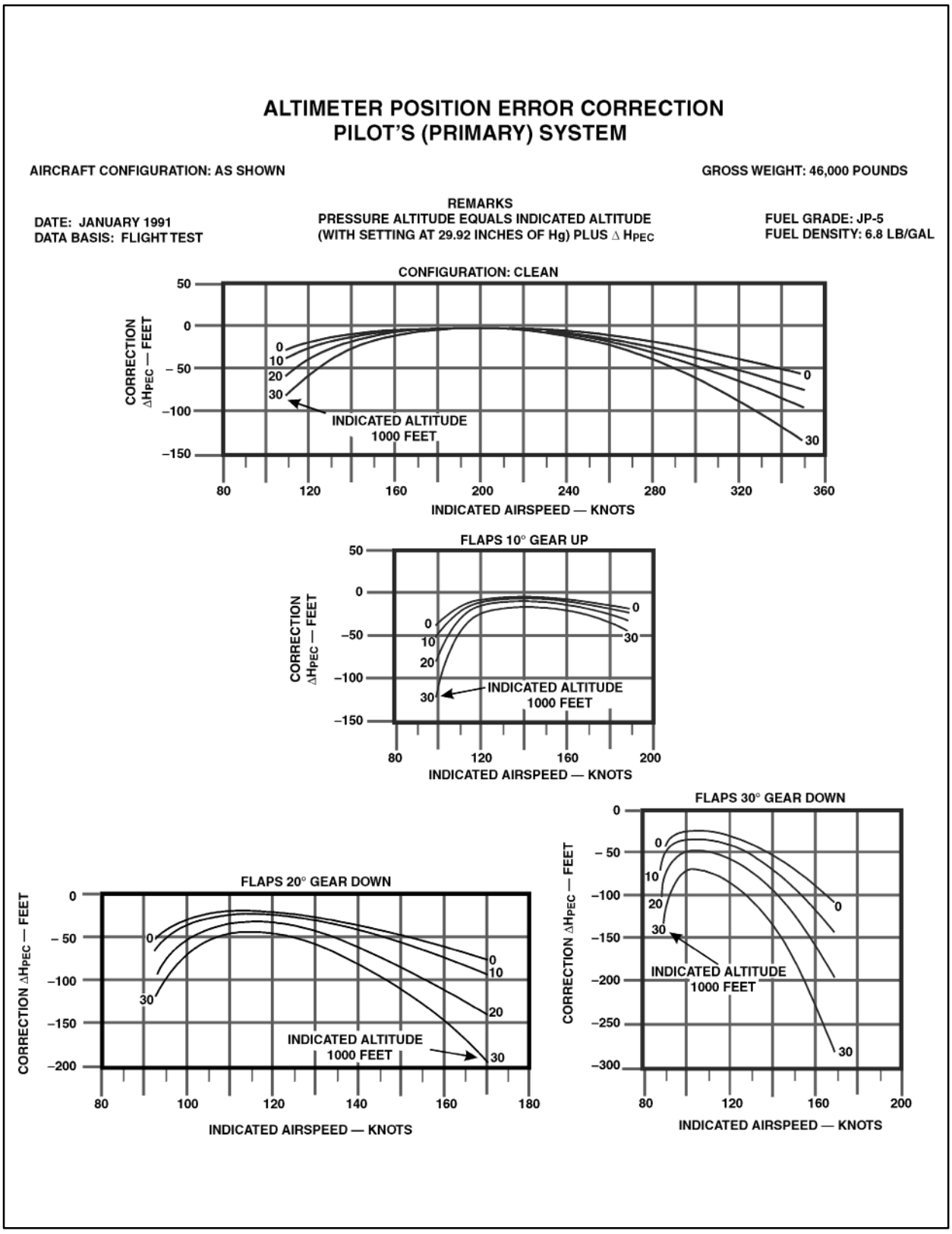


FIGURE 33. Example of altimeter position error correction chart.

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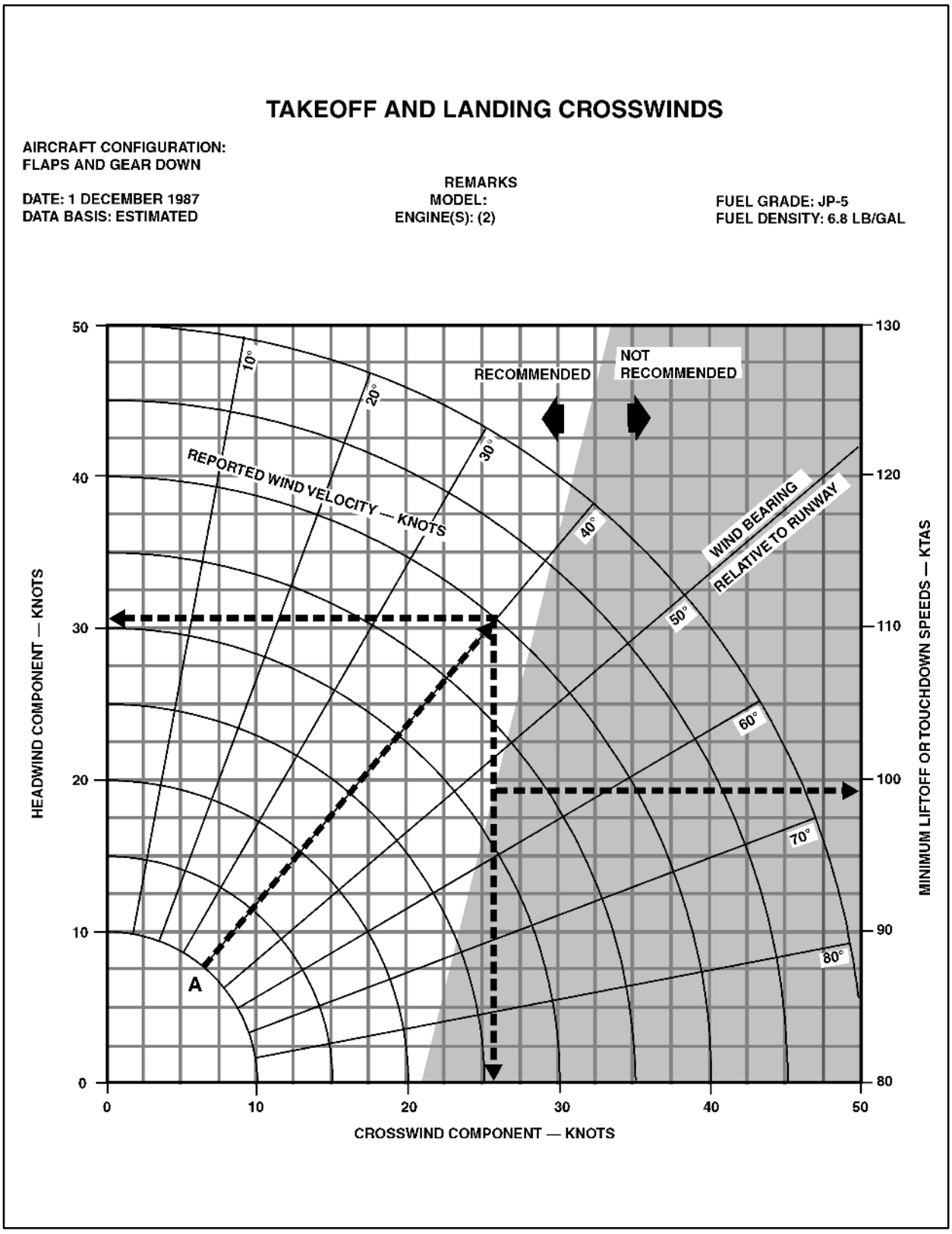


FIGURE 34. Example of takeoff/landing crosswind chart.

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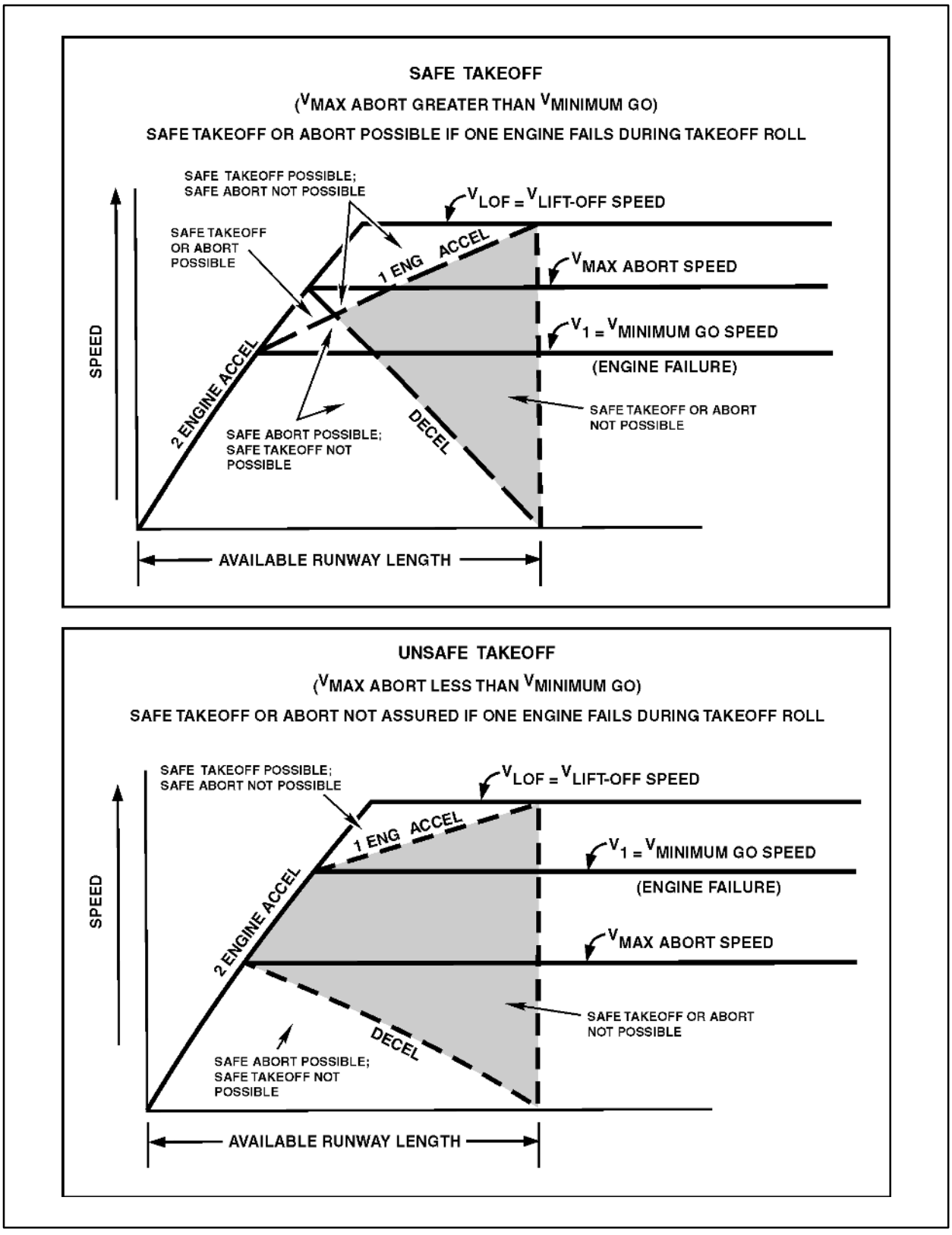


FIGURE 35. Example of takeoff illustrations.

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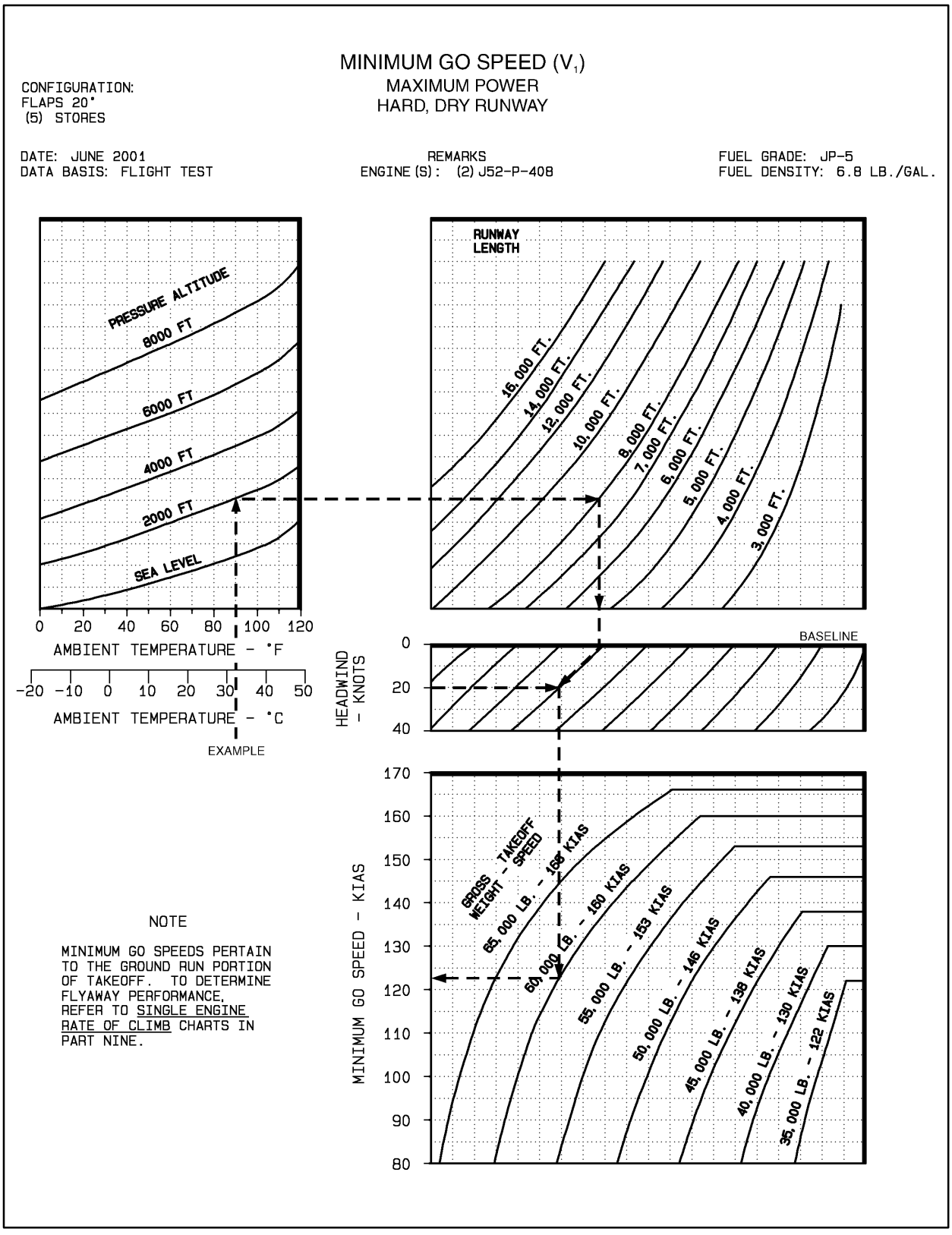


FIGURE 36. Example of minimum go speed (V_1) chart.

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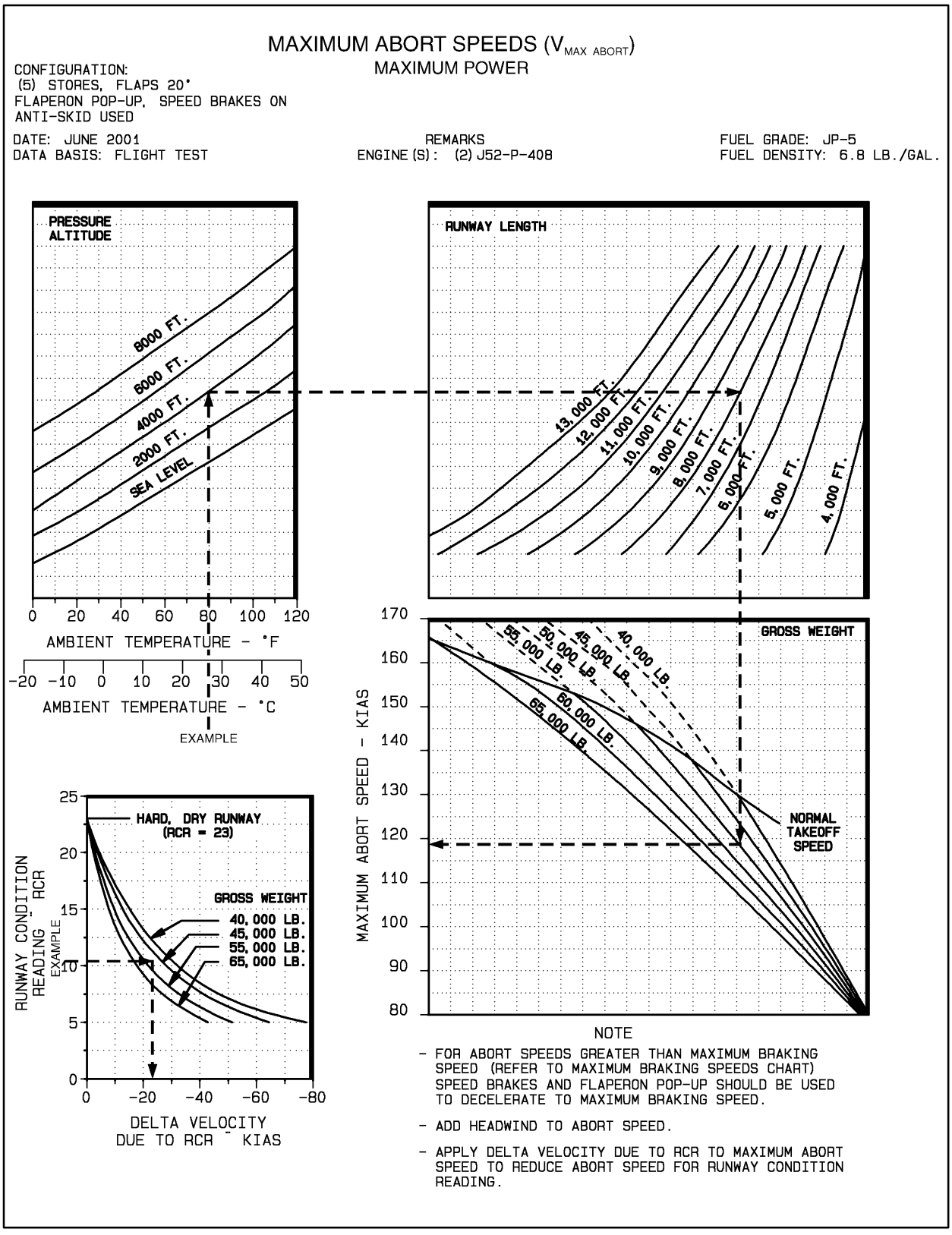


FIGURE 37. Example of maximum abort speed ($V_{max\ abort}$) chart.

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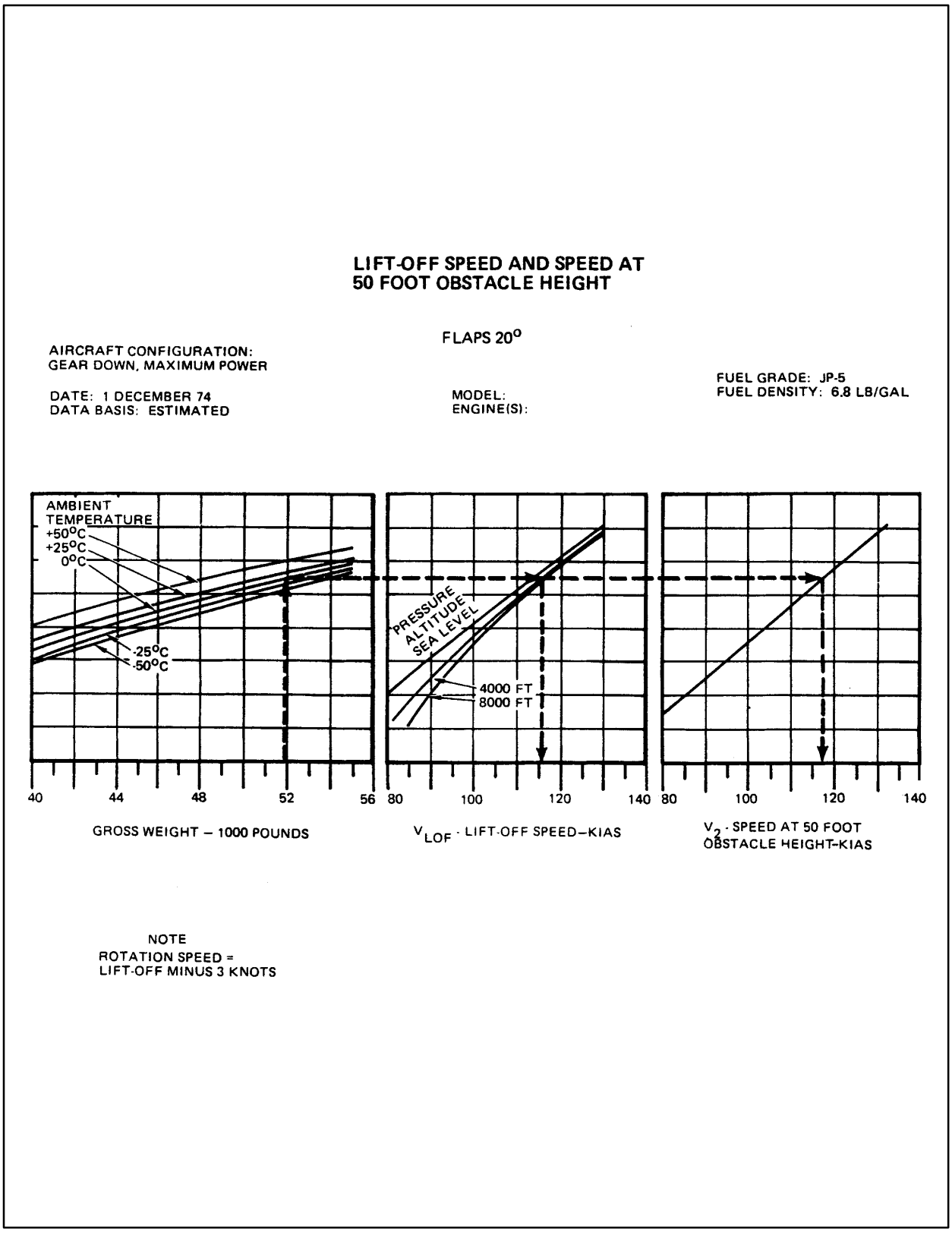


FIGURE 38. Example of lift–off speed (V_{lof}) and speed at 50–foot obstacle height (V_2) chart.

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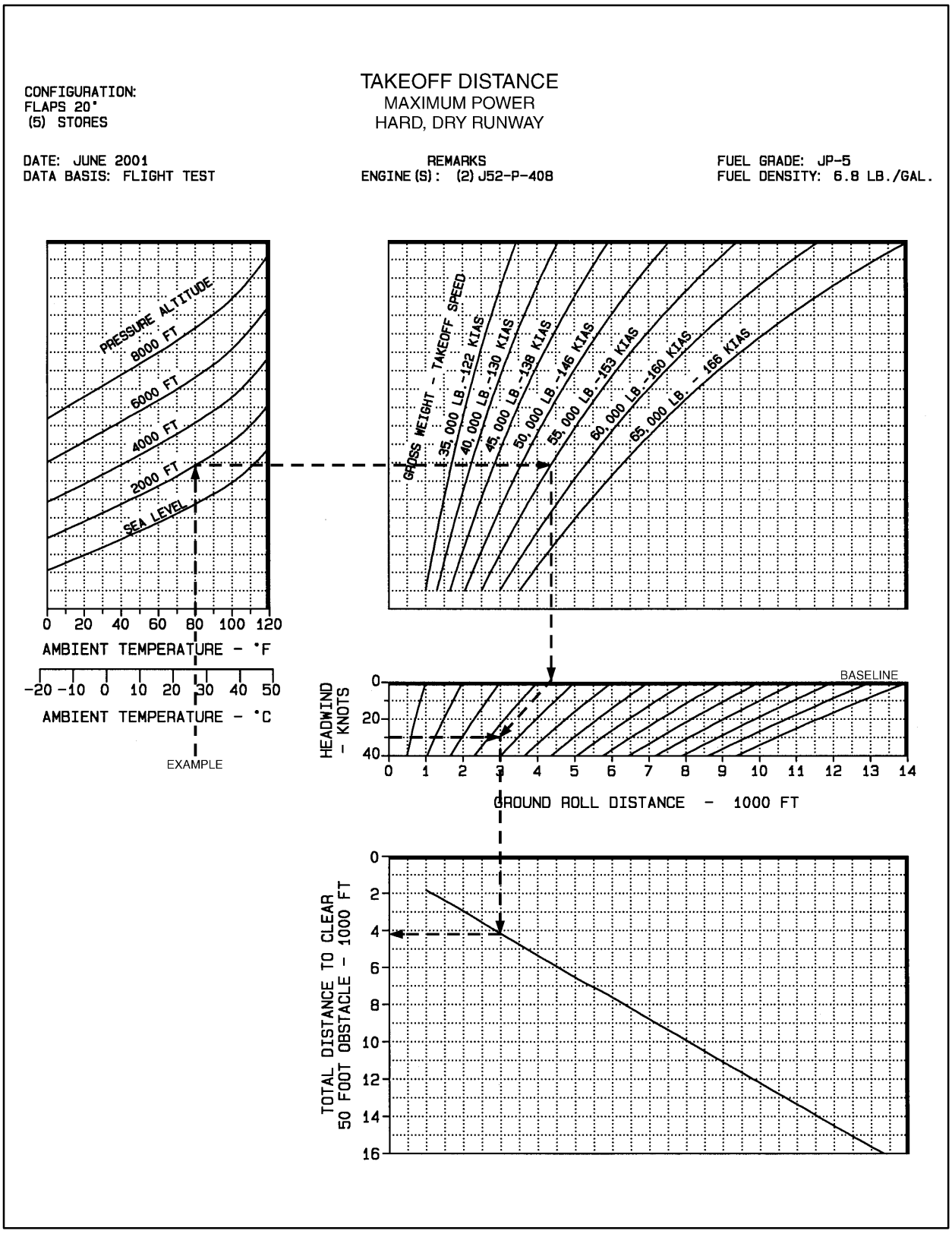


FIGURE 39. Example of takeoff distance chart.

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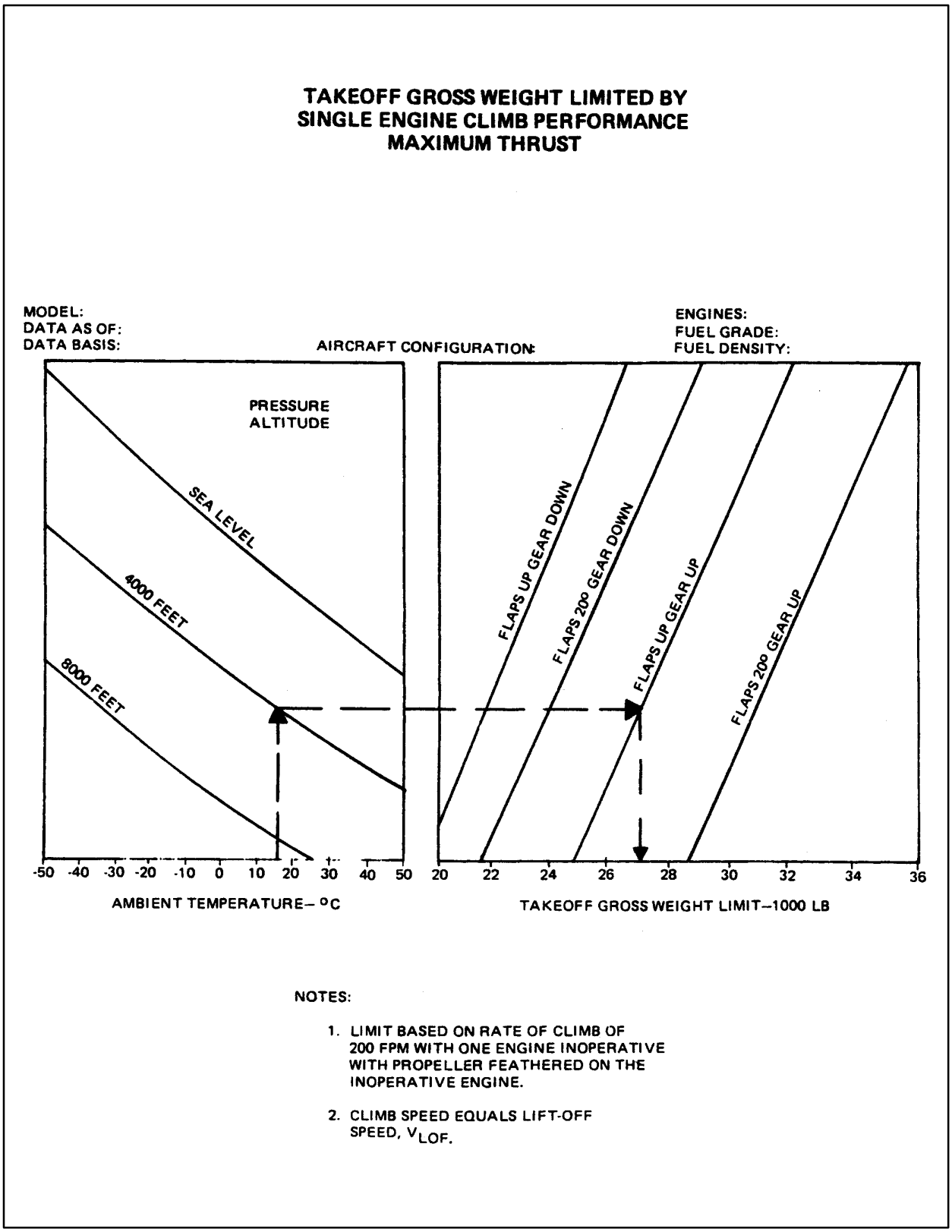


FIGURE 40. Example of takeoff gross weight limit chart.

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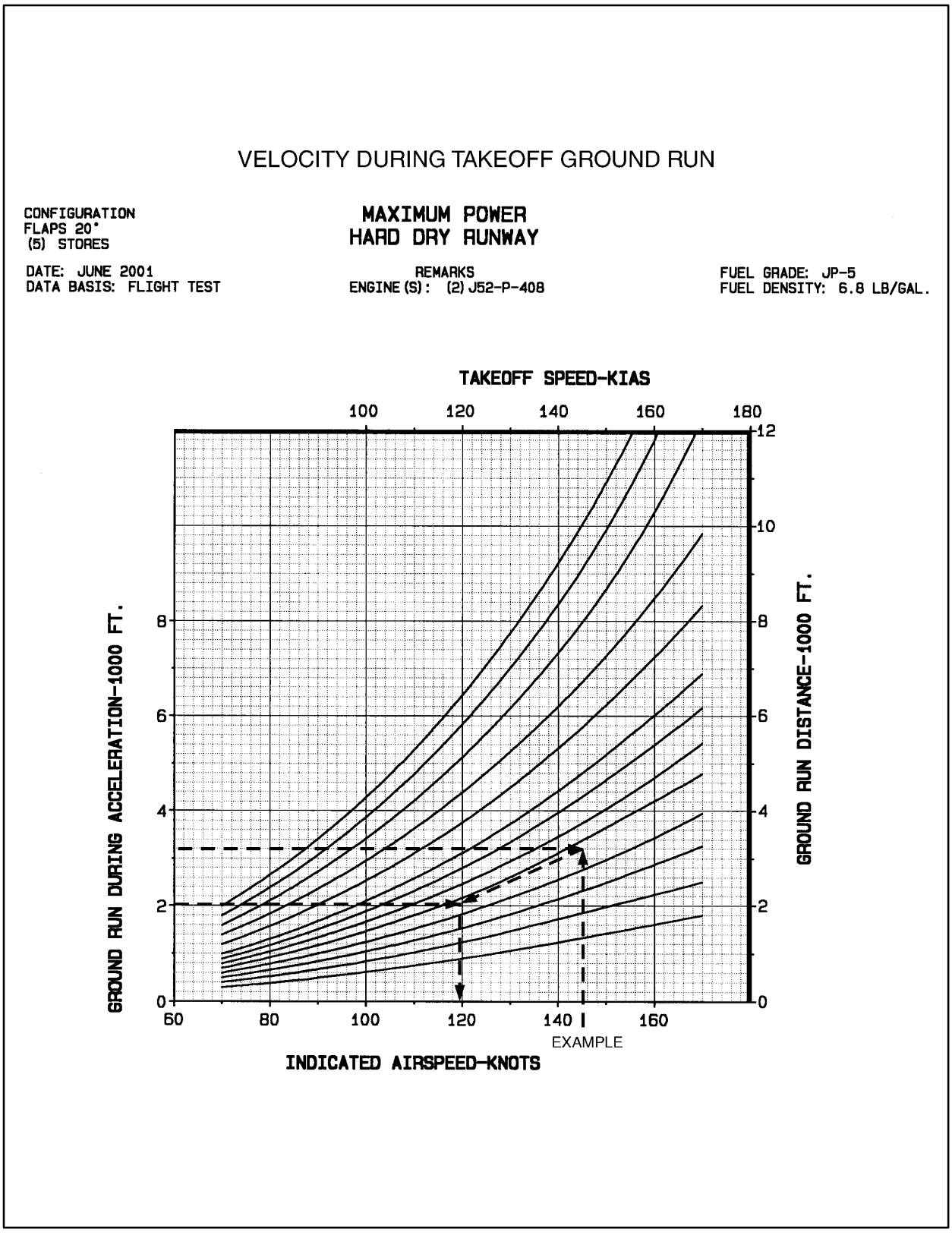


FIGURE 41. Example of velocity during takeoff ground run chart.

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Check the source to verify that this is the current version before use.

CLIMB PERFORMANCE
CLIMB SPEED SCHEDULE
MILITARY POWER

STORES CONFIGURATION:
ALL DRAG INDEXES

DATE:
DATA BASIS: ESTIMATED BASED ON FLIGHT TEST

REMARKS:
ENGINE(S): (2) F110-GE-400
ICAO STANDARD DAY

FUEL GRADE: JP-5
FUEL DENSITY: 6.8 LB/GAL

DRAG INDEX	INDICATED MACH NUMBER
0	0.745
10	0.743
20	0.740
30	0.737
40	0.735
50	0.732
60	0.729
70	0.726

DRAG INDEX	INDICATED MACH NUMBER
80	0.724
90	0.721
100	0.718
110	0.715
120	0.713
130	0.710
140	0.707
150	0.705

NOTE
TEMPERATURE DEVIATION FROM STANDARD DAY OF ±20°C DOES NOT HAVE AN EFFECT ON CLIMB SPEED SCHEDULE.

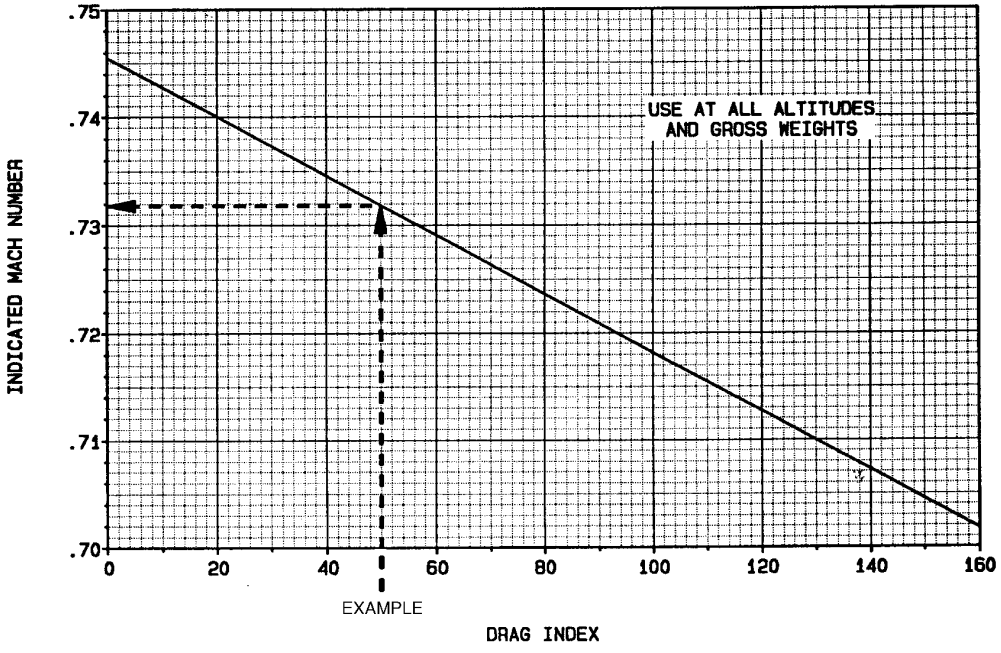


FIGURE 42. Example of climb performance chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

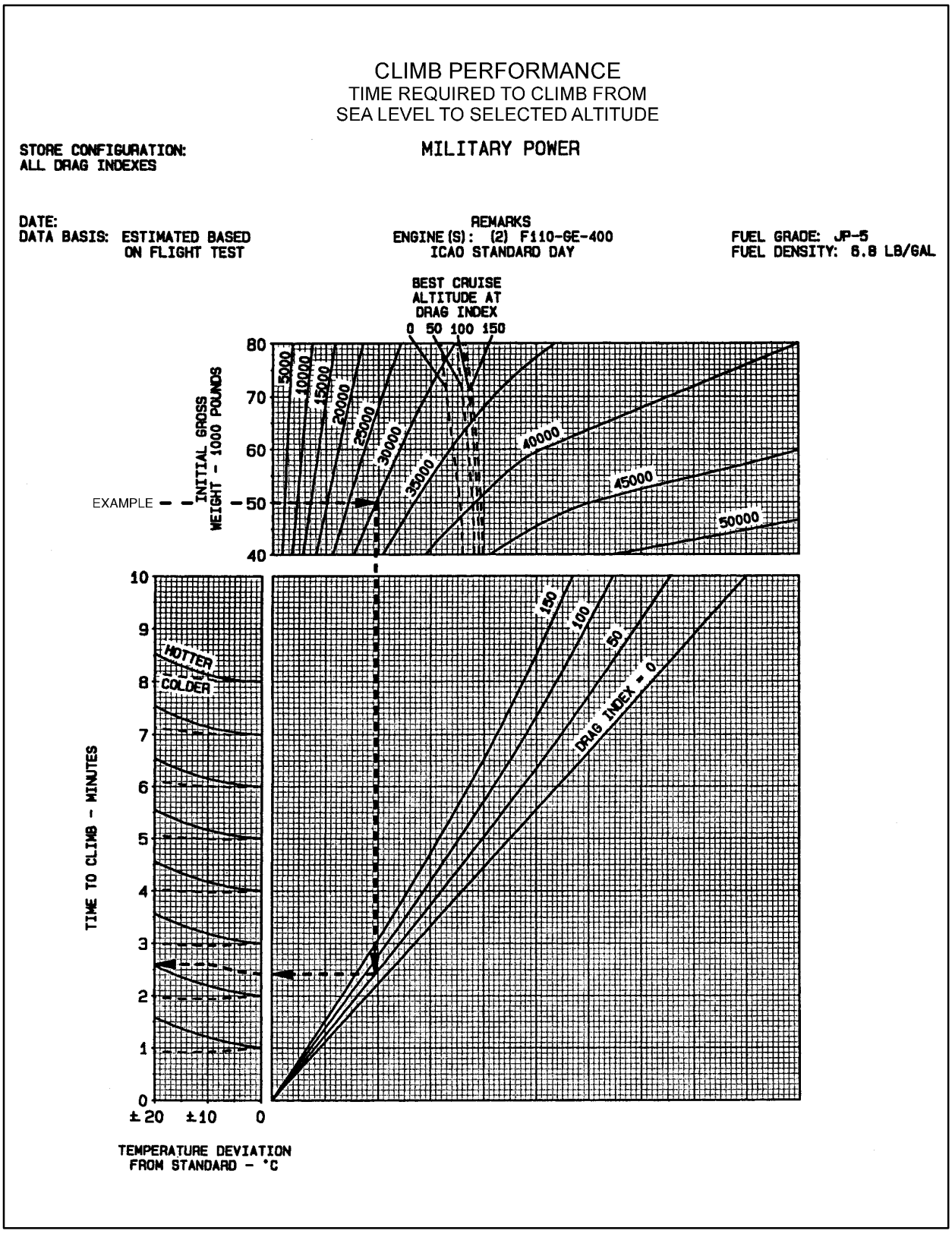


FIGURE 42. Example of climb performance chart — Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

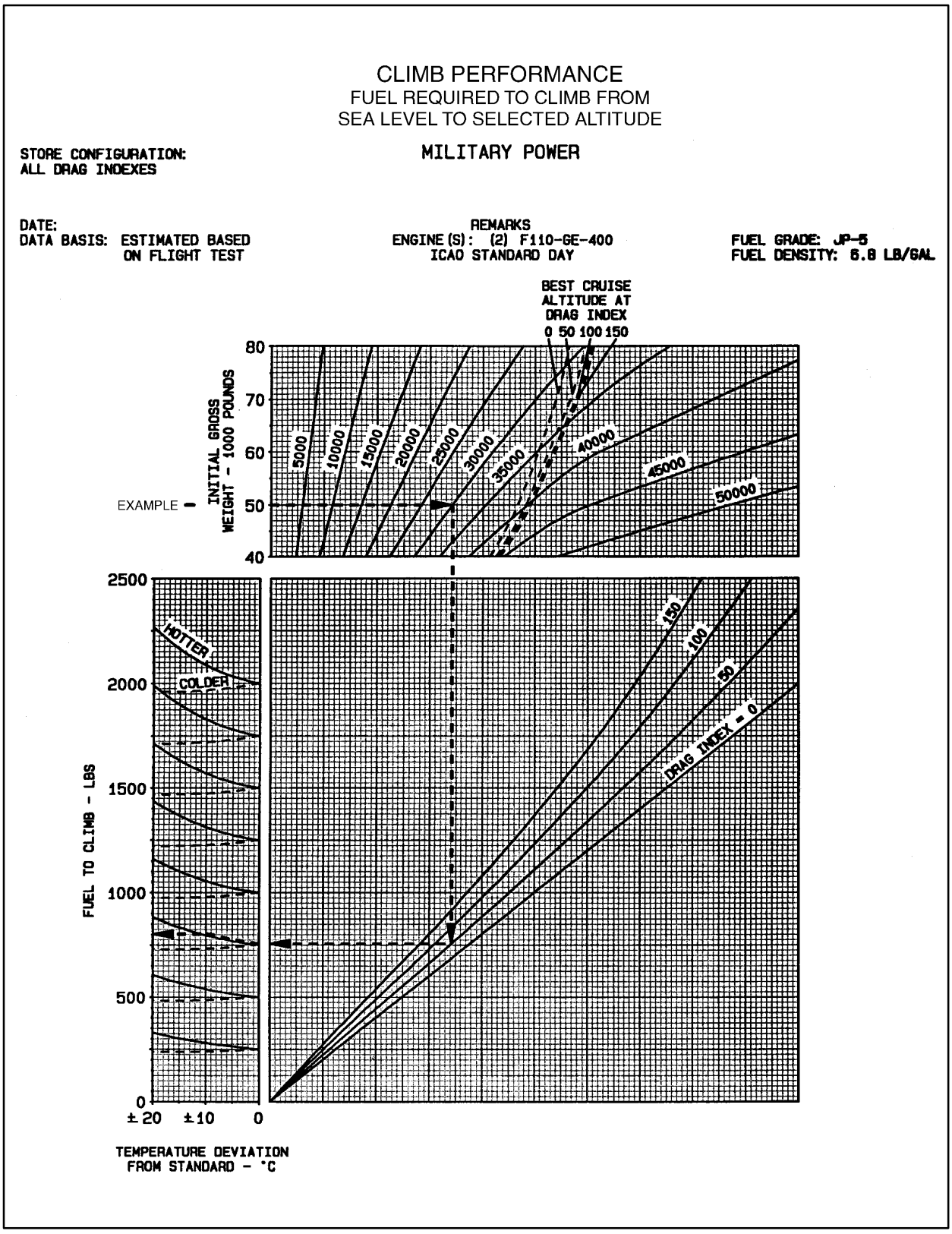


FIGURE 42. Example of climb performance chart — Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

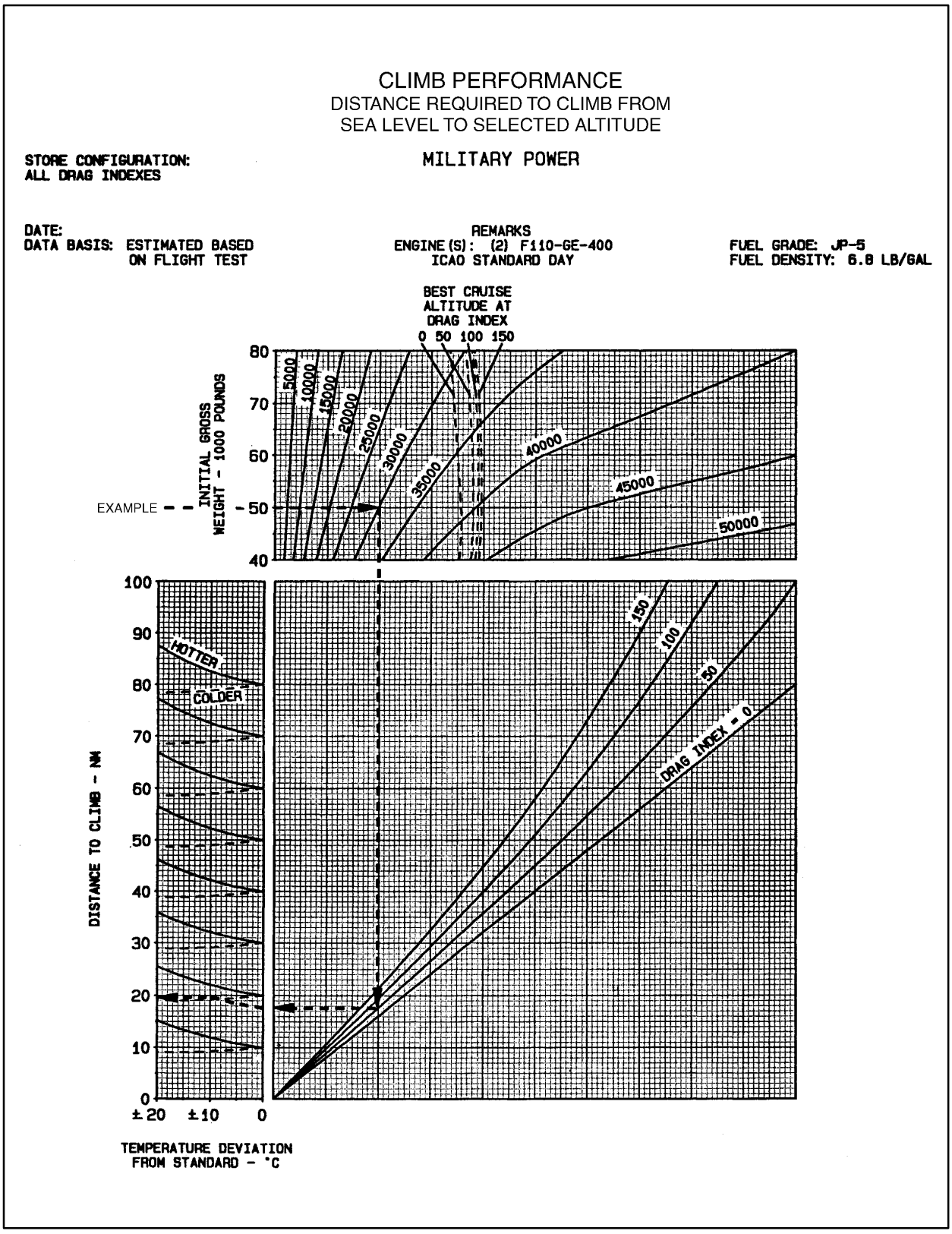


FIGURE 42. Example of climb performance chart — Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

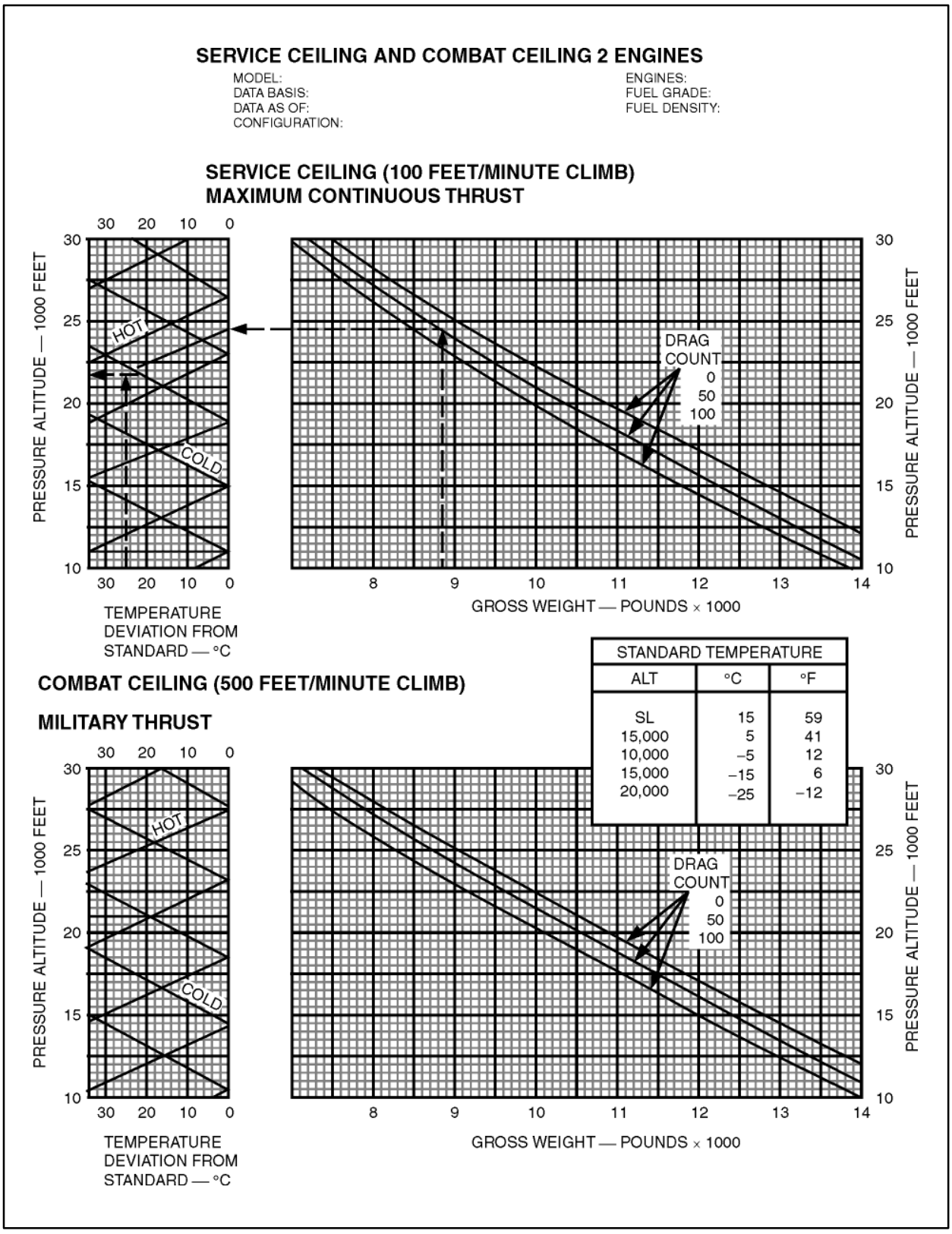


FIGURE 43. Example of service ceiling and combat ceiling chart.

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Check the source to verify that this is the current version before use.

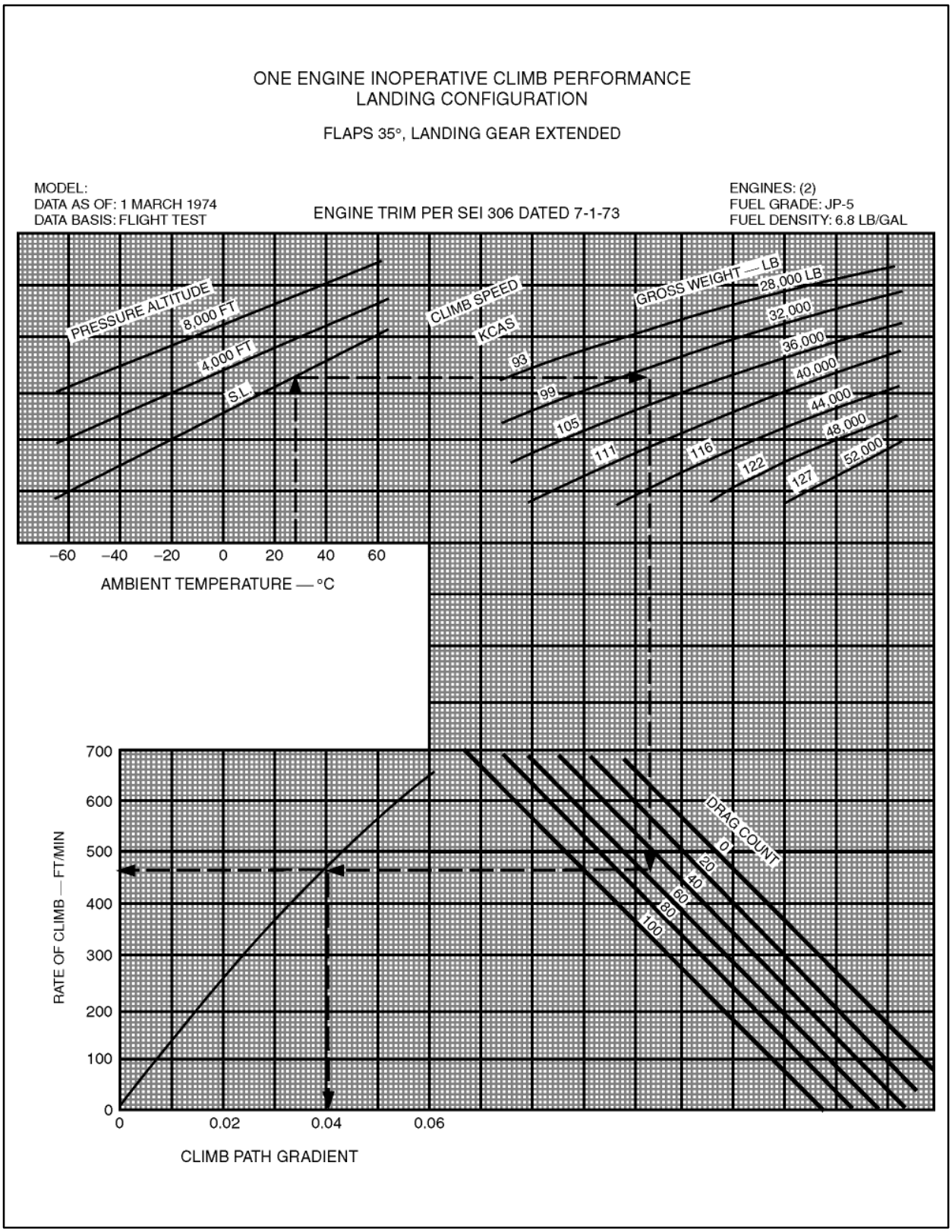


FIGURE 44. Example of one engine inoperative climb performance chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

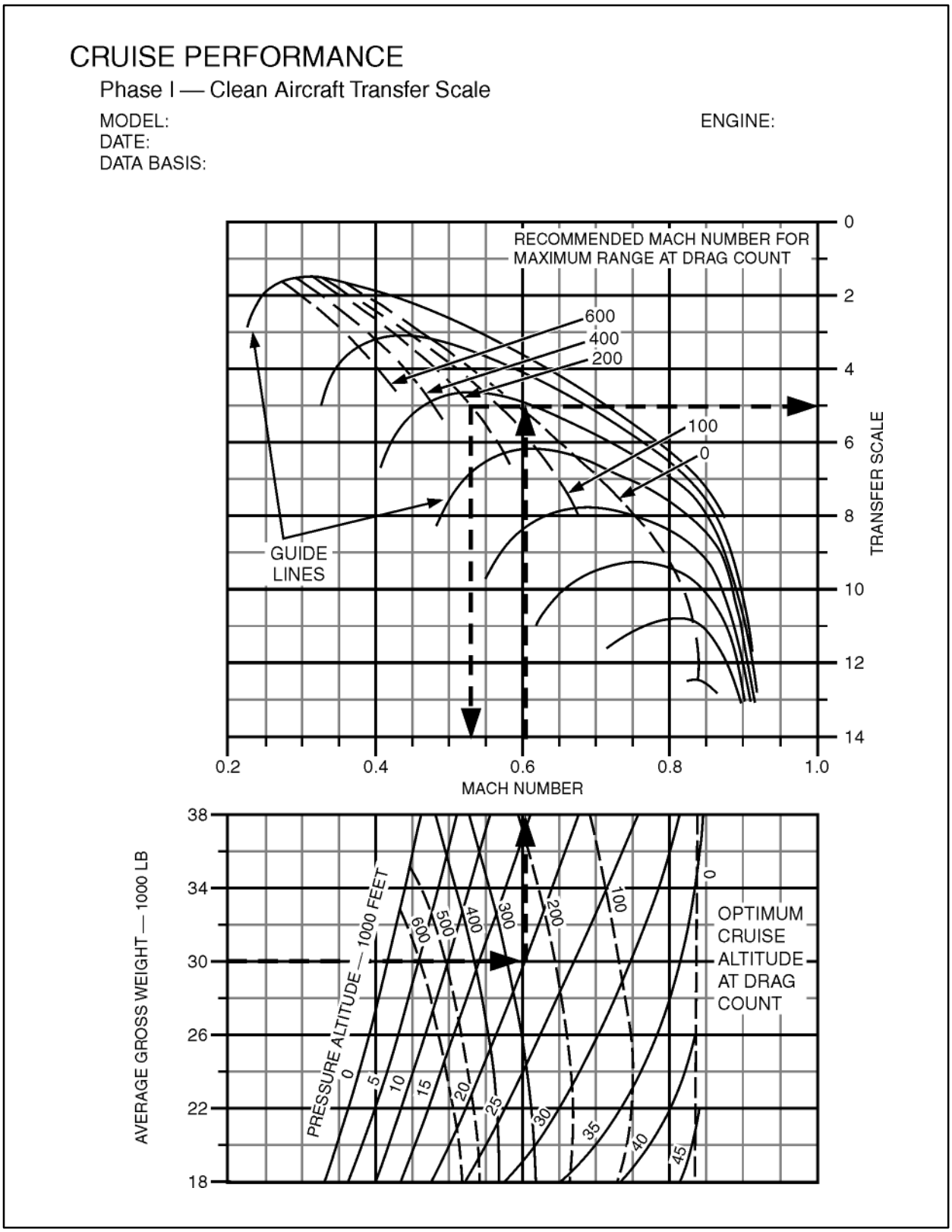


FIGURE 45. Example of cruise performance chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

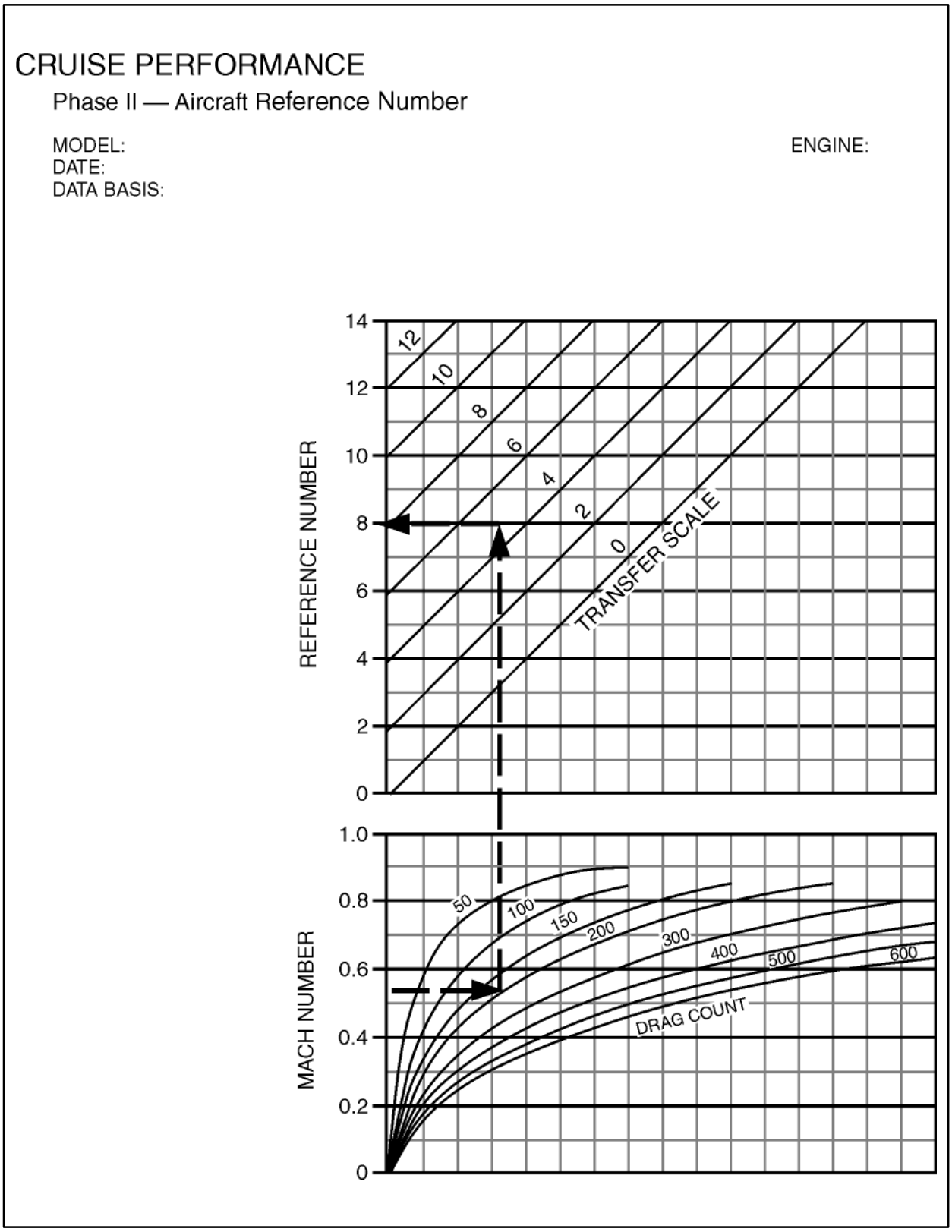


FIGURE 45. Example of cruise performance chart — Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

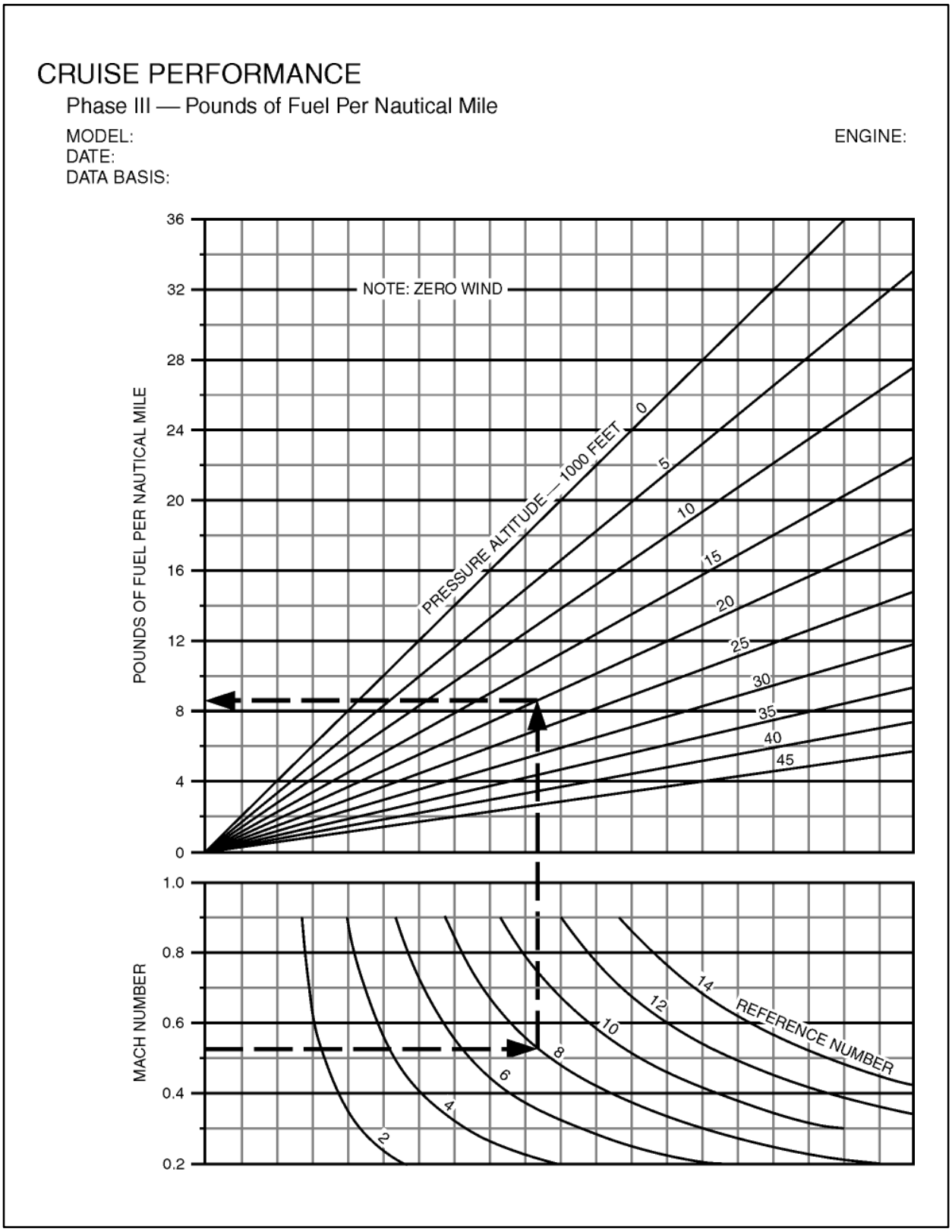


FIGURE 45. Example of cruise performance chart — Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

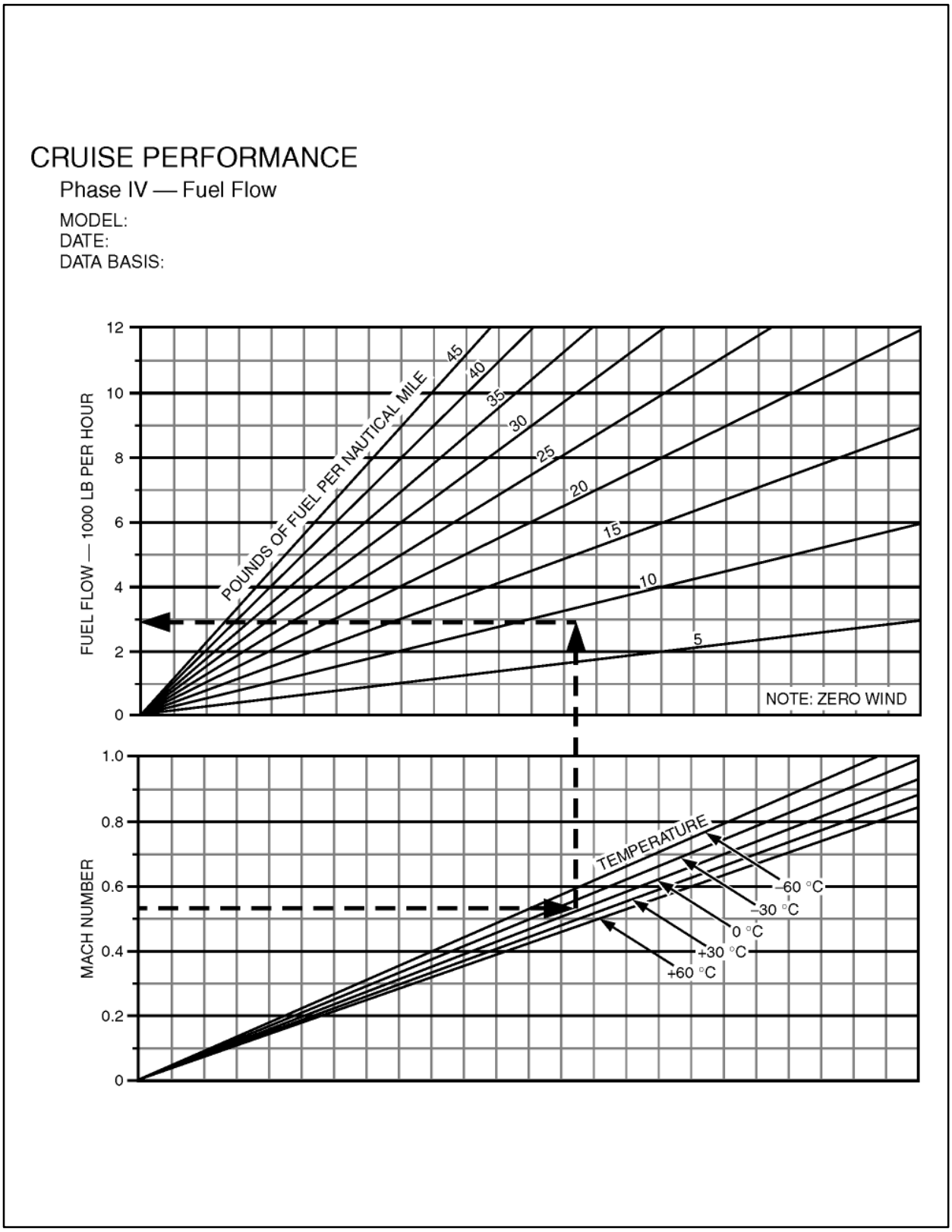


FIGURE 45. Example of cruise performance chart — Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

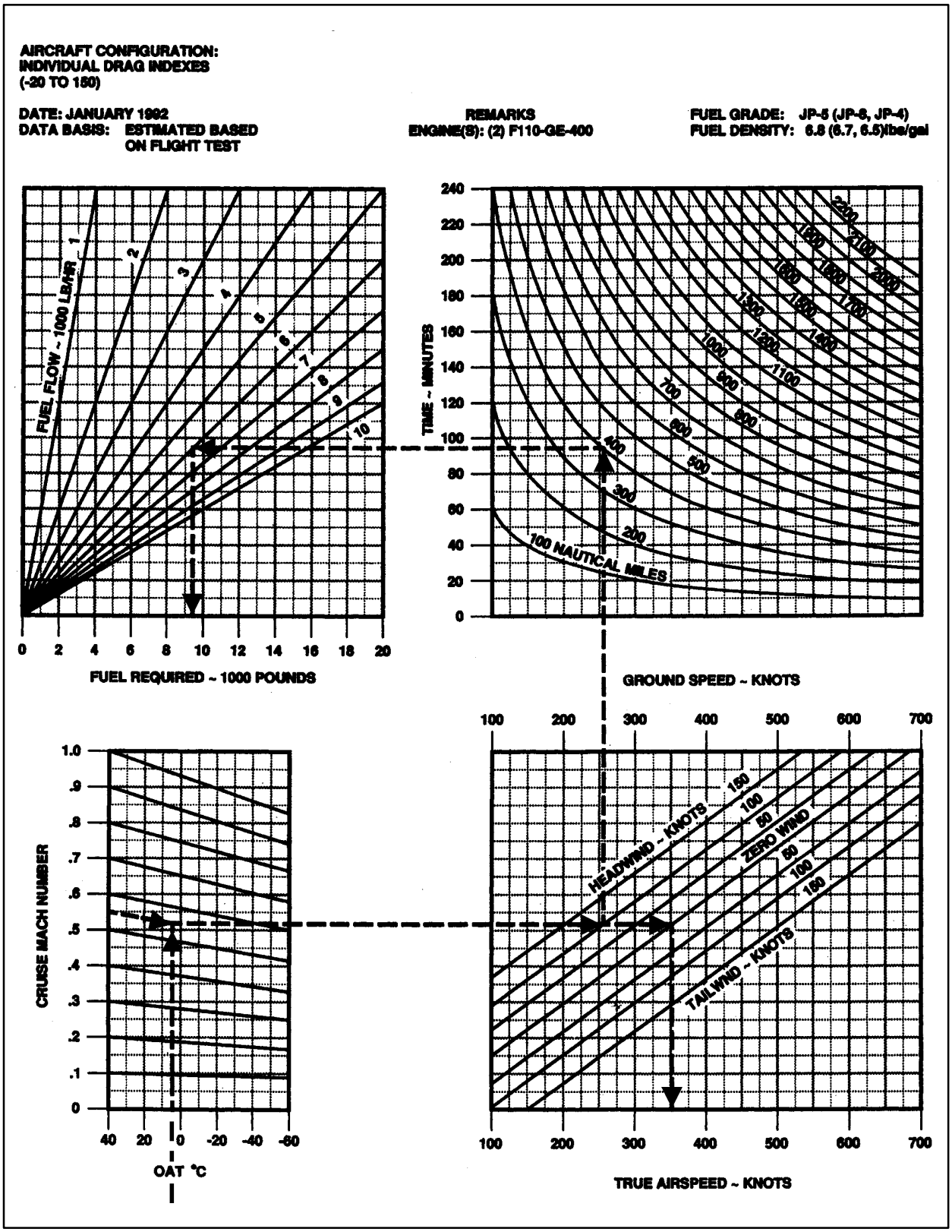


FIGURE 46. Example of maximum range cruise at constant altitude chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

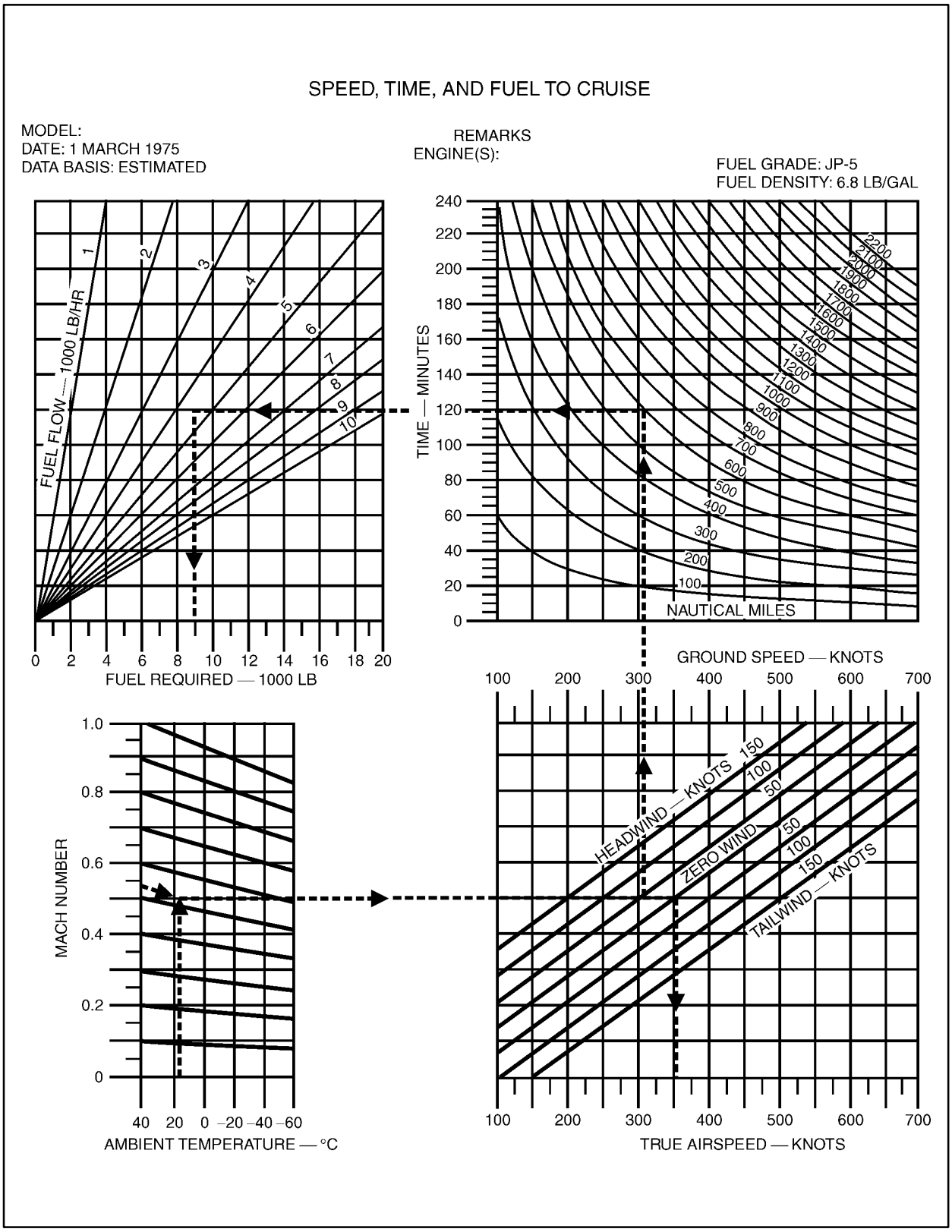


FIGURE 47. Example of speed, time, and fuel to cruise chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

LOW ALTITUDE CRUISE

GROSS WEIGHT – 45,000 POUNDS

AIRPLANE CONFIGURATION
INDIVIDUAL DRAG COUNTS

REMARKS
ENGINE(S): (2) J79-GE-10

MODEL:
DATE:
DATA BASIS:

FUEL GRADE:
FUEL DENSITY:

MULTIPLY TOTAL FUEL FLOW FROM CHART BY THE TEMPERATURE EFFECTS FACTOR TO OBTAIN ACTUAL FUEL FLOW

	KTAS	DRAG COUNT 0	TOTAL FUEL FLOW LB/HR 20	40	60	80	100	120	TEMP EFFECTS °C	FACTOR
SEA LEVEL (15 °C)	360	7672	8475	9277	10102	10953	11788	12678		
	420	9401	10559	11774	12930	14247	15564	16917	–40	0.899
	480	11681	13256	14932	16712	18536	20431	22830	–20	0.937
	540	14587	16681	18780	21421	24796			0	0.973
	600	20463	23565	27573					20	1.008
									40	1.042
4,000 FEET (7 °C)	360	6988	7689	8423	9175	9918	10678	11456		
	420	8493	9527	10571	11634	12711	13963	15355	–40	0.913
	480	10438	11856	13233	14861	16770	18705	20556	–20	0.949
	540	12987	14854	16787	19700	22525			0	0.987
	600	19082	22676						20	1.022
									40	1.057
8,000 FEET (–1 °C)	360	6365	7002	7639	8292	8969	9635	10333		
	420	7654	8573	9500	10431	11413	12389	13586	–40	0.925
	480	9308	10541	11778	13079	14681	16604	18734	–20	0.963
	540	11602	13139	14830	17299	20406			0	1.001
	600	17447	21064						20	1.037
									40	1.072
12,000 FEET (–9 °C)	360	5829	6384	6934	7532	8100	8709	9317		
	420	6930	7715	8526	9345	10213	11093	11990	–40	0.939
	480	8327	9394	10458	11600	12858	14376	16265	–20	0.978
	540	10357	11740	13170	15113	17750	19442		0	1.016
	600	15788	19093						20	1.052
									40	1.088
16,000 FEET (–16 °C)	360	5420	5900	6395	6876	7420	7929	8494		
	420	6300	6975	7706	8404	9186	9996	10891	–40	0.953
	480	7447	8383	9325	10374	11488	12730	14218	–20	0.993
	540	9335	10513	11884	13642	15551	17586		0	1.032
	600	14687	17797						20	1.069
									40	1.105

FIGURE 48. Example of low altitude cruise chart.

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Check the source to verify that this is the current version before use.

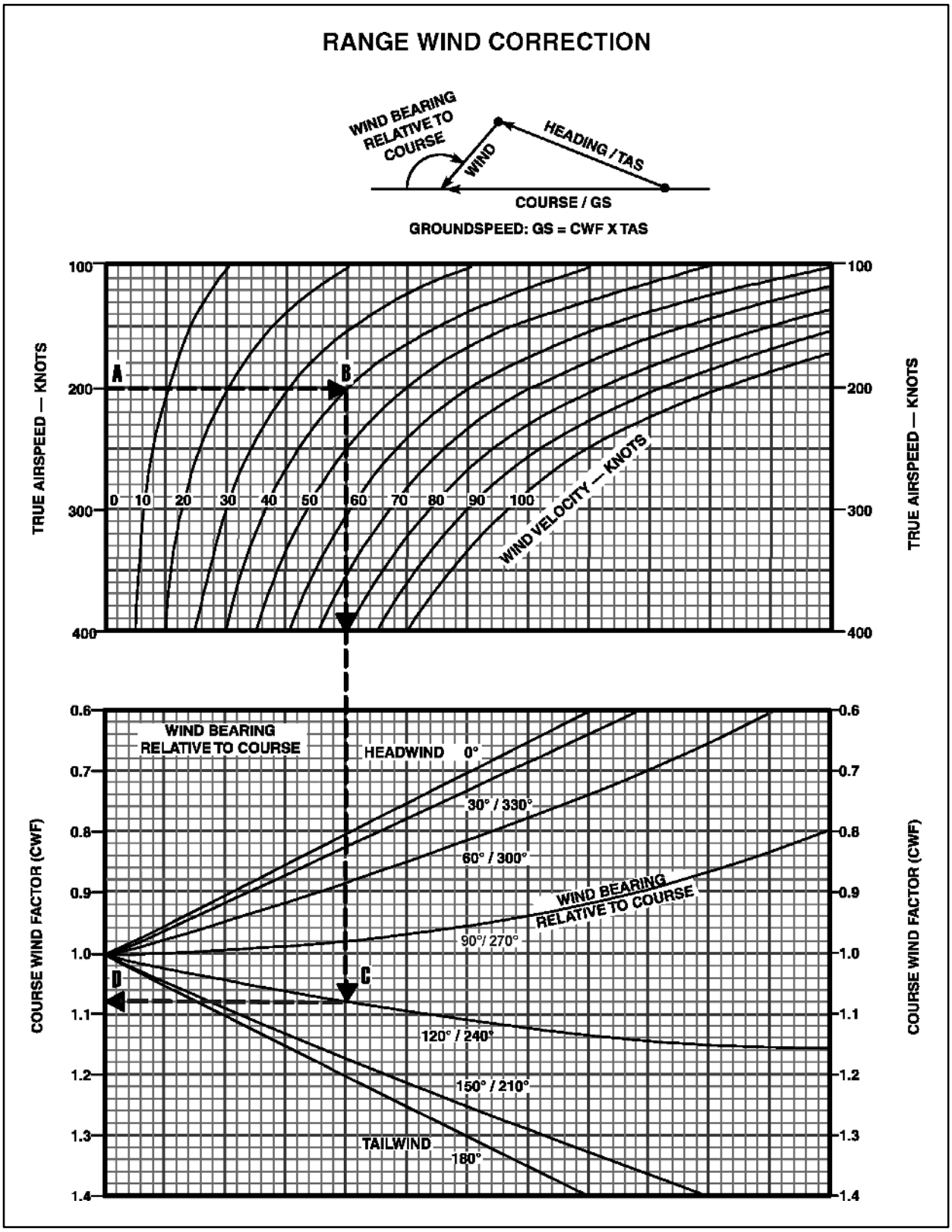


FIGURE 49. Example of range wind correction chart.

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TWO-ENGINE BINGO RANGE CHART

CONFIGURATION- _____

STANDARD DAY

MODEL:
DATA AS OF: 1 APRIL 1975
DATA BASIS: FLIGHT TEST

ENGINES:
FUEL GRADE: JP-5
FUEL DENSITY: 6.8 LB/GAL

- FUEL REQUIRED INCLUDES 1000 LB RESERVE FUEL
- INITIAL ALTITUDE IS SEA LEVEL
- CLIMB IS AT INTERMEDIATE POWER AT 220 KNOTS TO 0.6 MACH AND 0.6 MACH TO DESIRED ALTITUDE
- FLIGHT IDLE DESCENT IS AT 0.6 MACH TO 25,000 FT AND 250 KNOTS BELOW 25,000 FT

MAXIMUM RANGE CRUISE						SEA LEVEL CRUISE	
DIST TO BASE N M	FUEL REQ LB	TIME REQ MIN	CRUISE ALT FEET	CRUISE CAS KNOTS	DIST TO DESC N M	FUEL REQ LB	TIME REQ MIN
25	1180	6	5000	207	12	1210	7
50	1360	12	10,500	203	12	1420	14
75	1510	17	15,000	202	26	1630	22
100	1660	21	20,000	200	38	1840	29
125	1790	26	24,000	196	49	2050	36
150	1920	30	28,500	193	71	2250	43
175	2030	35	33,500	188	83	2460	50
200	2140	37	39,000	179	112	2670	57
225	2230	42	40,000	177	116	2880	64
250	2330	46	40,000	177	116	3090	71

ALT FEET	CRUISE KCAS
SL	212
5000	207
20,000	200

NOTES:

1. CHART VALID FOR GROSS WEIGHTS BELOW 36,000 LB
2. AT 5000 FT CRUISE FUEL REQUIRED IS 0 TO 200 LB LESS THAN SEA LEVEL CRUISE
3. AT 20,000 FT CRUISE FUEL REQUIRED IS 0 TO 100 LB MORE THAN MAXIMUM RANGE CRUISE

FIGURE 50. Example of bingo chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

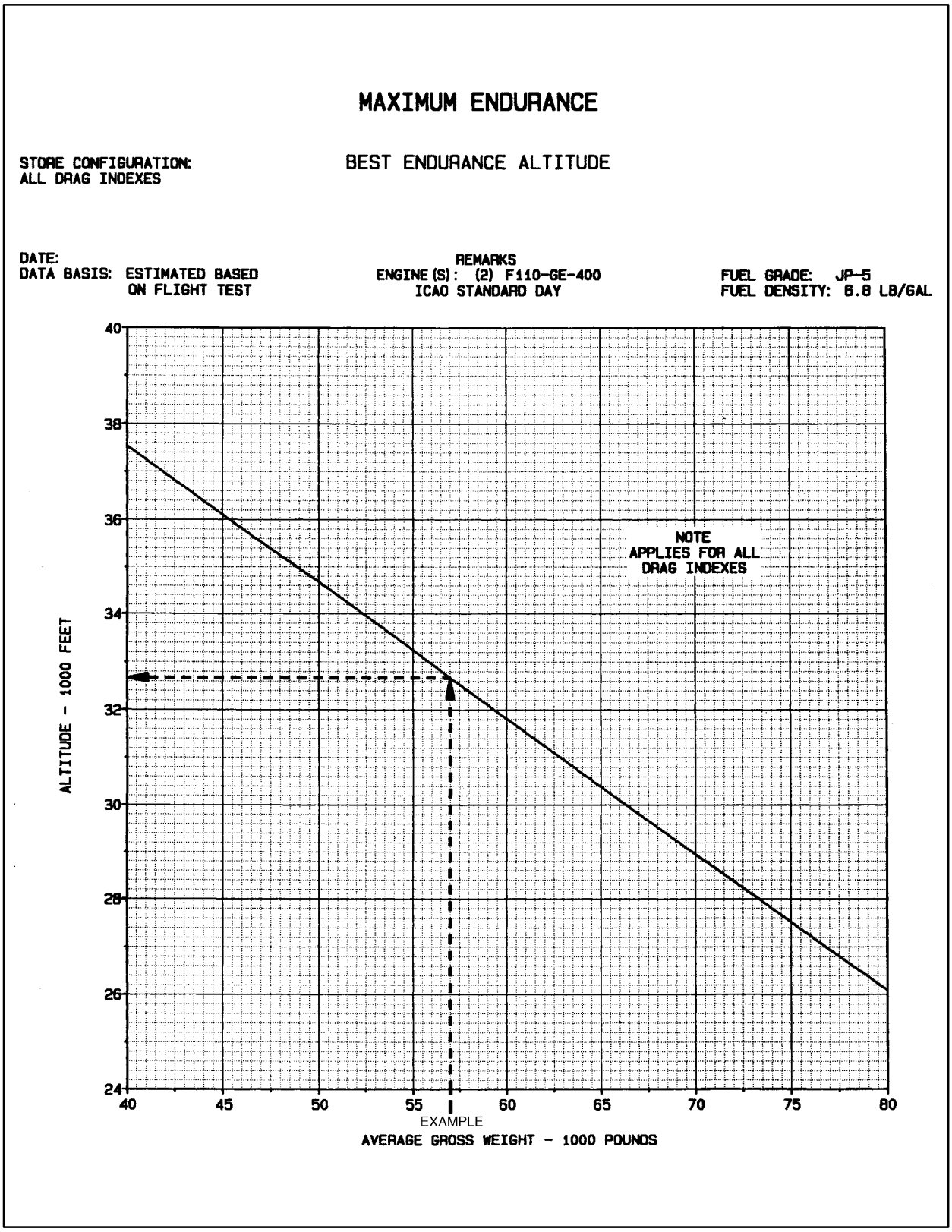


FIGURE 51. Example of maximum endurance chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

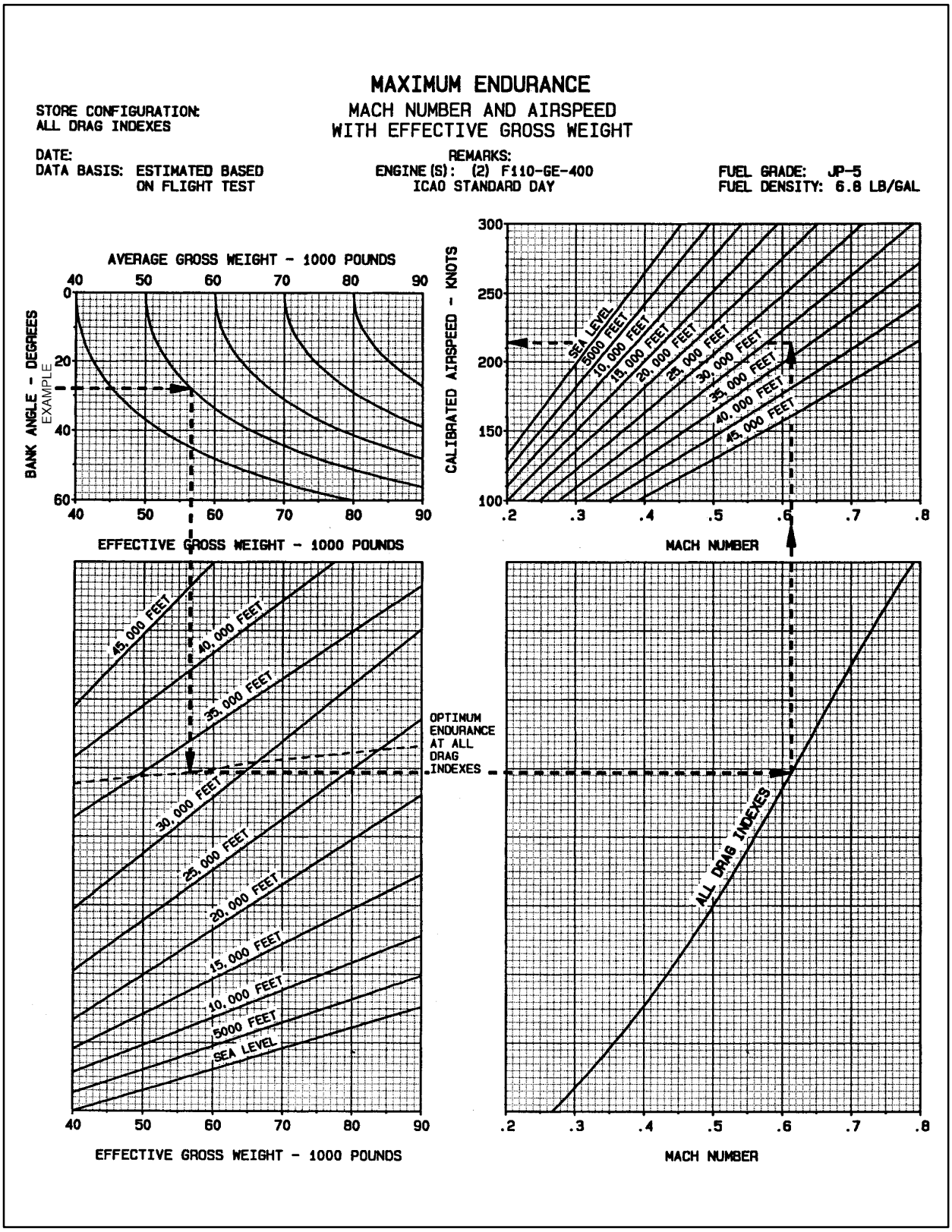


FIGURE 51. Example of maximum endurance chart — Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

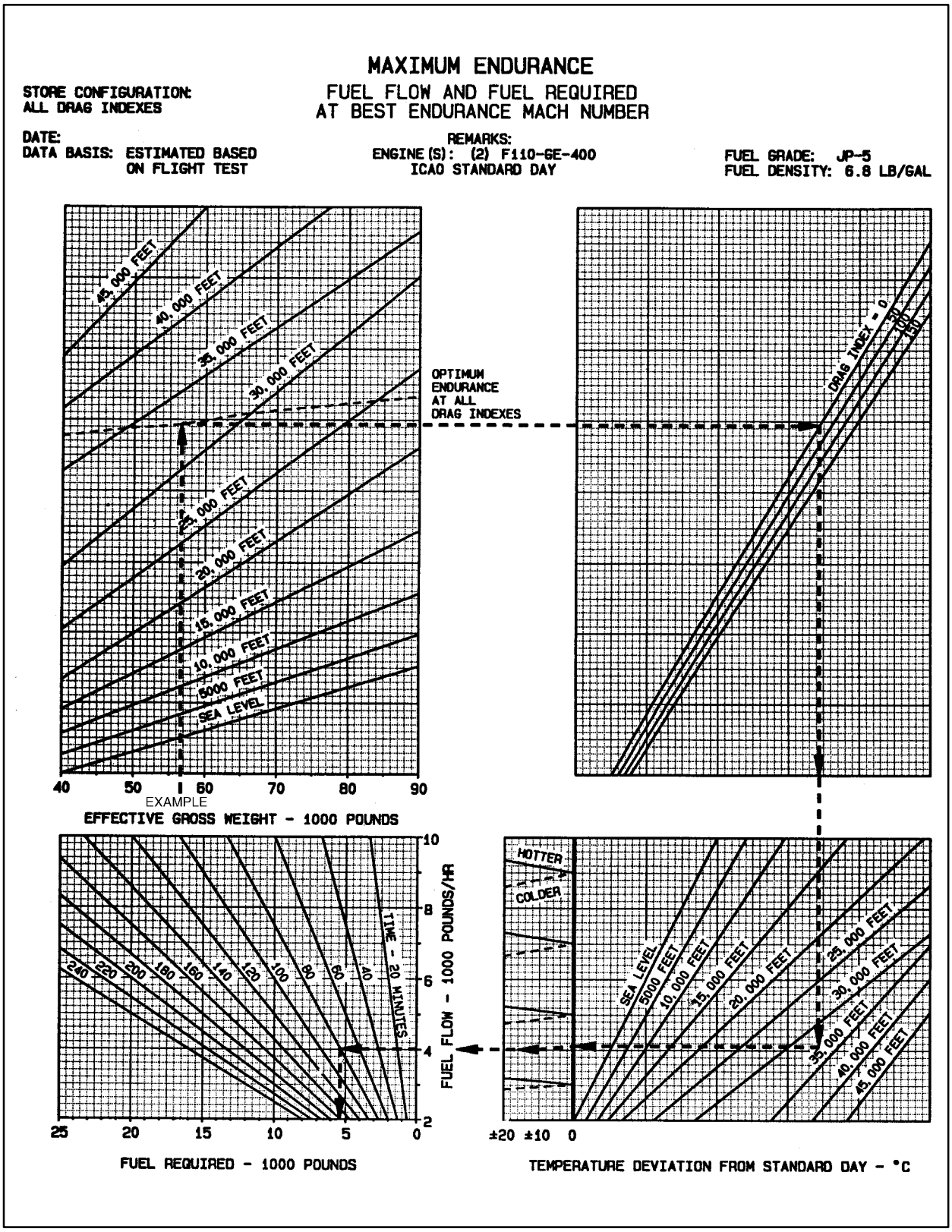


FIGURE 51. Example of maximum endurance chart — Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

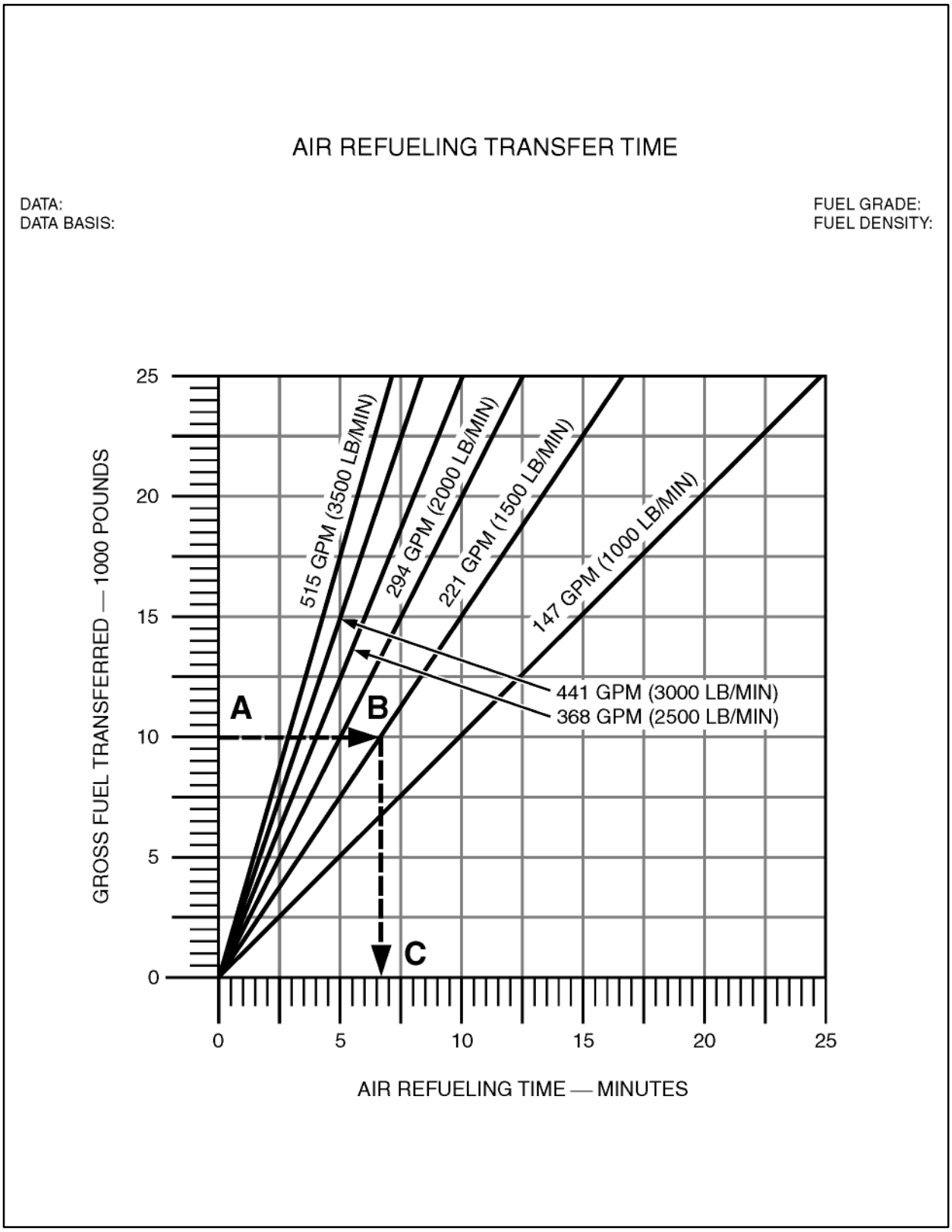


FIGURE 52. Example of air refueling transfer time chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

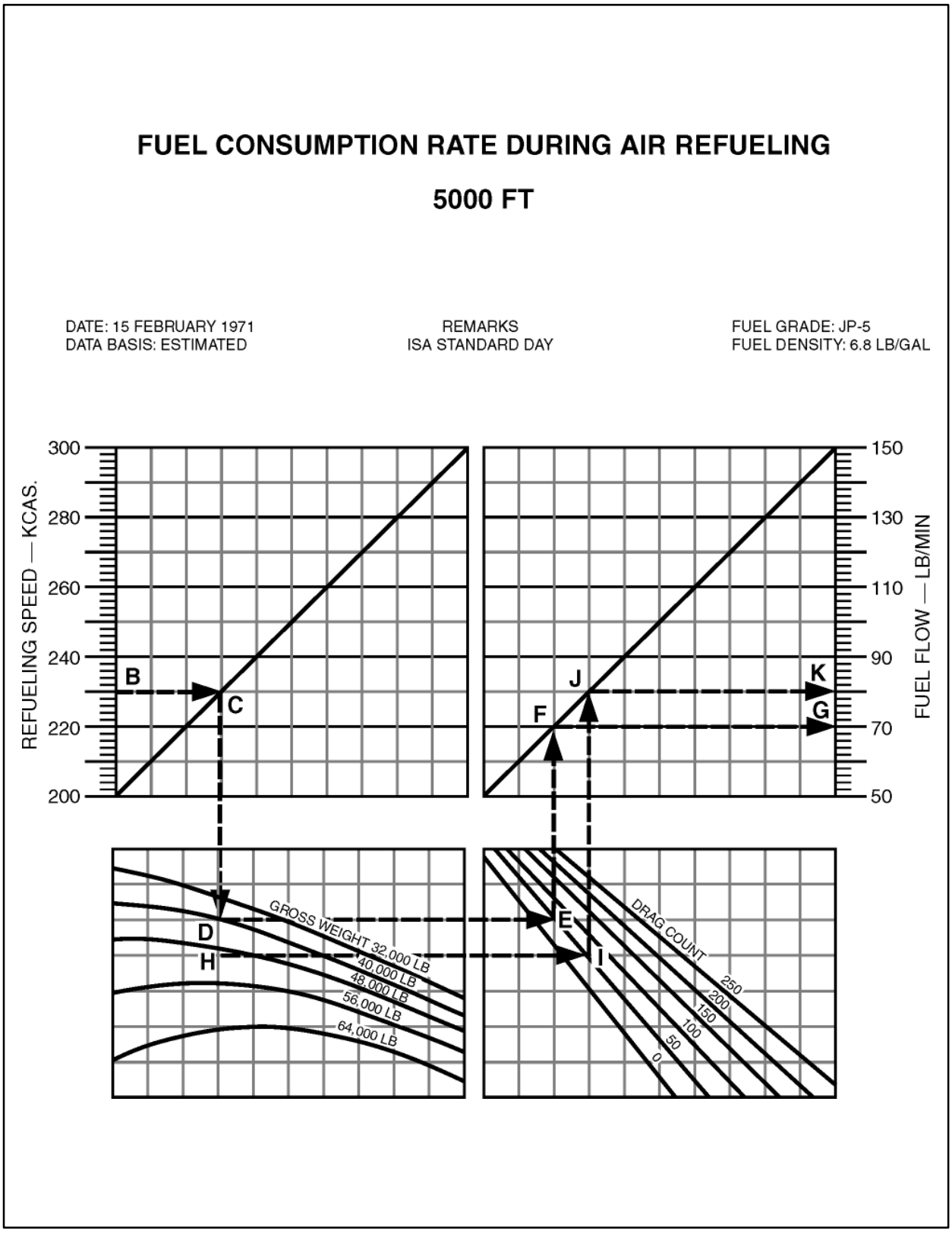


FIGURE 53. Example of fuel consumption rate during air refueling chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

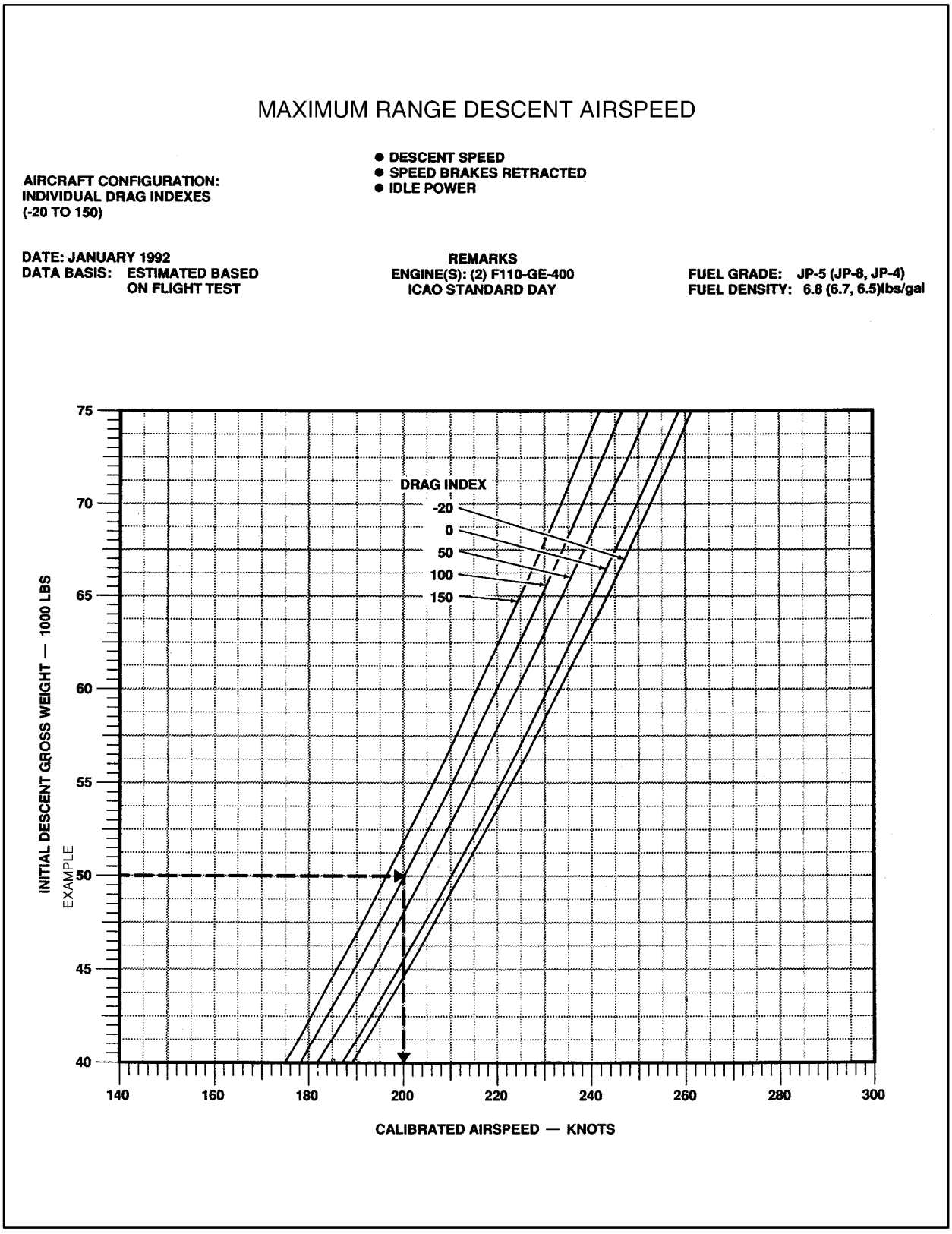


FIGURE 54. Example of maximum range descent chart

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

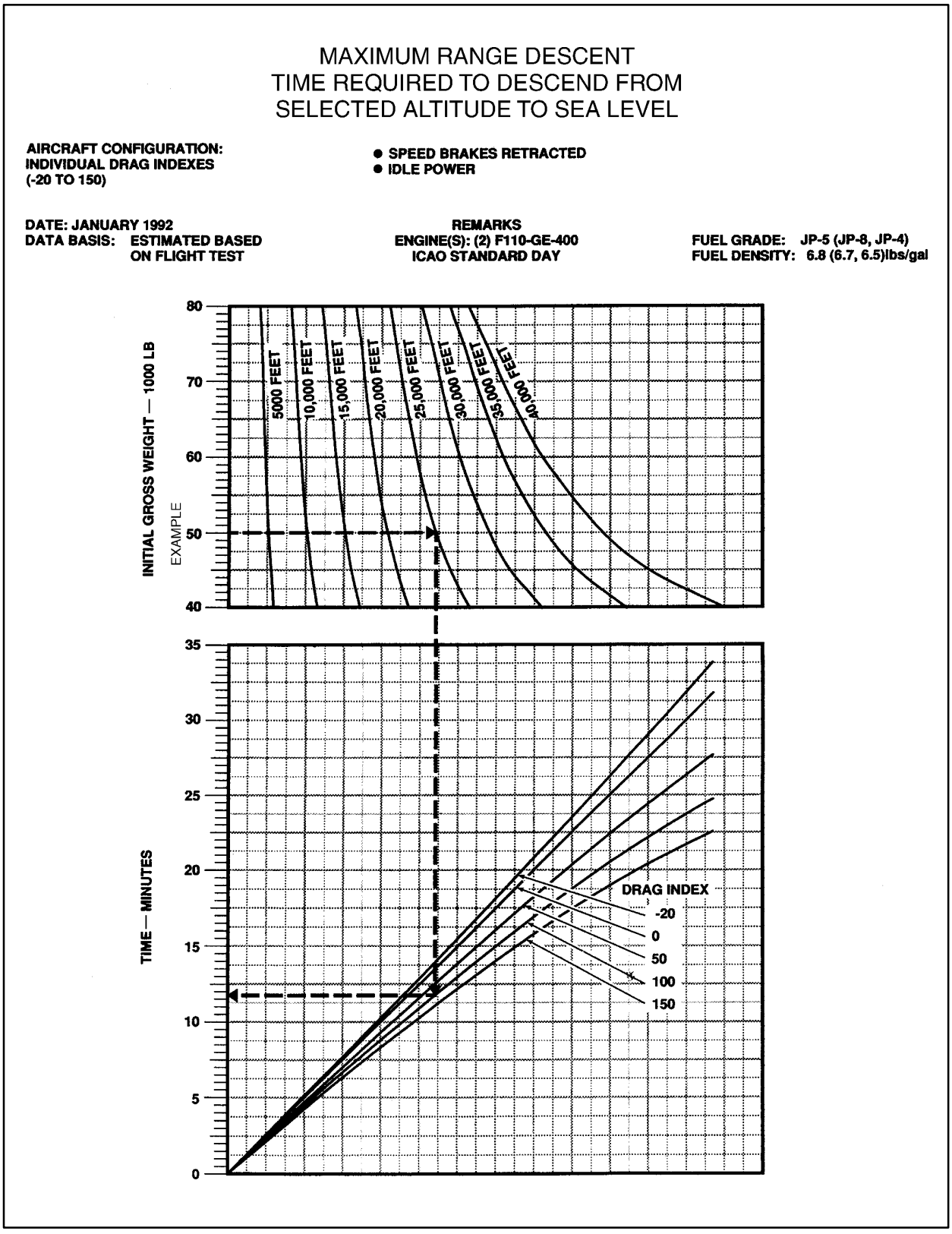


FIGURE 54. Example of maximum range descent chart — Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

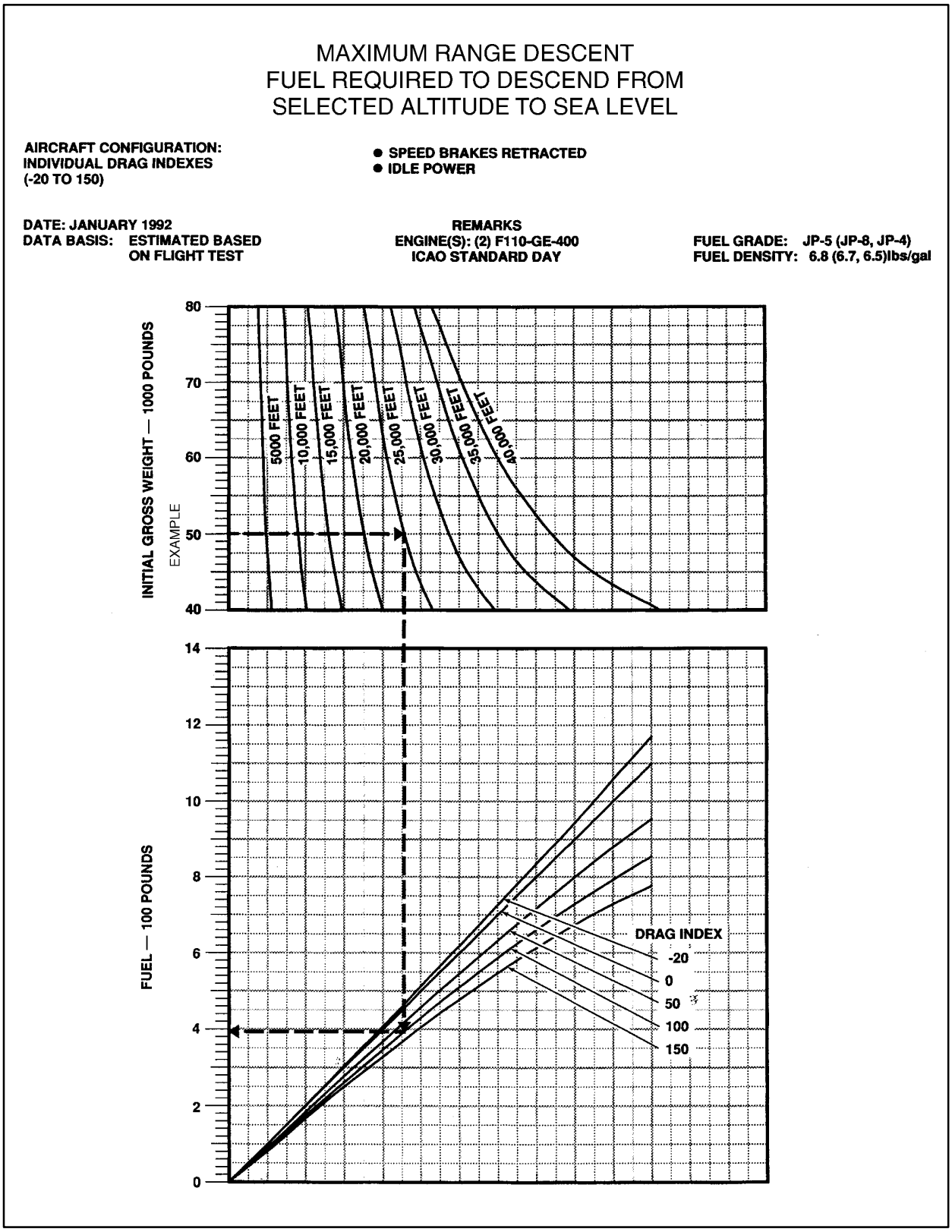


FIGURE 54. Example of maximum range descent chart — Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

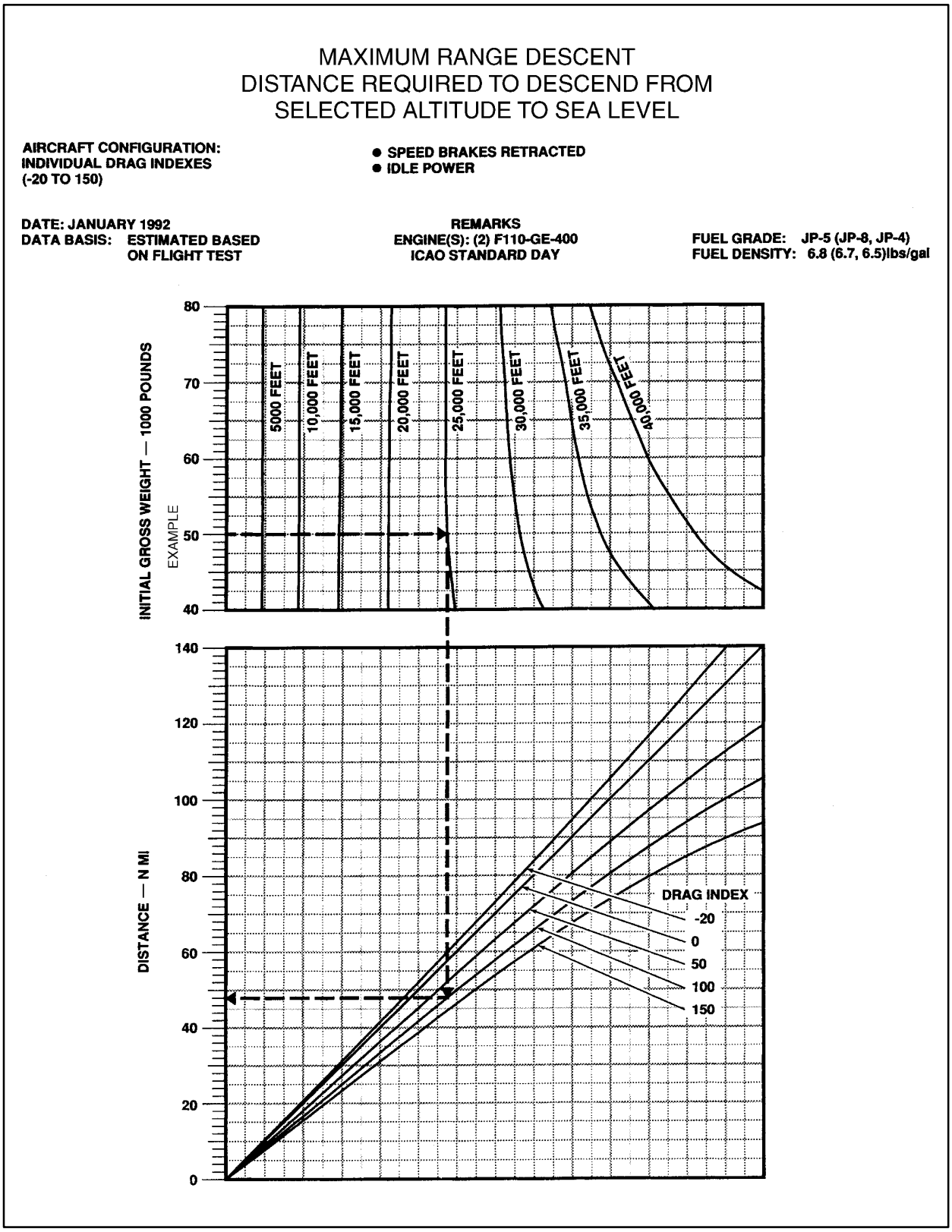


FIGURE 54. Example of maximum range descent chart — Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

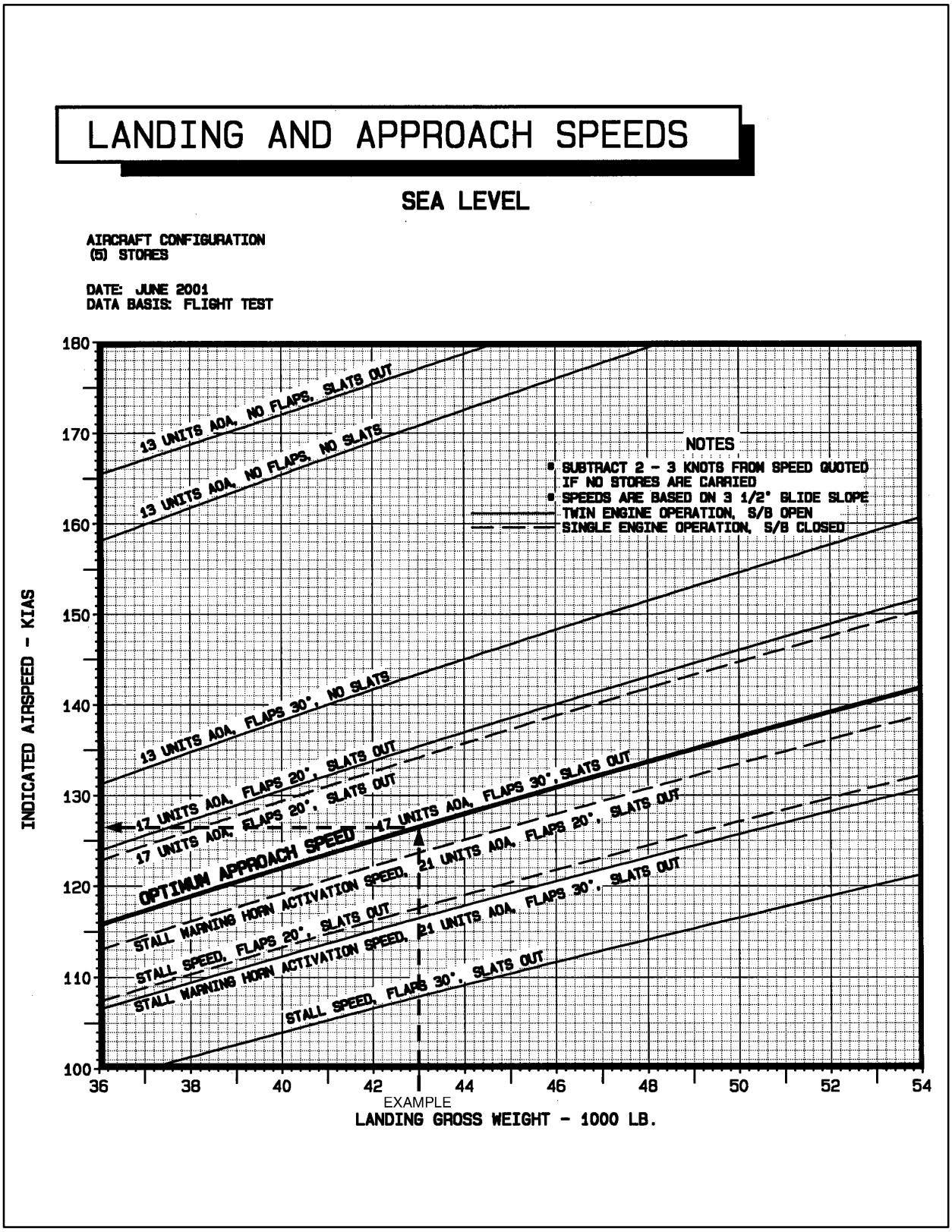


FIGURE 55. Example of landing speeds chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

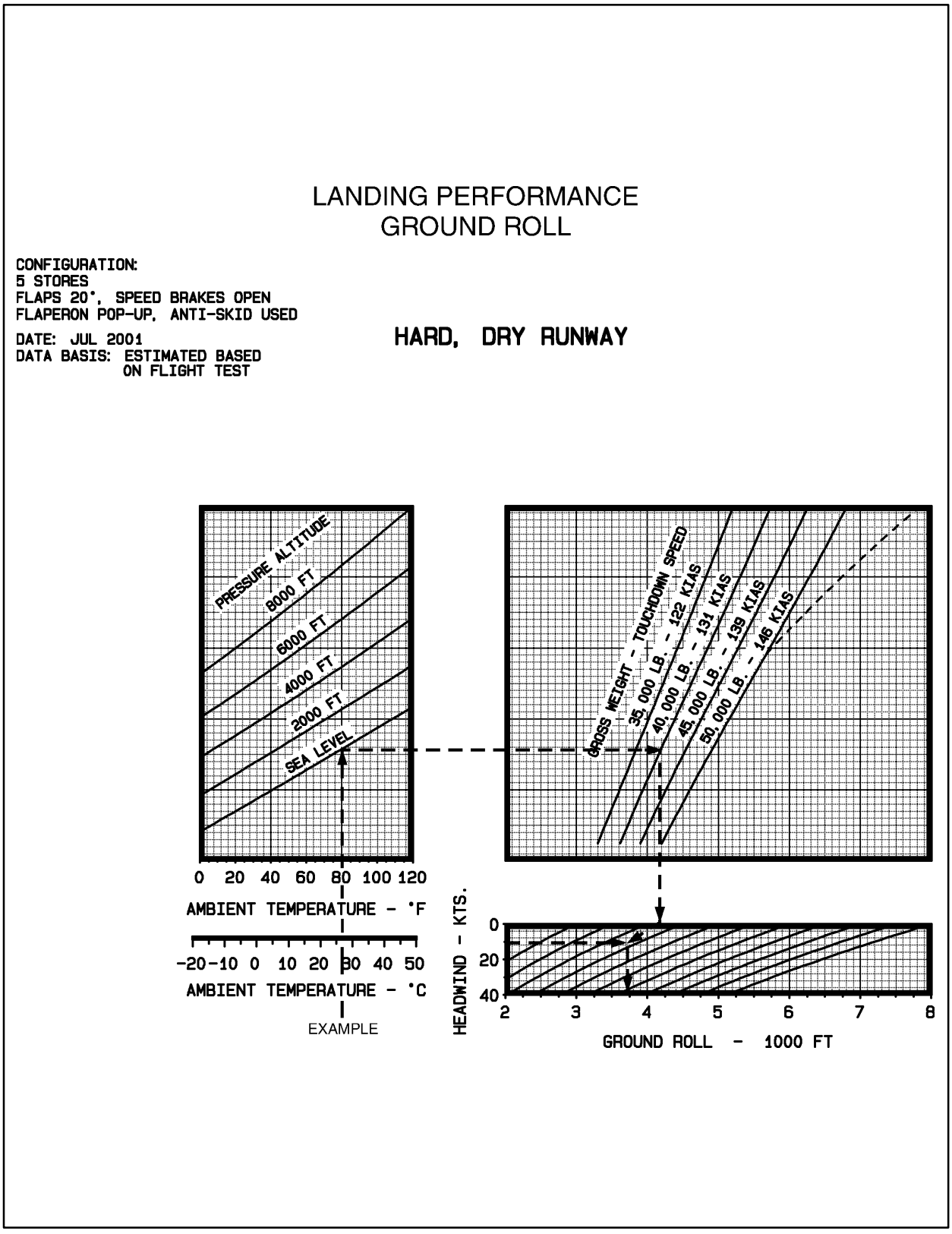


FIGURE 56. Example of landing performance — ground roll chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

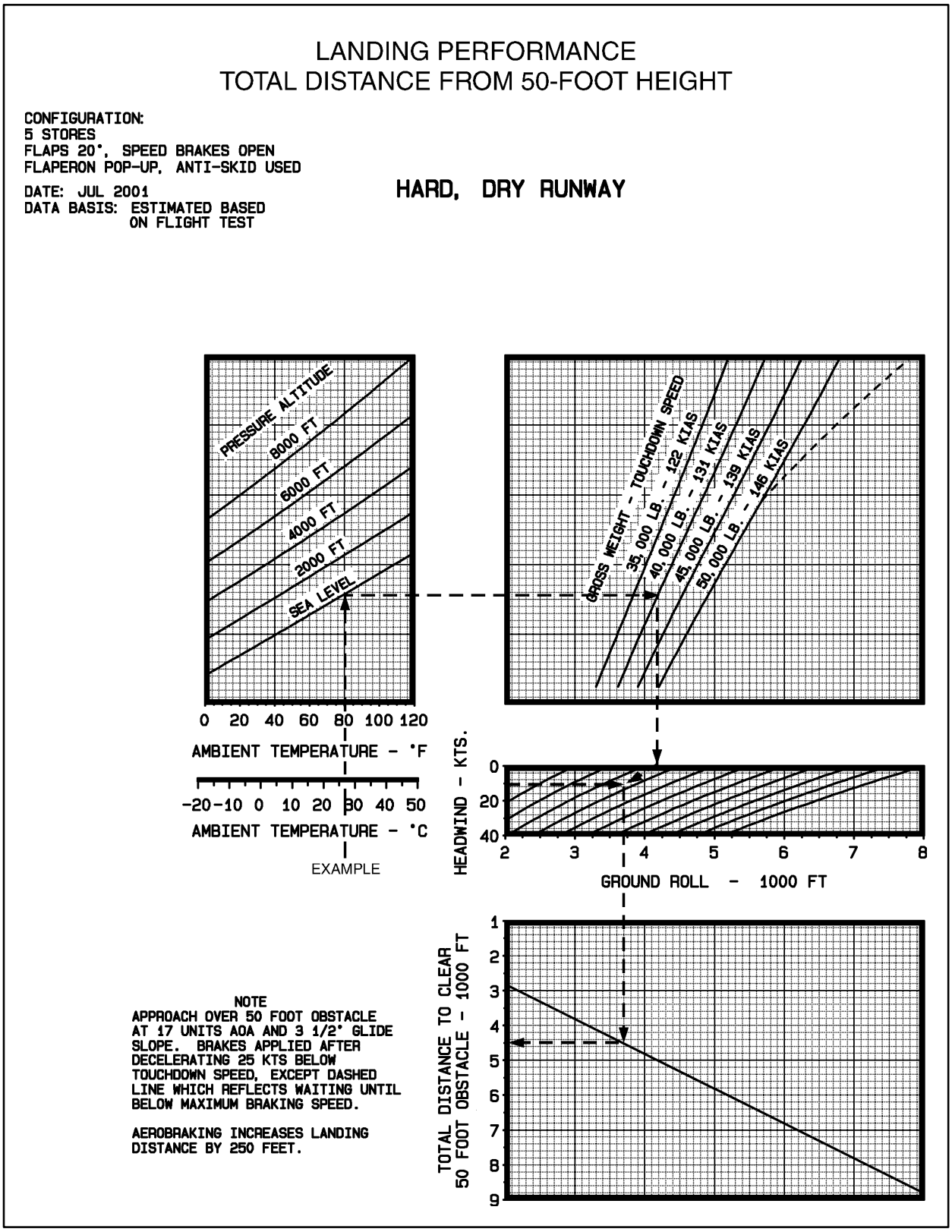


FIGURE 57. Example of landing performance — total distance from 50−foot height chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

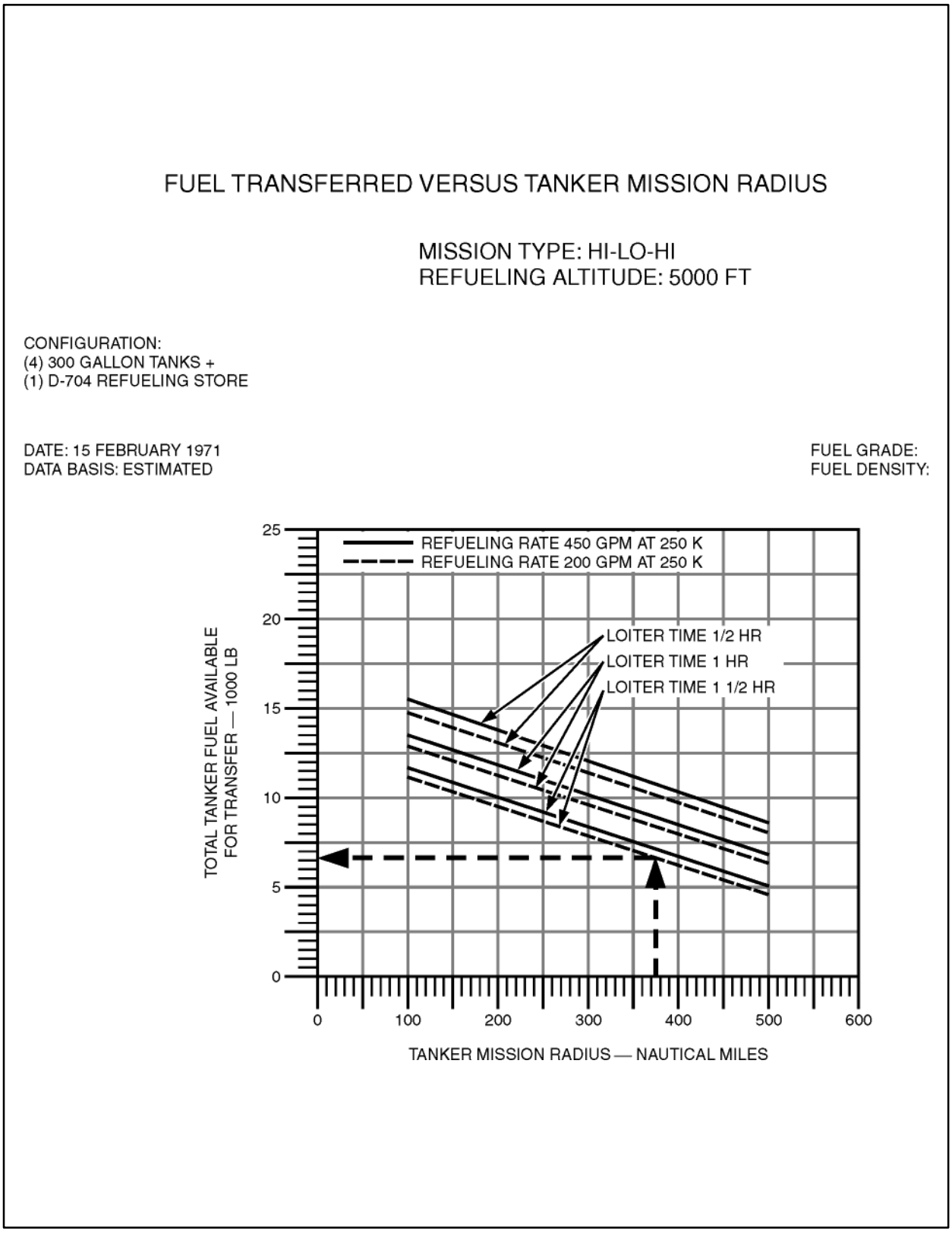


FIGURE 58. Example of fuel transferred versus tanker mission radius chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

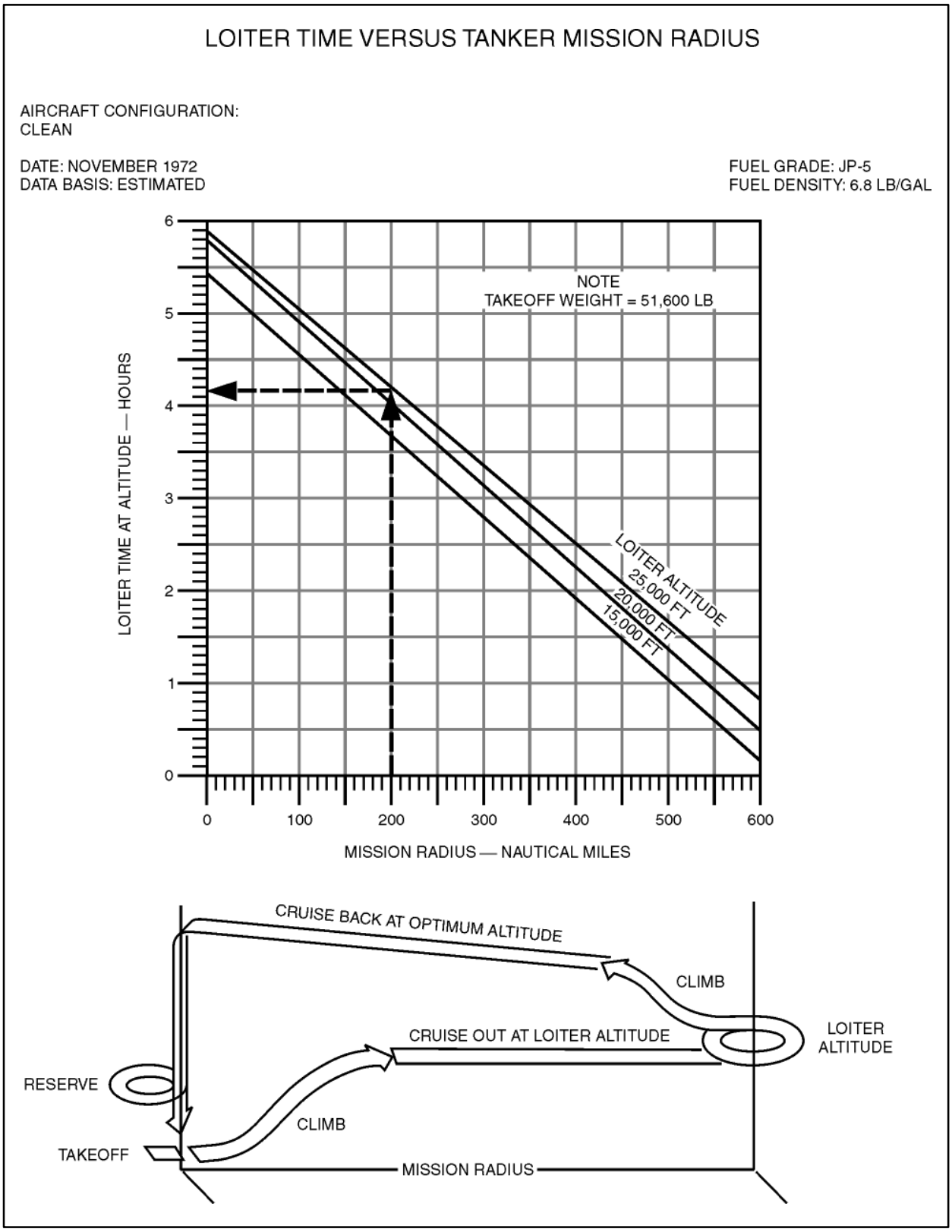


FIGURE 59. Example of loiter time versus tanker mission radius chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

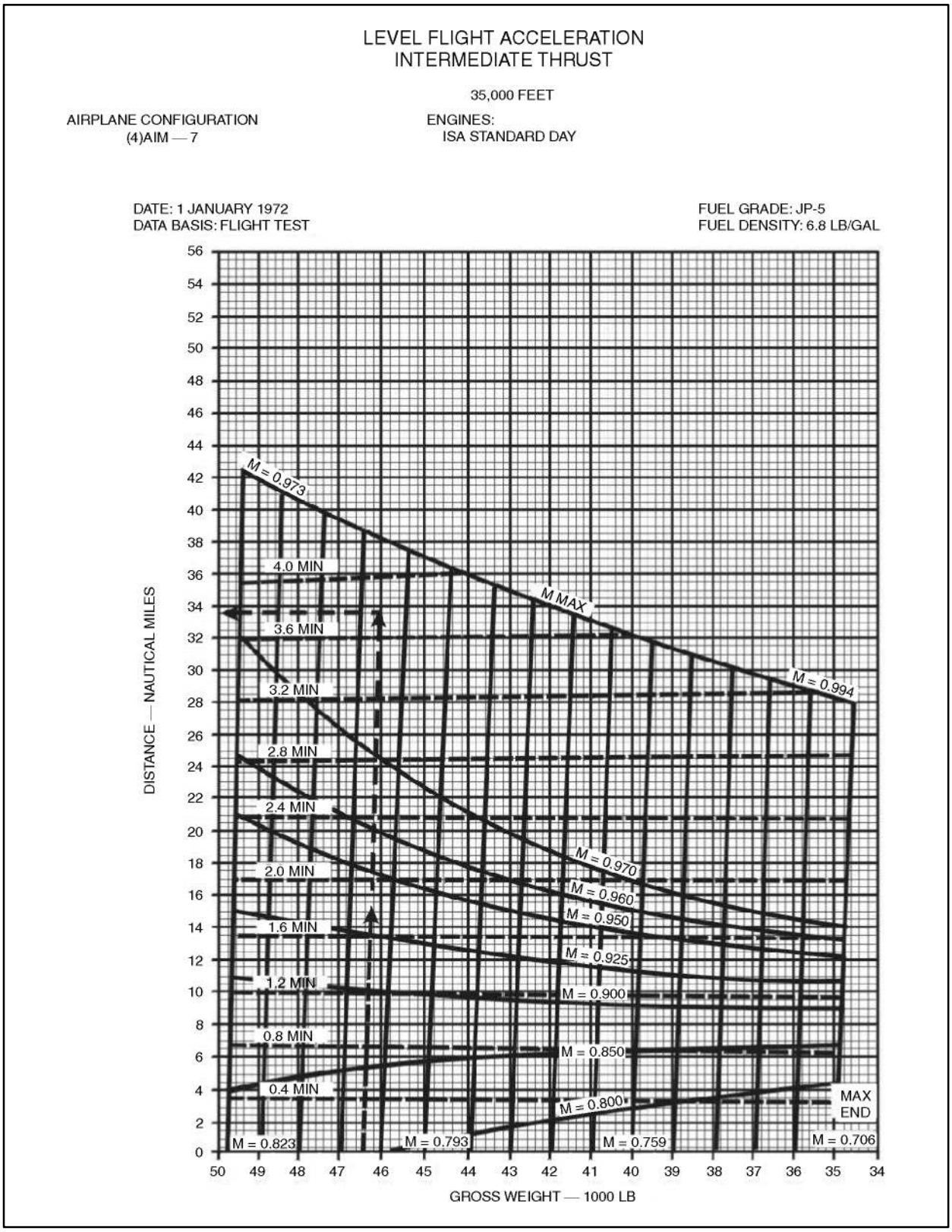


FIGURE 60. Example of level flight acceleration chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

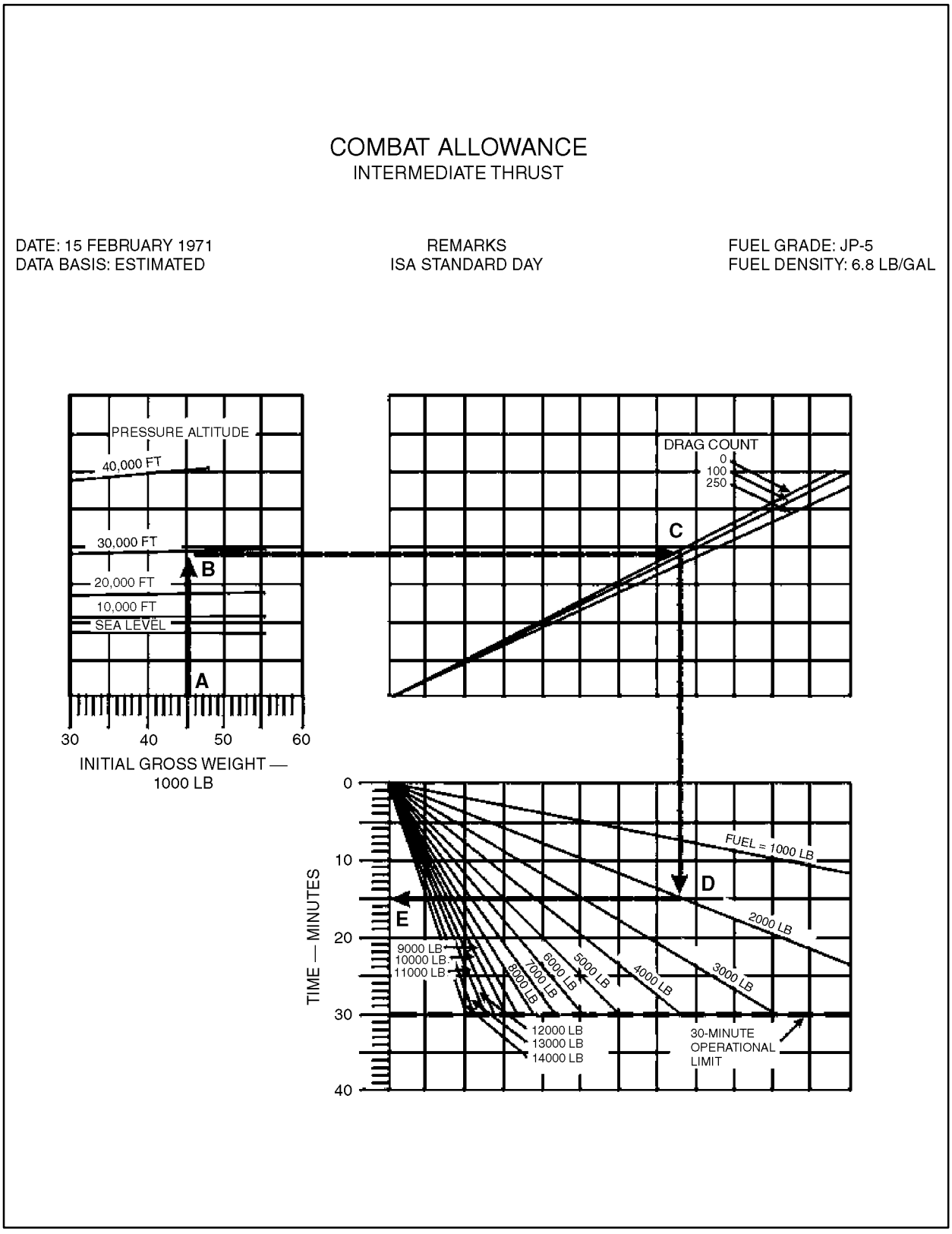


FIGURE 61. Example of combat allowance chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

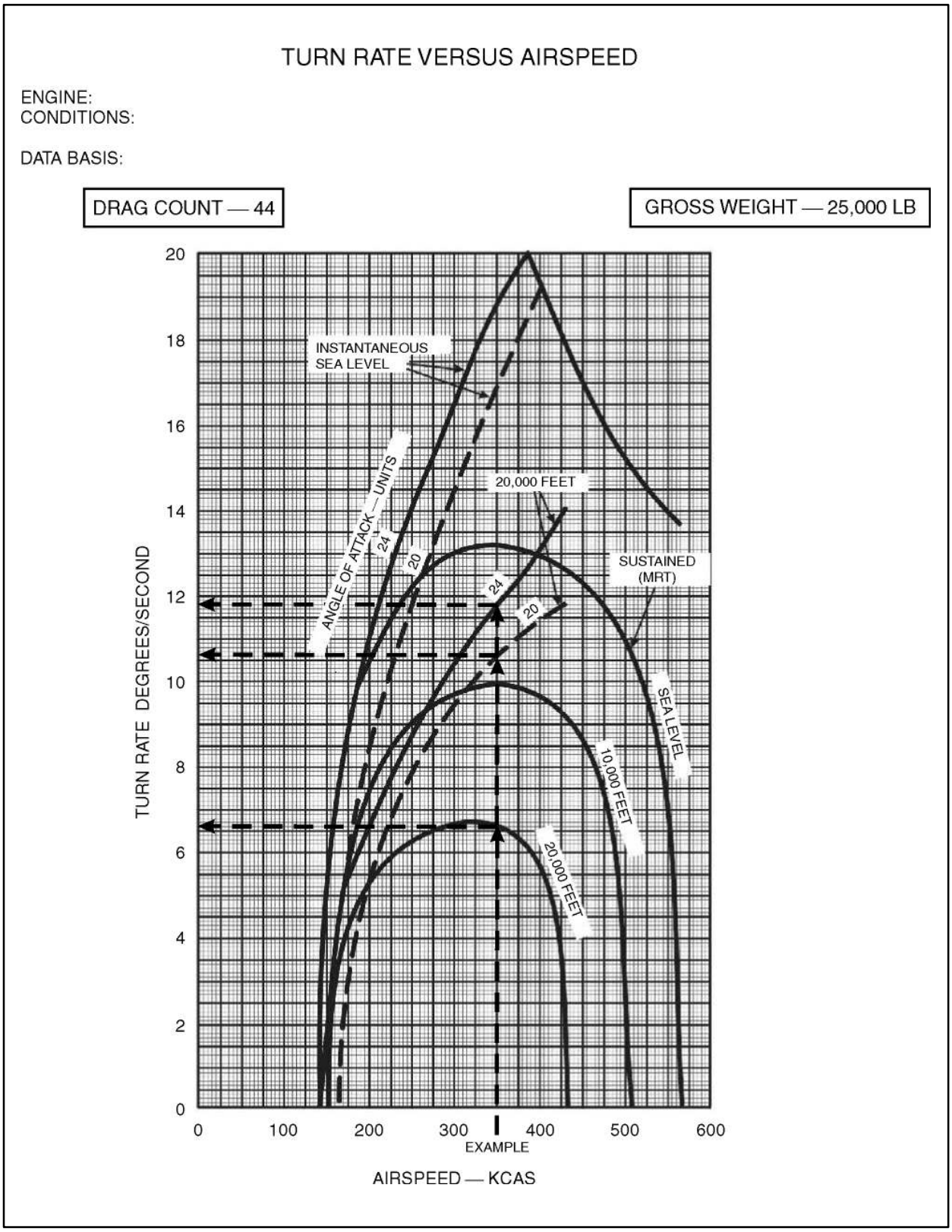


FIGURE 62. Example of turn rate versus airspeed chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

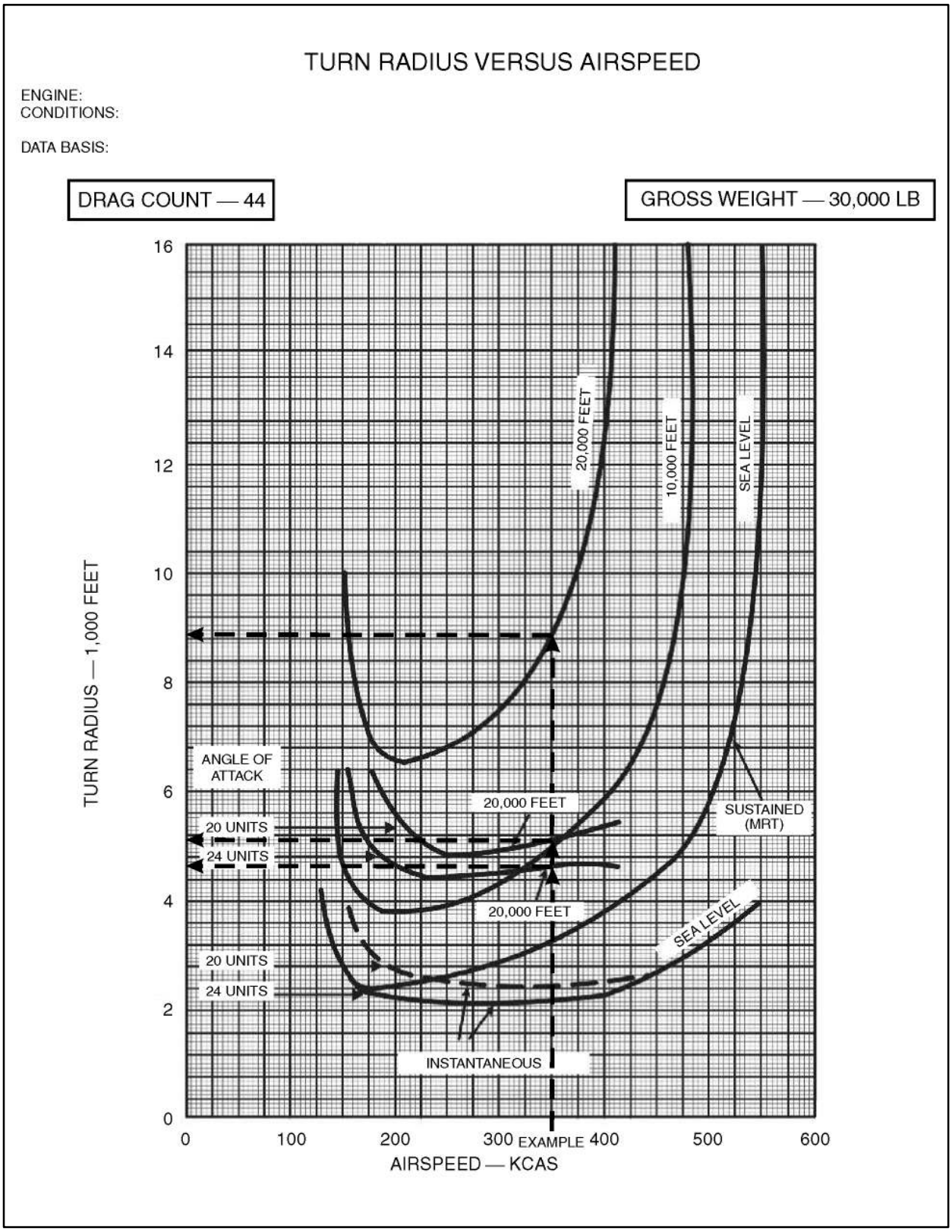


FIGURE 63. Example of turn radius versus airspeed chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

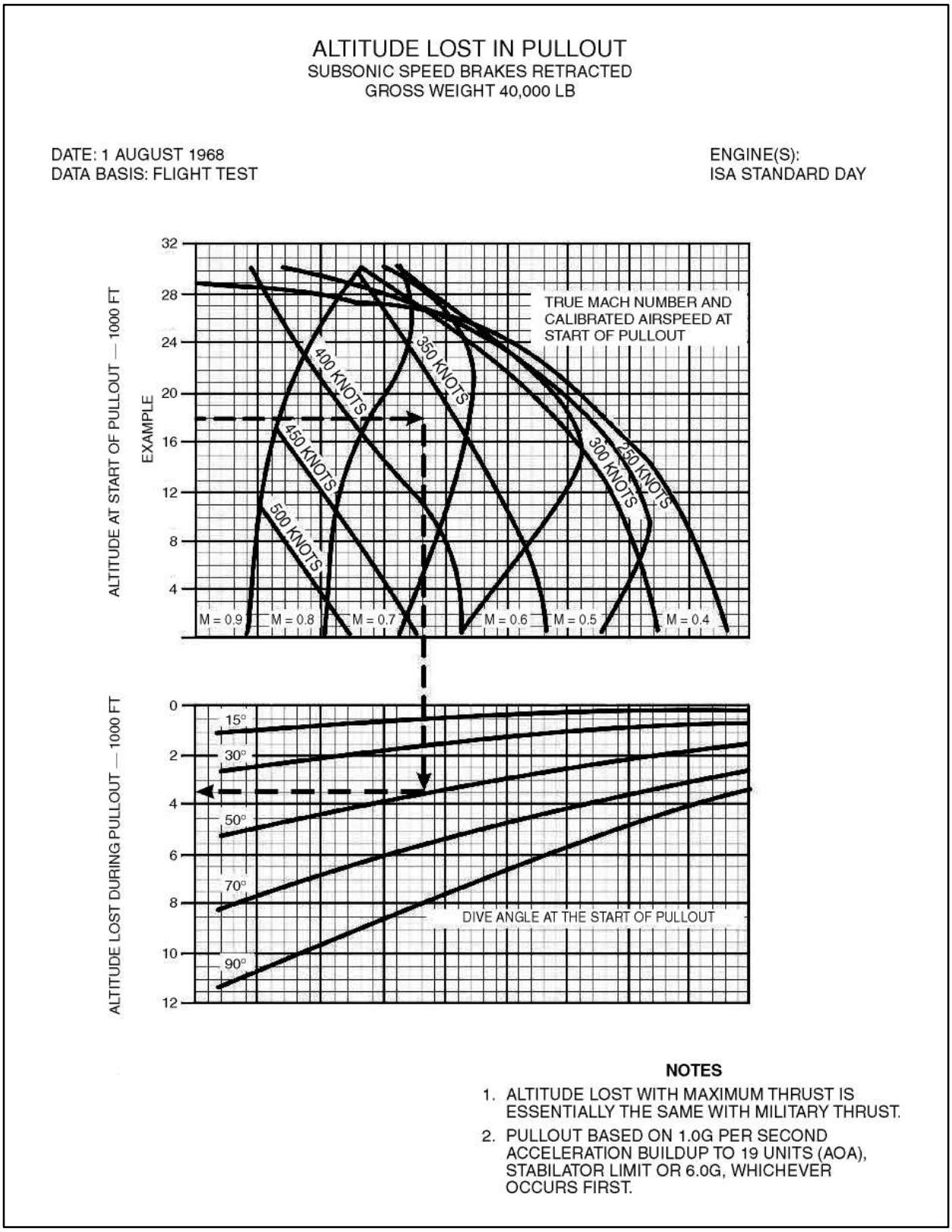


FIGURE 64. Example of altitude lost in pullout chart.

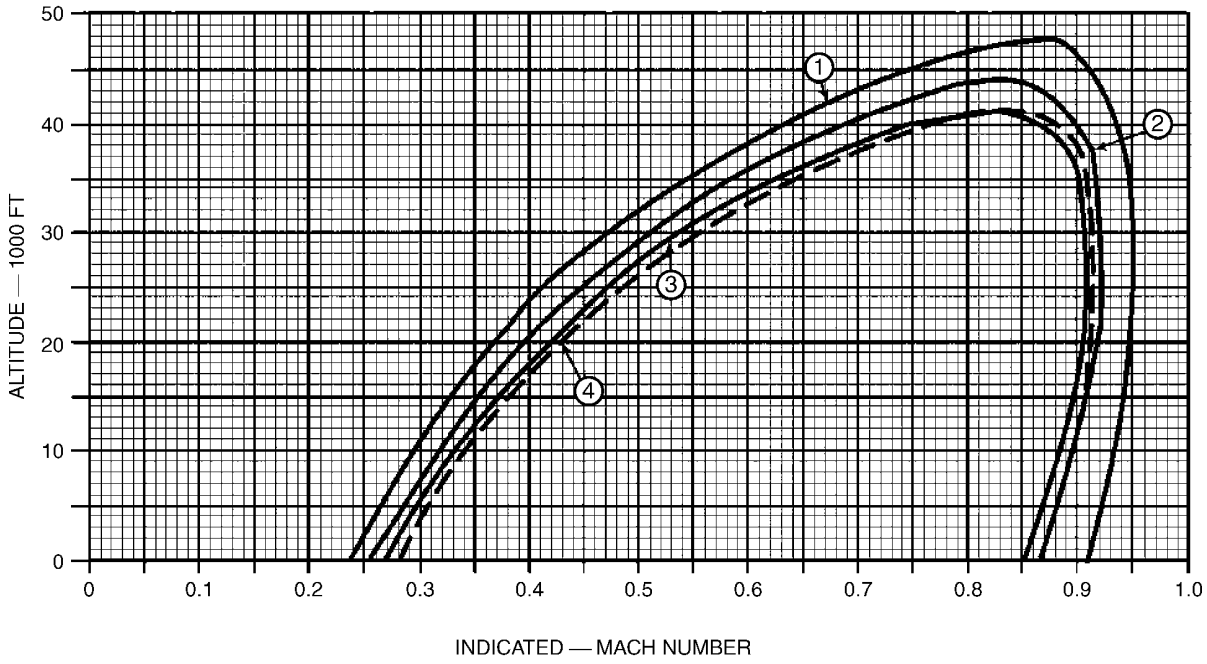
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

CONFIGURATION: SEE NOTES BELOW

REMARKS
ENGINE:
ISA STANDARD DAY

DATE: 1 MAY 1976
DATA BASIS: WIND TUNNEL AND FLIGHT EXPERIENCE

FUEL GRADE: JP-5
FUEL DENSITY: 6.8 LB/GAL



CURVE NO.	CONFIGURATION	GROSS WEIGHT	DRAG INDEX
①	CLEAN	15,275 LB	0
②	(2) AIM-9 (2) GUN PODS AND C_L RACK	17,090 LB	14.9
③	(2) 120-GAL WING TANKS (2) LAU-10A LAUNCHERS AND (2) GUN PODS	18,917 LB	19.9
④	(2) MK-82 SNAKEYES AND (2) MK-83 LDGP	19,584 LB	15.7

FIGURE 65. Example of level flight envelope chart.

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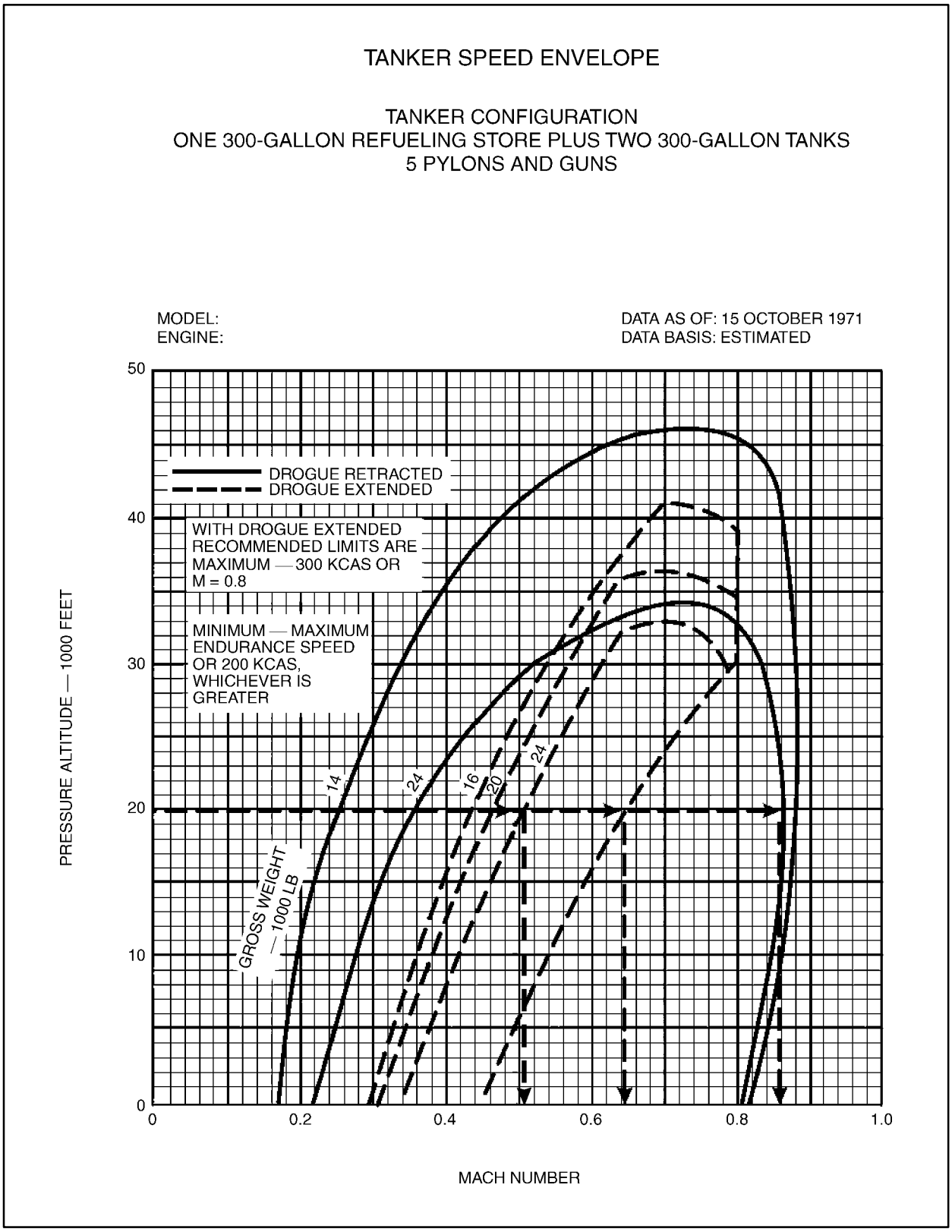


FIGURE 66. Example of tanker speed envelope chart.

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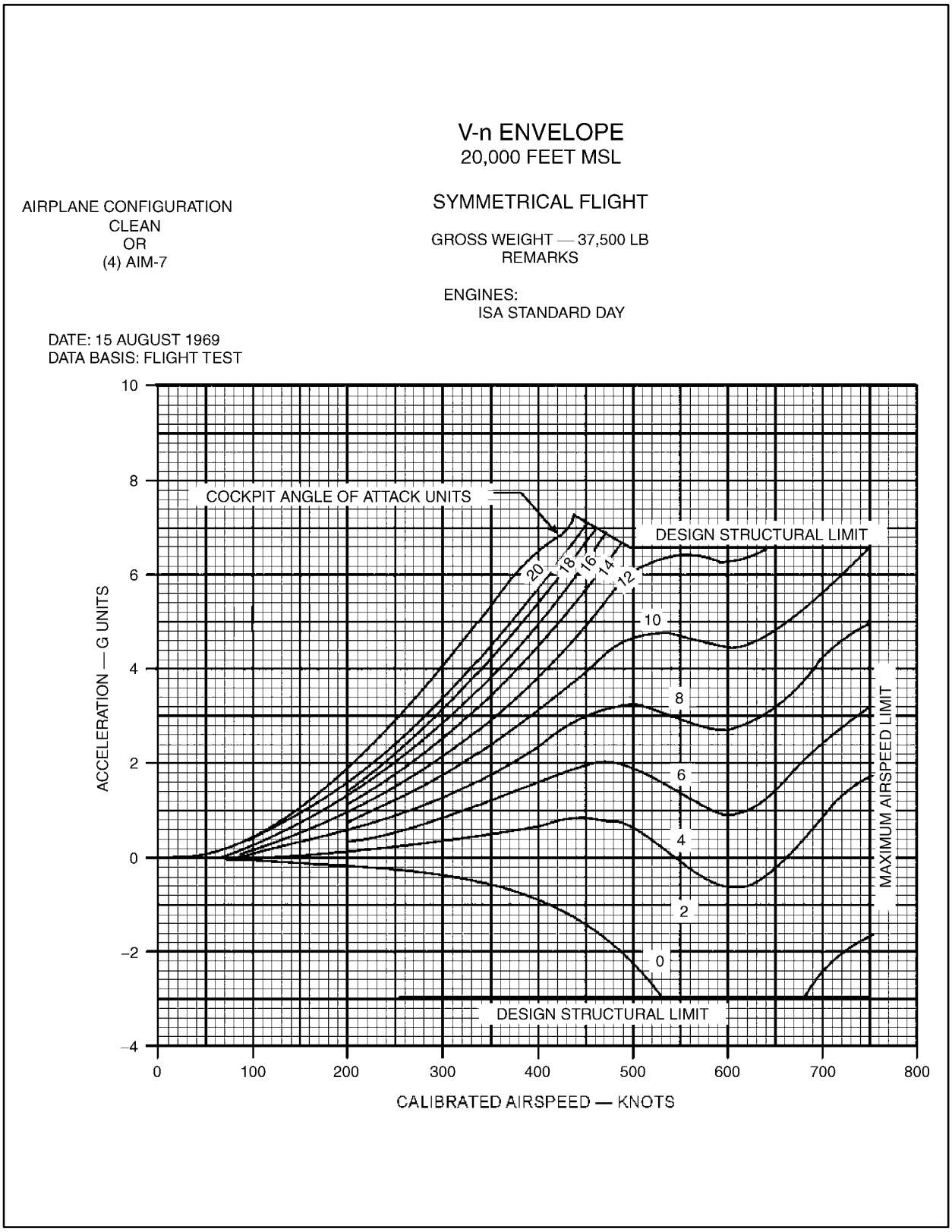


FIGURE 67. Example of V–n envelope chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

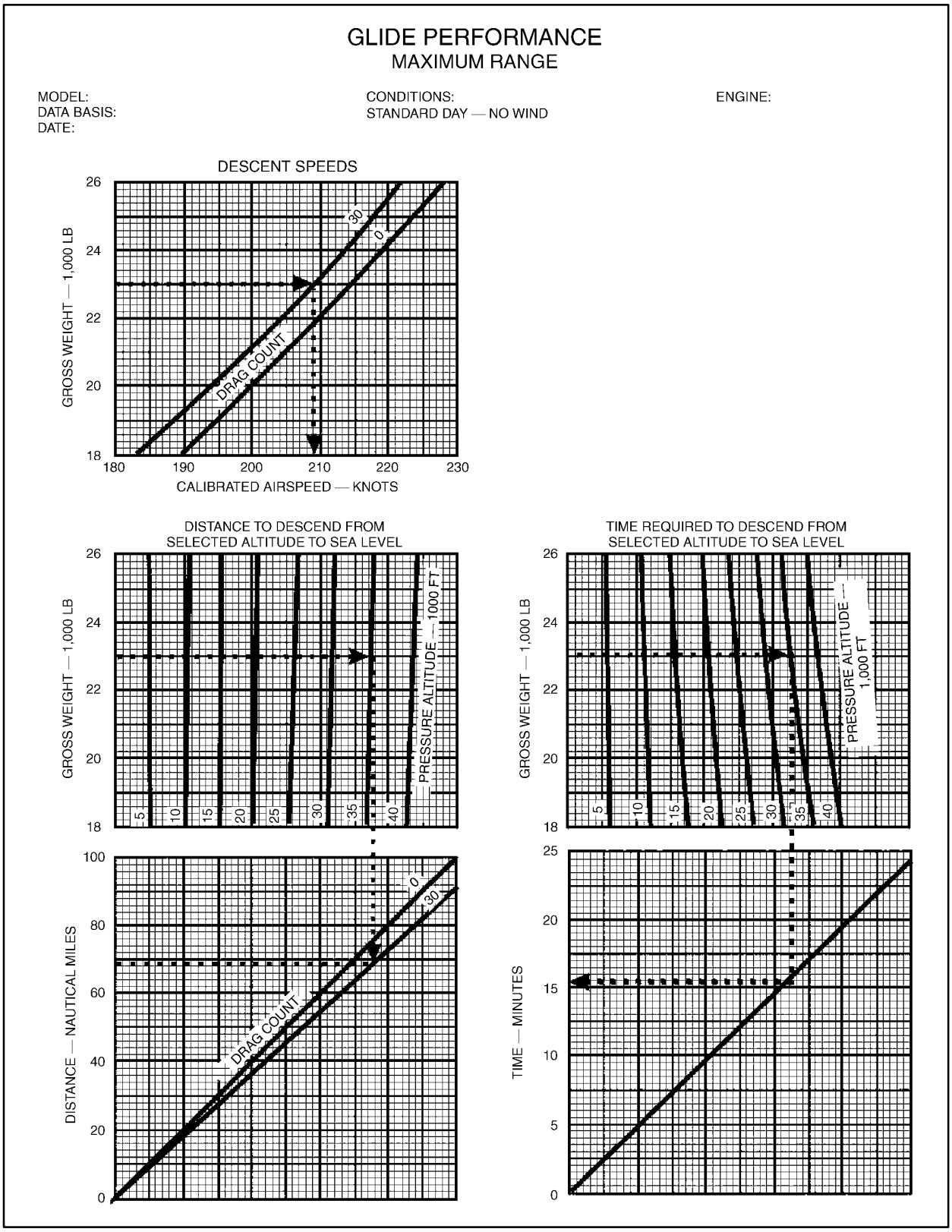


FIGURE 68. Example of glide performance chart.

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Check the source to verify that this is the current version before use.

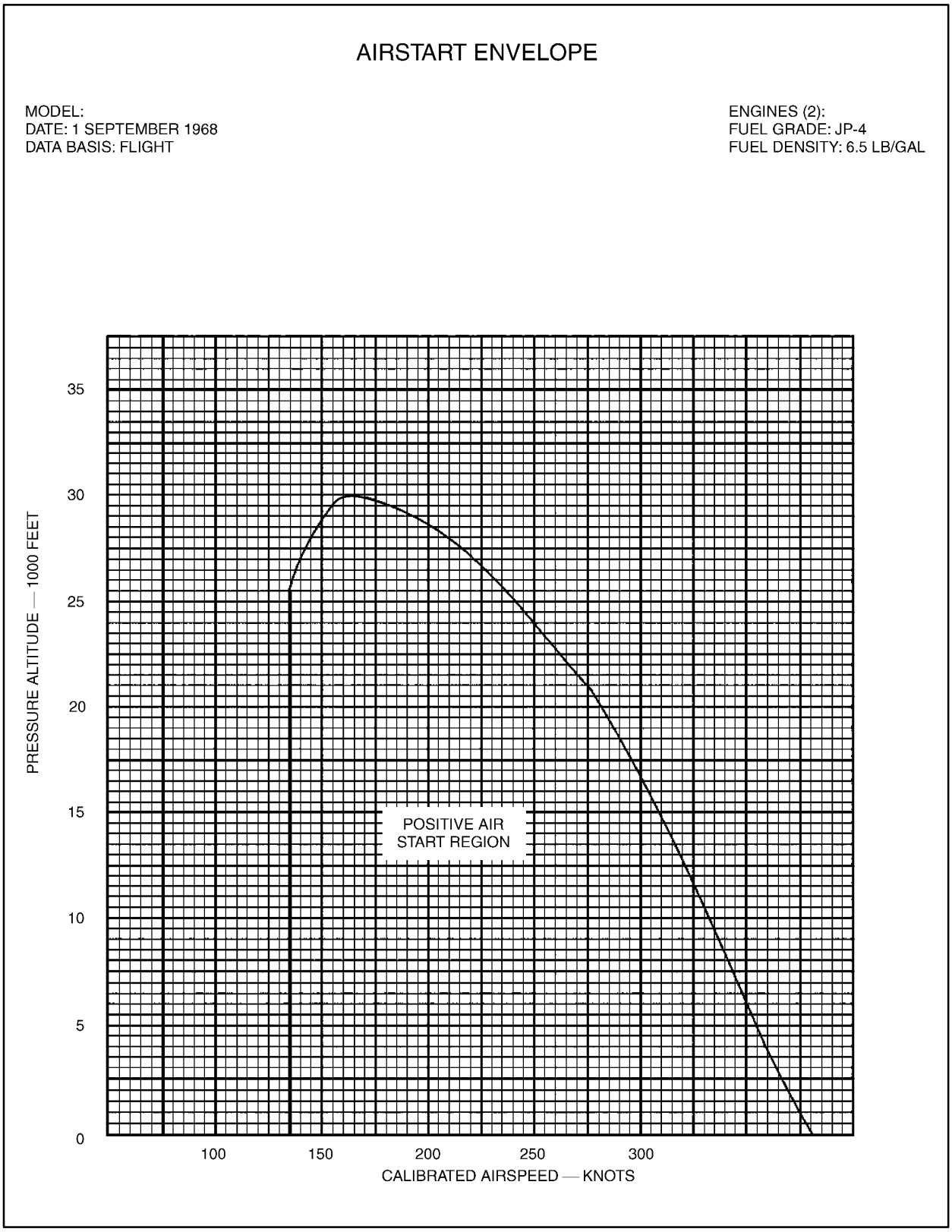


FIGURE 69. Example of airstart envelope chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

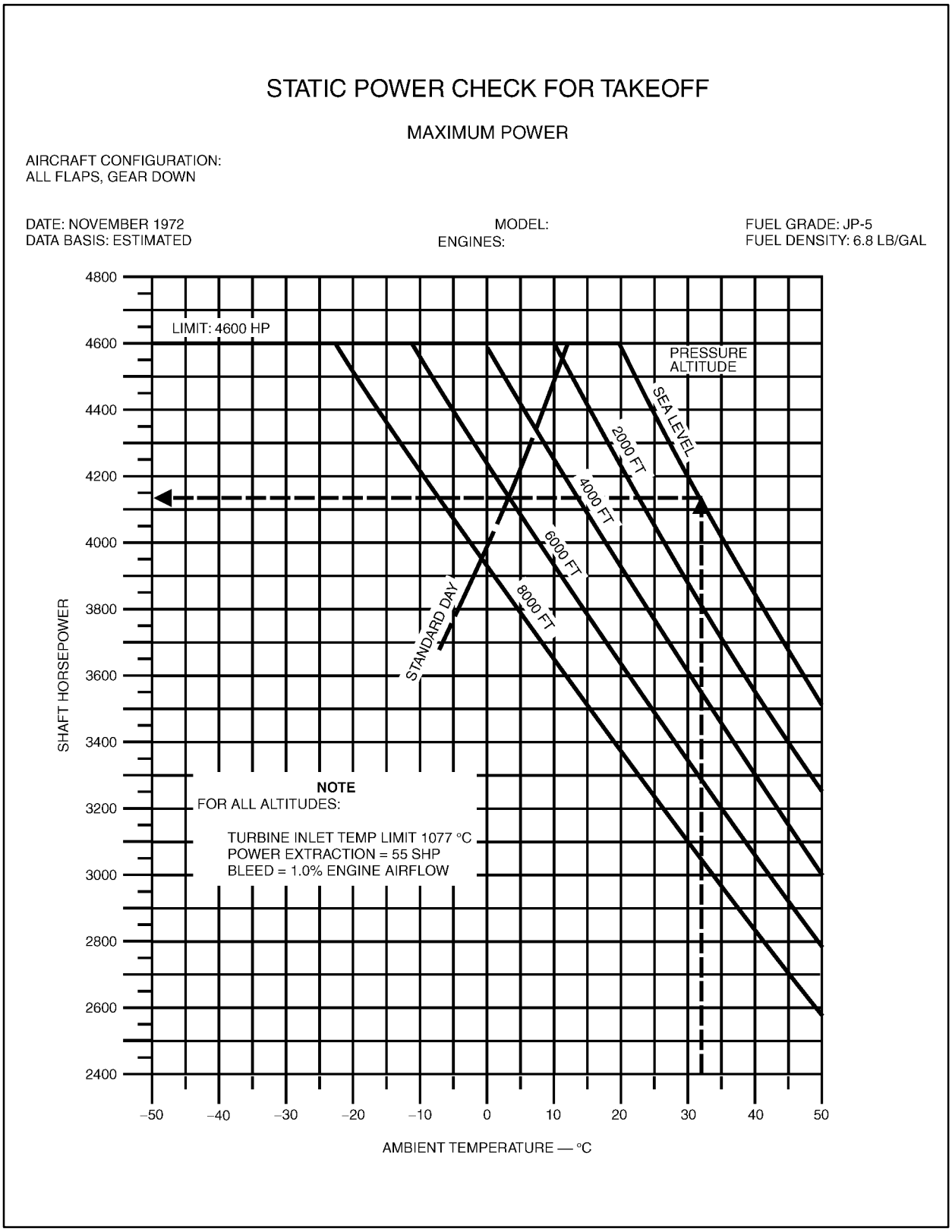


FIGURE 70. Example of static power check for takeoff chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

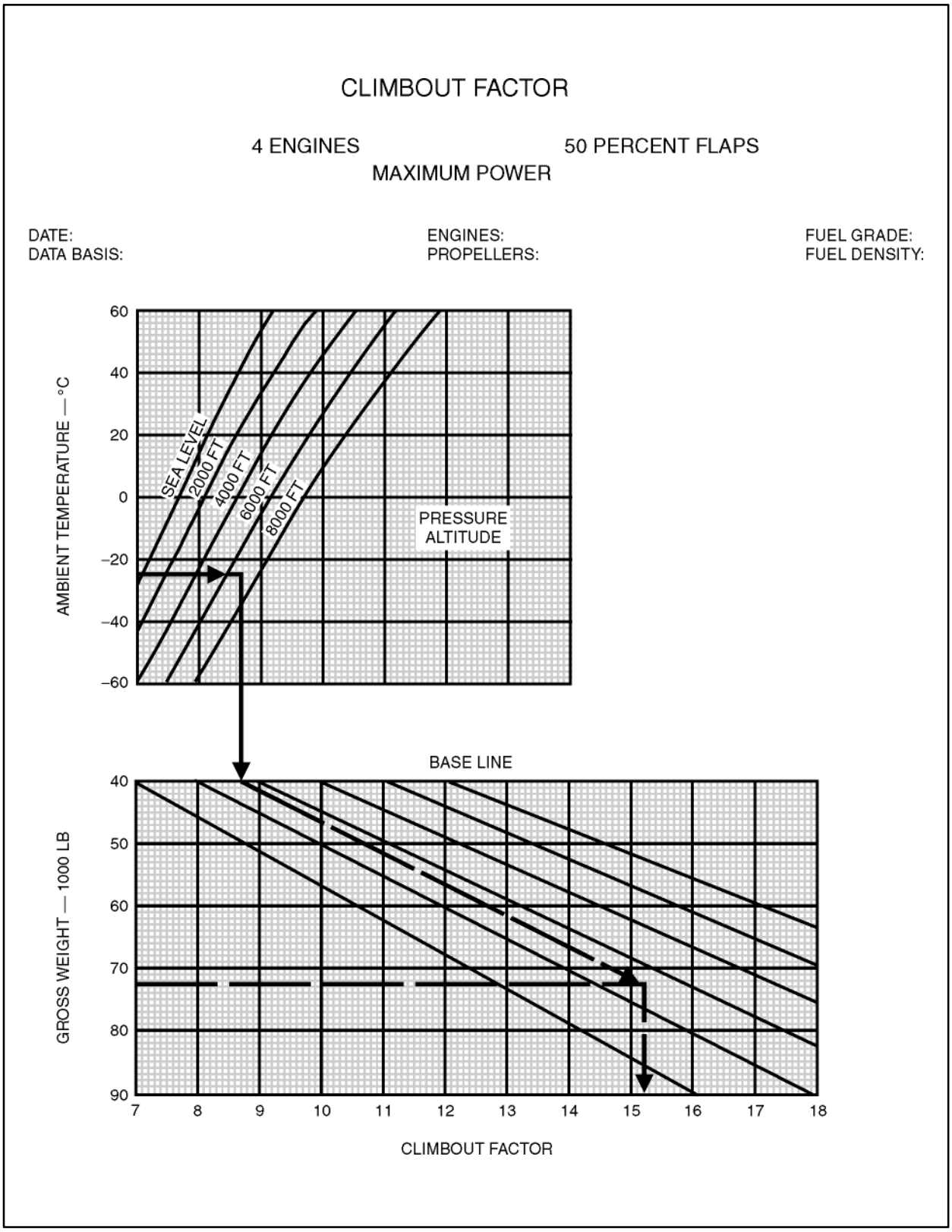


FIGURE 71. Example of climbout factor chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

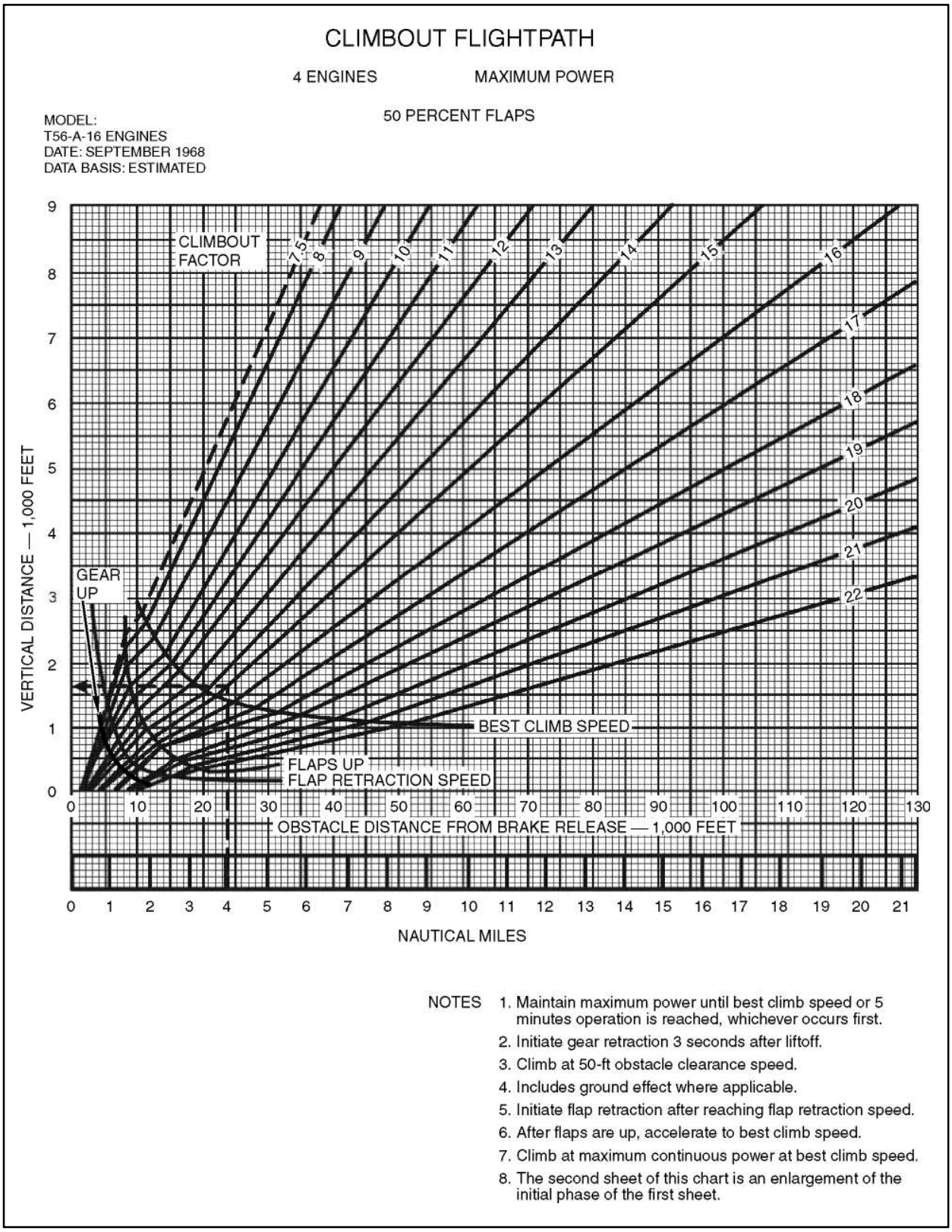


FIGURE 72. Example of climbout flightpath chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

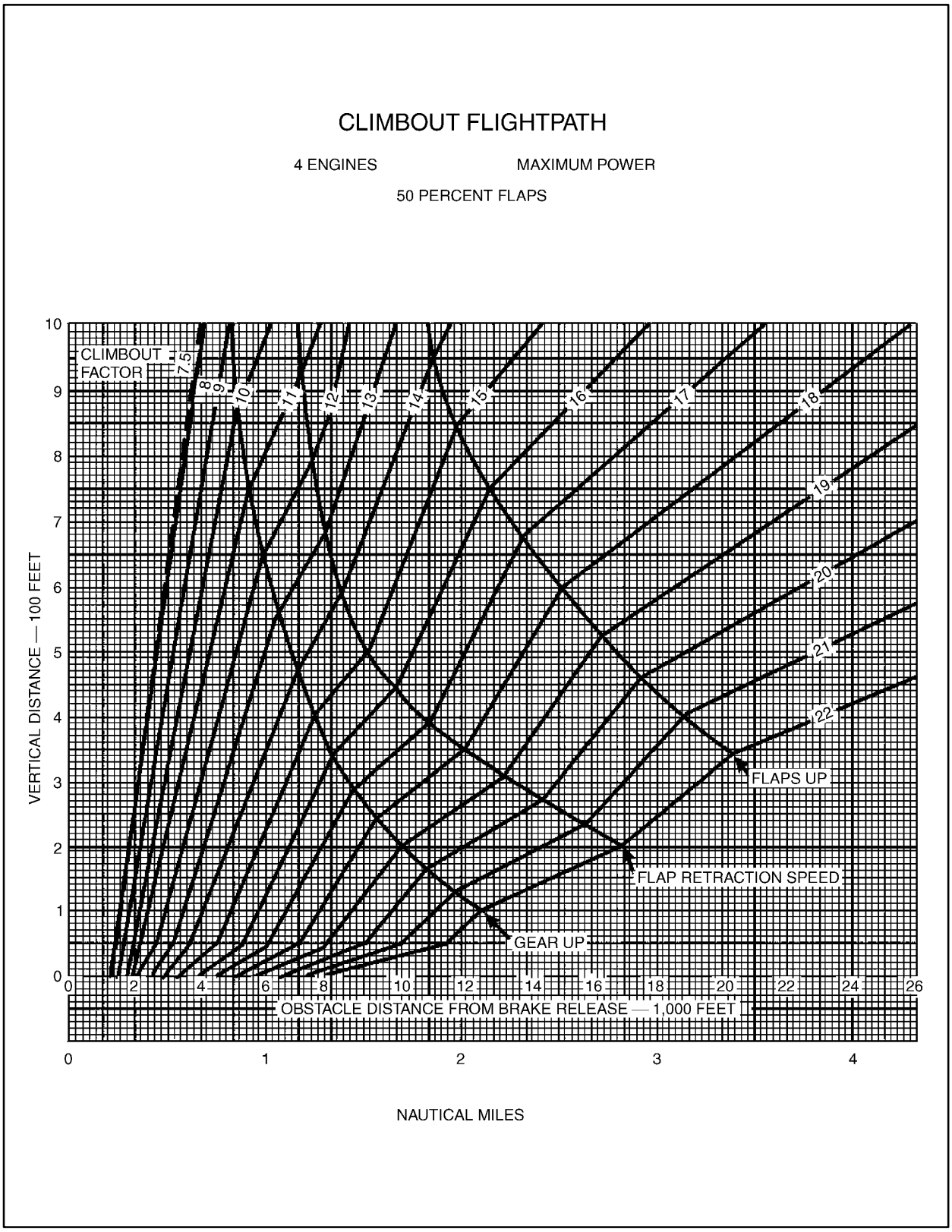


FIGURE 72. Example of climbout flightpath chart — Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

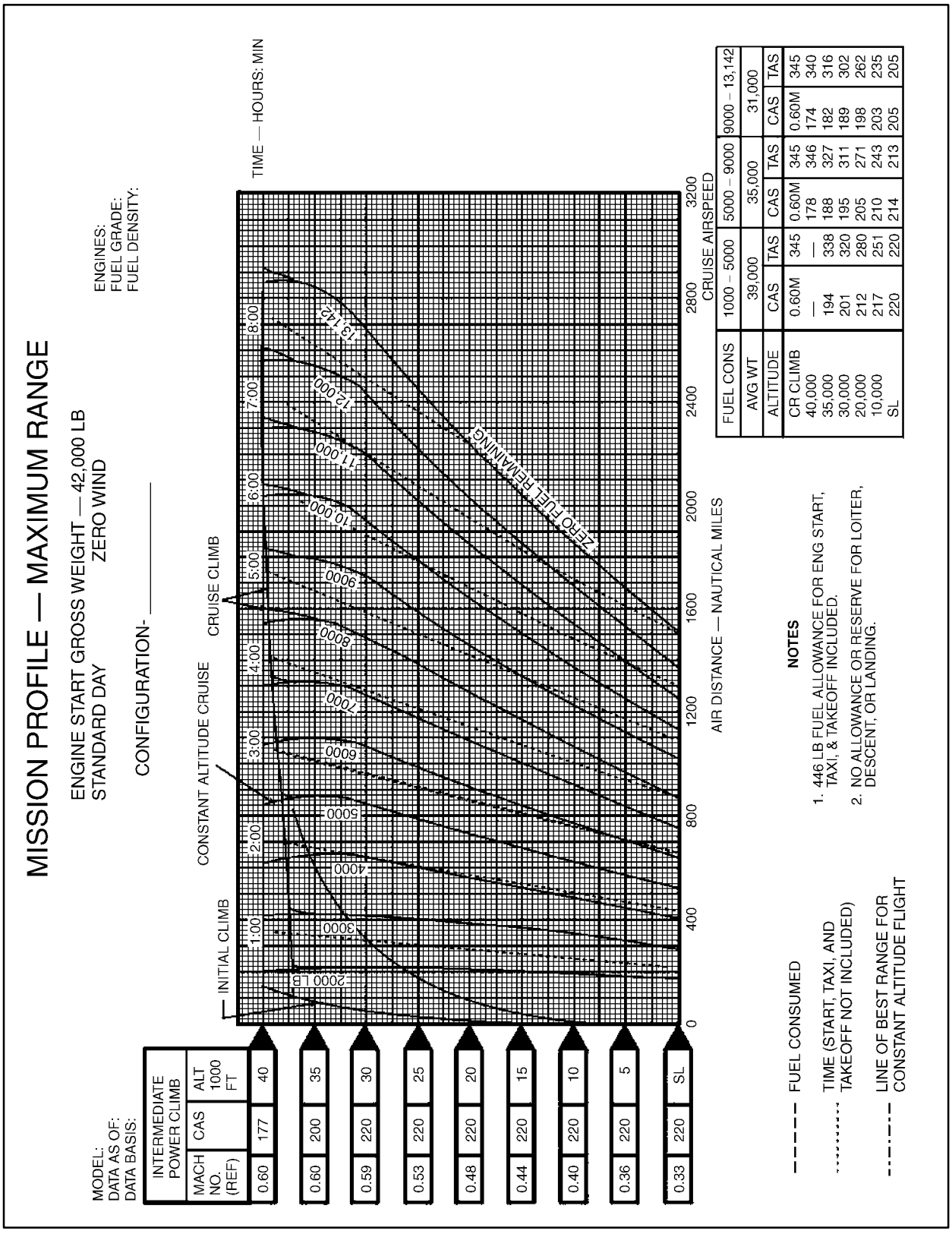


FIGURE 73. Example of mission profile — maximum range chart.

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Check the source to verify that this is the current version before use.

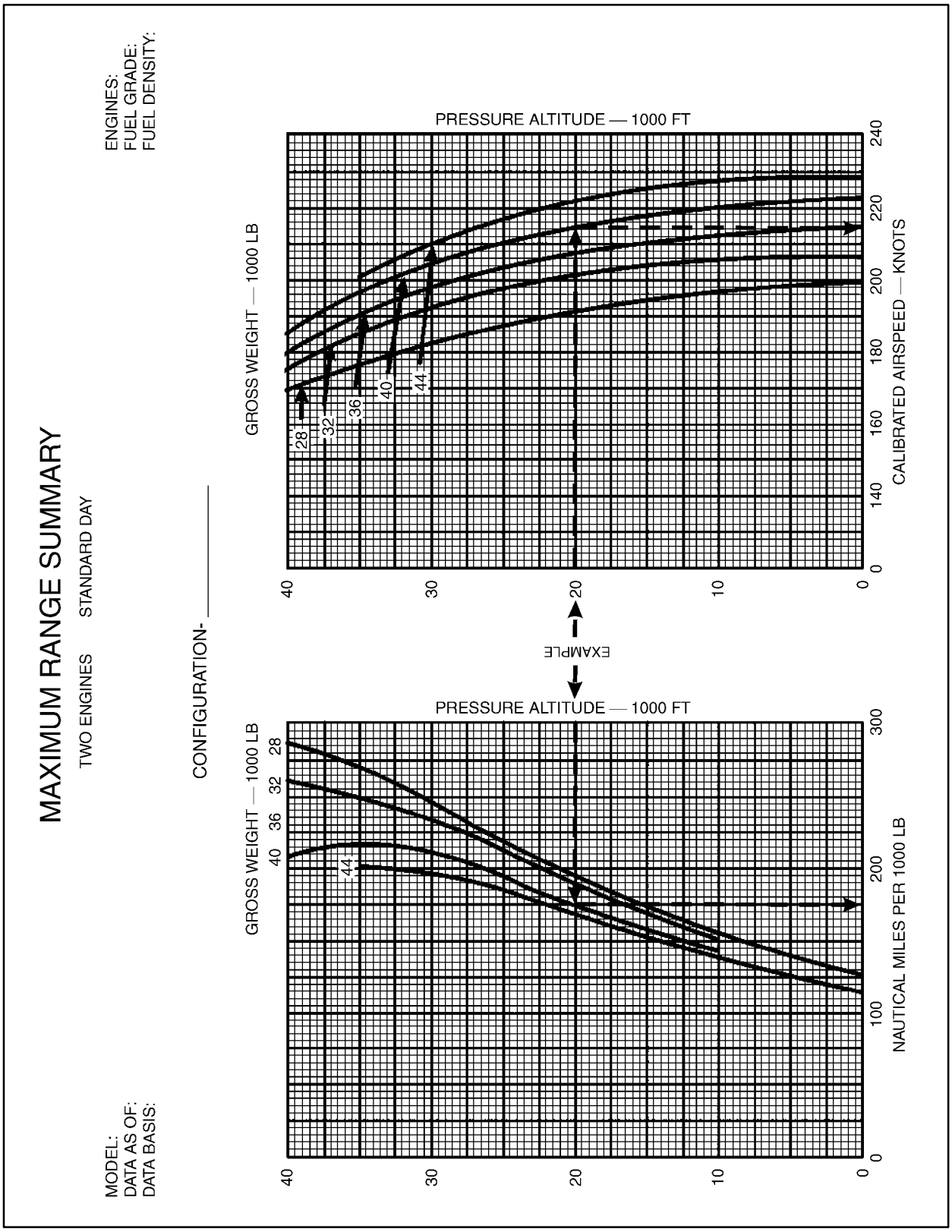
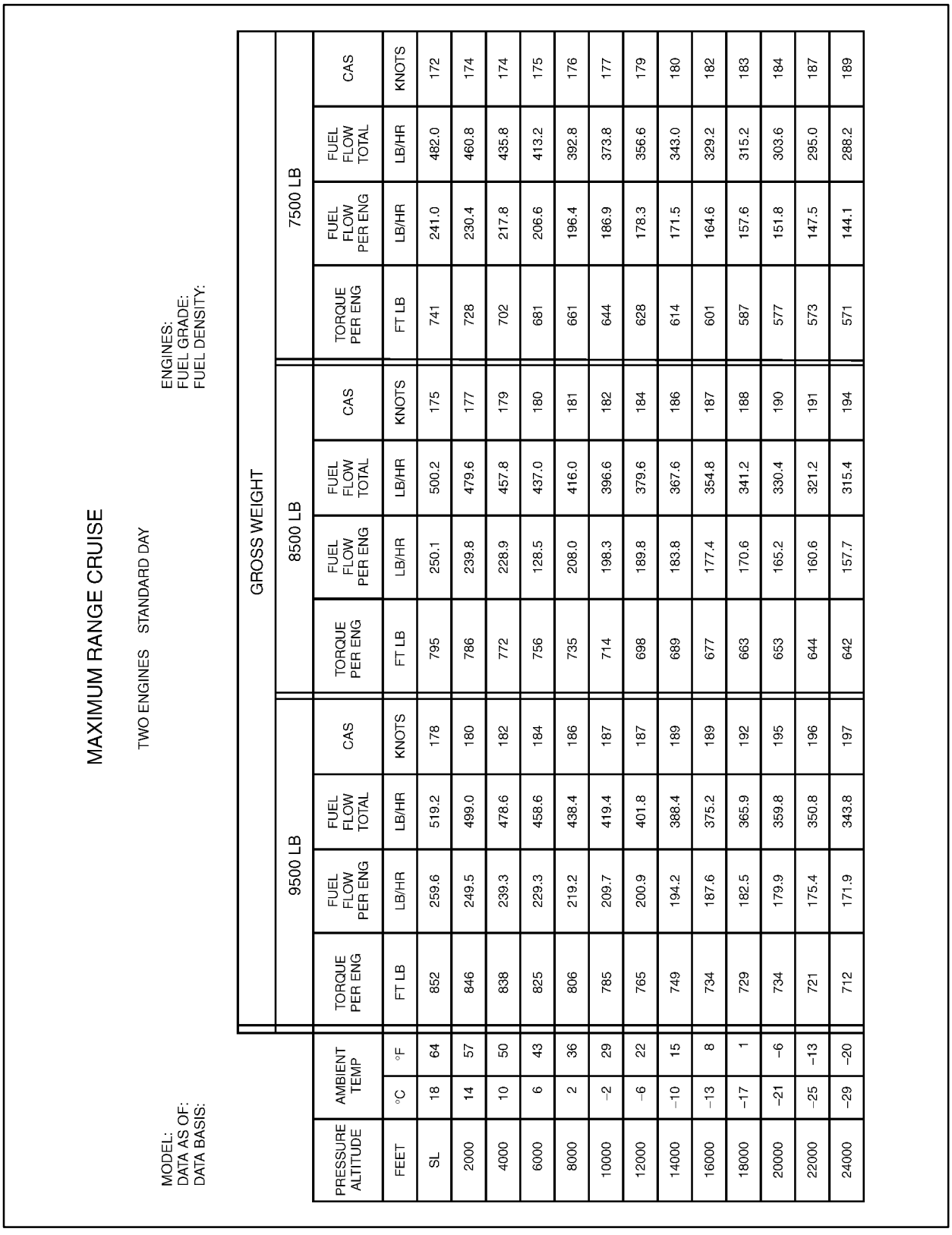


FIGURE 74. Example of maximum range summary chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.



MAXIMUM RANGE CRUISE

TWO ENGINES STANDARD DAY

MODEL:
DATA AS OF:
DATA BASIS:

ENGINES:
FUEL GRADE:
FUEL DENSITY:

PRESSURE ALTITUDE	AMBIENT TEMP		GROSS WEIGHT											
			9500 LB				8500 LB				7500 LB			
			TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	CAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	CAS	TORQUE PER ENG	FUEL FLOW PER ENG	FUEL FLOW TOTAL	CAS
FEET	°C	°F	FT LB	LB/HR	LB/HR	KNOTS	FT LB	LB/HR	LB/HR	KNOTS	FT LB	LB/HR	LB/HR	KNOTS
SL	18	64	852	259.6	519.2	178	795	250.1	500.2	175	741	241.0	482.0	172
2000	14	57	846	249.5	499.0	180	786	239.8	479.6	177	728	230.4	460.8	174
4000	10	50	838	239.3	478.6	182	772	228.9	457.8	179	702	217.8	435.8	174
6000	6	43	825	229.3	458.6	184	756	128.5	437.0	180	681	206.6	413.2	175
8000	2	36	806	219.2	438.4	186	735	208.0	416.0	181	661	196.4	392.8	176
10000	–2	29	785	209.7	419.4	187	714	198.3	396.6	182	644	186.9	373.8	177
12000	–6	22	765	200.9	401.8	187	698	189.8	379.6	184	628	178.3	356.6	179
14000	–10	15	749	194.2	388.4	189	689	183.8	367.6	186	614	171.5	343.0	180
16000	–13	8	734	187.6	375.2	189	677	177.4	354.8	187	601	164.6	329.2	182
18000	–17	1	729	182.5	365.9	192	663	170.6	341.2	188	587	157.6	315.2	183
20000	–21	–6	734	179.9	359.8	195	653	165.2	330.4	190	577	151.8	303.6	184
22000	–25	–13	721	175.4	350.8	196	644	160.6	321.2	191	573	147.5	295.0	187
24000	–29	–20	712	171.9	343.8	197	642	157.7	315.4	194	571	144.1	288.2	189

FIGURE 75. Example of maximum range cruise chart.

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Check the source to verify that this is the current version before use.

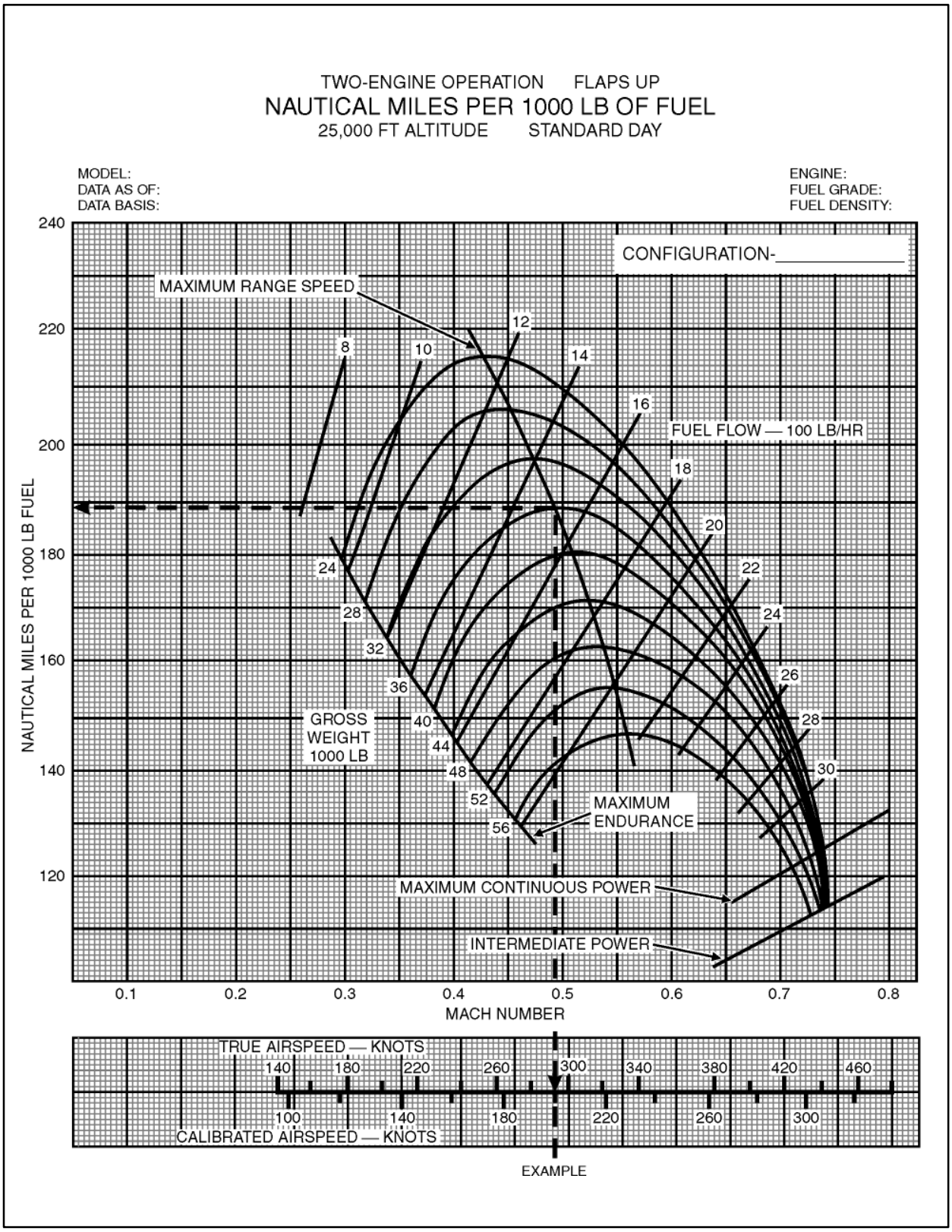


FIGURE 76. Example of nautical miles per 1,000 pounds of fuel chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

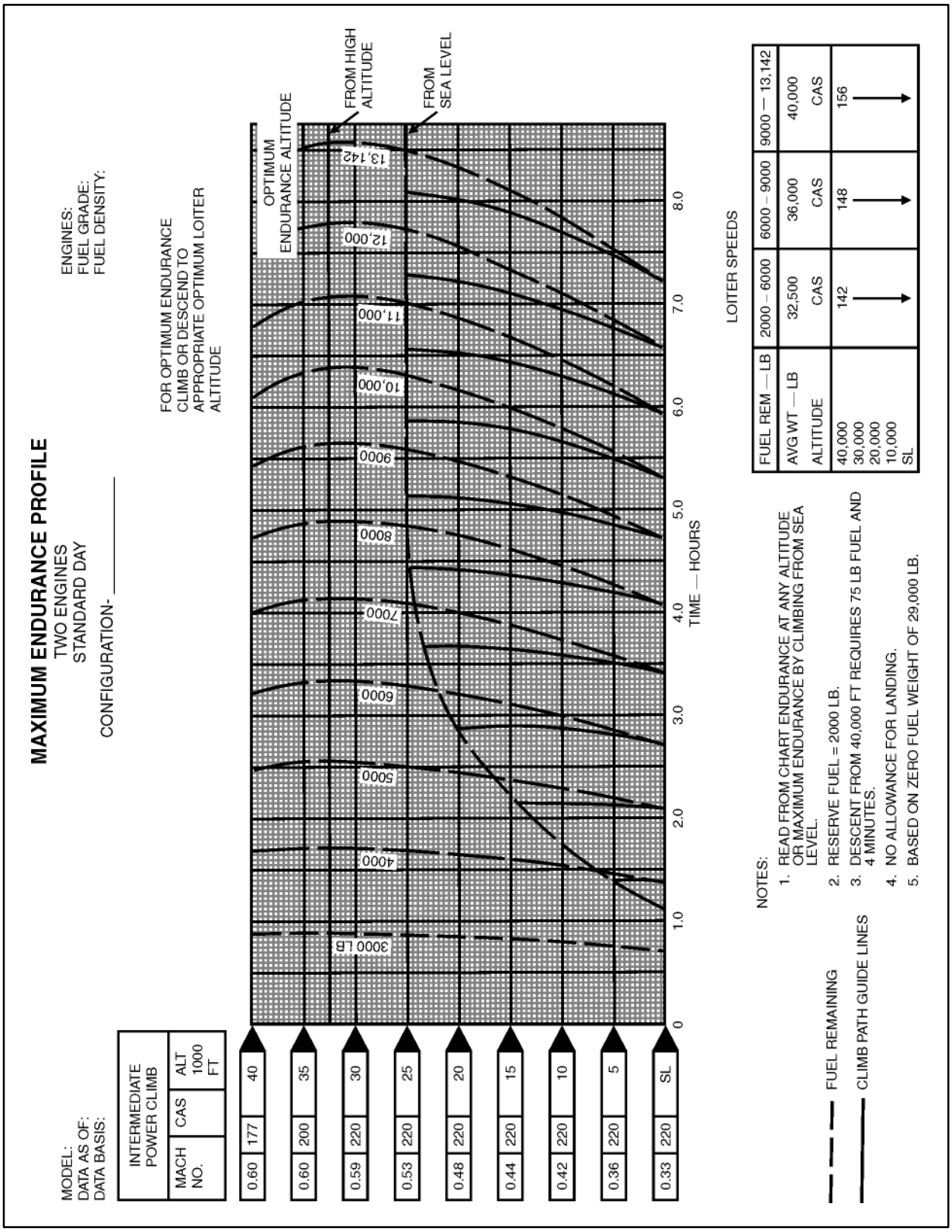


FIGURE 77. Example of maximum endurance profile chart.

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Check the source to verify that this is the current version before use.

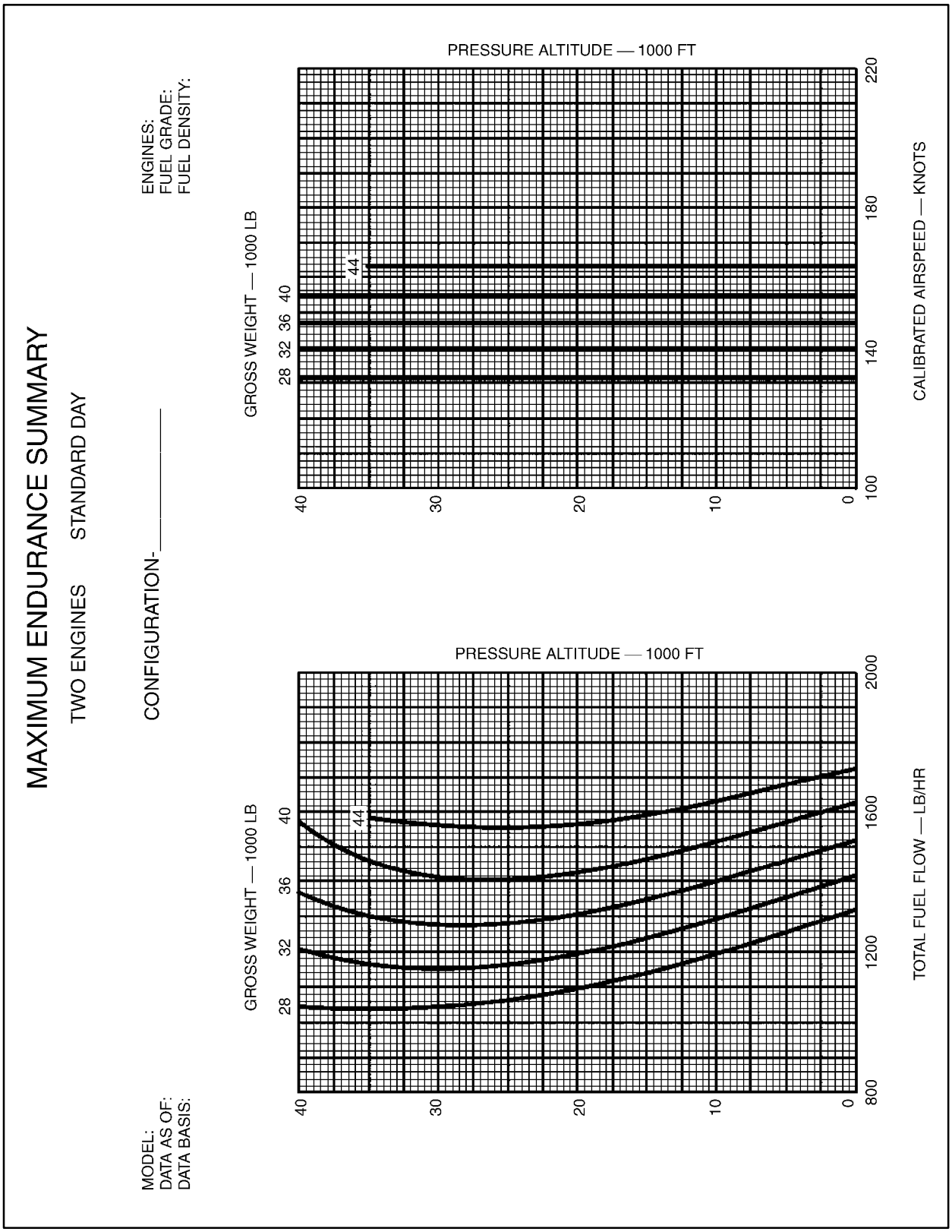


FIGURE 78. Example of maximum endurance summary chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

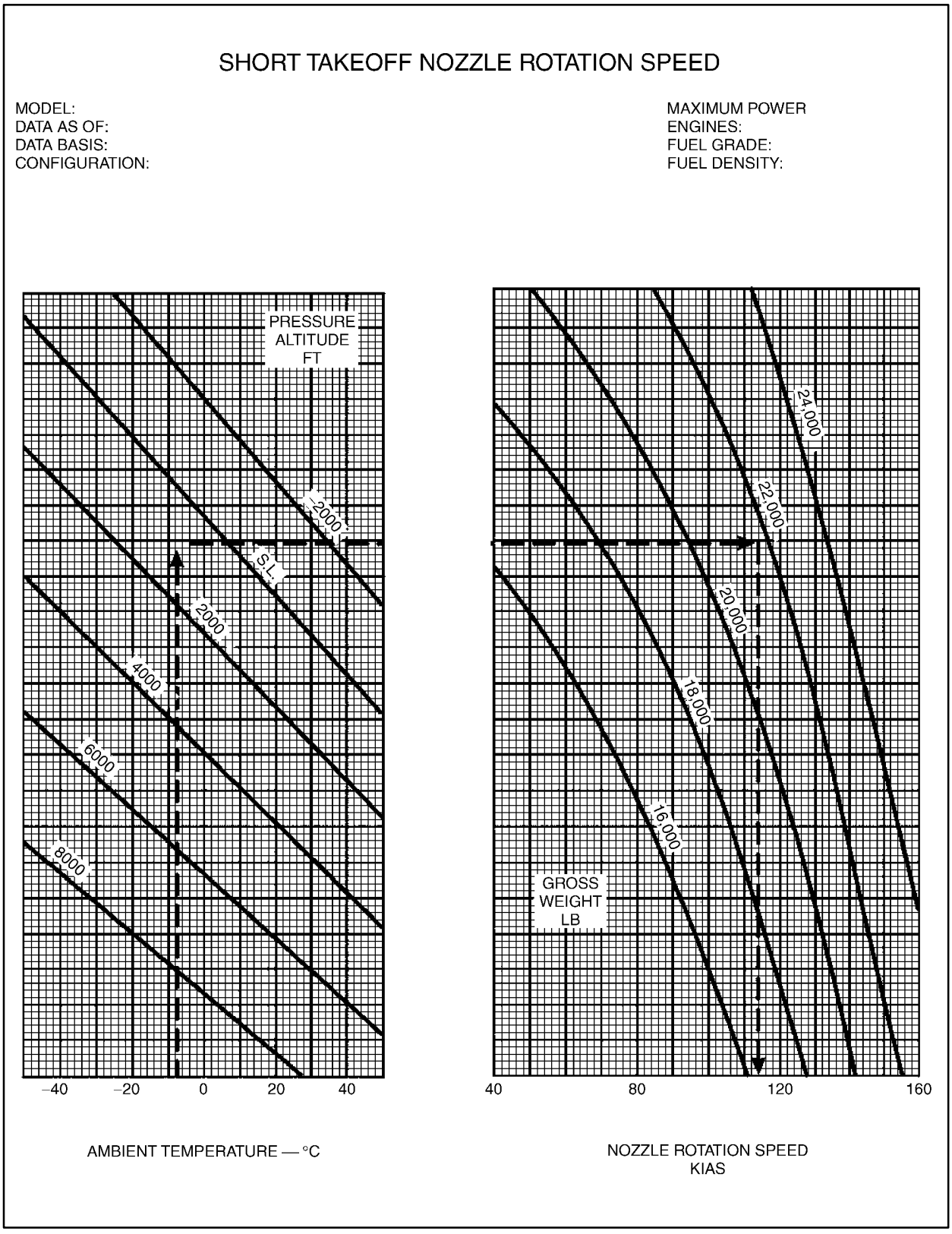


FIGURE 79. Example of short takeoff nozzle rotation speed chart.

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Check the source to verify that this is the current version before use.

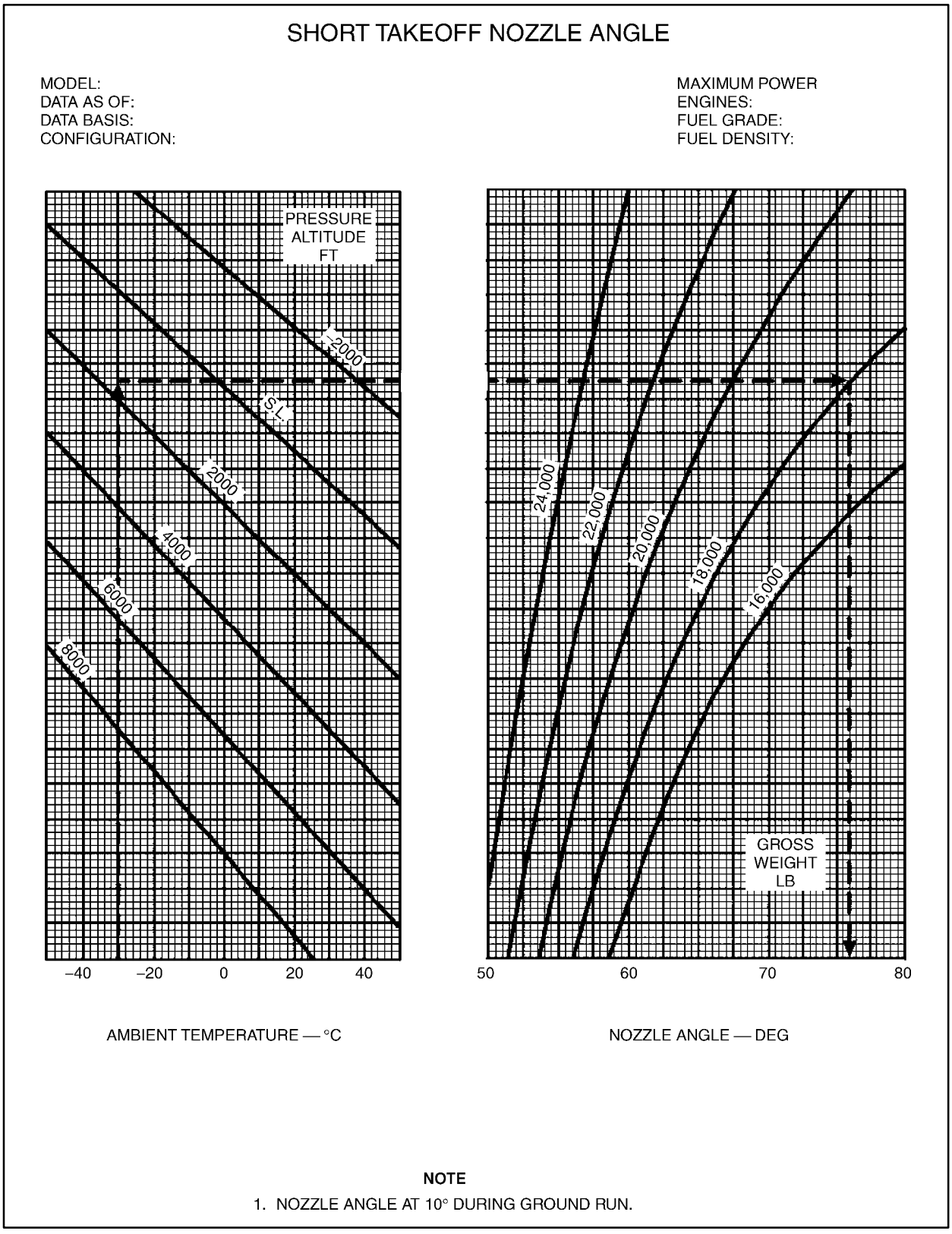


FIGURE 80. Example of short takeoff nozzle angle chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

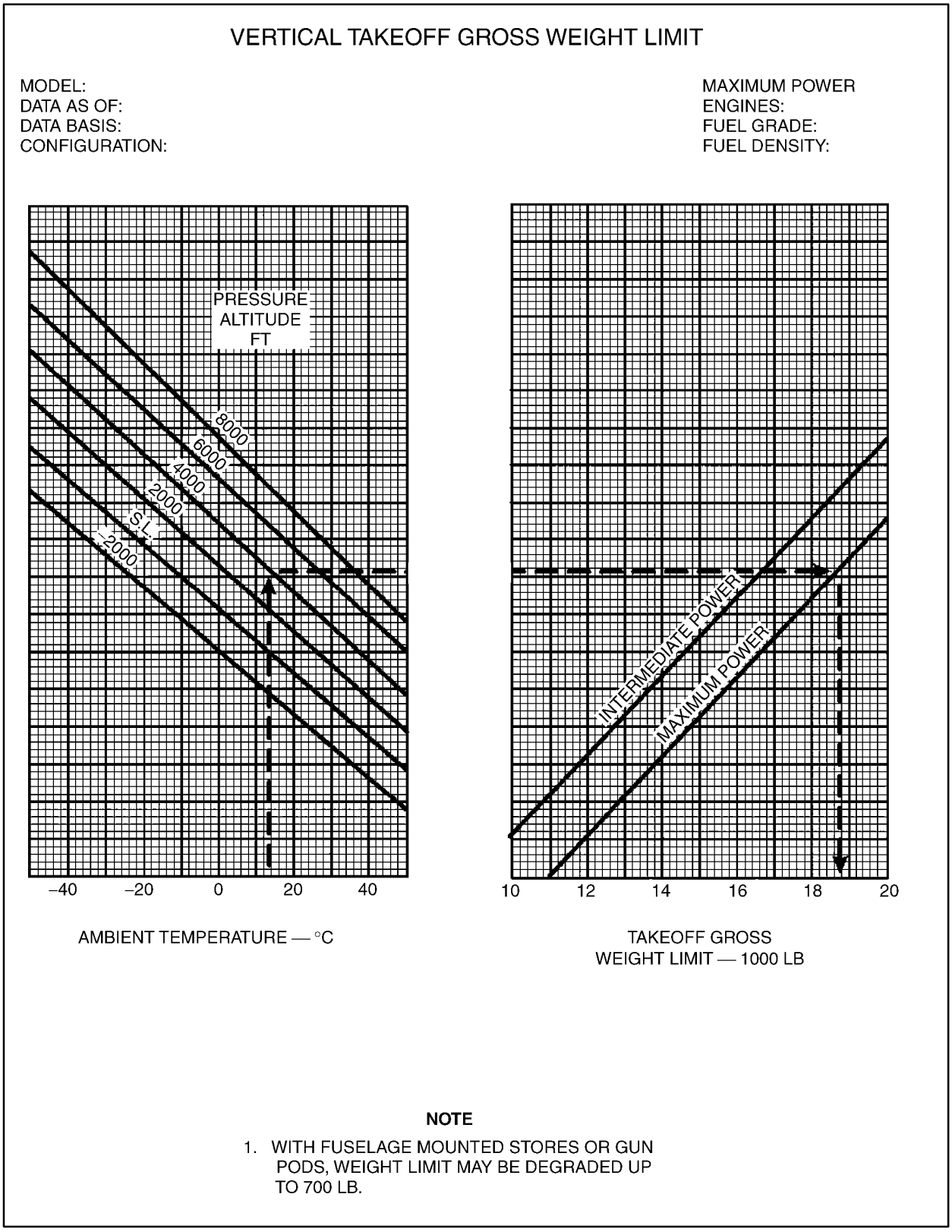


FIGURE 81. Example of vertical takeoff gross weight limit chart.

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Check the source to verify that this is the current version before use.

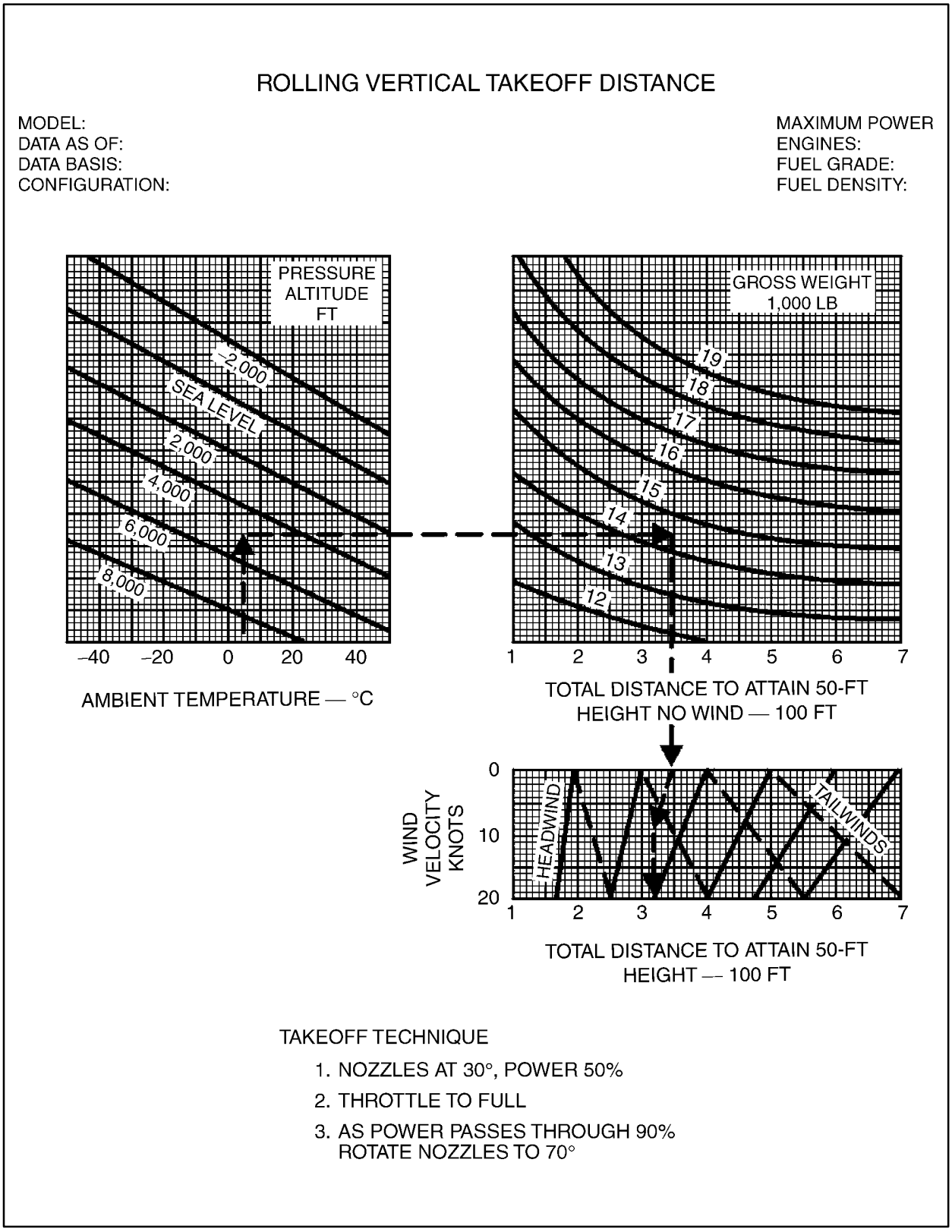


FIGURE 82. Example of rolling vertical takeoff distance chart.

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Check the source to verify that this is the current version before use.

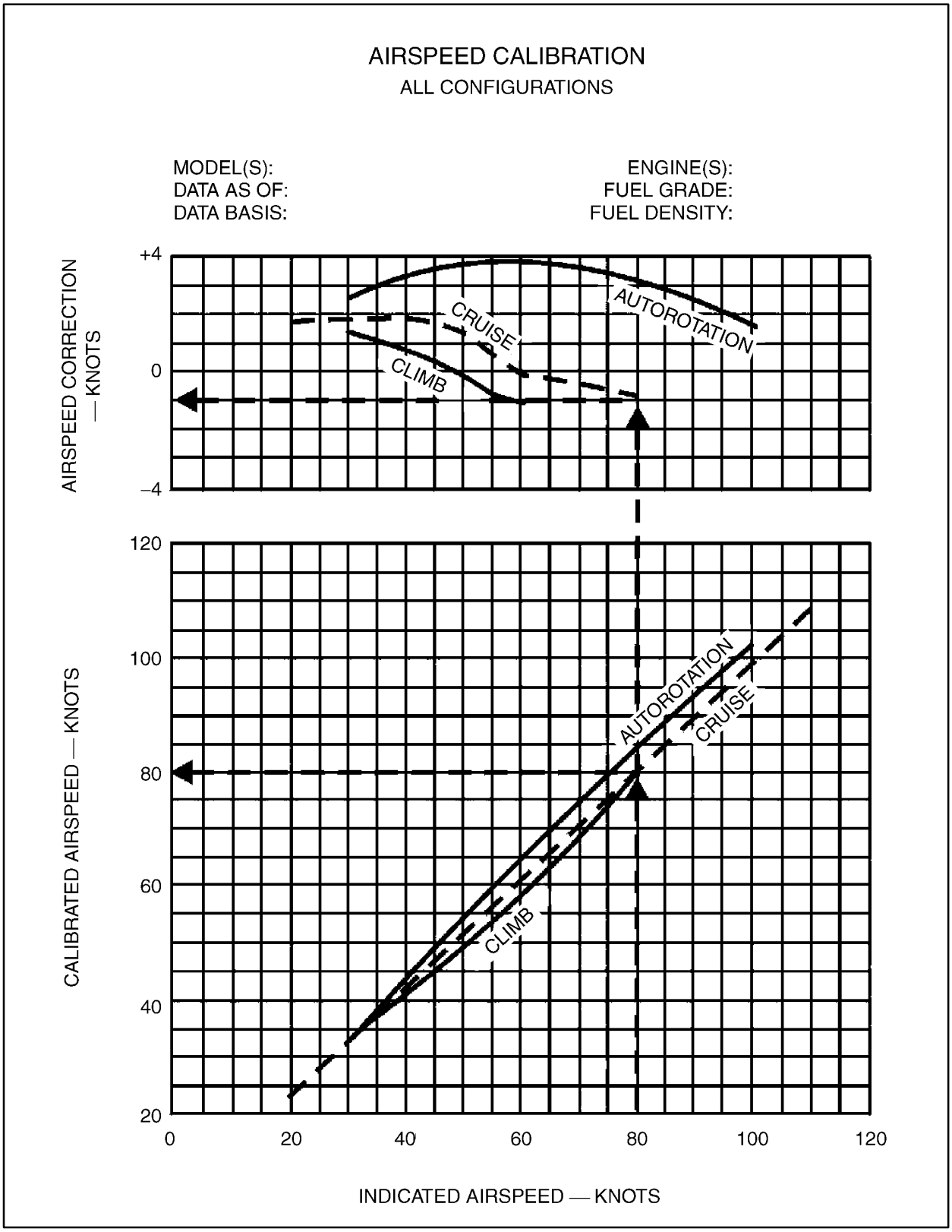


FIGURE 83. Example of airspeed calibration chart (helicopter).

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Check the source to verify that this is the current version before use.

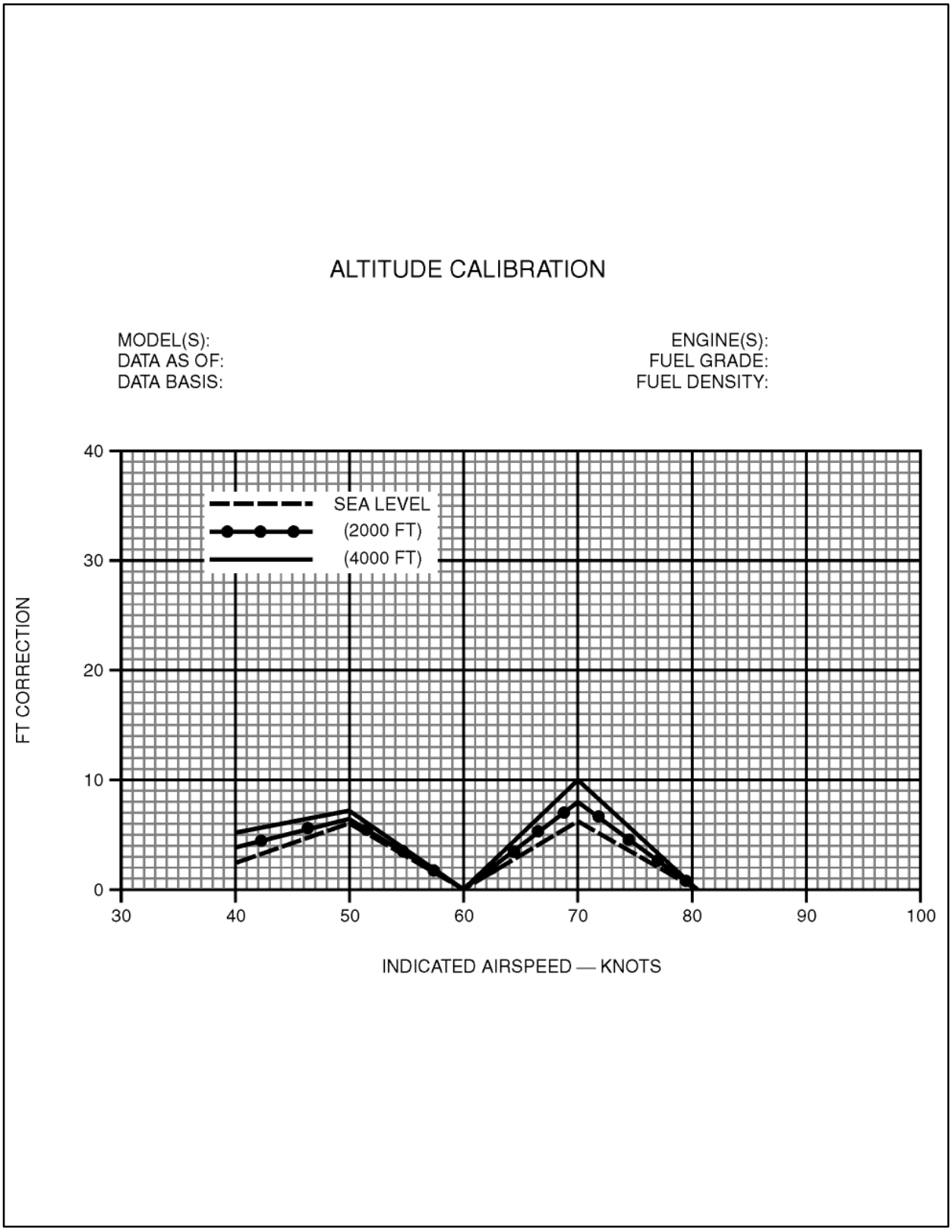


FIGURE 84. Example of altitude calibration chart (helicopter).

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Check the source to verify that this is the current version before use.

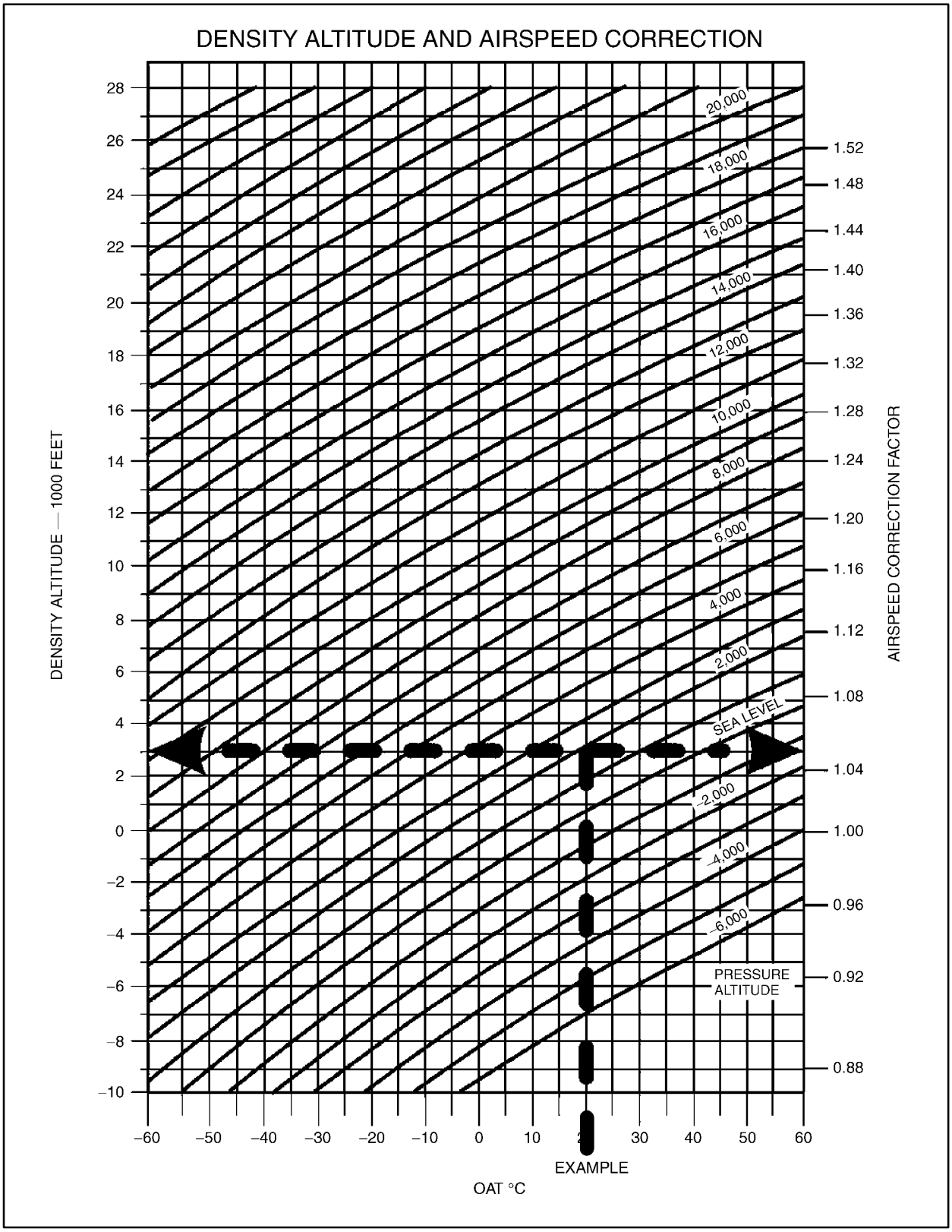


FIGURE 85. Example of density altitude/airspeed correction chart (helicopter).

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Check the source to verify that this is the current version before use.

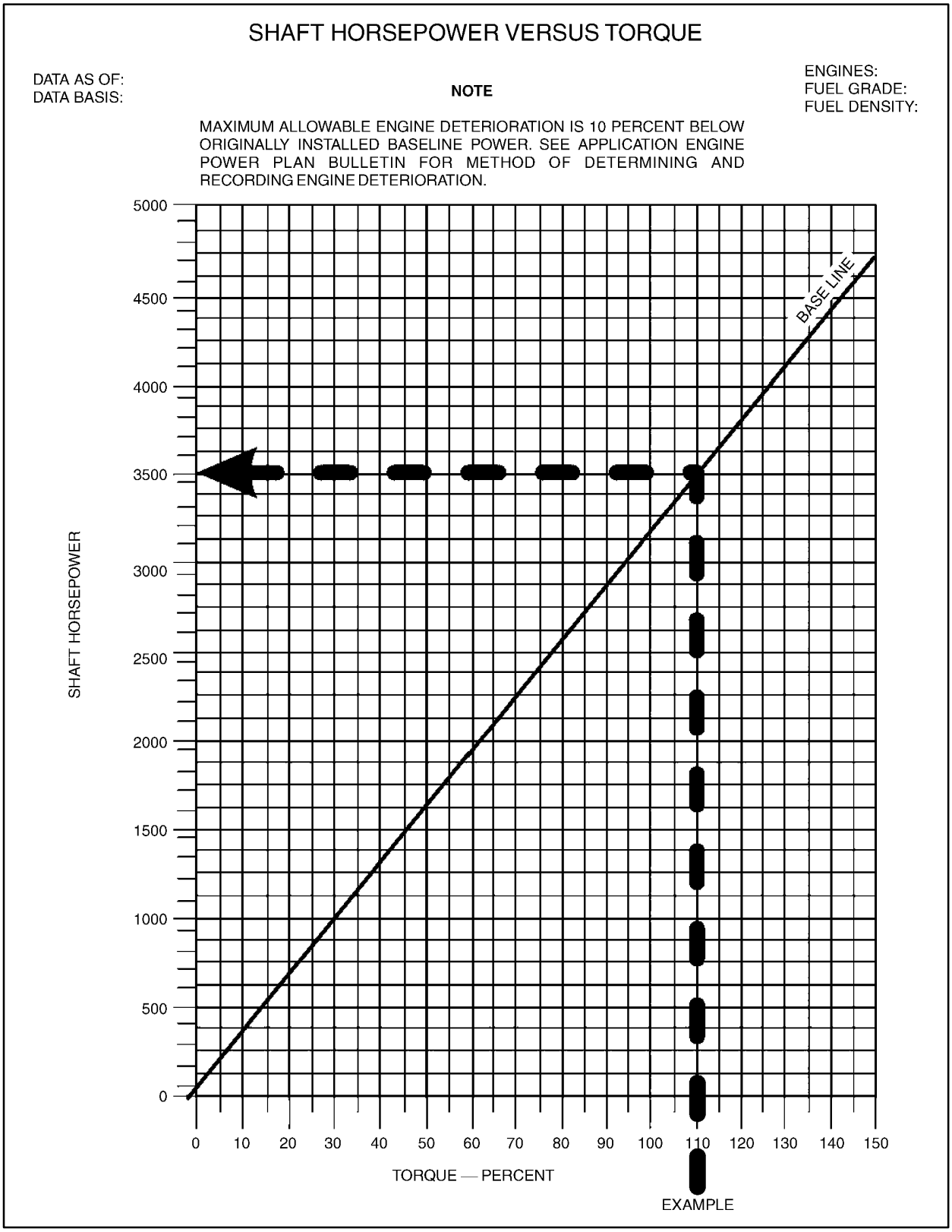


FIGURE 86. Example of shaft horsepower versus torque chart (helicopter).

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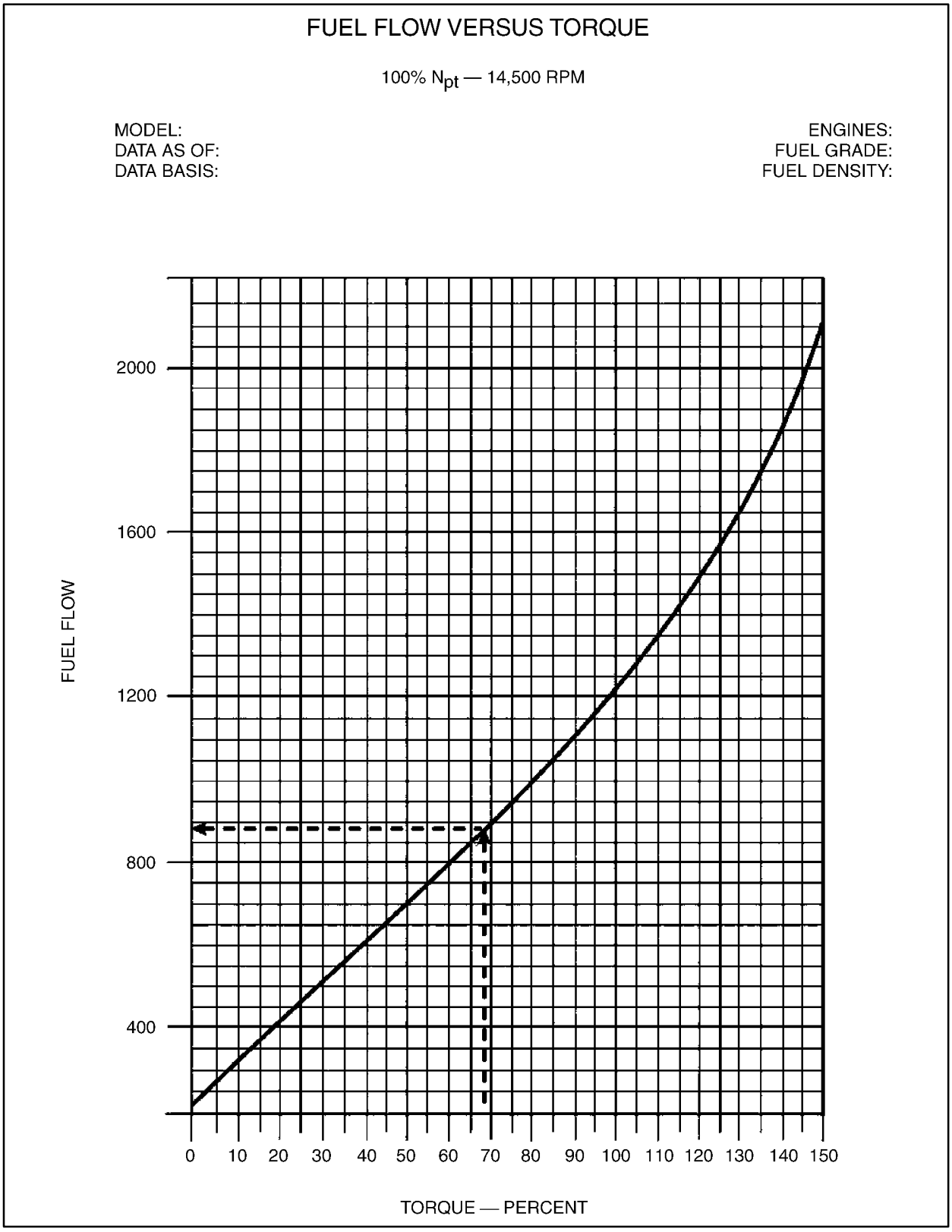


FIGURE 87. Example of fuel flow versus torque chart (helicopter).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

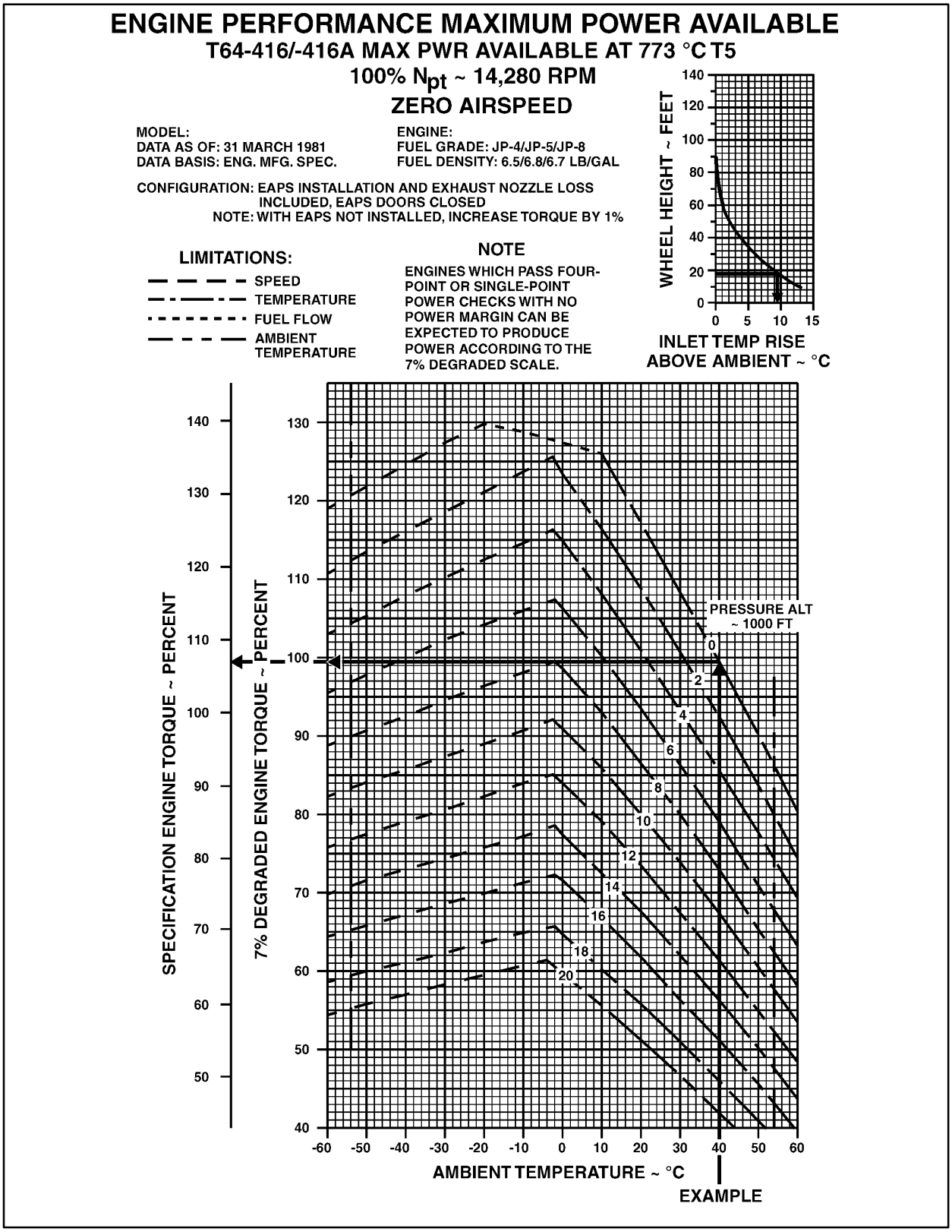


FIGURE 88. Example of power available chart (helicopter).

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Check the source to verify that this is the current version before use.

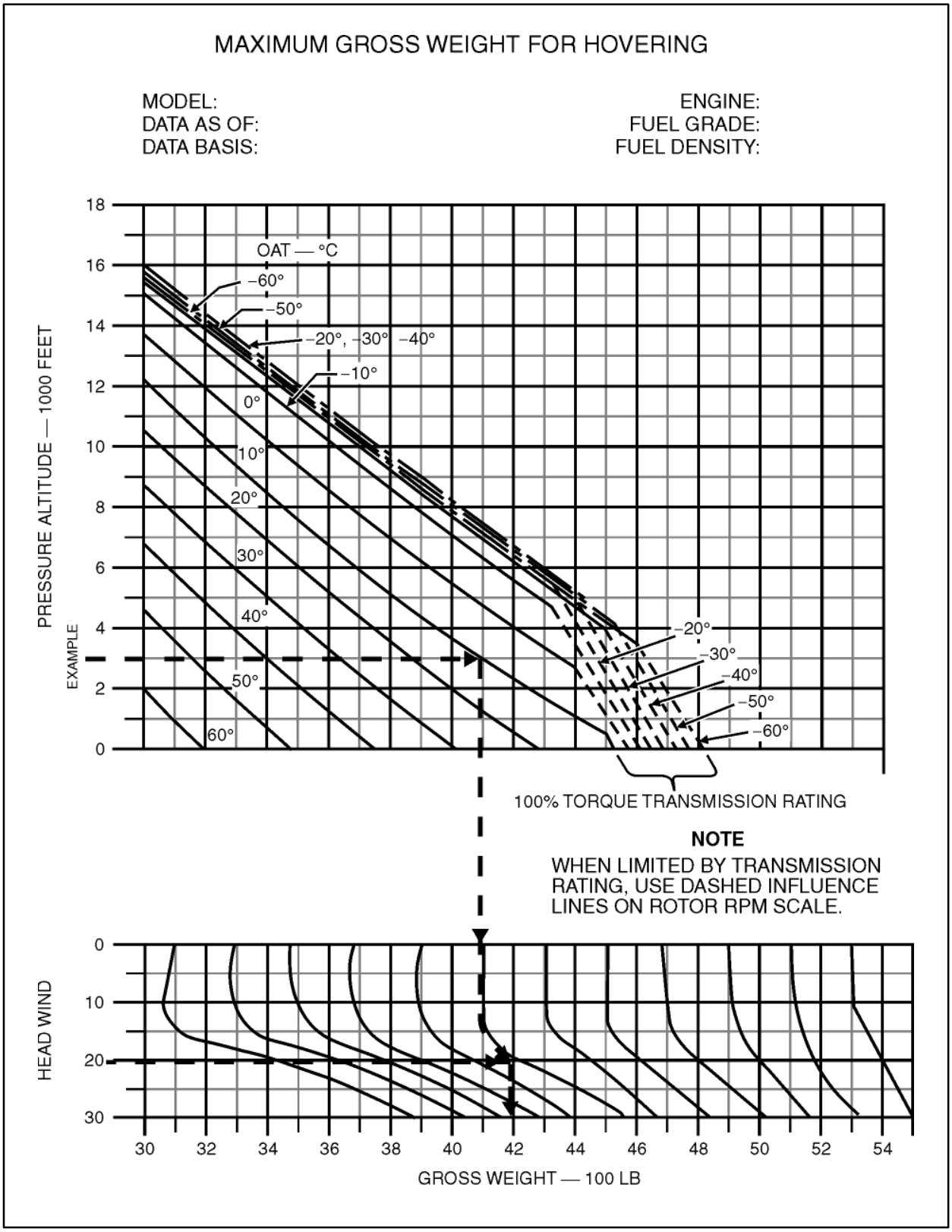


FIGURE 89. Example of maximum gross weight for hovering chart (helicopter).

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Check the source to verify that this is the current version before use.

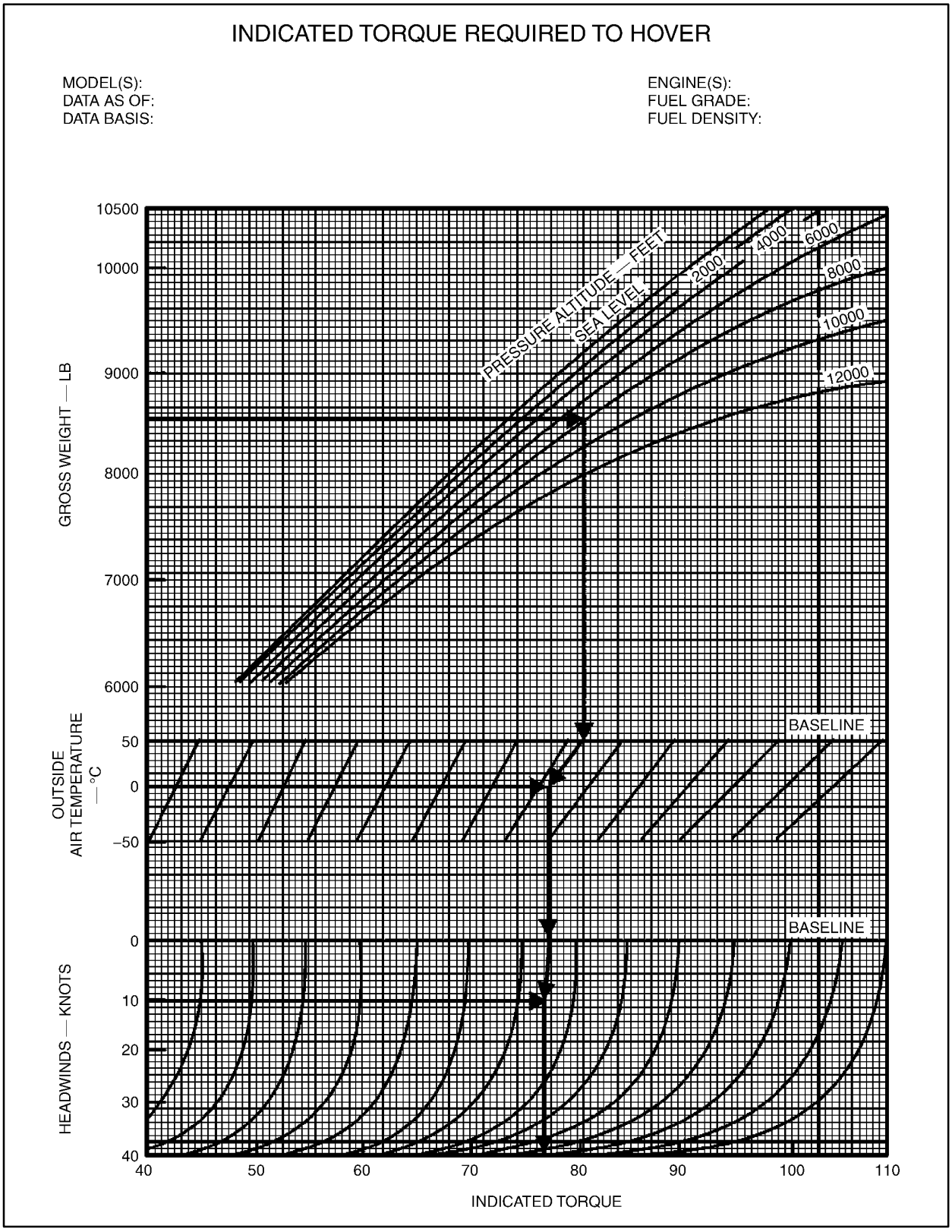


FIGURE 90. Example of indicated torque required to hover chart (helicopter).

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Check the source to verify that this is the current version before use.

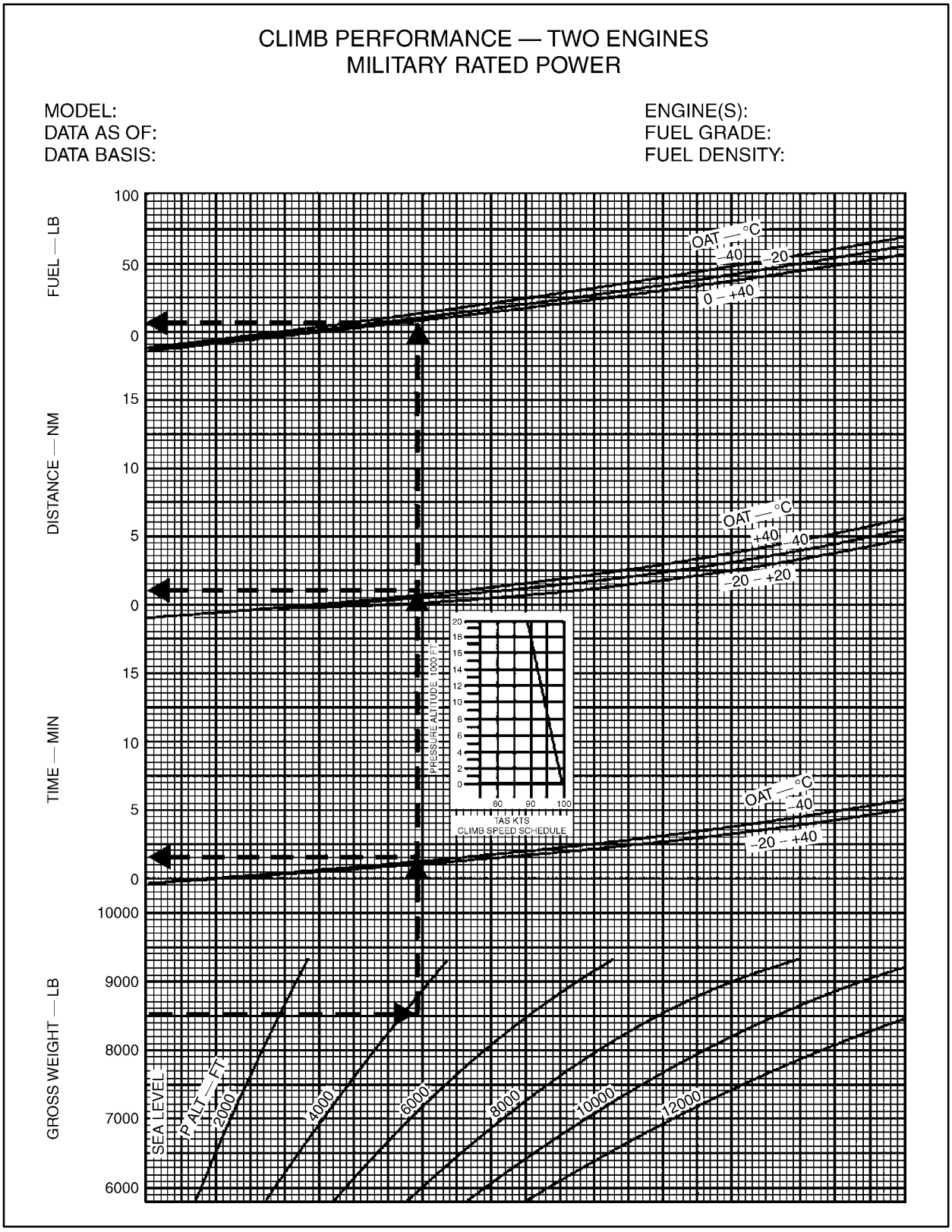


FIGURE 91. Example of climb performance chart (helicopter).

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Check the source to verify that this is the current version before use.

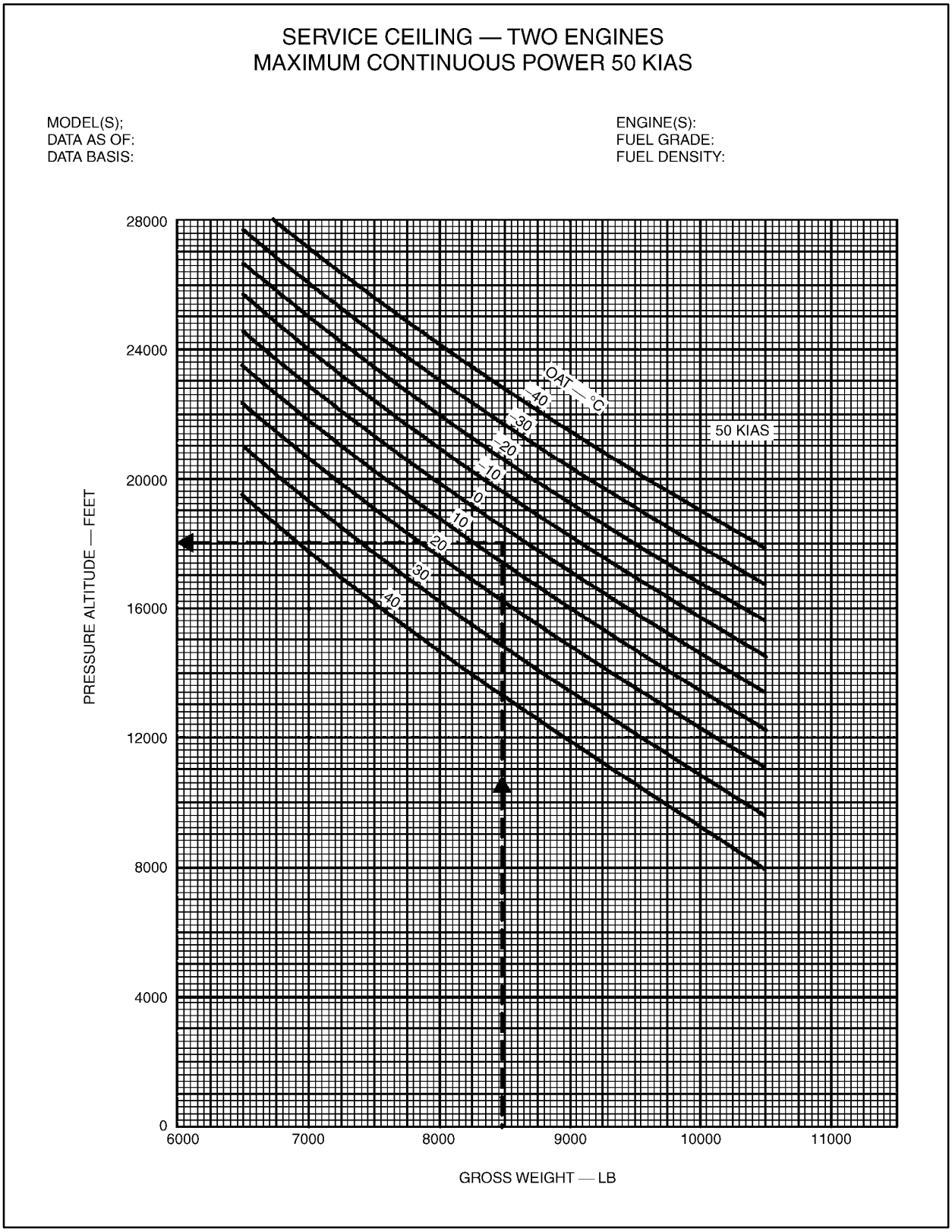


FIGURE 92. Example of service ceiling chart (helicopter).

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Check the source to verify that this is the current version before use.

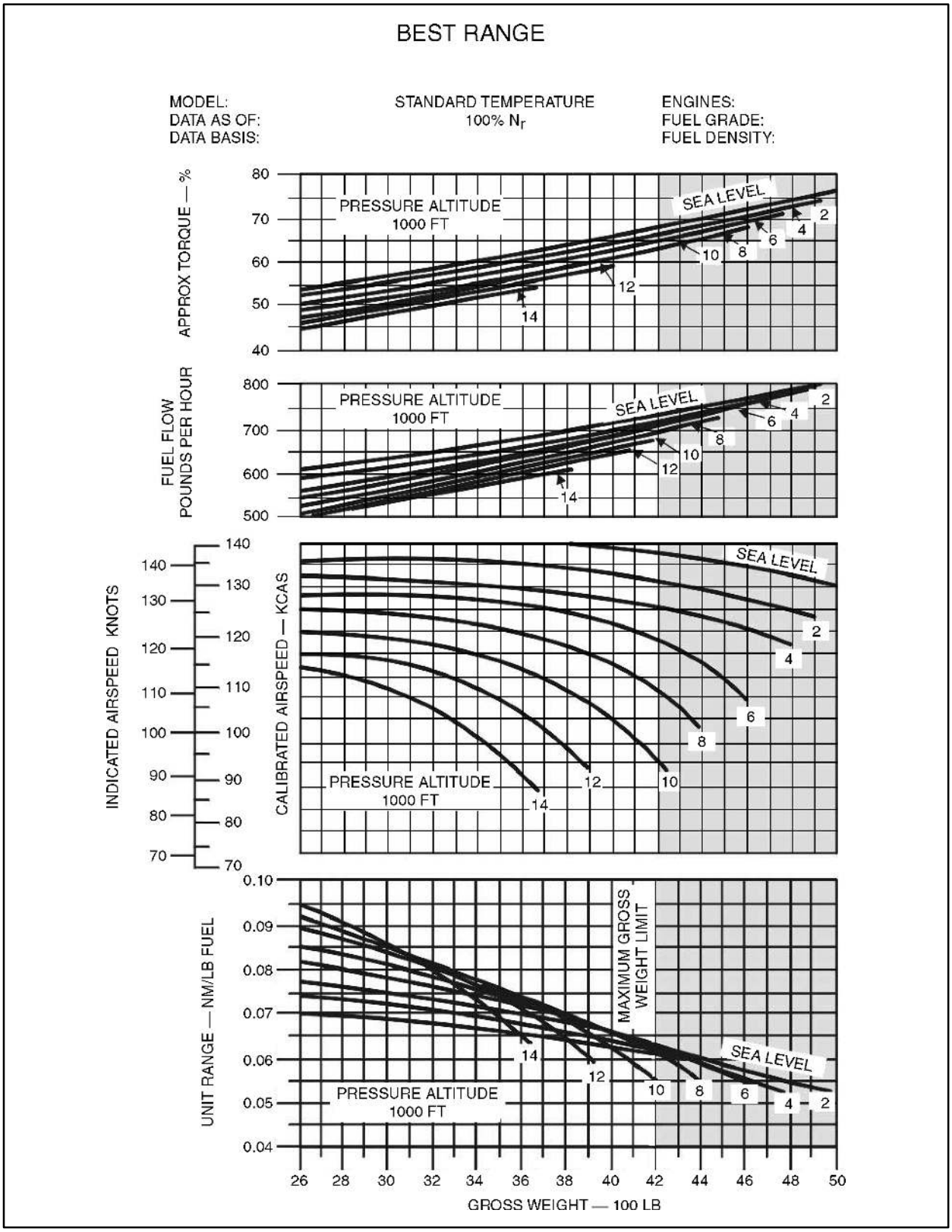


FIGURE 93. Example of best range chart (helicopter).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

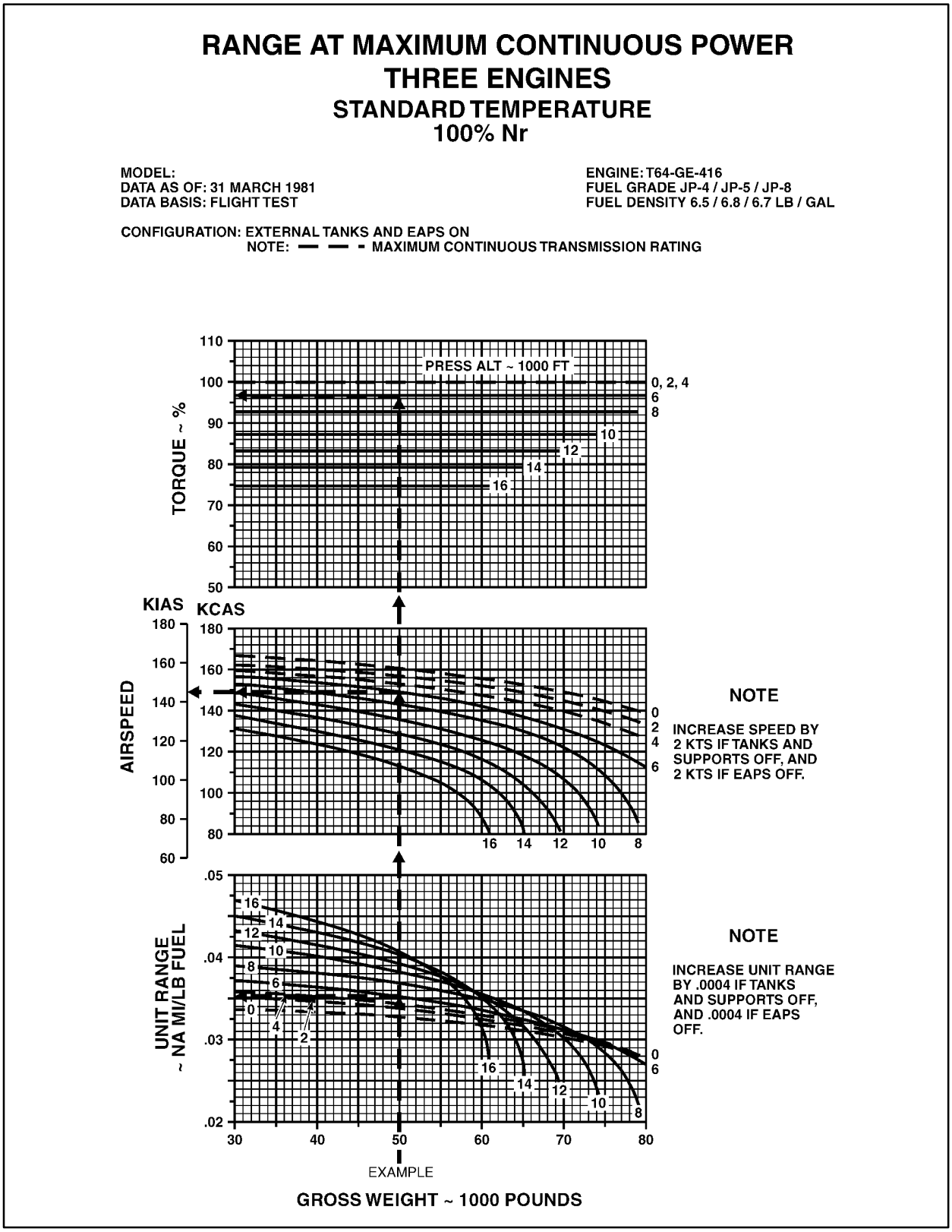


FIGURE 94. Example of range at maximum continuous power chart (helicopter).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

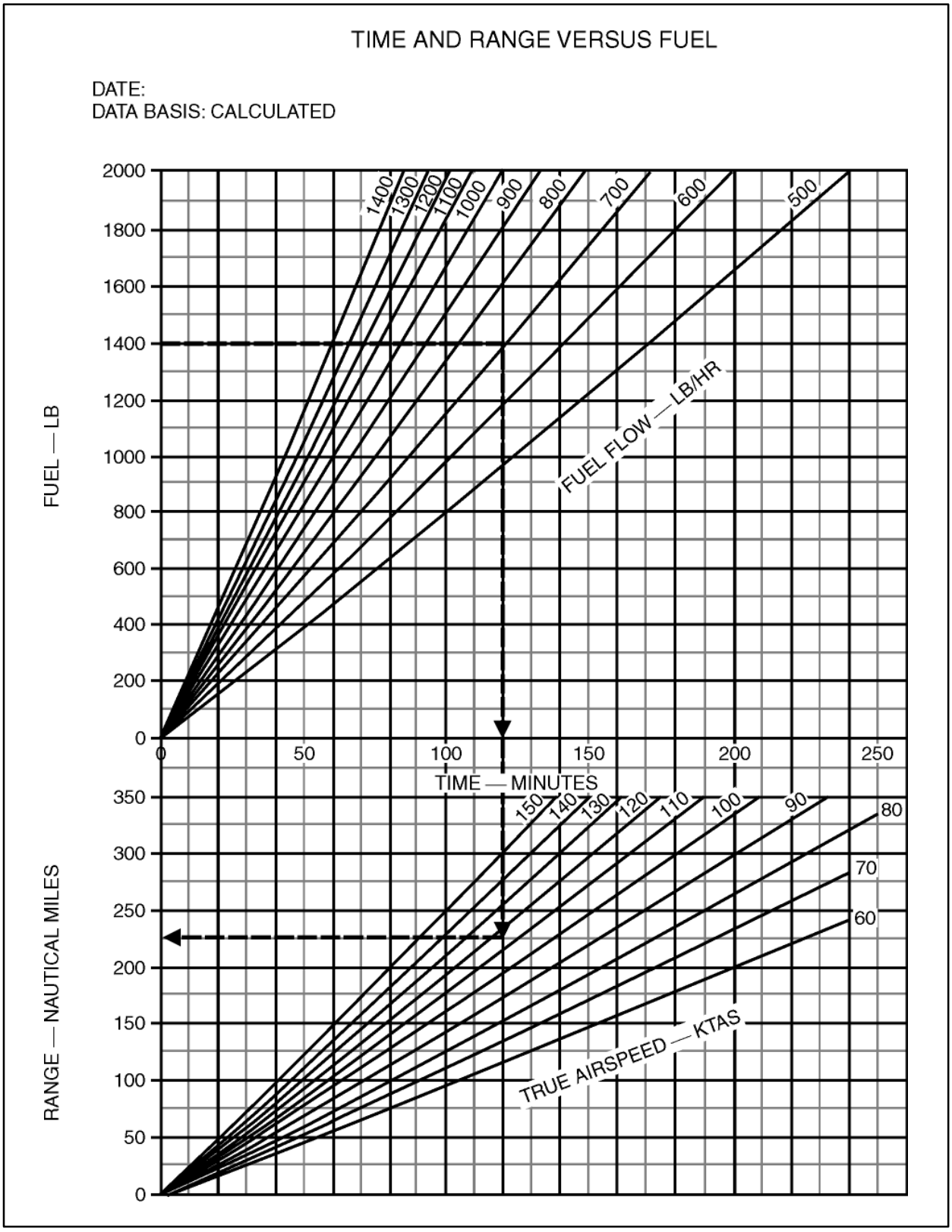


FIGURE 95. Time and range versus fuel chart (helicopter).

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Check the source to verify that this is the current version before use.

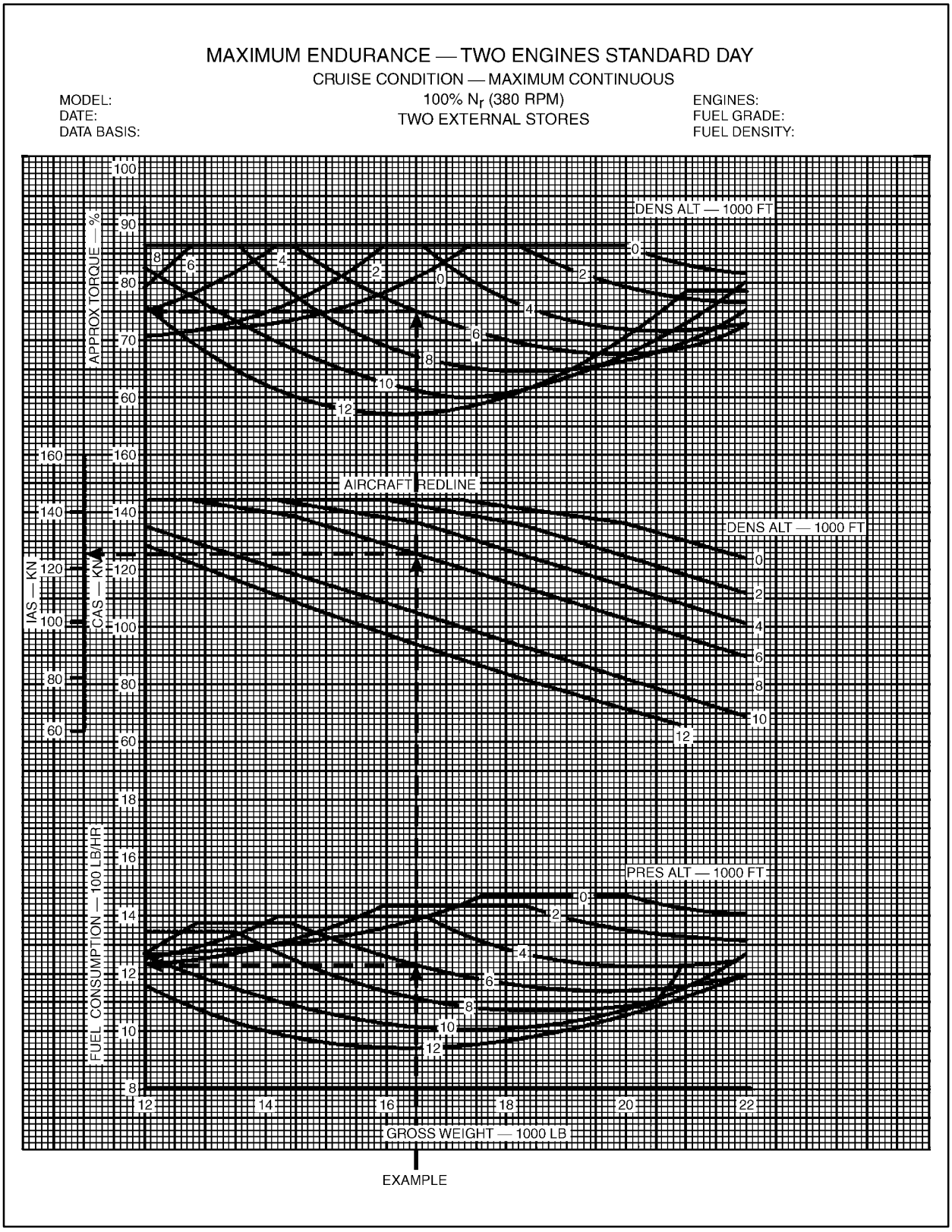


FIGURE 96. Example of maximum endurance chart (helicopter).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

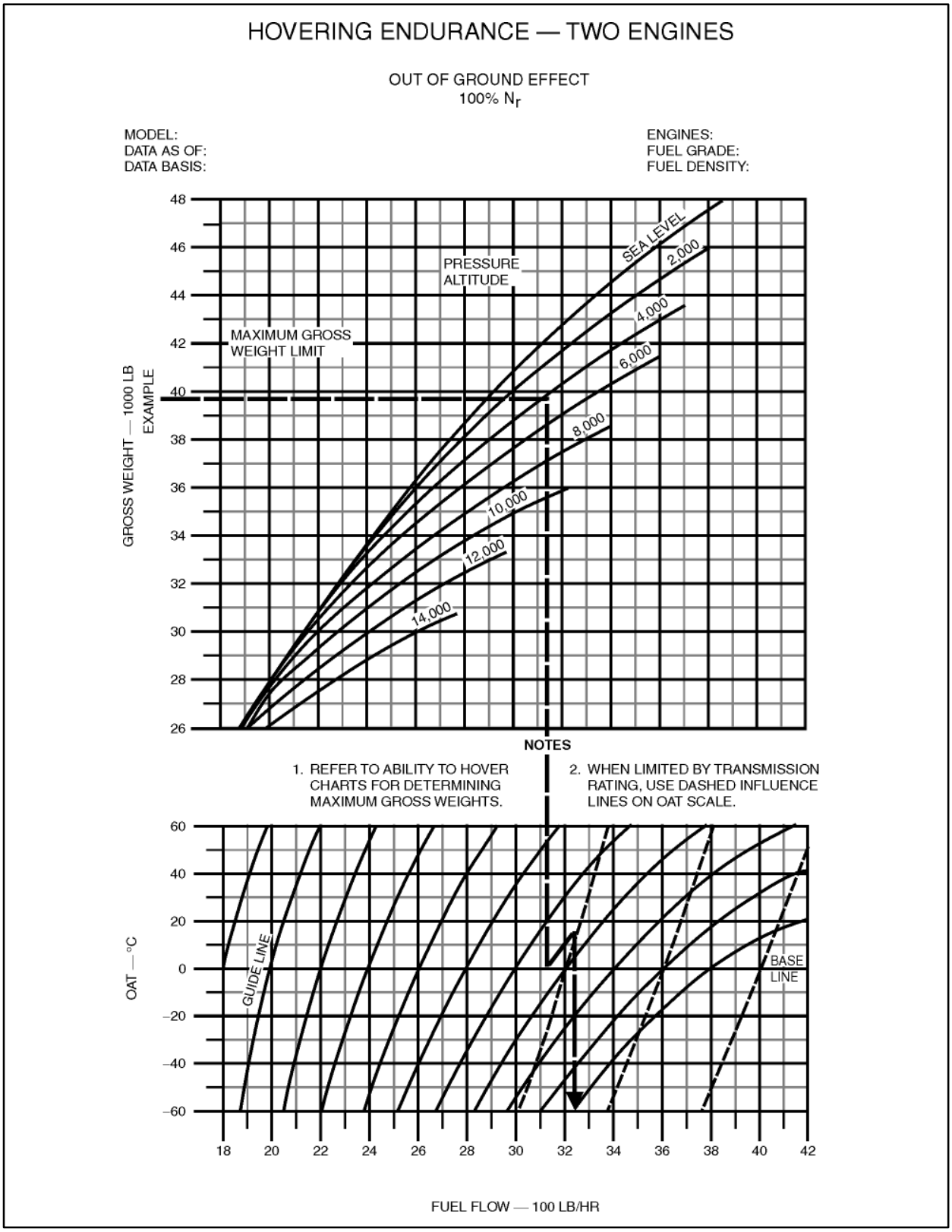


FIGURE 97. Example of hovering endurance chart (helicopter).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

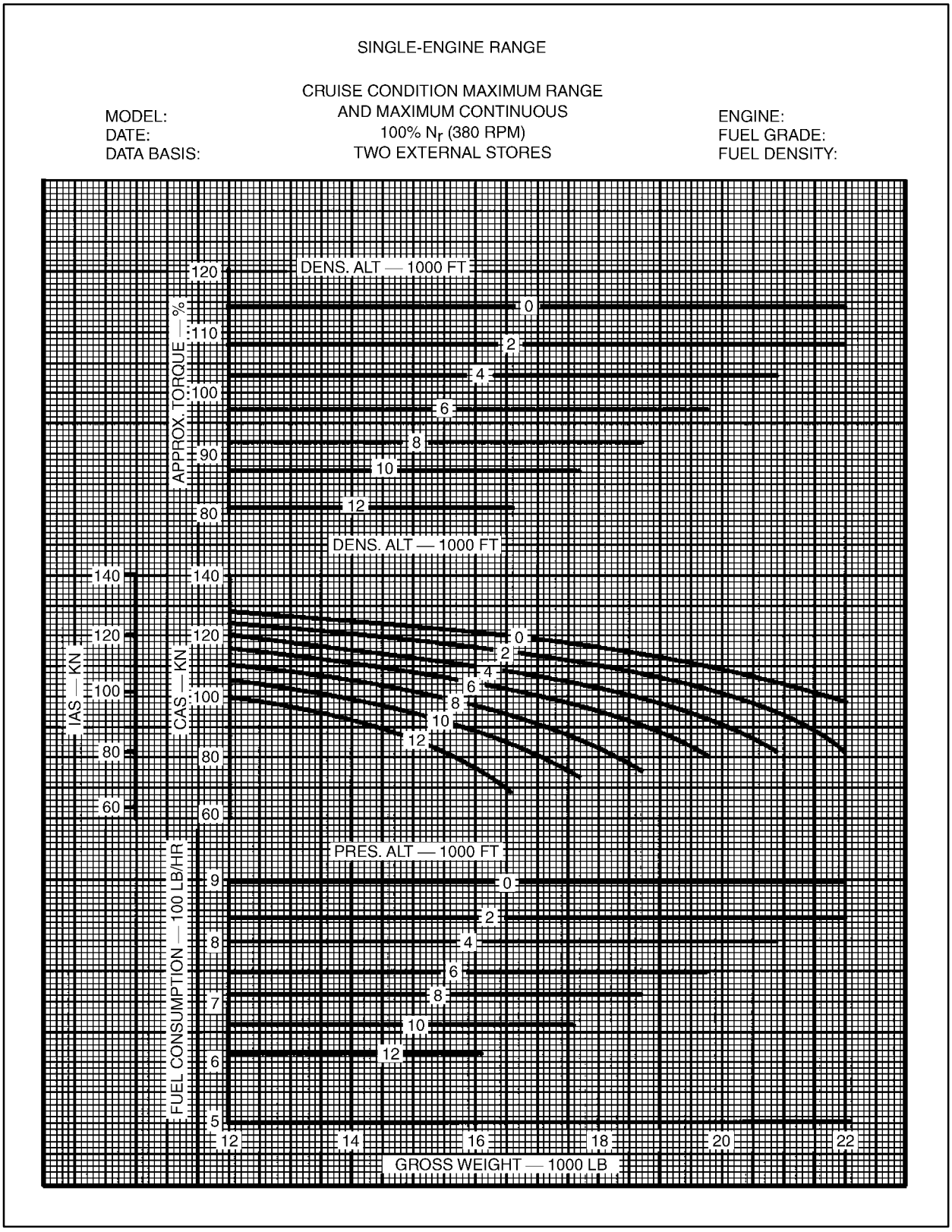


FIGURE 98. Example of single–engine range chart (helicopter).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

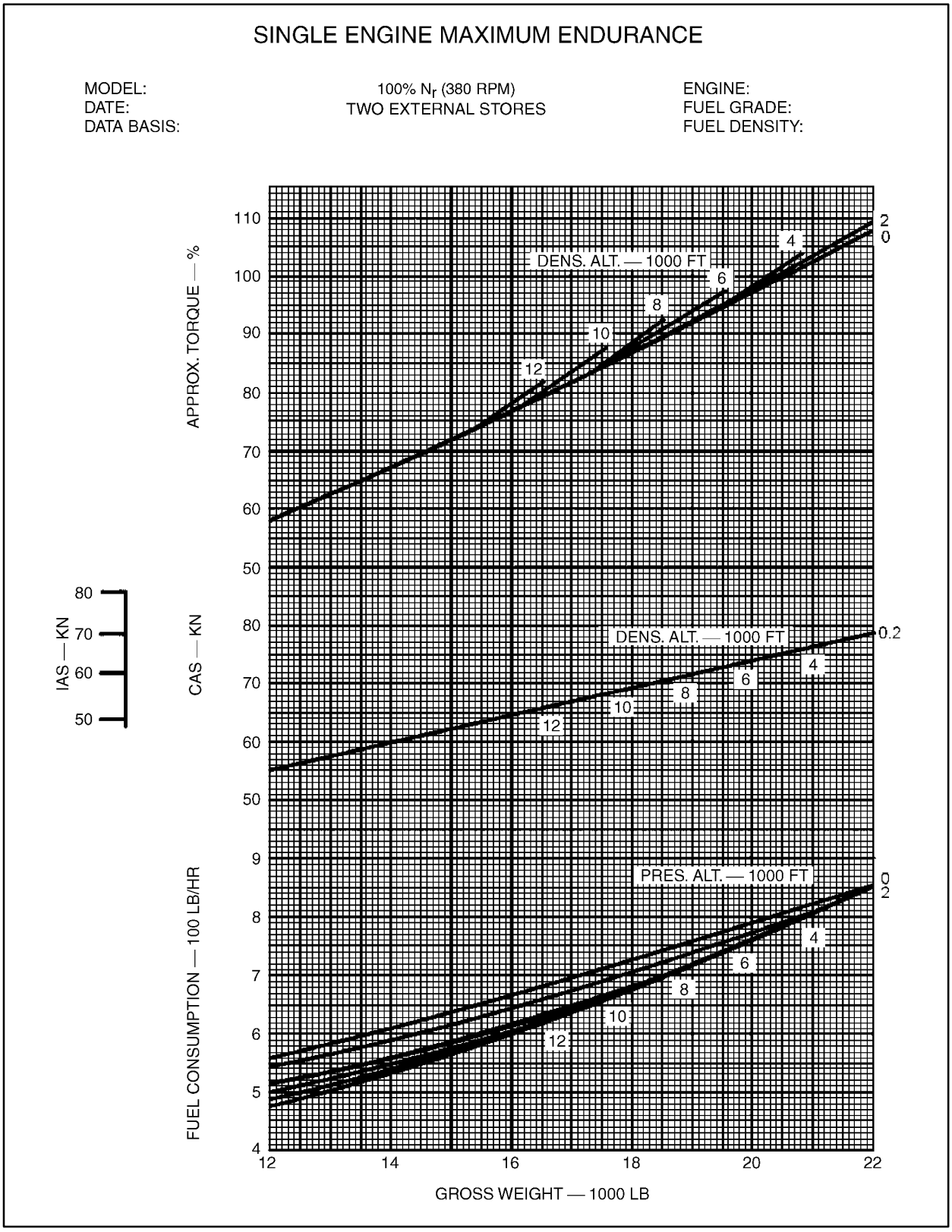


FIGURE 99. Example of single–engine endurance chart (helicopter).

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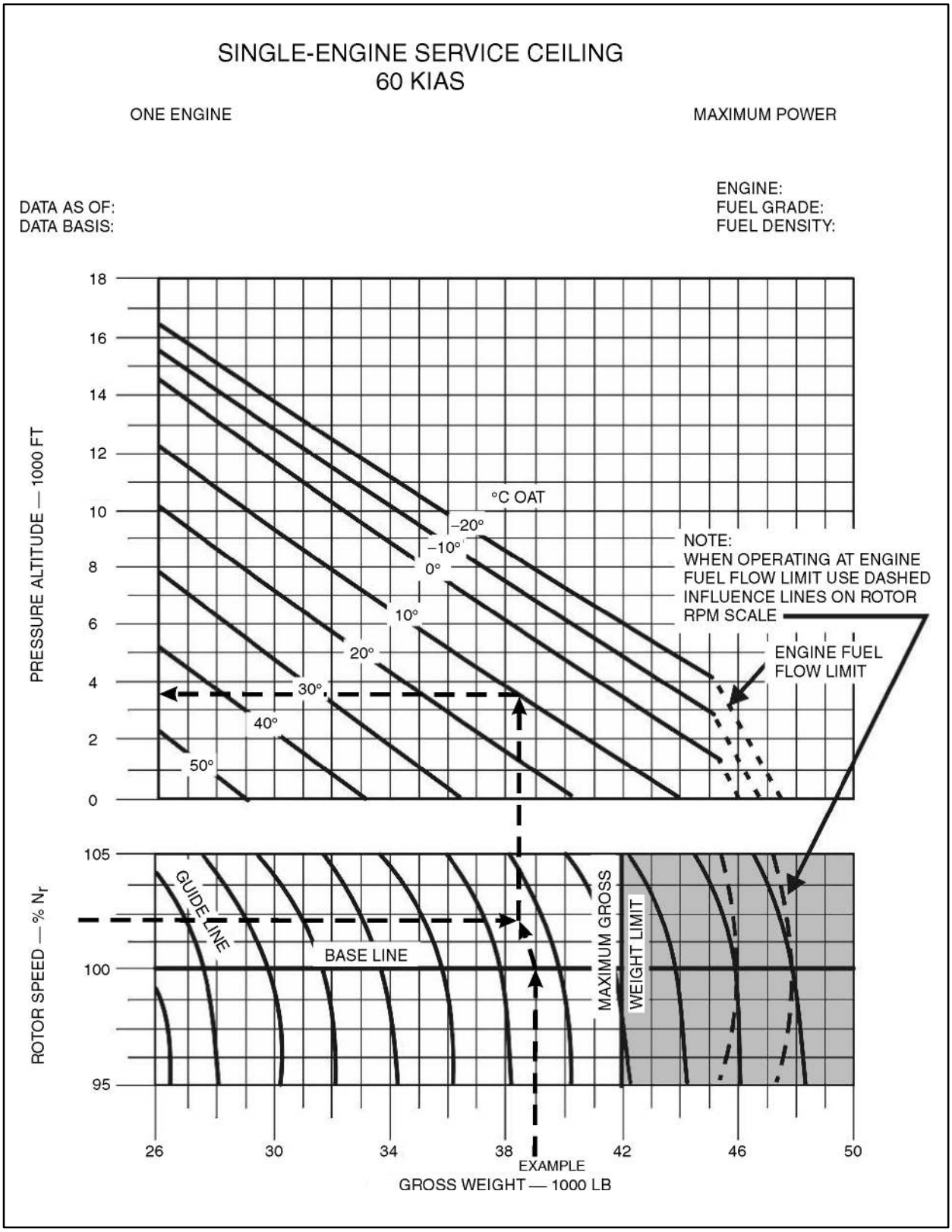


FIGURE 100. Example of single–engine service ceiling chart (helicopter).

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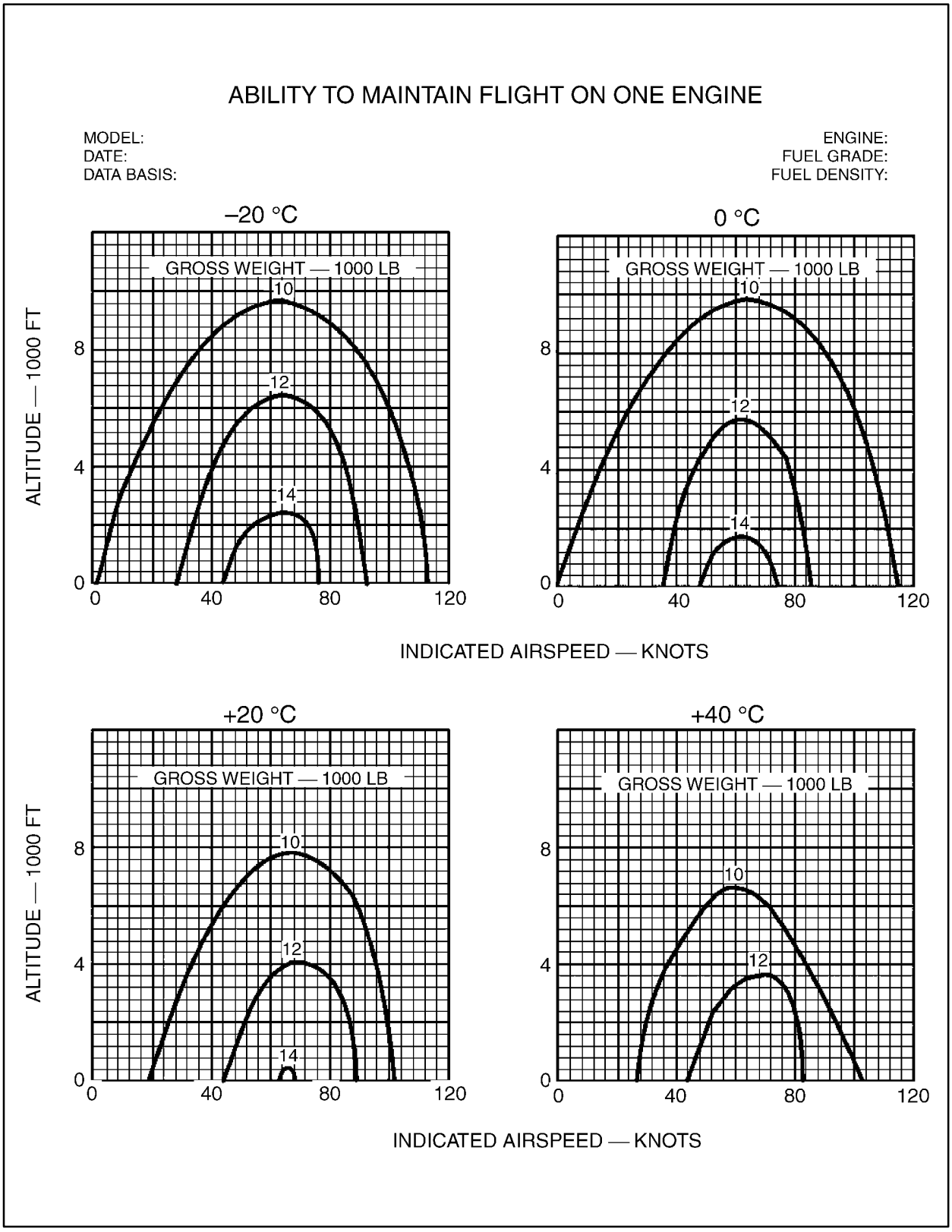


FIGURE 101. Examples of ability to maintain flight on one engine chart (helicopter).

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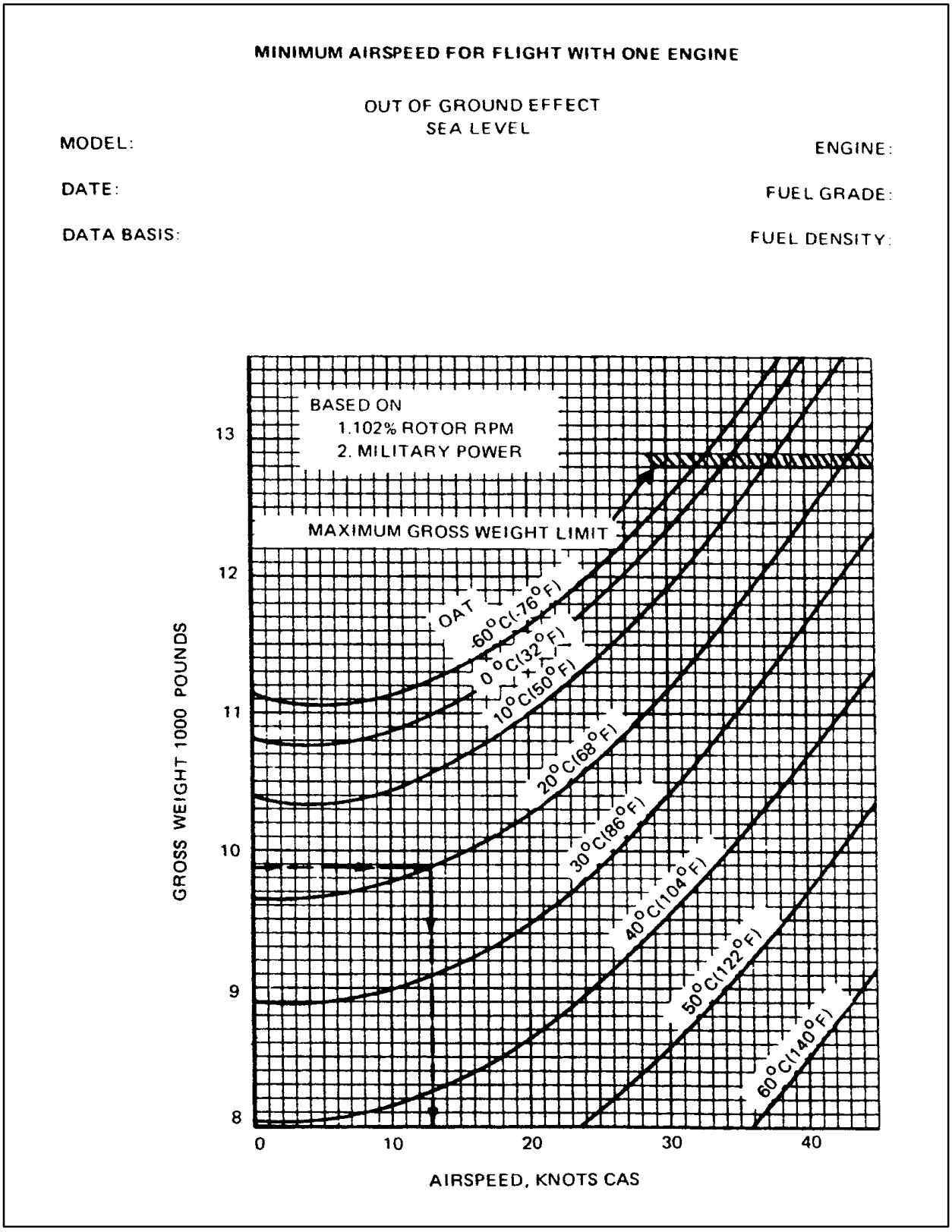


FIGURE 102. Example of minimum airspeed for flight with one engine chart (helicopter).

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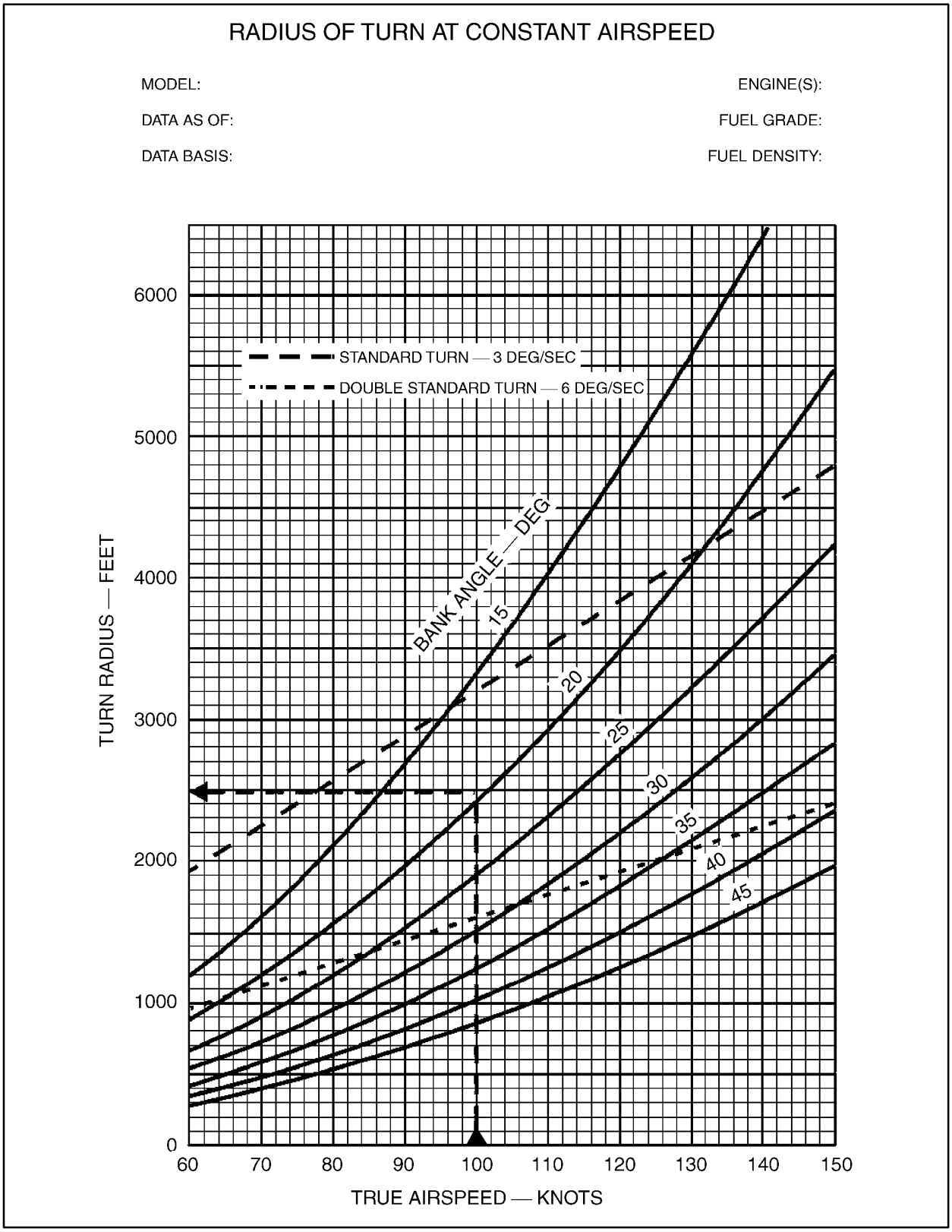


FIGURE 103. Example of radius of turn at constant airspeed chart (helicopter).

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FIGURE 104. Sample NATOPS flight manual alphabetical index.

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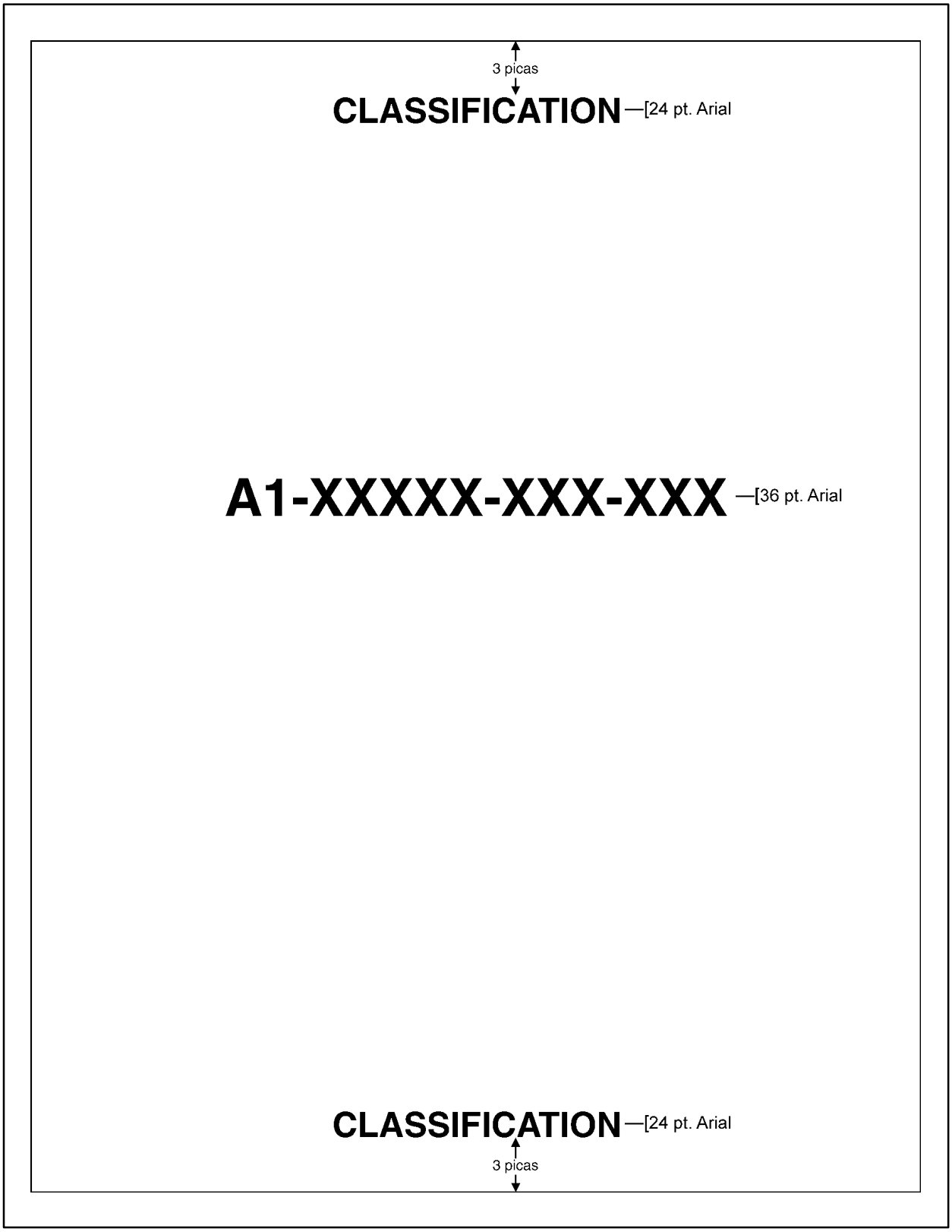


FIGURE 105. Sample NATOPS last page.

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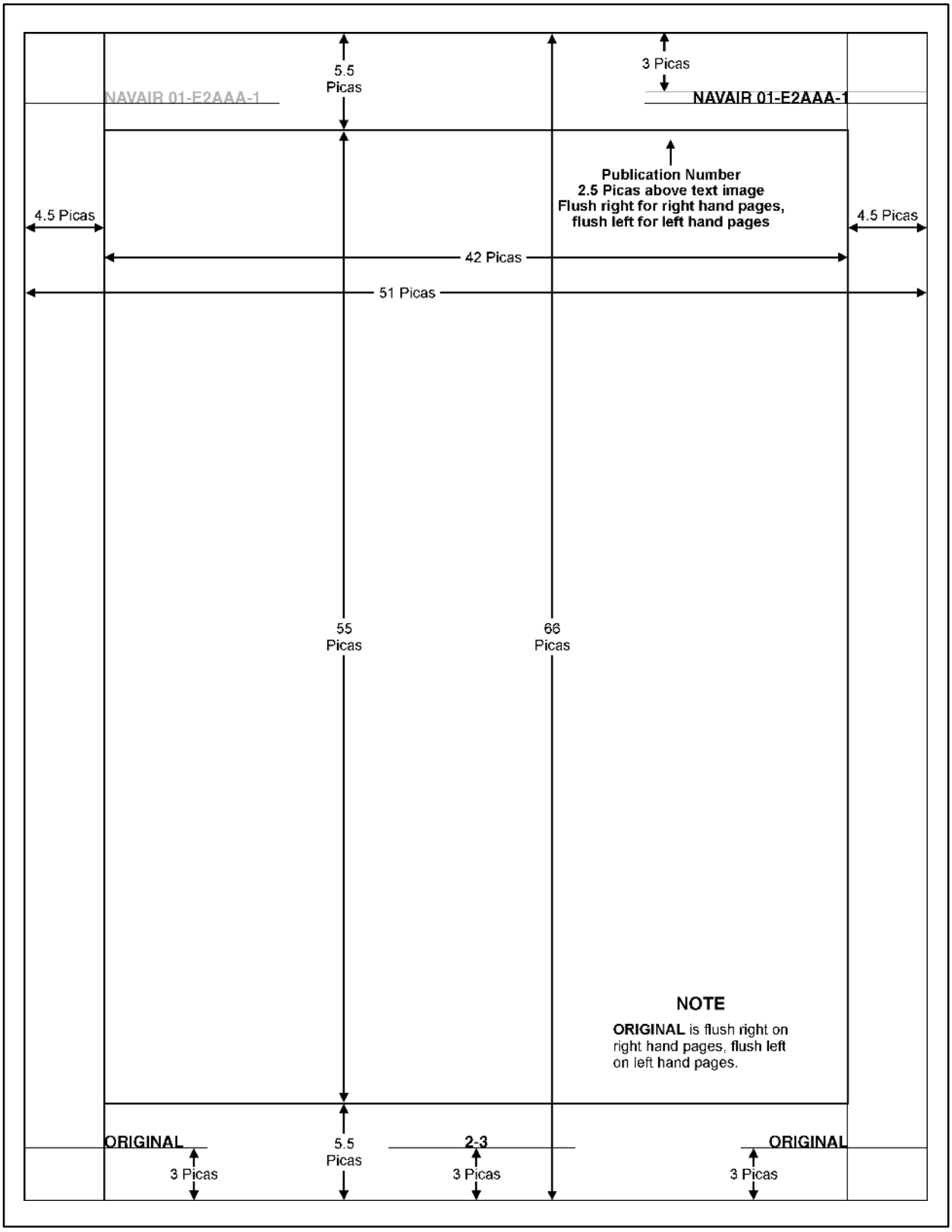


FIGURE 106. NATOPS flight manual page layout.

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PART III

Normal Procedures

Chapter 6 — Flight Preparation

Chapter 7 — Shore-Based/General Procedures — Pilots

Chapter 8 — Shore- or Carrier-Based Procedures — CICO, ACO, RO

Chapter 9 — Carrier-Based Procedures — Pilots

Chapter 10 — Formation Flying

Chapter 11 — Functional Checkflight Procedures

FIGURE 107. Sample NATOPS flight manual part page.

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CHAPTER 7

Shore-Based/General Procedures — Pilots

7.1 GENERAL

Briefing, debriefing, and mission planning should be accomplished in accordance with Chapter 6.

Commanding officers shall ensure that all flight operations are conducted in accordance with the rules and regulations promulgated by higher authority.

7.2 SCHEDULING

7.2.1 Daily Flight Schedule

Normally, all flights will be scheduled through the medium of a daily flight schedule. Most syllabus training flights will be scheduled for 2 or 3 hours duration, allowing an adequate briefing/debriefing period for proper evaluation. In order for a flight schedule to be useful, it should be derived from the master or long-range training plan or employment schedule, as approved by the commanding officer, and should include the following information:

1. Event number.
2. Briefing time, takeoff time, and recovery time.
3. Mission and alternate mission.
4. Names of flight crewmembers.
5. Operating areas, units involved, and communications information.
6. Other applicable notes (ADIZ information, clearance, flight rations, fuel, etc.).

7.2.2 Unscheduled Flights

Unscheduled flights will be handled in accordance with instructions promulgated by local authority.

7.3 LINE OPERATIONS

The pilot-in-command shall examine the Aircraft Discrepancy Book and shall review a record of aircraft discrepancies and corrective actions for the ten previous flights. The pilot-in-command shall then sign the Aircraft Inspection and Acceptance Record, assuming full responsibility for the safe operation of the aircraft and the safety of all individuals aboard.

Conduct the exterior inspection in accordance with Figure 7-1. In addition, ensure that a qualified person inspects the topside of the aircraft for security (see General Checks, Figure 7-1) and electronic integrity.

WARNING

- Radar should not be operated on the flight line (except with a dummy load), because of radiation hazard. See Figure 3-14 for the radar transmission danger areas.
- The CEC system should not be radiated in the line area due to a radiation hazard.

FIGURE 108. Sample NATOPS flight manual chapter introduction page.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
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11 pt Arial Bold or equivalent — NAVAIR 01-E2AAC-1

1-pt rules

10 pt Arial Bold or equivalent

Gross Weights	Maximum Brake Application Speed
53,001 to 55,000 pounds	93 knots
46,501 to 53,000 pounds	95 knots
46,500 pounds or less	100 knots

10 pt Arial or equivalent

(5) Check that emergency oxygen system pressure gauge needle is in the black range. Figure 7-3.

WARNING

6-pt rule

If the green apple is accidentally pulled with the dustcap on the emergency oxygen system hose, the system might not bleed down immediately, giving false indication of system readiness several hours after the system has been activated.

(6) Check that AAD aneroid port is GREEN and the AAD armed port is GREEN. Figure 7-3.

13. CAINS GROUND BY-PASS switch — CLOSED.

14. Vapor cycle scoop — CLEAR.

15. Emergency fuel caps — SECURE.

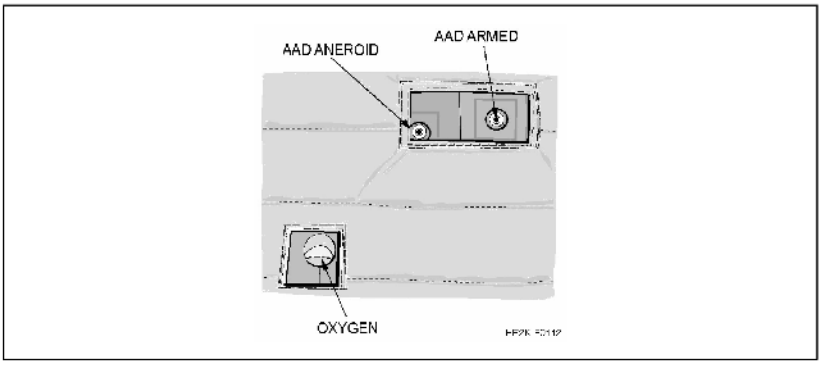
16. Nose wheel inspection window — CHECK CLEAR.

11 pt Arial Bold or equivalent — **7.7.1.2 Forward Equipment Compartment**

See FO-1 and FO-2 for equipment location and nomenclature.

11 pt Times New Roman or equivalent — 1. All electronic equipment will be checked for:

a. Proper cabling.



11 pt Times New Roman or equivalent — Figure 7-3. Emergency Oxygen System Pressure Gauge and AAD Inspection Windows

FIGURE 109. Font requirements for NFM paragraphs, figures, tables, and headers.

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APPENDIX D

General Aircraft Handling Signals

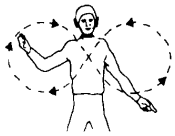
SIGNAL	DAY	NIGHT	REMARKS
 HOT BRAKES S-0001	Makes rapid fanning motion with one hand in front of face and points to wheel with other hand.	Same as day except with wands.	Signal: U.S. Navy
 ENGINE FIRE S-0002	Describes a large figure eight with one hand and points to the fire area with the other hand.	Same as day except with wands.	Signal is meant for information only. Pilot should be given a cut engine or continuous turn-up signal, as appropriate. Signal: U.S. Navy
 AFFIRMATIVE (ALL CLEAR) S-0003	Hand raised, thumb up.	Same as day except with wands.	Conforms to ICAO signal. Signal: U.S. Navy/NATO
 NEGATIVE (NOT CLEAR) S-0004	Arm held out, hand below waist level, thumb turned downwards.	Same as day except with wands.	Signal: U.S. Navy/NATO
 PERSONNEL APPROACHING AIRCRAFT S-0005	Left hand raised vertically overhead, palm toward aircraft The other hand indicates to personnel concerned and gestures towards aircraft.	Same as day except with wands.	Signal: U.S. Navy/NATO
 I HAVE COMMAND S-0006	Hold one hand open, motionless and high above head, with palm forward.	Same as day except with wands.	Signal: U.S. Navy

Figure D-1. General Aircraft Handling Signals (Sheet 1 of 4)

FIGURE 110. Sample NATOPS flight manual appendix page.

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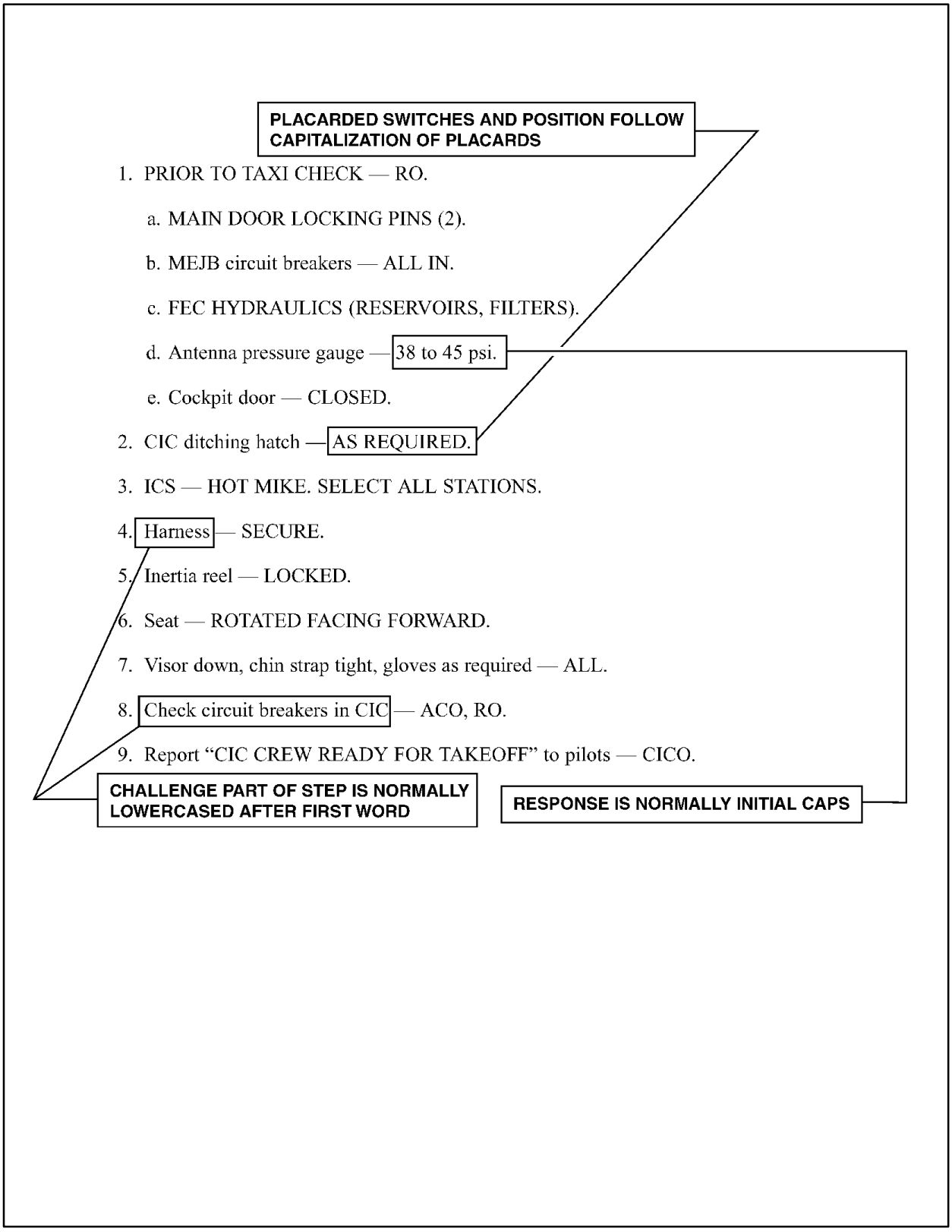


FIGURE 111. Sample NATOPS checklist format.

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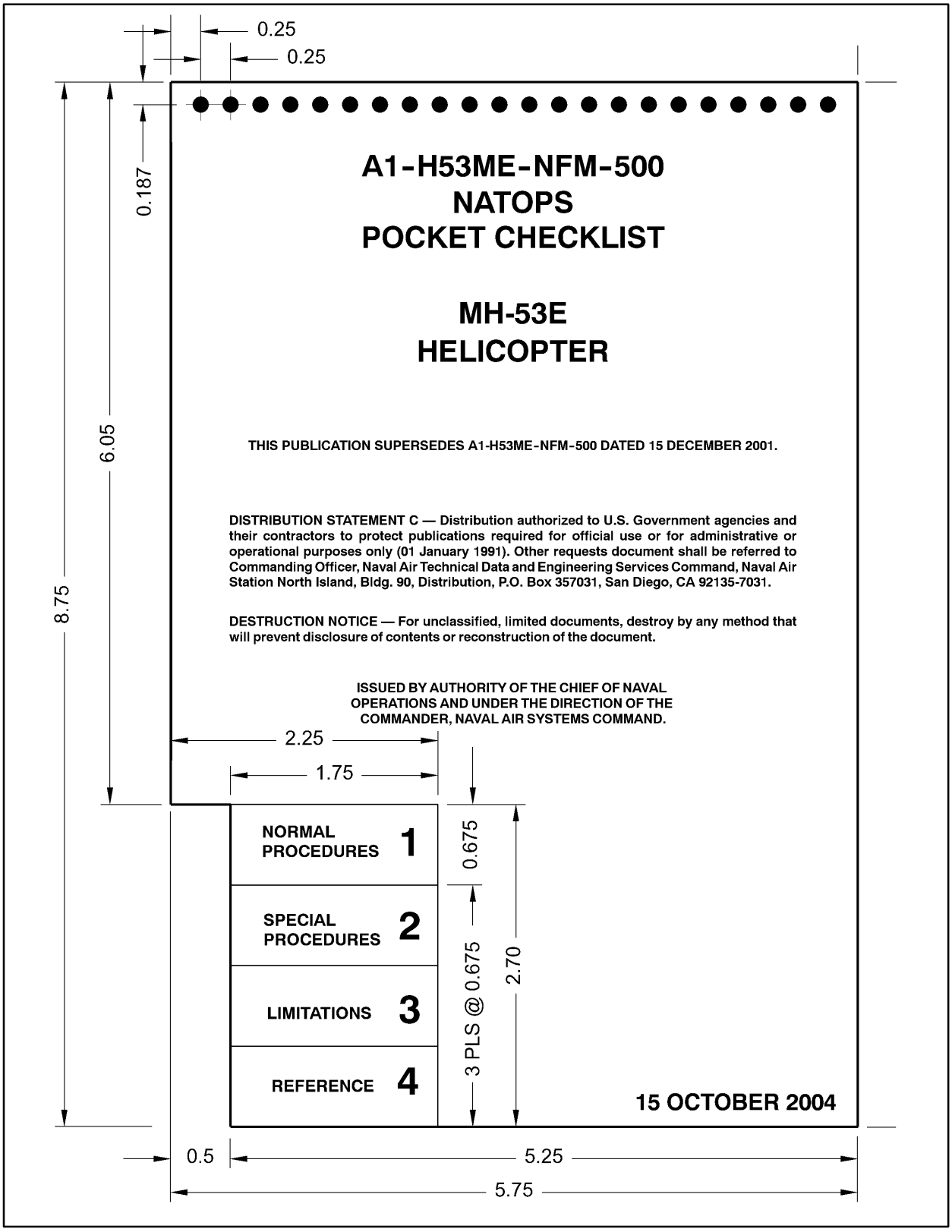


FIGURE 112. Typical pocket checklist cover page.

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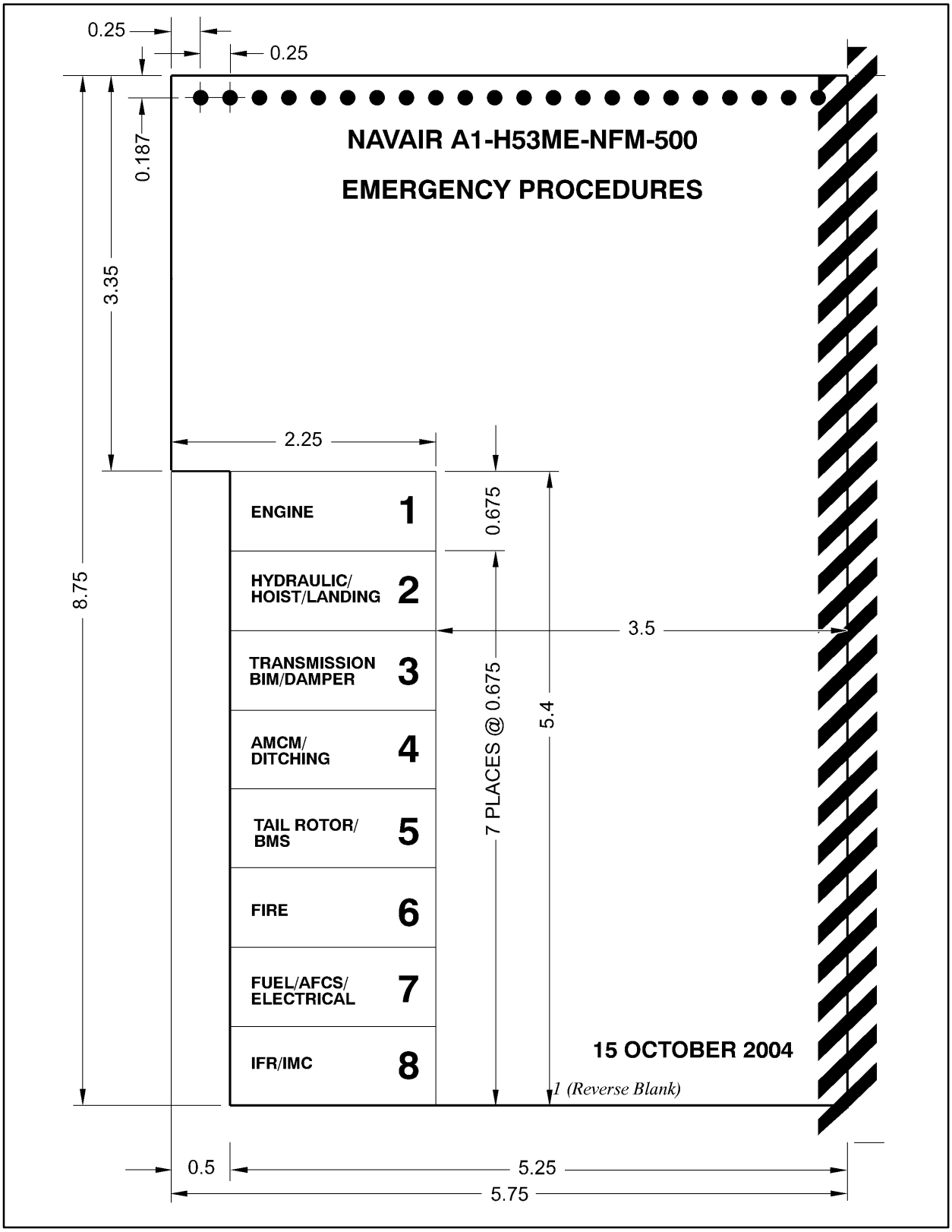


FIGURE 112. Typical Pocket checklist cover page — Continued.

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A

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LIST OF EFFECTIVE PAGES

NOTE: Text affected by current change indicated by vertical line in outer margin.

DATES OF ISSUE FOR ORIGINAL AND CHANGED PAGES ARE:

Original . . . . . 0 . . . . 15 OCTOBER 2004

TOTAL NUMBER OF PAGES IN THIS PUBLICATION IS 244 CONSISTING OF THE FOLLOWING:

Page No.	*Issue*
Title thru S	0
1 thru 39/(40 Blank)	0
41 thru 136	0
EP Title/(Reverse Blank)	0
Index-1, Index-2	0
E1 thru E49/(E50 Blank)	0
E51/(E52 Blank)	0
E53 thru E59/(E60 Blank)	0
E61 thru E81/(E82 Blank)	0

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FIGURE 113. Pocket checklist list of effective pages ("A" page).

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INTERIM CHANGE SUMMARY

The following Interim Changes have been cancelled or previously incorporated into this manual.

INTERIM CHANGE NUMBER(S)	REMARKS/PURPOSE
1 THRU 16	Previously Incorporated

The following Interim Changes have been cancelled or incorporated into this Change/Revision.

INTERIM CHANGE NUMBER(S)	REMARKS/PURPOSE
17	Single-Point Performance Check Data for 419 Engine
18	Engine Chip Light Procedures and Bearing Monitor System

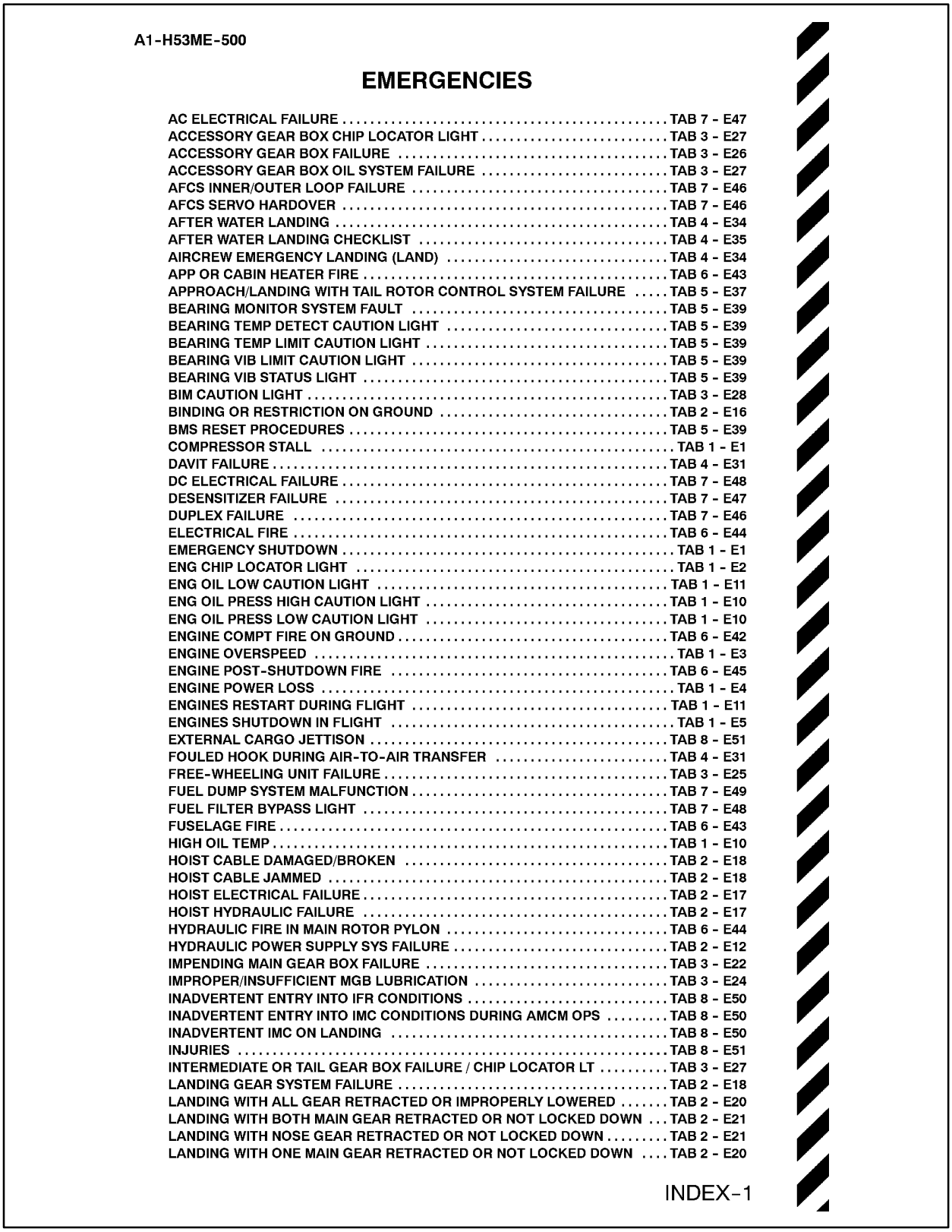
Interim Changes Outstanding — To be maintained by the custodian of this manual.

INTERIM CHANGE NUMBER(S)	ORIGINATOR/DATE (or DATE/TIME GROUP)	PAGES AFFECTED	REMARKS/ PURPOSE

B

FIGURE 114. Pocket checklist interim change summary (“B” page).

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INDEX-1

FIGURE 115. Pocket checklist emergency procedures index page.

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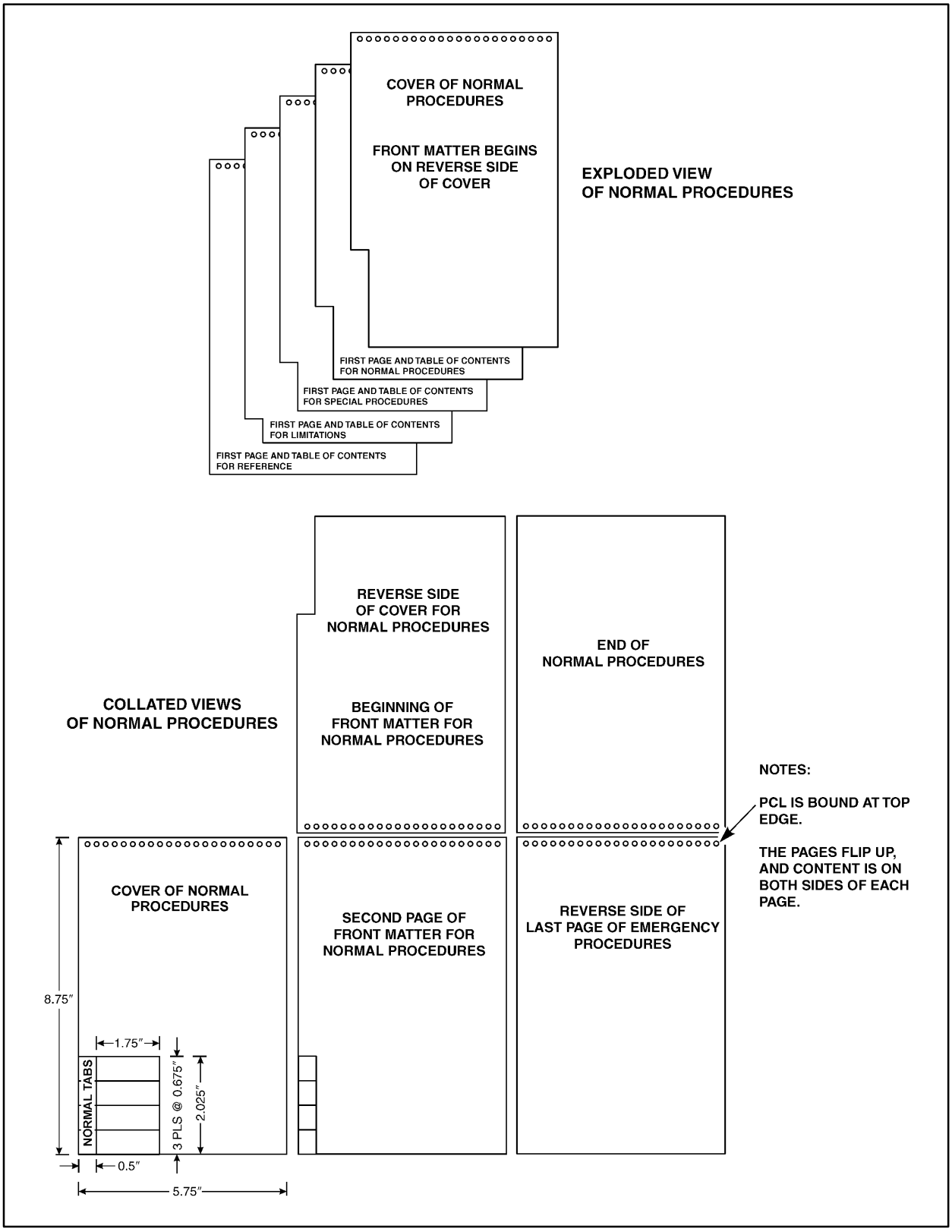


FIGURE 116. Pocket checklist general arrangement and divider (tab) pages.

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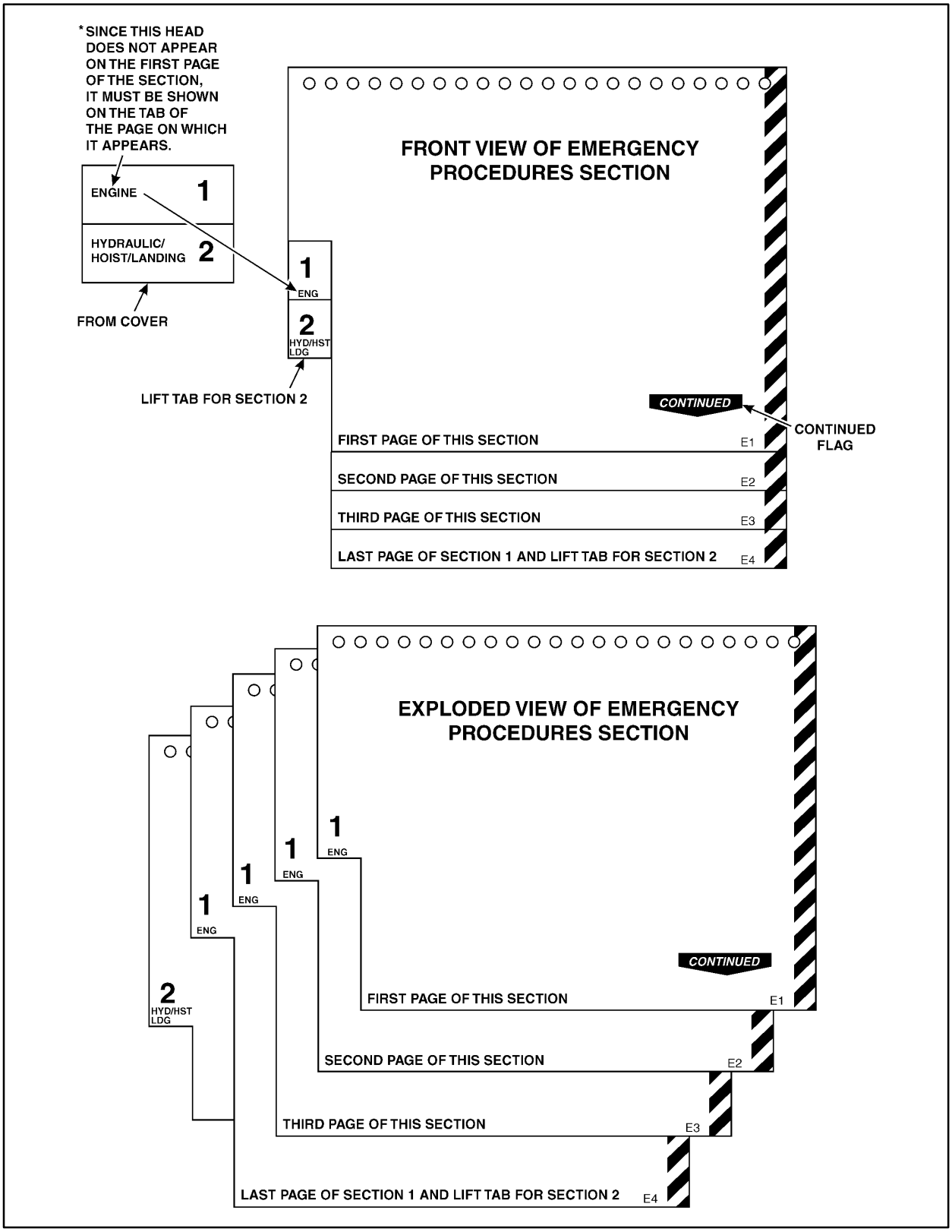


FIGURE 116. Pocket checklist general arrangement and divider (tab) pages — Continued.

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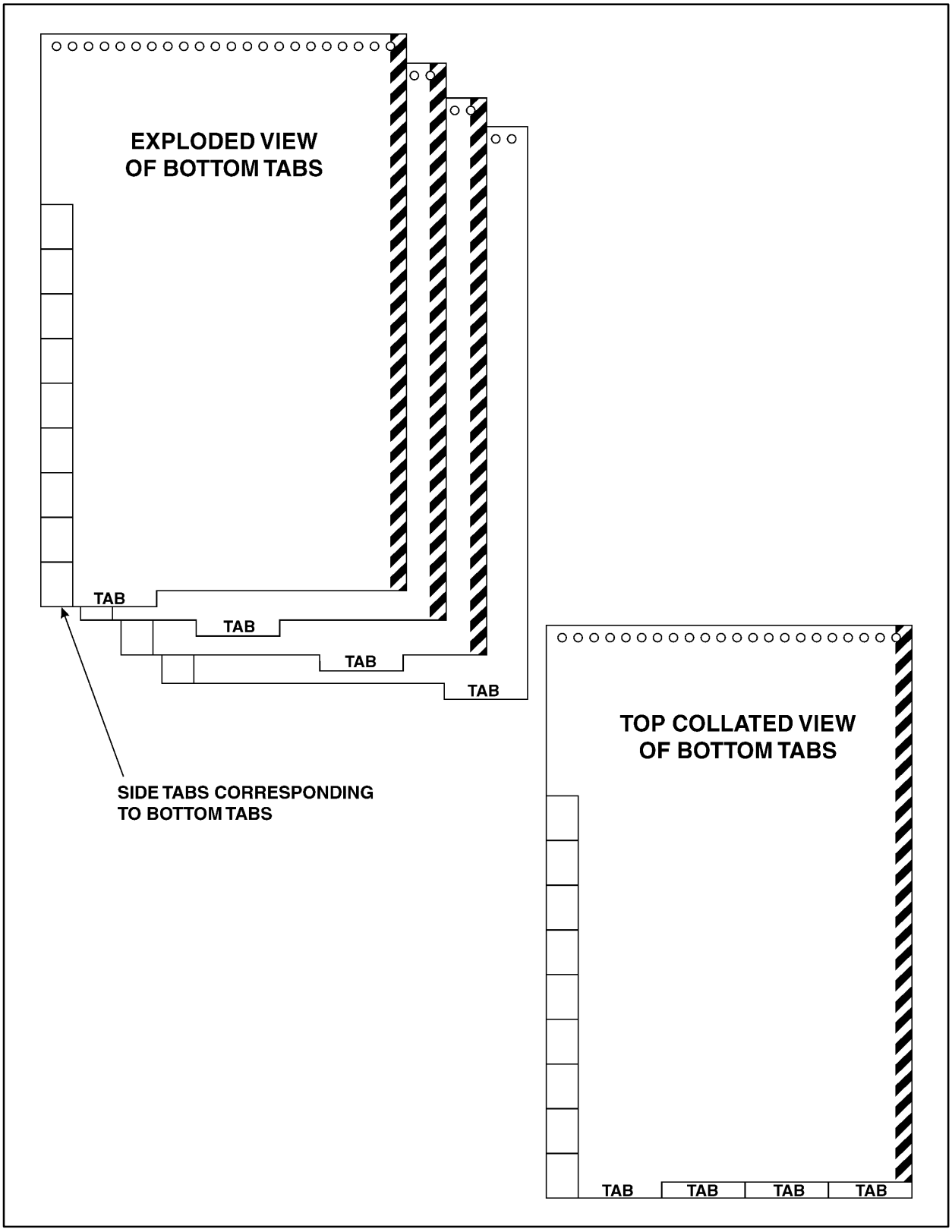


FIGURE 116. Pocket checklist general arrangement and divider (tab) pages — Continued.

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Emergency Procedures Pages Sizes

Plastic Cover	9 x 5.75
EP Title Page	8.75 x 4.75
EP Index Page	8.75 x 5.75
EP Left Tab 1, Step 1	3.06 x 5.75
EP Left Tab 1, Step 2	3.49 x 5.75
EP Left Tab 1, Step 3	3.93 x 5.75
EP Left Tab 1, Step 4	4.37 x 5.75
EP Left Tab 1, Step 5	4.81 x 5.75
EP Left Tab 1, Step 6	5.25 x 5.75
EP Left Tab 1, Step 7	5.68 x 5.75
EP Left Tab 1, Step 8	6.12 x 5.75
EP Left Tab 1, Step 9	6.56 x 5.75
EP Left Tab 1, Step 10	7.0 x 5.75
EP Left Tab 1, Step 11	7.43 x 5.75
EP Left Tab 1, Step 12	7.87 x 5.75
EP Left Tab 1, Step 13	8.31 x 5.75
EP Left Tab 1, Step 14	9 x 5.75
EP Left Tab 2, Step 1	3.93 x 5.75
EP Left Tab 2, Step 2	4.37 x 5.75
EP Left Tab 2, Step 3	4.81 x 5.75
EP Left Tab 2, Step 4	5.25 x 5.75
EP Left Tab 2, Step 5	5.68 x 5.75
EP Left Tab 2, Step 6	6.12 x 5.75
EP Left Tab 2, Step 7	6.56 x 5.75
EP Left Tab 2, Step 8	7.0 x 5.75
EP Left Tab 2, Step 9	7.43 x 5.75
EP Left Tab 2, Step 10	7.87 x 5.75
EP Left Tab 2, Step 11	8.31 x 5.75
EP Left Tab 2, Step 12	8.75 x 5.75
EP Left Tab 2, Step 12 Cont 1	9 x 5.75
EP Left Tab 2, Step 12 Cont 2	9 x 5.75
EP Left Tab 2, Step 12 Cont 3	9 x 5.75
EP Left Tab 2, Step 12 Cont 4	9 x 5.75
EP Left Tab 3, Step 1	4.37 x 5.75
EP Left Tab 3, Step 2	4.81 x 5.75
EP Left Tab 3, Step 3	5.25 x 5.75
EP Left Tab 3, Step 4	5.68 x 5.75
EP Left Tab 3, Step 5	6.12 x 5.75
EP Left Tab 3, Step 6	6.56 x 5.75
EP Left Tab 3, Step 7	7.0 x 5.75
EP Left Tab 3, Step 8	7.43 x 5.75
EP Left Tab 3, Step 9	7.87 x 5.75
EP Left Tab 3, Step 10	8.31 x 5.75
EP Left Tab 3, Step 11	8.75 x 5.75
EP Left Tab 3, Step 11 Cont 1	9 x 5.75
EP Left Tab 3, Step 11 Cont 2	9 x 5.75
EP Left Tab 3, Step 11 Cont 3	9 x 5.75
EP Left Tab 3, Step 11, Cont 4	9 x 5.75
EP Left Tab 4, Step 1	5.25 x 5.75
EP Left Tab 4, Step 2	5.68 x 5.75
EP Left Tab 4. Step 3	6.12 x 5.75
EP Left Tab 4, Step 4	6.56 x 5.75
EP Left Tab 4, Step 5	7.0 x 5.75
EP Left Tab 4, Step 6	7.43 x 5.75
EP Left Tab 4, Step 7	7.87 x 5.75
EP Left Tab 4, Step 8	8.31 x 5.75
EP Left Tab 4, Step 9	8.75 x 5.75
EP Left Tab 4, Step 9 Cont 1	9 x 5.75
EP Left Tab 4, Step 9 Cont 2	9 x 5.75
EP Left Tab 4, Step 9 Cont 3	9 x 5.75
EP Left Tab 4, Step 9 Cont 4	9 x 5.75
EP Left Tab 5, Step 1	5.68 x 5.75
EP Left Tab 5, Step 2	6.12 x 5.75
EP Left Tab 5, Step 3	6.56 x 5.75
EP Left Tab 5, Step 4	7.0 x 5.75
EP Left Tab 5, Step 5	7.43 x 5.75
EP Left Tab 5, Step 6	7.87 x 5.75
EP Left Tab 5, Step 7	8.31 x 5.75
EP Left Tab 5, Step 8	8.75 x 5.75
EP Left Tab 5, Step 8 Cont 1	9 x 5.75
EP Left Tab 5, Step 8 Cont 2	9 x 5.75
EP Left Tab 5, Step 8 Cont 3	9 x 5.75
EP Left Tab 5, Step 8 Cont 4	9 x 5.75
EP Left Tab 6, Step 1	6.56 x 5.75
EP Left Tab 6, Step 2	7.0 x 5.75
EP Left Tab 6, Step 3	7.43 x 5.75
EP Left Tab 6, Step 4	7.87 x 5.75
EP Left Tab 6, Step 5	8.31 x 5.75
EP Left Tab 6, Step 6	8.75 x 5.75
EP Left Tab 6, Step 6 Cont 1	9 x 5.75
EP Left Tab 6, Step 6 Cont 2	9 x 5.75
EP Left Tab 6, Step 6 Cont 3	9 x 5.75
EP Left Tab 6, Step 6 Cont 4	9 x 5.75
EP Left Tab 7, Step 1	7.0 x 5.75
EP Left Tab 7, Step 2	7.43 x 5.75
EP Left Tab 7, Step 3	7.87 x 5.75
EP Left Tab 7, Step 4	8.31 x 5.75
EP Left Tab 7, Step 5	8.75 x 5.75
EP Left Tab 7, Step 5 Cont 1	9 x 5.75
EP Left Tab 7, Step 5 Cont 2	9 x 5.75
EP Left Tab 7, Step 5 Cont 3	9 x 5.75
EP Left Tab 7, Step 5 Cont 4	9 x 5.75
EP Left Tab 8, Step 1	7.87 x 5.75
EP Left Tab 8, Step 2	8.31 x 5.75
EP Left Tab 8, Step 3	8.75 x 5.75
EP Left Tab 8, Step 3 Cont 1	9 x 5.75
EP Left Tab 8, Step 3 Cont 2	9 x 5.75
EP Left Tab 8, Step 3 Cont 3	9 x 5.75
EP Left Tab 8, Step 3 Cont 4	9 x 5.75
EP Left Tab 9, Step 1	8.31 x 5.75
EP Left Tab 9, Step 2	8.75 x 5.75
EP Left Tab 9, Step 2 Cont 1	9 x 5.75
EP Left Tab 9, Step 2 Cont 2	9 x 5.75
EP Left Tab 9, Step 2 Cont 3	9 x 5.75
EP Left Tab 9, Step 2 Cont 4	9 x 5.75
EP Left Tab 10, Step 1	8.75 x 5.75
EP Left Tab 10, Step 1 Cont 1	9 x 5.75
EP Left Tab 10, Step 1 Cont 2	9 x 5.75
EP Left Tab 10, Step 1 Cont 3	9 x 5.75
EP Left Tab 10, Step 2 Cont 4	9 x 5.75

All left/Bottom tab pages sizes are 9 x 5.75

FIGURE 117. PCL tab/step page sizes.

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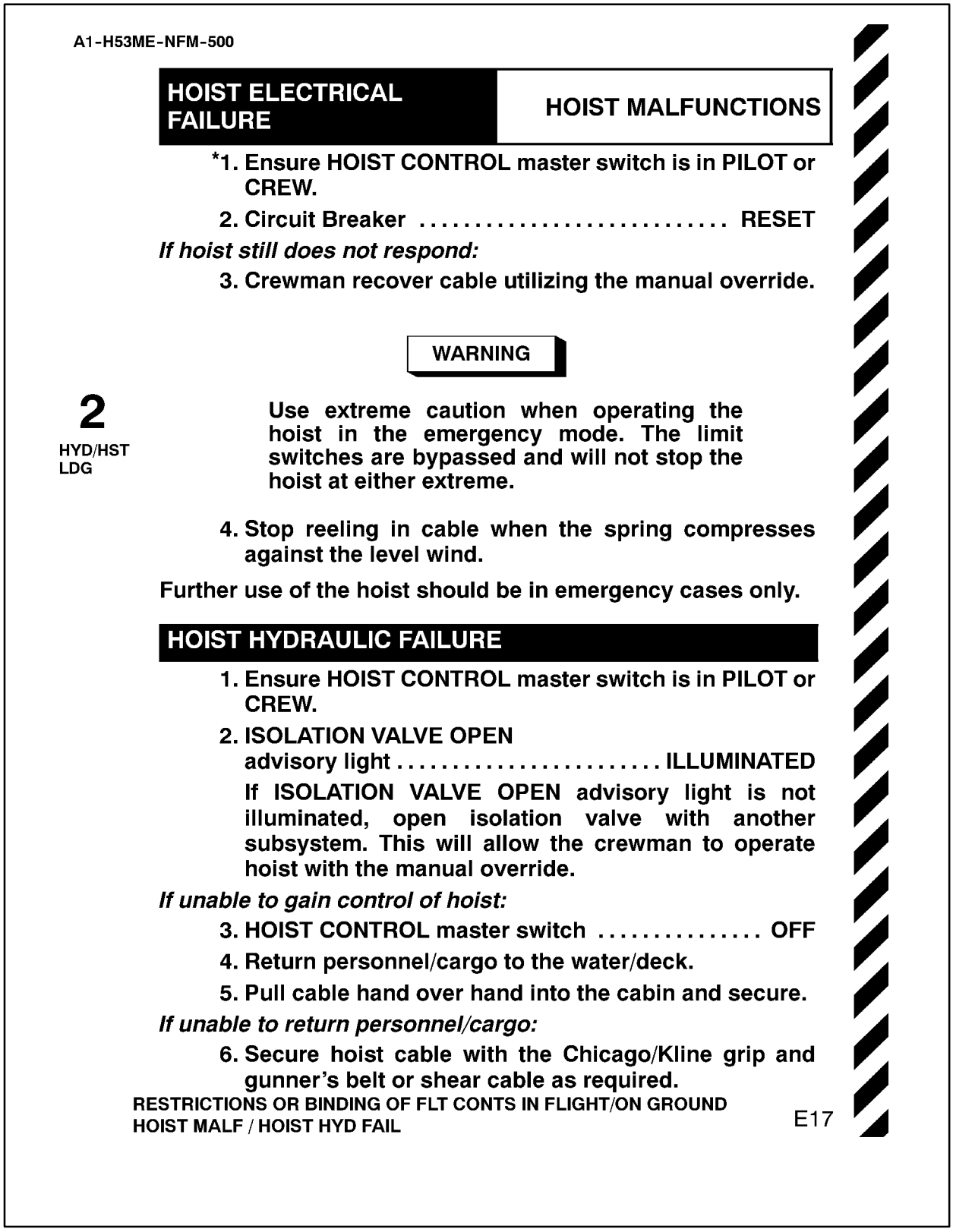


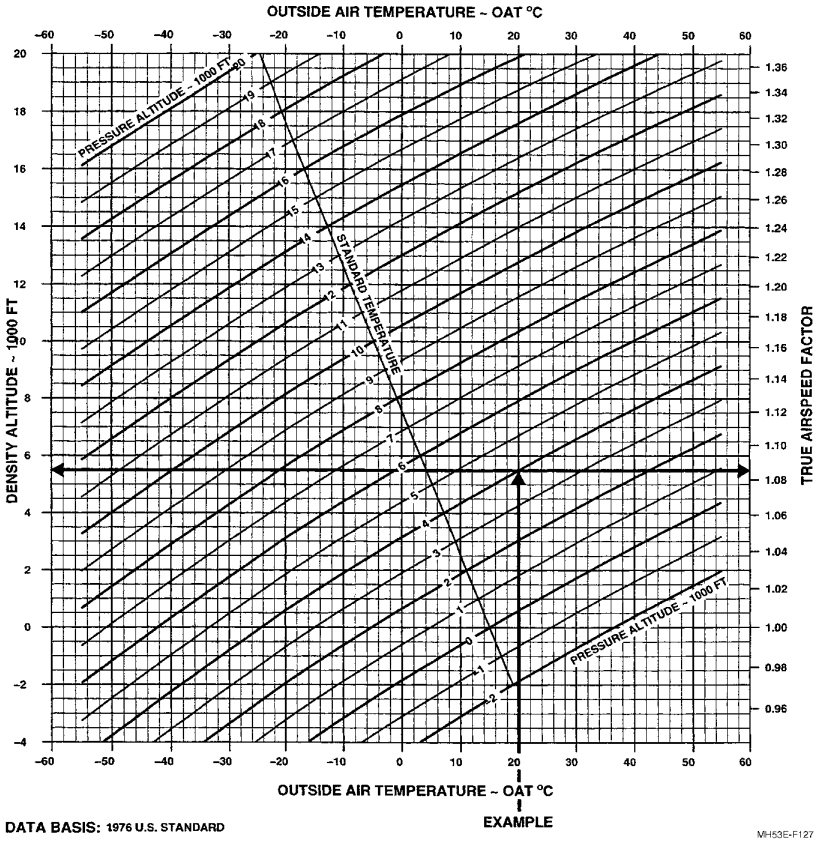
FIGURE 118. Typical pocket checklist emergency procedures page.

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A1-H53ME-NFM-500

DENSITY ALTITUDE CHART

DENS ALT



67

FIGURE 119. Typical pocket checklist reference data page.

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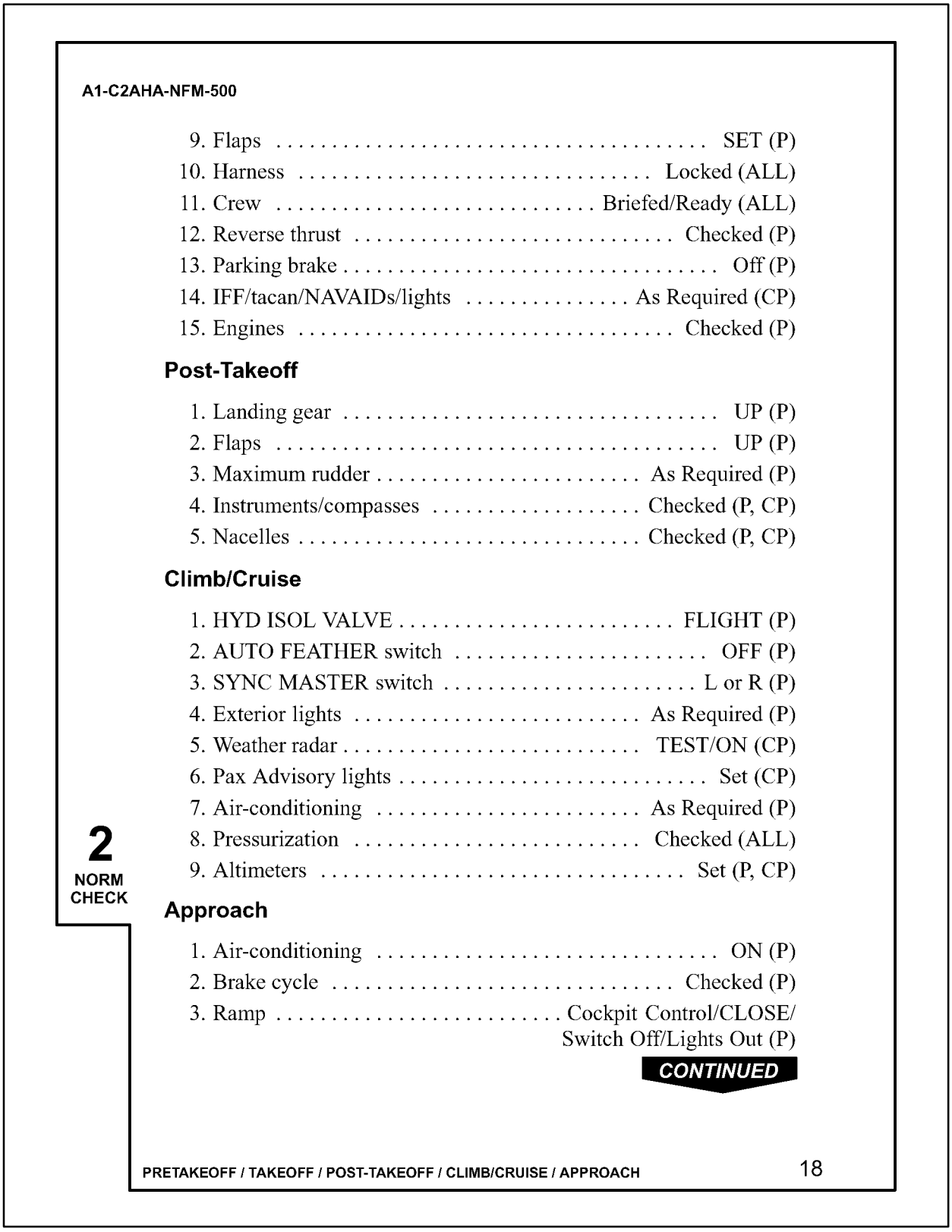


FIGURE 120. Typical pocket checklist normal procedures page.

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A1-C2AHA-NFM-500

TANK	NO.	APPROXIMATE USABLE FUEL IN LEVEL FLIGHT	
		GALLONS	POUNDS
LEFT WING TANK	1	912	6,200
RIGHT WING TANK	1	912	6,200
TOTAL	—	1,824	12,400

NOTE

WEIGHTS ARE BASED ON 6.8 POUNDS PER GALLON OF JP-5 FUEL FOR STANDARD DAY CONDITIONS.

FUEL QUANTITY DATA

FUEL	MIL-T-56724 (JP-5, JP-8, OR JP-4)
HYDRAULIC FLUID	*MIL-PRF-83282
LIQUID OXYGEN	MIL-O-27210
ENGINE OIL	MIL-L-23669 NATO NO. 0-156
PROPELLER OIL	MIL-PRF-83282 HYDRAULIC FLUID
EXTINGUISHING AGENT	MIL-B-12218

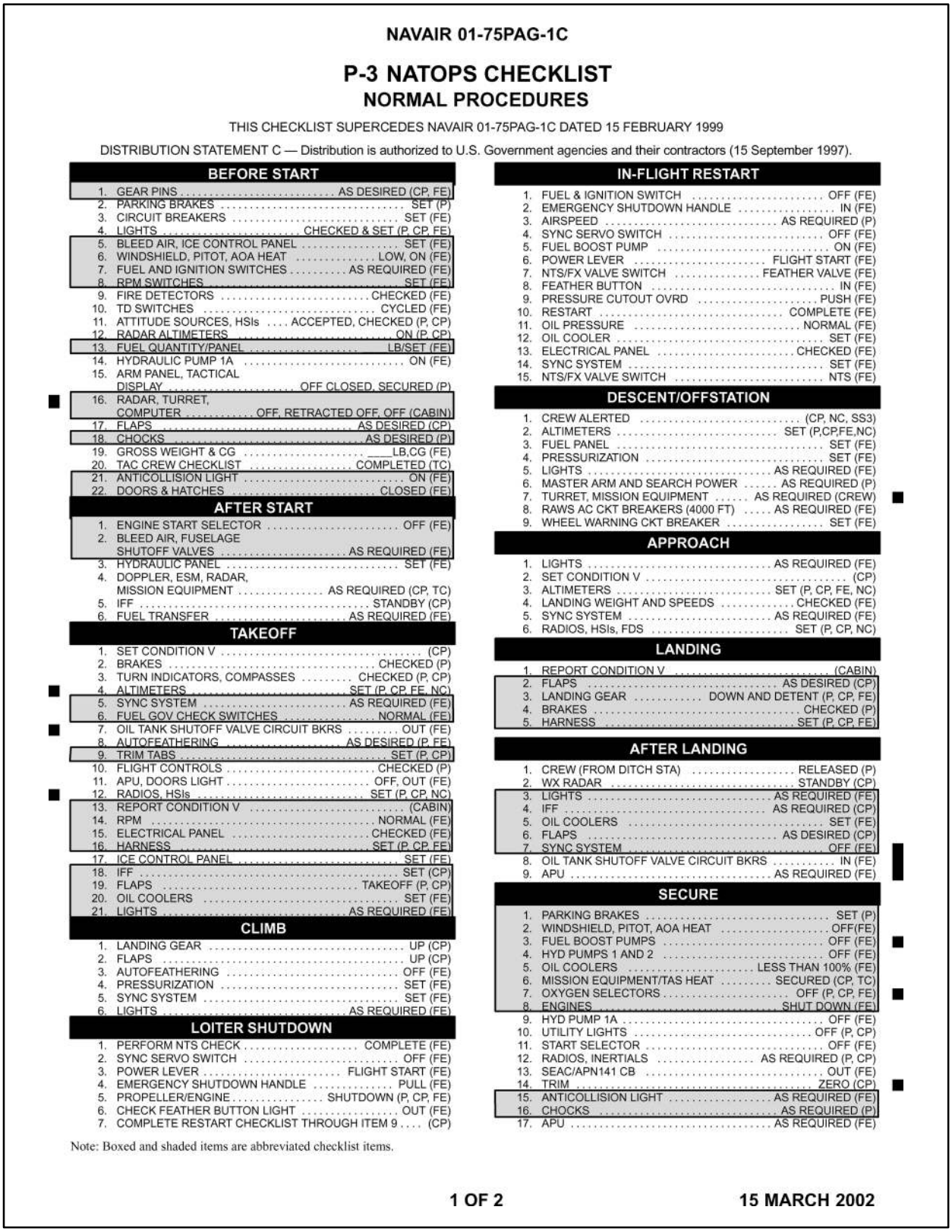
* IN AN EMERGENCY, MIL-H-5606B (NATO NO. H-515) MAY BE USED FOR SERVICING. MAINTENANCE SHALL BE NOTIFIED OF QUANTITY ADDED TO EACH SYSTEM WHEN AIRCRAFT RETURNS TO BASE.

AIRCRAFT SERVICING DATA

CONTINUED

FIGURE 120. Typical pocket checklist normal procedures page — Continued.

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NAVAIR 01-75PAG-1C

P-3 NATOPS CHECKLIST
NORMAL PROCEDURES

THIS CHECKLIST SUPERCEDES NAVAIR 01-75PAG-1C DATED 15 FEBRUARY 1999

DISTRIBUTION STATEMENT C — Distribution is authorized to U.S. Government agencies and their contractors (15 September 1997).

BEFORE START

1.	GEAR PINS	AS DESIRED (CP, FE)
2.	PARKING BRAKES	SET (P)
3.	CIRCUIT BREAKERS	SET (FE)
4.	LIGHTS	CHECKED & SET (P, CP, FE)
5.	BLEED AIR, ICE CONTROL PANEL	SET (FE)
6.	WINDSHIELD, PITOT, AOA HEAT	LOW, ON (FE)
7.	FUEL AND IGNITION SWITCHES	AS REQUIRED (FE)
8.	RPM SWITCHES	SET (FE)
9.	FIRE DETECTORS	CHECKED (FE)
10.	TD SWITCHES	CYCLED (FE)
11.	ATTITUDE SOURCES, HSIs	ACCEPTED, CHECKED (P, CP)
12.	RADAR ALTIMETERS	ON (P, CP)
13.	FUEL QUANTITY/PANEL	LB/SET (FE)
14.	HYDRAULIC PUMP 1A	ON (FE)
15.	ARM PANEL, TACTICAL DISPLAY	OFF CLOSED, SECURED (P)
16.	RADAR, TURRET, COMPUTER	OFF, RETRACTED OFF, OFF (CABIN)
17.	FLAPS	AS DESIRED (CP)
18.	CHOCKS	AS DESIRED (P)
19.	GROSS WEIGHT & CG	____LB,CG (FE)
20.	TAC CREW CHECKLIST	COMPLETED (TC)
21.	ANTICOLLISION LIGHT	ON (FE)
22.	DOORS & HATCHES	CLOSED (FE)

AFTER START

1.	ENGINE START SELECTOR	OFF (FE)
2.	BLEED AIR, FUSELAGE SHUTOFF VALVES	AS REQUIRED (FE)
3.	HYDRAULIC PANEL	SET (FE)
4.	DOPPLER, ESM, RADAR, MISSION EQUIPMENT	AS REQUIRED (CP, TC)
5.	IFF	STANDBY (CP)
6.	FUEL TRANSFER	AS REQUIRED (FE)

TAKEOFF

1.	SET CONDITION V	(CP)
2.	BRAKES	CHECKED (P)
3.	TURN INDICATORS, COMPASSES	CHECKED (P, CP)
4.	ALTIMETERS	SET (P, CP, FE, NC)
5.	SYNC SYSTEM	AS REQUIRED (FE)
6.	FUEL GOV CHECK SWITCHES	NORMAL (FE)
7.	OIL TANK SHUTOFF VALVE CIRCUIT BKRS	OUT (FE)
8.	AUTOFEATHERING	AS DESIRED (P, FE)
9.	TRIM TABS	SET (P, CP)
10.	FLIGHT CONTROLS	CHECKED (P)
11.	APU, DOORS LIGHT	OFF, OUT (FE)
12.	RADIOS, HSIs	SET (P, CP, NC)
13.	REPORT CONDITION V	(CABIN)
14.	RPM	NORMAL (FE)
15.	ELECTRICAL PANEL	CHECKED (FE)
16.	HARNESS	SET (P, CP, FE)
17.	ICE CONTROL PANEL	SET (FE)
18.	IFF	SET (CP)
19.	FLAPS	TAKEOFF (P, CP)
20.	OIL COOLERS	SET (FE)
21.	LIGHTS	AS REQUIRED (FE)

CLIMB

1.	LANDING GEAR	UP (CP)
2.	FLAPS	UP (CP)
3.	AUTOFEATHERING	OFF (FE)
4.	PRESSURIZATION	SET (FE)
5.	SYNC SYSTEM	SET (FE)
6.	LIGHTS	AS REQUIRED (FE)

LOITER SHUTDOWN

1.	PERFORM NTS CHECK	COMPLETE (FE)
2.	SYNC SERVO SWITCH	OFF (FE)
3.	POWER LEVER	FLIGHT START (FE)
4.	EMERGENCY SHUTDOWN HANDLE	PULL (FE)
5.	PROPELLER/ENGINE	SHUTDOWN (P, CP, FE)
6.	CHECK FEATHER BUTTON LIGHT	OUT (FE)
7.	COMPLETE RESTART CHECKLIST THROUGH ITEM 9	(CP)

Note: Boxed and shaded items are abbreviated checklist items.

IN-FLIGHT RESTART

1.	FUEL & IGNITION SWITCH	OFF (FE)
2.	EMERGENCY SHUTDOWN HANDLE	IN (FE)
3.	AIRSPEED	AS REQUIRED (P)
4.	SYNC SERVO SWITCH	OFF (FE)
5.	FUEL BOOST PUMP	ON (FE)
6.	POWER LEVER	FLIGHT START (FE)
7.	NTS/FX VALVE SWITCH	FEATHER VALVE (FE)
8.	FEATHER BUTTON	IN (FE)
9.	PRESSURE CUTOUT OVRD	PUSH (FE)
10.	RESTART	COMPLETE (FE)
11.	OIL PRESSURE	NORMAL (FE)
12.	OIL COOLER	SET (FE)
13.	ELECTRICAL PANEL	CHECKED (FE)
14.	SYNC SYSTEM	SET (FE)
15.	NTS/FX VALVE SWITCH	NTS (FE)

DESCENT/OFFSTATION

1.	CREW ALERTED	(CP, NC, SS3)
2.	ALTIMETERS	SET (P,CP,FE,NC)
3.	FUEL PANEL	SET (FE)
4.	PRESSURIZATION	SET (FE)
5.	LIGHTS	AS REQUIRED (FE)
6.	MASTER ARM AND SEARCH POWER	AS REQUIRED (P)
7.	TURRET, MISSION EQUIPMENT	AS REQUIRED (CREW)
8.	RAWS AC CKT BREAKERS (4000 FT)	AS REQUIRED (FE)
9.	WHEEL WARNING CKT BREAKER	SET (FE)

APPROACH

1.	LIGHTS	AS REQUIRED (FE)
2.	SET CONDITION V	(CP)
3.	ALTIMETERS	SET (P, CP, FE, NC)
4.	LANDING WEIGHT AND SPEEDS	CHECKED (FE)
5.	SYNC SYSTEM	AS REQUIRED (FE)
6.	RADIOS, HSIs, FDS	SET (P, CP, NC)

LANDING

1.	REPORT CONDITION V	(CABIN)
2.	FLAPS	AS DESIRED (CP)
3.	LANDING GEAR	DOWN AND DETENT (P, CP, FE)
4.	BRAKES	CHECKED (P)
5.	HARNESS	SET (P, CP, FE)

AFTER LANDING

1.	CREW (FROM DITCH STA)	RELEASED (P)
2.	WX RADAR	STANDBY (CP)
3.	LIGHTS	AS REQUIRED (FE)
4.	IFF	AS REQUIRED (CP)
5.	OIL COOLERS	SET (FE)
6.	FLAPS	AS DESIRED (CP)
7.	SYNC SYSTEM	OFF (FE)
8.	OIL TANK SHUTOFF VALVE CIRCUIT BKRS	IN (FE)
9.	APU	AS REQUIRED (FE)

SECURE

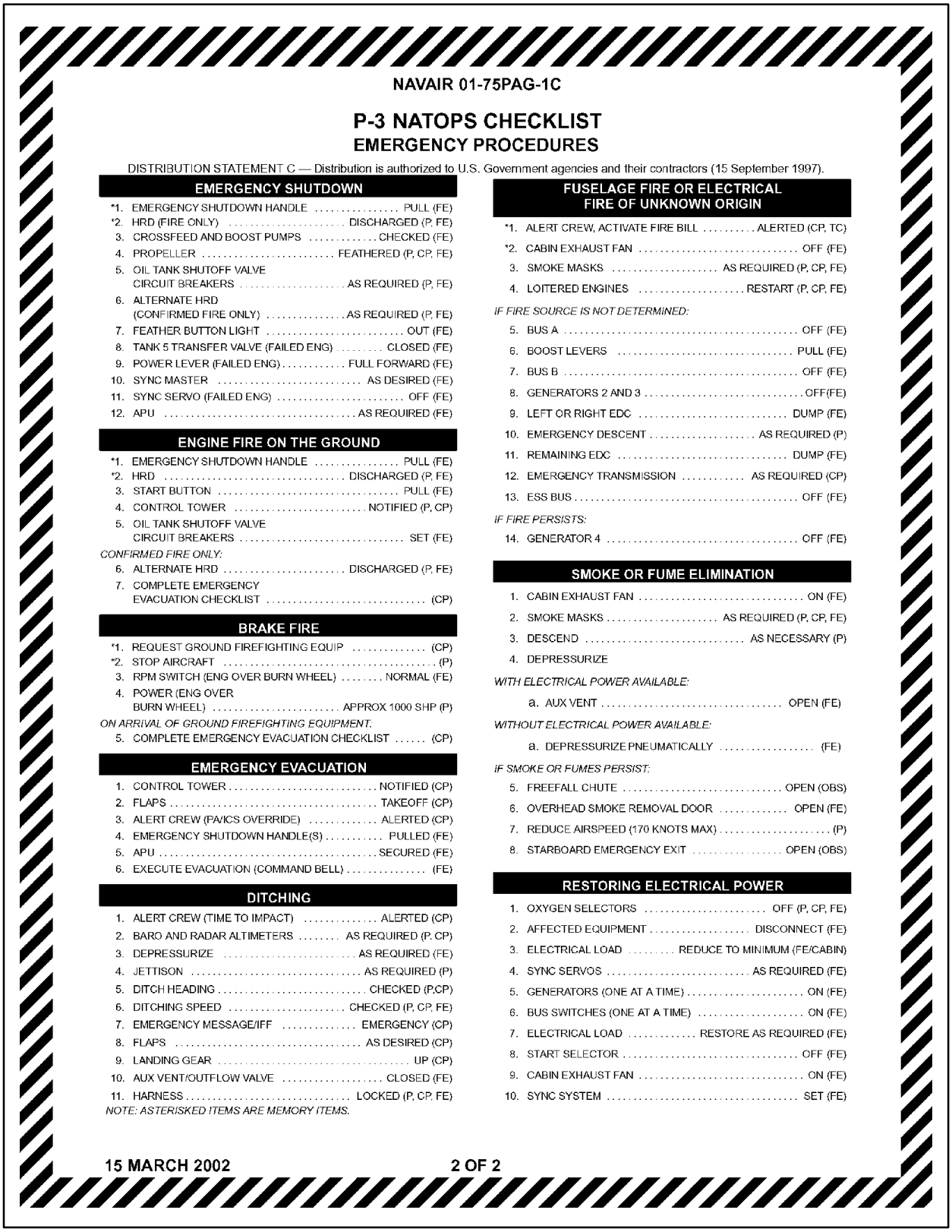
1.	PARKING BRAKES	SET (P)
2.	WINDSHIELD, PITOT, AOA HEAT	OFF(FE)
3.	FUEL BOOST PUMPS	OFF (FE)
4.	HYD PUMPS 1 AND 2	OFF (FE)
5.	OIL COOLERS	LESS THAN 100% (FE)
6.	MISSION EQUIPMENT/TAS HEAT	SECURED (CP, TC)
7.	OXYGEN SELECTORS	OFF (P, CP, FE)
8.	ENGINES	SHUT DOWN (FE)
9.	HYD PUMP 1A	OFF (FE)
10.	UTILITY LIGHTS	OFF (P, CP)
11.	START SELECTOR	OFF (FE)
12.	RADIOS, INERTIALS	AS REQUIRED (P, CP)
13.	SEAC/APN141 CB	OUT (FE)
14.	TRIM	ZERO (CP)
15.	ANTICOLLISION LIGHT	AS REQUIRED (FE)
16.	CHOCKS	AS REQUIRED (P)
17.	APU	AS REQUIRED (FE)

1 OF 2

15 MARCH 2002

FIGURE 121. Sample card checklist.

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NAVAIR 01-75PAG-1C

P-3 NATOPS CHECKLIST
EMERGENCY PROCEDURES

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EMERGENCY SHUTDOWN

*1. EMERGENCY SHUTDOWN HANDLE ..... PULL (FE)
*2. HRD (FIRE ONLY) ..... DISCHARGED (P, FE)
3. CROSSFEED AND BOOST PUMPS ..... CHECKED (FE)
4. PROPELLER ..... FEATHERED (P, CP, FE)
5. OIL TANK SHUTOFF VALVE CIRCUIT BREAKERS ..... AS REQUIRED (P, FE)
6. ALTERNATE HRD (CONFIRMED FIRE ONLY) ..... AS REQUIRED (P, FE)
7. FEATHER BUTTON LIGHT ..... OUT (FE)
8. TANK 5 TRANSFER VALVE (FAILED ENG) ..... CLOSED (FE)
9. POWER LEVER (FAILED ENG) ..... FULL FORWARD (FE)
10. SYNC MASTER ..... AS DESIRED (FE)
11. SYNC SERVO (FAILED ENG) ..... OFF (FE)
12. APU ..... AS REQUIRED (FE)

ENGINE FIRE ON THE GROUND

*1. EMERGENCY SHUTDOWN HANDLE ..... PULL (FE)
*2. HRD ..... DISCHARGED (P, FE)
3. START BUTTON ..... PULL (FE)
4. CONTROL TOWER ..... NOTIFIED (P, CP)
5. OIL TANK SHUTOFF VALVE CIRCUIT BREAKERS ..... SET (FE)

CONFIRMED FIRE ONLY:

6. ALTERNATE HRD ..... DISCHARGED (P, FE)
7. COMPLETE EMERGENCY EVACUATION CHECKLIST ..... (CP)

BRAKE FIRE

*1. REQUEST GROUND FIREFIGHTING EQUIP ..... (CP)
*2. STOP AIRCRAFT ..... (P)
3. RPM SWITCH (ENG OVER BURN WHEEL) ..... NORMAL (FE)
4. POWER (ENG OVER BURN WHEEL) ..... APPROX 1000 SHP (P)

ON ARRIVAL OF GROUND FIREFIGHTING EQUIPMENT:

5. COMPLETE EMERGENCY EVACUATION CHECKLIST ..... (CP)

EMERGENCY EVACUATION

1. CONTROL TOWER ..... NOTIFIED (CP)
2. FLAPS ..... TAKEOFF (CP)
3. ALERT CREW (PA/ICS OVERRIDE) ..... ALERTED (CP)
4. EMERGENCY SHUTDOWN HANDLE(S) ..... PULLED (FE)
5. APU ..... SECURED (FE)
6. EXECUTE EVACUATION (COMMAND BELL) ..... (FE)

DITCHING

1. ALERT CREW (TIME TO IMPACT) ..... ALERTED (CP)
2. BARO AND RADAR ALTIMETERS ..... AS REQUIRED (P, CP)
3. DEPRESSURIZE ..... AS REQUIRED (FE)
4. JETTISON ..... AS REQUIRED (P)
5. DITCH HEADING ..... CHECKED (P,CP)
6. DITCHING SPEED ..... CHECKED (P, CP, FE)
7. EMERGENCY MESSAGE/IFF ..... EMERGENCY (CP)
8. FLAPS ..... AS DESIRED (CP)
9. LANDING GEAR ..... UP (CP)
10. AUX VENT/OUTFLOW VALVE ..... CLOSED (FE)
11. HARNESS ..... LOCKED (P, CP, FE)

NOTE: ASTERISKED ITEMS ARE MEMORY ITEMS.

FUSELAGE FIRE OR ELECTRICAL FIRE OF UNKNOWN ORIGIN

*1. ALERT CREW, ACTIVATE FIRE BILL ..... ALERTED (CP, TC)
*2. CABIN EXHAUST FAN ..... OFF (FE)
3. SMOKE MASKS ..... AS REQUIRED (P, CP, FE)
4. LOITERED ENGINES ..... RESTART (P, CP, FE)

IF FIRE SOURCE IS NOT DETERMINED:

5. BUS A ..... OFF (FE)
6. BOOST LEVERS ..... PULL (FE)
7. BUS B ..... OFF (FE)
8. GENERATORS 2 AND 3 ..... OFF(FE)
9. LEFT OR RIGHT EDC ..... DUMP (FE)
10. EMERGENCY DESCENT ..... AS REQUIRED (P)
11. REMAINING EDC ..... DUMP (FE)
12. EMERGENCY TRANSMISSION ..... AS REQUIRED (CP)
13. ESS BUS ..... OFF (FE)

IF FIRE PERSISTS:

14. GENERATOR 4 ..... OFF (FE)

SMOKE OR FUME ELIMINATION

1. CABIN EXHAUST FAN ..... ON (FE)
2. SMOKE MASKS ..... AS REQUIRED (P, CP, FE)
3. DESCEND ..... AS NECESSARY (P)
4. DEPRESSURIZE

WITH ELECTRICAL POWER AVAILABLE:

a. AUX VENT ..... OPEN (FE)

WITHOUT ELECTRICAL POWER AVAILABLE:

a. DEPRESSURIZE PNEUMATICALLY ..... (FE)

IF SMOKE OR FUMES PERSIST:

5. FREEFALL CHUTE ..... OPEN (OBS)
6. OVERHEAD SMOKE REMOVAL DOOR ..... OPEN (FE)
7. REDUCE AIRSPEED (170 KNOTS MAX) ..... (P)
8. STARBOARD EMERGENCY EXIT ..... OPEN (OBS)

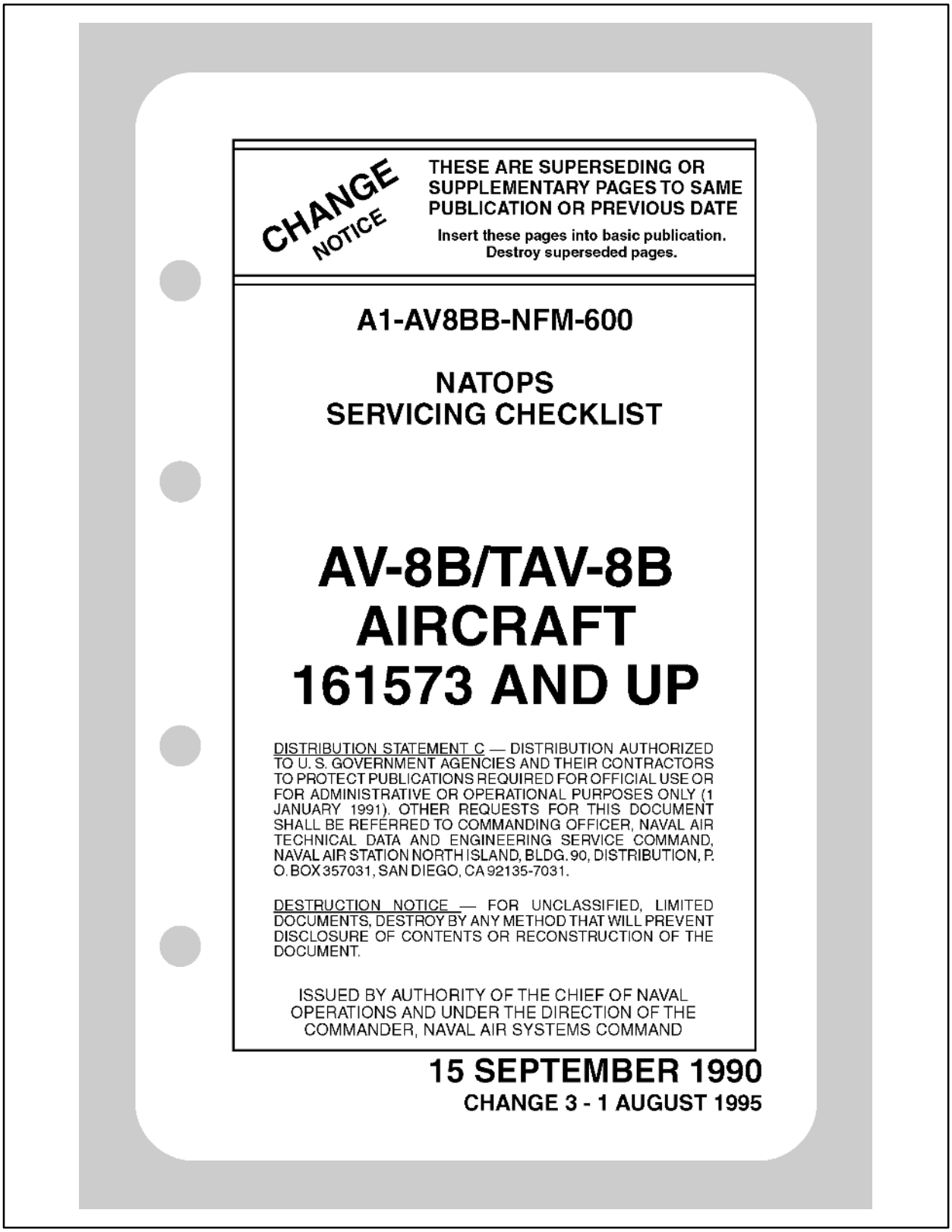
RESTORING ELECTRICAL POWER

1. OXYGEN SELECTORS ..... OFF (P, CP, FE)
2. AFFECTED EQUIPMENT ..... DISCONNECT (FE)
3. ELECTRICAL LOAD ..... REDUCE TO MINIMUM (FE/CABIN)
4. SYNC SERVOS ..... AS REQUIRED (FE)
5. GENERATORS (ONE AT A TIME) ..... ON (FE)
6. BUS SWITCHES (ONE AT A TIME) ..... ON (FE)
7. ELECTRICAL LOAD ..... RESTORE AS REQUIRED (FE)
8. START SELECTOR ..... OFF (FE)
9. CABIN EXHAUST FAN ..... ON (FE)
10. SYNC SYSTEM ..... SET (FE)

15 MARCH 2002 2 OF 2

FIGURE 121. Sample card checklist — Continued.

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A1-AV8BB-NFM-600

NATOPS
SERVICING CHECKLIST

AV-8B/TAV-8B
AIRCRAFT
161573 AND UP

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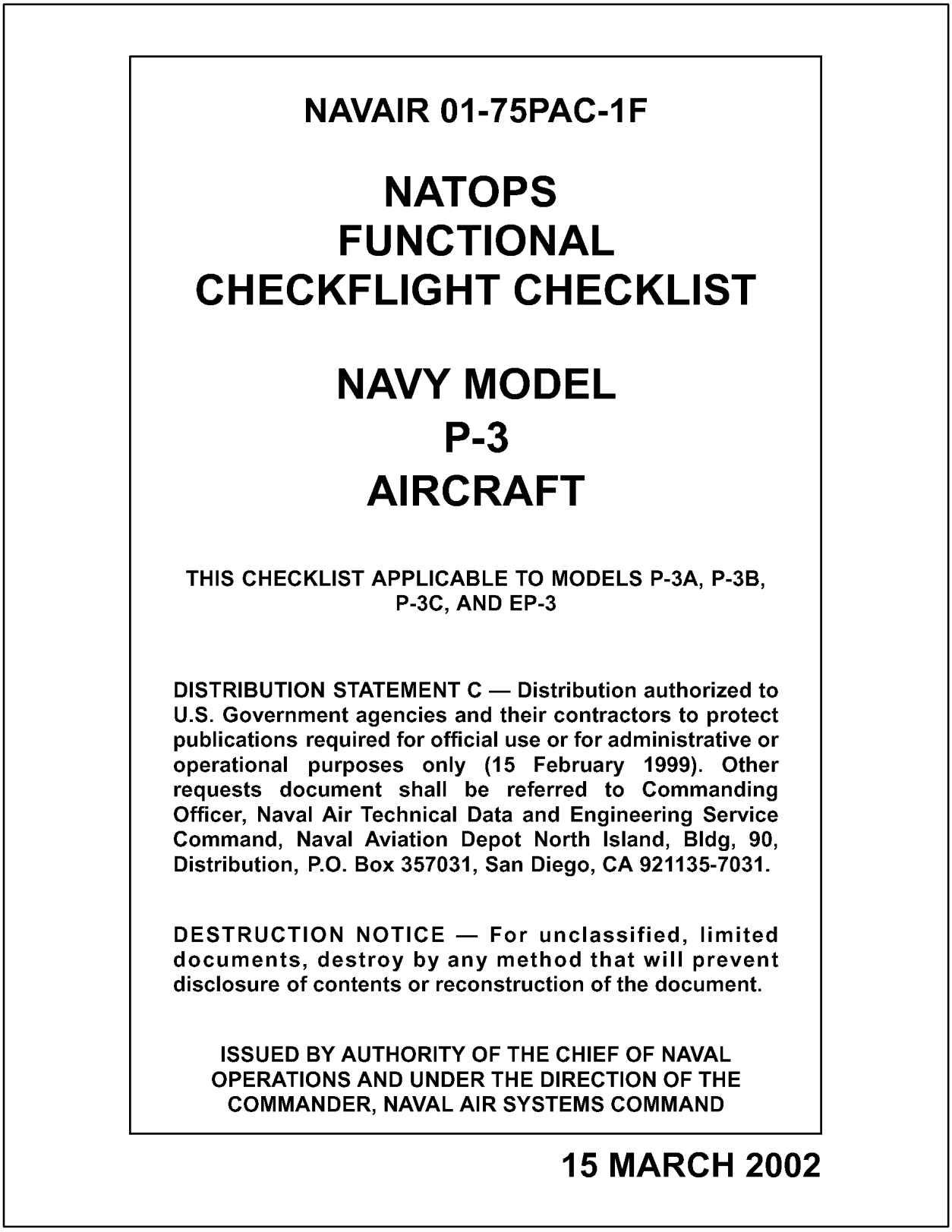
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15 SEPTEMBER 1990
CHANGE 3 - 1 AUGUST 1995

FIGURE 122. Servicing checklist (SCL) title page.

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NAVAIR 01-75PAC-1F

NATOPS FUNCTIONAL CHECKFLIGHT CHECKLIST

NAVY MODEL P-3 AIRCRAFT

THIS CHECKLIST APPLICABLE TO MODELS P-3A, P-3B, P-3C, AND EP-3

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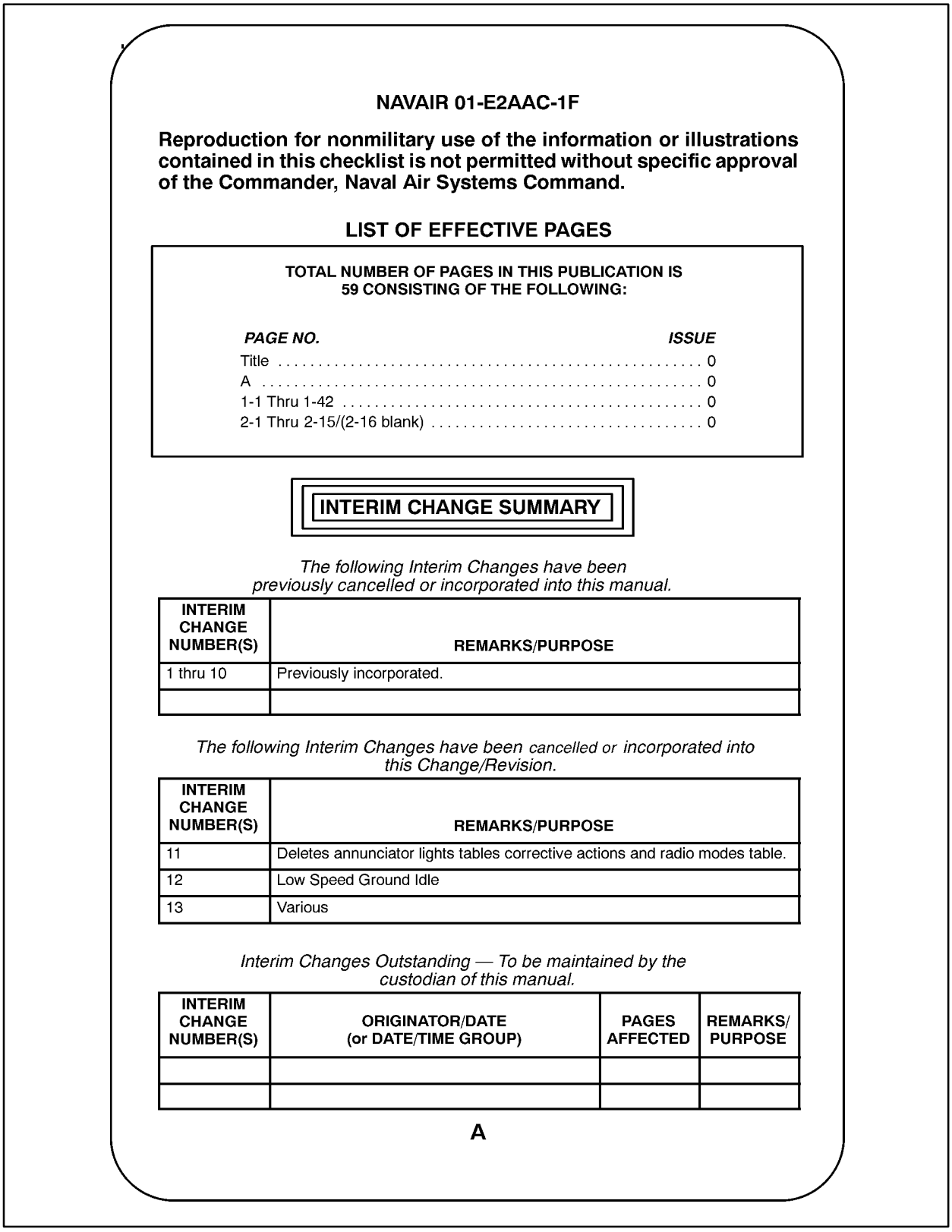
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15 MARCH 2002

FIGURE 123. Functional checkflight checklist front cover and title page.

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NAVAIR 01-E2AAC-1F

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LIST OF EFFECTIVE PAGES

TOTAL NUMBER OF PAGES IN THIS PUBLICATION IS 59 CONSISTING OF THE FOLLOWING:

PAGE NO.	*ISSUE*
Title	0
A	0
1-1 Thru 1-42	0
2-1 Thru 2-15/(2-16 blank)	0

INTERIM CHANGE SUMMARY

The following Interim Changes have been previously cancelled or incorporated into this manual.

INTERIM CHANGE NUMBER(S)	REMARKS/PURPOSE
1 thru 10	Previously incorporated.

The following Interim Changes have been cancelled or incorporated into this Change/Revision.

INTERIM CHANGE NUMBER(S)	REMARKS/PURPOSE
11	Deletes annunciator lights tables corrective actions and radio modes table.
12	Low Speed Ground Idle
13	Various

Interim Changes Outstanding — To be maintained by the custodian of this manual.

INTERIM CHANGE NUMBER(S)	ORIGINATOR/DATE (or DATE/TIME GROUP)	PAGES AFFECTED	REMARKS/ PURPOSE

A

FIGURE 124. Functional checkflight checklist “A” page (combined LEP and interim change summary).

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NAVAIR 01-E2AAD-1F

PRELIMINARY NATOPS
FUNCTIONAL CHECKFLIGHT CHECKLIST

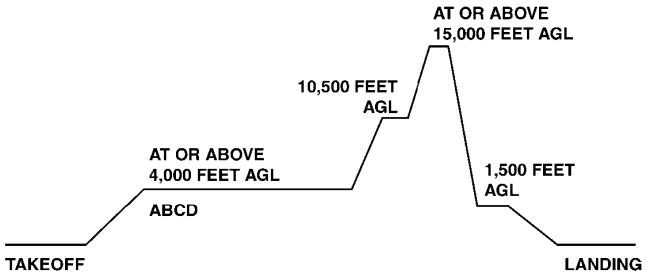
NAVY MODEL
E-2C PLUS NP2000
PART 1 — PILOTS

BUNO _____

Check Aircrew Signature(s) Date of Flight

_____

Record time, temperature or pressure where indicated by a line (_____). Where a box ☐ is shown, use a ✓ for satisfactory or an X for unsatisfactory.



FLIGHT PROFILES

Perform applicable flight profile and associated checks in accordance with the following checkflight conditions.

A. At completion of aircraft rework. When an aircraft has not flown in more than 30 days and is returned to flight status. Minimum checks required are prefixed by letter A.

B. After installation of engine, propeller, engine fuel control, or any components that cannot be checked in ground operation. Minimum checks required are prefixed by letter B.

C. When fixed or movable flight surfaces, or flight control system components have been installed, reinstalled, adjusted or rerigged, and improper adjustment or replacement of such components could cause an unsafe operating condition. Minimum checks required are prefixed by letter C.

D. After installation of ground frames, plenums, or major avionics repair that cannot be checked in ground operation. Minimum checks required are prefixed by letter D.

1-1

FIGURE 125. First page of functional checkflight checklist with profile.

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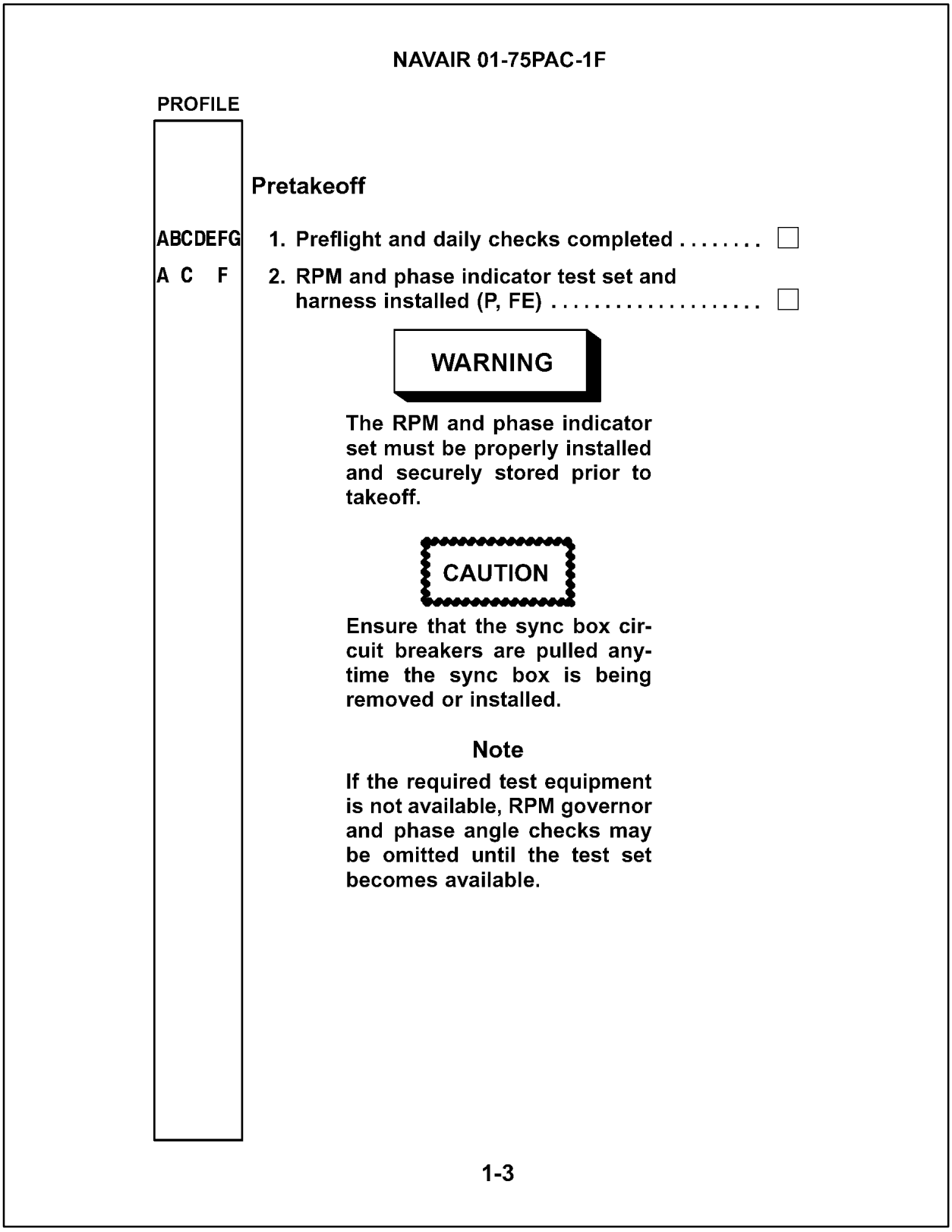


FIGURE 126. Typical functional checkflight checklist page.

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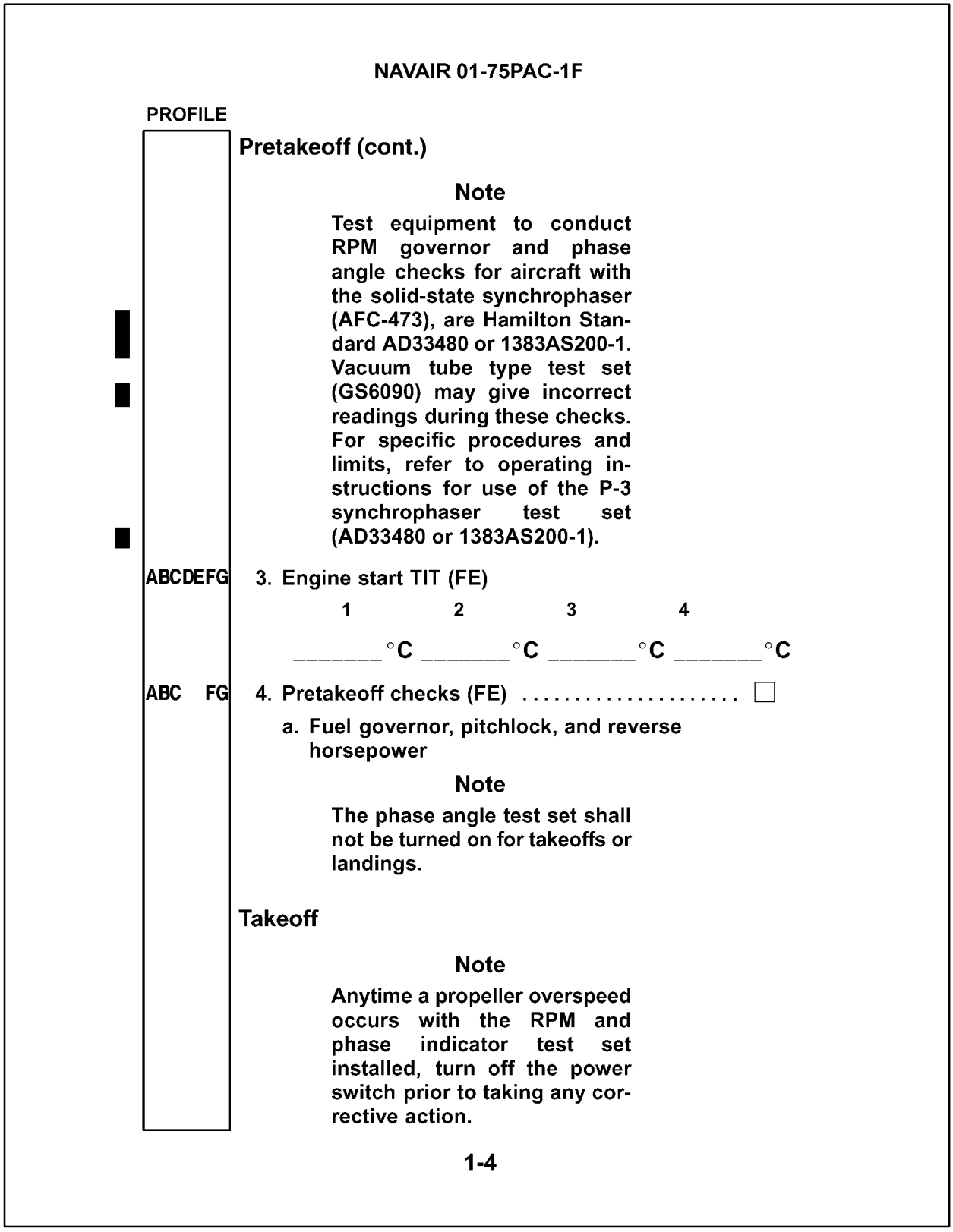


FIGURE 126. Typical functional checkflight checklist page — Continued.

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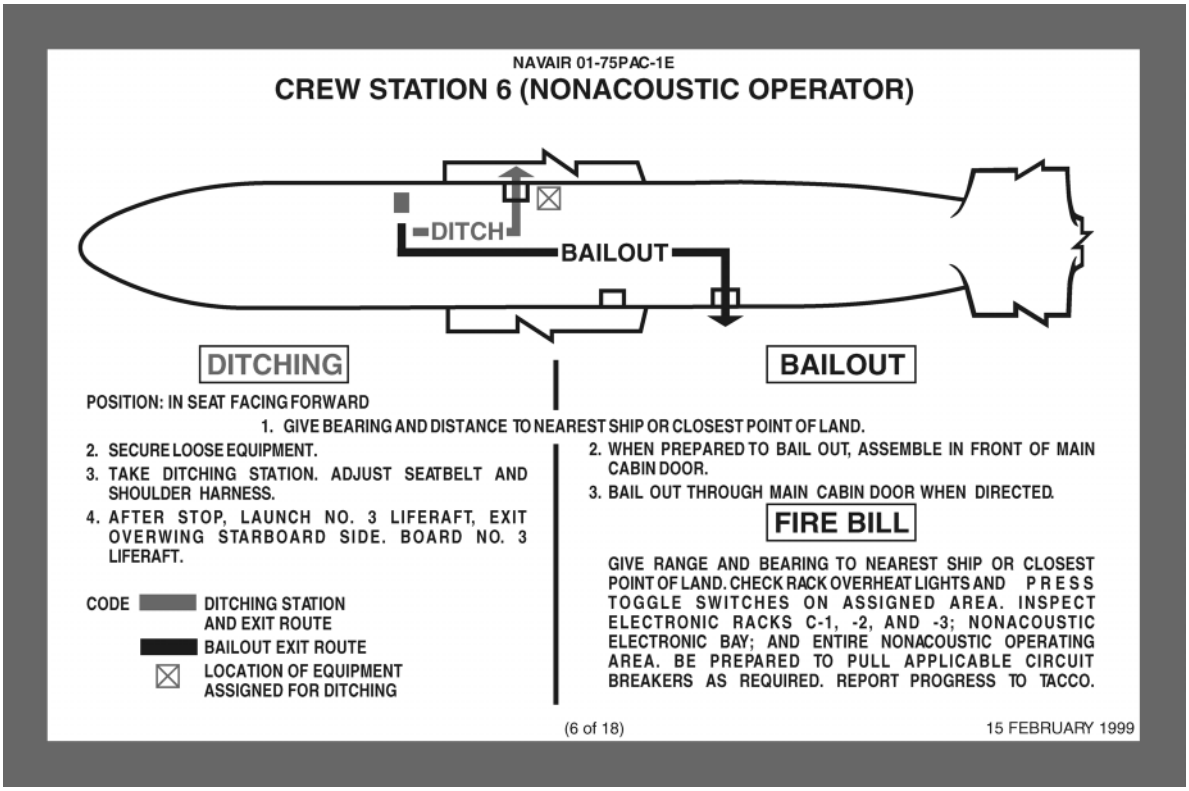
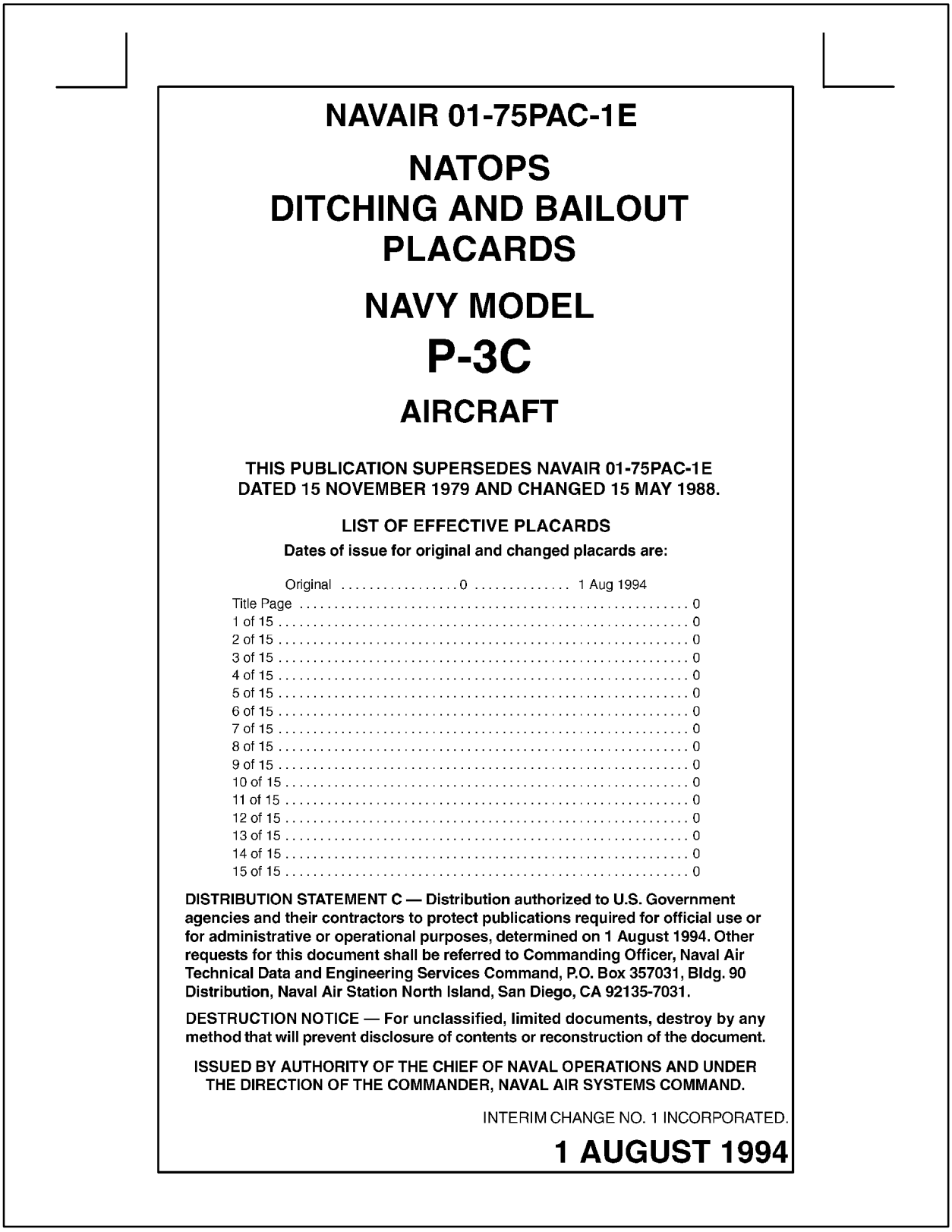


FIGURE 127. Sample ditching and bailout placard.

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NAVAIR 01-75PAC-1E

NATOPS DITCHING AND BAILOUT PLACARDS

NAVY MODEL

P-3C

AIRCRAFT

THIS PUBLICATION SUPERSEDES NAVAIR 01-75PAC-1E
DATED 15 NOVEMBER 1979 AND CHANGED 15 MAY 1988.

LIST OF EFFECTIVE PLACARDS

Dates of issue for original and changed placards are:

Original	0	1 Aug 1994
Title Page		0
1 of 15		0
2 of 15		0
3 of 15		0
4 of 15		0
5 of 15		0
6 of 15		0
7 of 15		0
8 of 15		0
9 of 15		0
10 of 15		0
11 of 15		0
12 of 15		0
13 of 15		0
14 of 15		0
15 of 15		0

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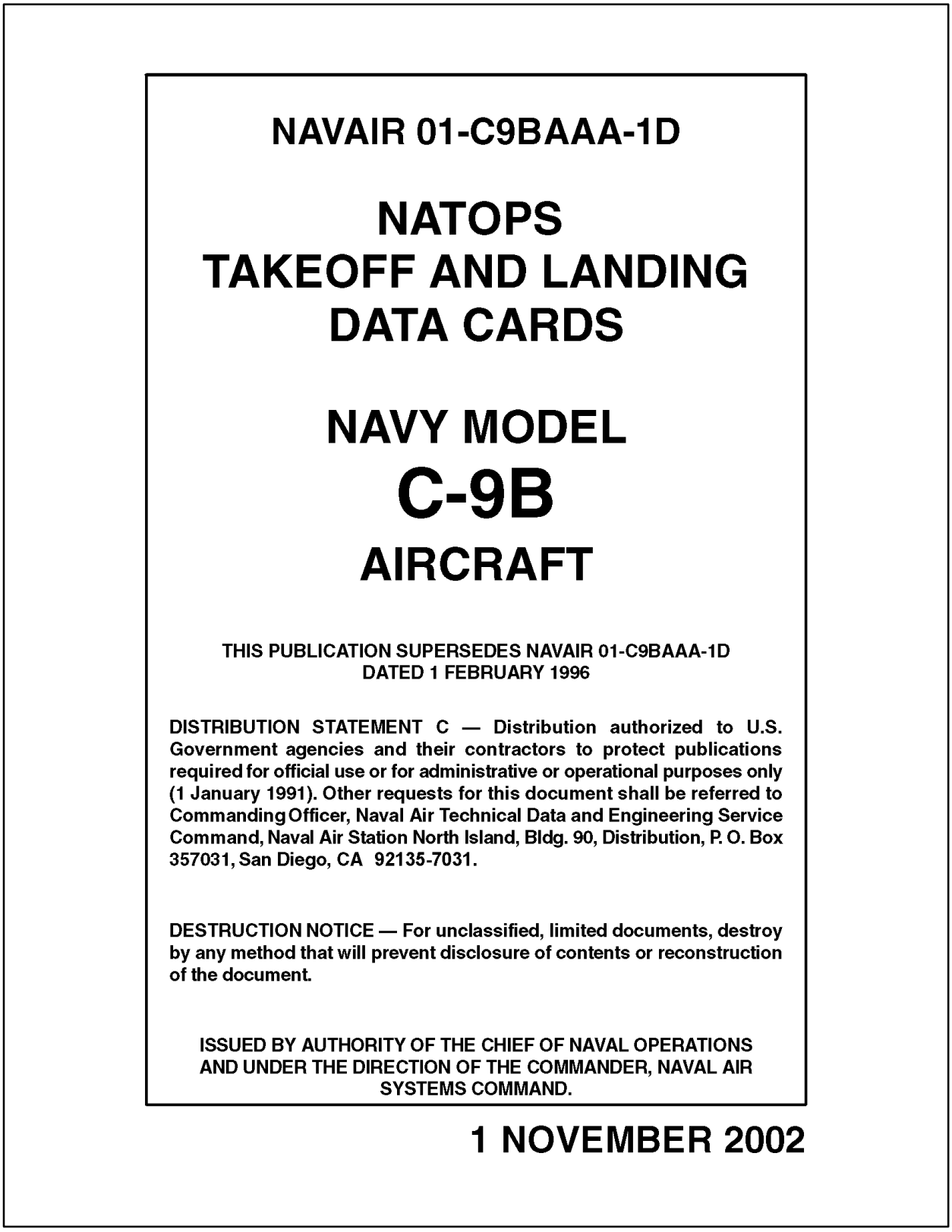
ISSUED BY AUTHORITY OF THE CHIEF OF NAVAL OPERATIONS AND UNDER THE DIRECTION OF THE COMMANDER, NAVAL AIR SYSTEMS COMMAND.

INTERIM CHANGE NO. 1 INCORPORATED.

1 AUGUST 1994

FIGURE 128. Sample ditching and bailout placard cover sheet.

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NAVAIR 01-C9BAAA-1D

NATOPS TAKEOFF AND LANDING DATA CARDS

NAVY MODEL C-9B AIRCRAFT

THIS PUBLICATION SUPERSEDES NAVAIR 01-C9BAAA-1D
DATED 1 FEBRUARY 1996

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1 NOVEMBER 2002

FIGURE 129. Sample takeoff and landing data (TOLD) card cover.

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NAVAIR 01-C9BAAA-1D

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LIST OF EFFECTIVE PAGES

TOTAL NUMBER OF PAGES IN THIS PUBLICATION IS 60 CONSISTING OF THE FOLLOWING:

PAGE NO.	*ISSUE*
Title	0
A	1
1 thru 60	1

INTERIM CHANGE SUMMARY

The following Interim Changes have been previously cancelled or incorporated into this manual.

INTERIM CHANGE NUMBER(S)	REMARKS/PURPOSE

The following Interim Changes have been cancelled or incorporated into this Change/Revision.

INTERIM CHANGE NUMBER(S)	REMARKS/PURPOSE
1	Derated thrust takeoff restrictions, flaps 40, V_{REF} speeds

Interim Changes Outstanding — To be maintained by the custodian of this manual.

INTERIM CHANGE NUMBER(S)	ORIGINATOR/DATE (or DATE/TIME GROUP)	PAGES AFFECTED	REMARKS/ PURPOSE

A

FIGURE 130. Sample takeoff and landing data (TOLD) card “A” page.

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NAVAIR 01-XXXXX-1D

MODEL:
ENGINES: (2) JT8D-9A

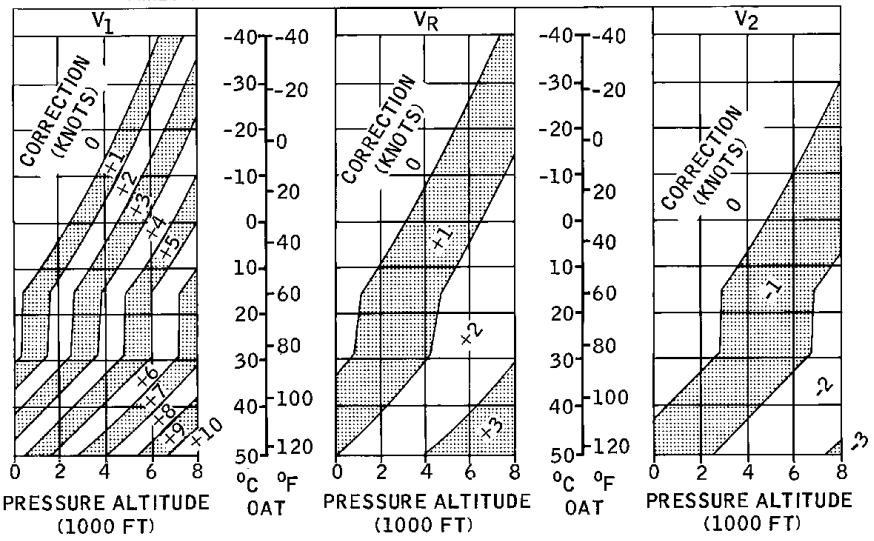
CONDITIONS:
SLATS EXTENDED
WITH OR WITHOUT ICE PROTECTION

DATA AS OF:
DATA BASIS: FLIGHT TEST

NOTE:
1. Minimum speeds must be checked at weights below 86,000 pounds due to V_{MCG}.
2. Flap retraction speed = V_2
3. Slat retraction speed = V_2 + 25 KIAS.
4. Final segment climb speed = 100 KIAS + 1 knot per 1000 lb. gross weight.

TAKEOFF GROSS WEIGHT (1000 LB)	70	75	80	85	90	95	100	105	110
V_1 (KIAS)	103	108	114	119	124	129	133	137	141
V_R (KIAS)	117	122	126	131	135	139	143	147	150
V_2 (KIAS)	128	132	136	140	142	147	149	154	157

CORRECTION TO SPEEDS
AMBIENT TEMPERATURE AND AIRPORT PRESSURE ALTITUDE



		V_1
SLOPE PER 1%	UPHILL	+ 1 KNOT
	DOWNHILL	- 1 KNOT
WIND PER 10 KNOTS	HEADWIND	+ 1 KNOT
	TAILWIND	- 1 KNOT

V_R
0
0
0
0

V_2
0
0
0
0

C9B0199

V_1, V_R, and V_2 Speeds — Flaps 5°

8

FIGURE 131. Sample takeoff and landing data (TOLD) card.

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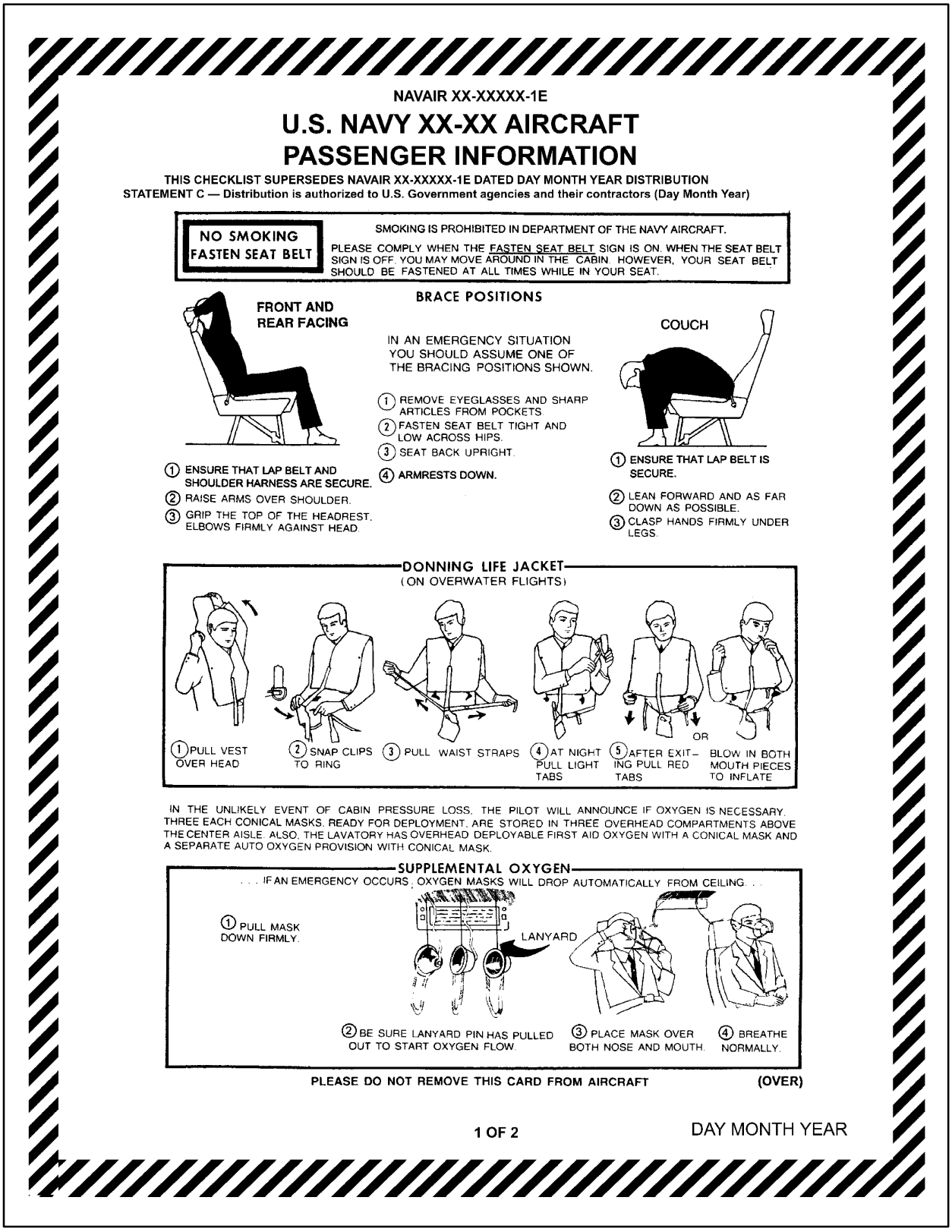


FIGURE 132. Sample passenger emergency data card.

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NAVAIR XX-XXXXX-1E

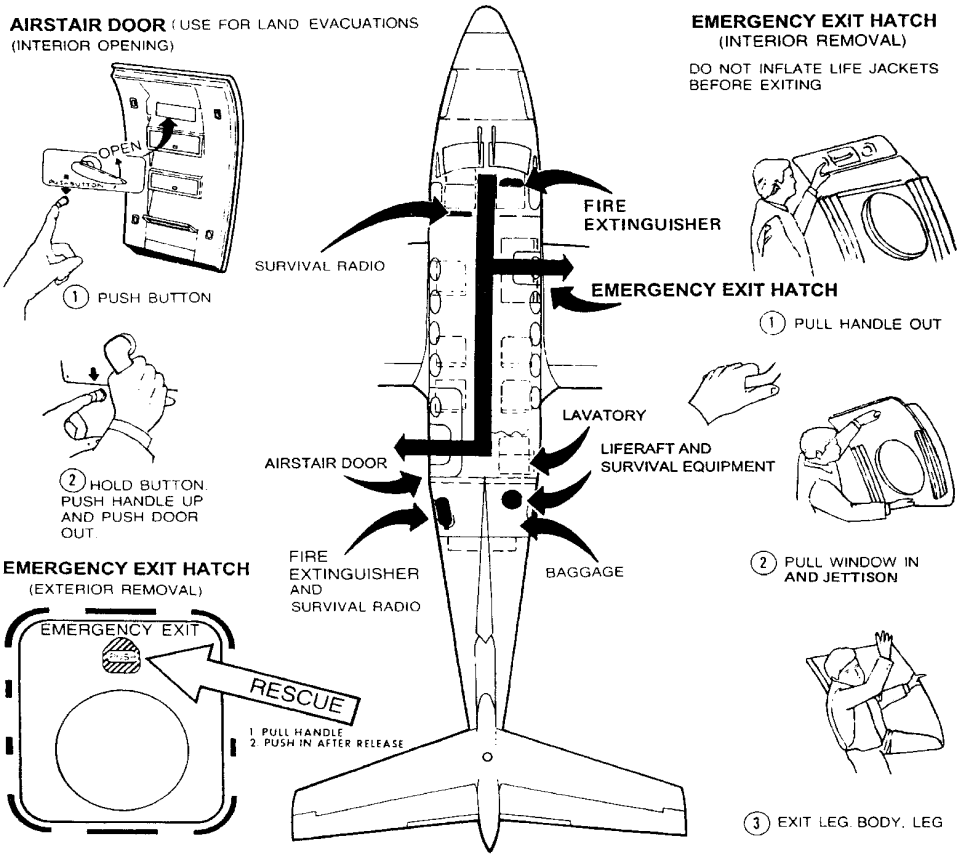
U.S. NAVY XX-XX AIRCRAFT
PASSENGER INFORMATION

THIS CHECKLIST SUPERSEDES NAVAIR XX-XXXXX-1E DATED DAY MONTH YEAR DISTRIBUTION
STATEMENT C — Distribution is authorized to U.S. Government agencies and their contractors (Day Month Year)

EMERGENCY EXIT INFORMATION

IT IS UNLIKELY THAT A SITUATION REQUIRING EMERGENCY PROCEDURES WILL BE ENCOUNTERED, BUT IT IS A GOOD PRACTICE TO BE ACQUAINTED WITH THE SAFETY FEATURES PROVIDED FOR YOU ON THIS AIRCRAFT.

WHEN THE AIRCRAFT HAS COME TO A COMPLETE STOP AFTER AN UNSCHEDULED LANDING, IT MAY BE ADVISABLE FOR PASSENGERS TO EVACUATE THE AIRCRAFT USING THE EXITS DESIGNATED BY THE CREW.



2 OF 2

DAY MONTH YEAR

FIGURE 132. Sample passenger emergency data card — Continued.

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APPENDIX A

PREFERRED USAGE IN NATOPS PUBLICATIONS

A.1 SCOPE

A.1.1 Scope. As addressed in Preferred Usage (see 3.1.7), Capitalization (see 3.1.8.4), Spelling (see 3.1.8.5), and Compound Words (see 3.1.8.6), this appendix standardizes terms for use in NATOPS publications and products. The abbreviations contained within the parentheses that follow some of the terms indicate they are applicable to the (n.) noun, (v.) verb, or (u.m) unit modifier (adjective/adverb preceding the word modified) forms of those terms. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 PREFERRED TERMS

A

about (rather than as to, in reference to, or relating to)
above ground level (use AGL)
aim point
aim-point (u.m.)
airblast
aircrewman
aircrewmember
airdoor
air-launch (v.)
air-launched (u.m.)
air-override (u.m.)
airstair
airstart
airsweep
airwing (u.m.)
all (hyphenate adjective compound)
all out
all-out (u.m.)
alphanumeric
amphibious operation area
and others (use et. al.)
and so forth (use etc.)
angels (altitude, feet)
angle of attack (AOA)
angle-of-attack (u.m.)
area of probability (use AOP)
auxiliary powerplant (use APP)

B

backlobe
bailout (n.)

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APPENDIX A

bail out (v.)
bandpass
bandwidth
baseline
beam width
bingo
brakerider
B-scan (GPO 6.51)
BuNo
burnoff (n., u.m.)
burnthrough (u.m.)
bus-tie (n., u.m.)
buses

C

cabin-conditioned (u.m.)
call sign
carryover
caster
centerline
centerpoint
chapstick
chemlight
climb out (v.)
climbout (u.m.)
close-in (u.m.)
co-altitude
coastdown
coder
code word
cold-weather (u.m.)
collocated
companionway
comparative lofar fixing (use COFIX)
Condition I or 1 (*but* condition if nonspecific)
cone-shaped (u.m.)
control-indicator
countermeasures
(*but* counter-countermeasures)
crewchief
crewman
crossbleed
cross-fix
cross-section

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crossturn
cueing

D

D drive (computer)
data (plural) or datum (singular); *but* datum and datums when reference is to a submarine contact
database (n.)
data-base (u.m.)
data link (n.)
data-link (u.m.)
daytime
decision-maker
decision-making (process)
deckman
deenergize
difar
Doppler
downlink
downrange
downwash
dual-channel (u.m.)
dual-engine (u.m.)
dual-fuzed (u.m.)

E

Earth (planet) or earth (soil)
engine Nos. 1, 2, and 4 (*not* engines one, two, and four)
et. al. (and others)
etc. (*not* and so forth)

F

facepiece
fixed-wing (u.m.)
flammable (do not use inflammable)
flaps-down (u.m.)
flaps-up (u.m.)
flapwell
flash blind (n.)
flash-blind (u.m.)
flightcrewmember
fly-by
fly-to
for example (use e.g., parenthetically)
forklift
freefall (n.)

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APPENDIX A

free-fall (u.m.)
fresh water (n.)
freshwater (u.m.)
fuse (electrical)
fuze (ordnance)

G

gangbar
gases
gearbox
generator Nos. 1, 2, and 3 (*not* generators one, two and three)
geostrobing
gigahertz (use GHz)
glareshield (n., u.m.)
glideslope
gridlock
groundcrew
groundcrewmember
Guard (frequency)
guided-missile (u.m.)
gunmount
gunpod
gun-target line (use GTL)
gustlock
gyroscope

H

half-speed
Halon
hand-held
Harpoon
hertz (use Hz)
high (hyphenate adjective compounds;
for example, high-altitude, high-g,
high-impact, high-speed, high-threat)
hookman
hover in ground effect (use HIGE)
hover out of ground effect (use HOGE)
hull-mounted (u.m.)

I

inflammable (do not use; use “flammable” for combustible,
“nonflammable” for noncombustible)
in-flight (u.m.)
in ground effect
in-line (u.m.)
inshore

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APPENDIX A

J

jackbox
Jezebel
joinup
Julie
jumpmaster

K

kilohertz (use kHz)

L

laser-guided (u.m.)
level-off (n., u.m.)
lifeboat
liferaft
light-off (n., u.m.)
light-out (n.)
line of sight (use LOS)
Link-11 (but link if nonspecific)
load-carrying (u.m.)
lock-on (n., u.m.)
lofar
long range
long-range (u.m.)
look-on (n., u.m.)
loran
low (hyphenate adjective compounds, for example, low-altitude, low-drag, low-g, low-speed)

M

mainframe
man-portable
manpower
megahertz (use MHz)
mid (prefix, nearly always closed; for example, midair)
mile-counter readings
mile counters
minefield
mine hunting
mission-essential (u.m.)
modem
multiple-bearing (u.m.)

N

narrowband

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APPENDIX A

naval (in reference to things pertaining to a navy)
Navy (in reference to the U.S. Navy)
nighttime
nonflammable (do not use inflammable)
non-real-time
nosecone
nosegear
nose-low (u.m.)
noseplug
nosestrut
nose wheelwell

O

off-axis
off board (prep. phrase)
off-board (u.m.)
off-line (u.m.)
off station (prep. phrase)
on board (prep. phrase)
on-board (u.m.)
one-half of the system
online (u.m.)
on station (prep. phrase)
on-station (time; u.m.)
open-ocean (u.m.)
observer-target line (use OTL)
out-of-ground effect (u.m.)
outside air temperature (use OAT)
own-force (u.m.)
own-ship (u.m.)

P

part-time (n.)
pathway
peacetime
penlight
percent (do not use % in text)
pillbox
pitchdown (n., u.m.)
pitch feel
pitchlock
pitch-off (n., u.m.)
power-up (n., u.m.)
PriFly
pullup point (aviation use) (use PUP)
pulse width

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APPENDIX A

pushplate
pushtile

Q

Q messages
Q-route

R

radarman
radio frequency
ram (air pressure)
random access memory (use RAM)
rate of climb (n.)
rate-of-climb (u.m.)
rate of descent
read only memory (use ROM)
real time
real-time (u.m.)
roll in (v.)
roll-in (n., u.m.)
roll-on/roll-off (use RO/RO)
rotary-wing (u.m.)
rotorwash

S

salt water (n.)
saltwater (u.m.)
searchlight
searchpath
searchplane
sea state
seawater
seeker head
self-protection (u.m.)
servoswitch
shore base
shore-based
short range
short-range (u.m.)
shutdown (u.m., n.)
shut down (v.)
sidelobe
signalman
single-axis (u.m.)
single-engine (u.m.)

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APPENDIX A

single sideband (use SSB)
SLAM in-video
slow-fly (v.)
small-boat (u.m.)
sonar
Sparrow
speedbrake
spotlight
spring loaded (pred. adj.)
spring-loaded (u.m.)
stand-alone (u.m.)
station-conditioned (u.m.)
stationkeeping
straight and level (n.)
straight-and-level (u.m.)
straight-path
sweep width

T

tacan
tac turn
tailchase
tailcone
tail rotor
tailskid
tailwalker
task force
tear gas
that is (use i.e., parenthetically)
three-dimensional
thrustpower
thumbrule
thumbwheel
tidewater
tie down (v.)
tiedown (n, u.m.)
timeline
tip-path
tip-plane
Tomahawk
toolbox
TOPGUN
towball
towboom
towcable
towcutters

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APPENDIX A

towed array
towhook
towline
tow ring
turbine inlet temperature (use TIT)
turn-on (n., u.m.)
turret deck
turret ship

U

uplatch
uplink
usable

W

warplane
waterline
watertight
weapon delivery impairment (use WDI)
weaponproof
weather map
weight on wheels (n.)
weight-on-wheels (u.m.)
well defined
well-defined (u.m.)
wetsuit
wheelbrake (n., u.m.)
wheelbraking
wheelwell
wideband
wing-fold
windline
winglock
windshear (n.)
wing sweep (n.)
wing-sweep (u.m.)
wingtip
wings-level (u.m.)
worldwide

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APPENDIX B

STANDARD ABBREVIATIONS/ACRONYMS IN NATOPS PUBLICATIONS

B.1 SCOPE

B.1.1 Scope. As addressed in Abbreviations and Acronyms (see 3.1.8.8), this appendix lists standardized abbreviations and acronyms for use in NATOPS publications and related products. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

B.2 STANDARD NATOPS-RELATED ABBREVIATIONS AND ACRONYMS

ac	alternating current
AC	air controller
a/c	aircraft
AFCS	auto flight control system
AGL	above ground level
a.m.	ante meridian (before noon)
AM	amplitude modulation
APCL	aircrew pocket checklist
APCS	autopilot control system
APLN	Airplane (flight mode for tiltrotor aircraft)
APU	auxiliary power unit
ASE/AFCS	automatic stabilization equipment/automatic flight control system
ATO	assisted takeoff
ATP	allied tactical publication
bbl	barrel (also plural)
bhp	brake horsepower
bps	bits per second
Btu	British thermal unit
BuNo	Bureau Number
°C	degrees Celsius (after number)
CAS	calibrated airspeed
CCA	carrier controlled approach
ccw	counterclockwise
CDA	commercial derivative aircraft
CD-ROM	compact disc – read only memory
cg	center of gravity; centigram(also plural)
ckt bkr or cb	circuit breaker (use either, but ckt bkr is preferred)
cm	centimeter (also plural)
cm^3	cubic centimeter (also plural)
CMYK	cyan magenta yellow black printing color method
CONV	conversion (flight mode for tiltrotor aircraft)
cps	cycles per second
CRM	crew resource management
CSD	constant speed drive
cw	clockwise
dB	decibel (also plural)

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APPENDIX B

dc	direct current
DF	direction finding or finder
dpi	dots per inch
Dplr	Doppler (effect)
DTD	document type definition
EHF	extremely high frequency
elec	electric
elek	electronic
elex	electronics
ELT	emergency locator transmitter
EPR	engine pressure ratio
°F	degrees Fahrenheit (after number)
FE	flight engineer
FM	frequency modulation
fpm	feet per minute
FRS	Fleet Readiness Squadron
ft	foot (also plural) (spell out in text)
ft/s	feet per second
g	acceleration of gravity (unit); (add s as g's) (6g's); gram (also plural)
GCA	ground controlled approach
GHz	gigahertz
gpm	gallons per minute
GPS	global positioning system
gw	gross weight
HF	high frequency
Hp	horsepower
hr	hour (also plural) (spell out in text)
HTML	hypertext markup language
Hz	hertz
IAS	indicated airspeed
ICAO	International Civil Aviation Organization
id	inside diameter
IFF/SIF	identification friend or foe/selective identification feature
IFR	instrument flight rules
in	inch (also plural) (spell out in text)
ISA	international standard atmosphere
JATO	jet-assisted takeoff
JP	joint publication
K	Kelvin (degree symbol improper)
kbps	kilobits per second
KCAS	knots calibrated airspeed (not knots CAS)
kg	kilogram (also plural)
kHz	kilohertz
KIAS	knots indicated airspeed (not IAS knots or knots IAS)
KL	kiloliter

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APPENDIX B

km	kilometer (also plural)
kPa	kilopascal (metric unit of pressure)
kt	knot (also plural) (spell out in text)
kT	kiloton (also plural)
KTAS	knots true airspeed (not knots TAS)
kVA	kilovolt-ampere
L	liter (also plural)
Lb	pound (also plural) (spell out in text)
LEP	List of effective pages
LF	low frequency
LOI	list of illustrations
lpi	lines per inch
lpm	liters per minute
m	meter (also plural)
m.	noon (meridian)
m^3	cubic meter (also plural)
Mach	Mach (physicists name)
mHz	millihertz
MHz	megahertz
mi	mile (also plural) (spell out in text)
mil	mil (measurement)
MIL	military
Min	minute (also plural) (spell out in text)
Mk	Mark (number follows)
mL	milliliter (also plural)
mm	millimeter (also plural)
mm^3	cubic millimeter (also plural)
Mod	Modification (number follows)
MSL	mean sea level
MT	megaton (also plural)
NATIP	Naval Aviation Technical Information Product
NFM	NATOPS Flight Manual
NLL	Naval Logistics Library
nm	nautical mile (also plural)
nmpp	nautical miles per pound
No. or Nos.	number(s) (number or series of numbers follow)
NOTAMs	Notices to Airmen
N_{pt}	power turbine rpm
N_r	rotor rpm
NTRP	Navy Tactical Reference Publication
OASIS	Organization for the Advancement of Structured Information Standards
OAT	outside air temperature
od	outside diameter
OFT/WST	operational flight trainer/weapons system trainer
ORM	operational risk management

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APPENDIX B

Pa	Pascals (metric unit of pressure)
Pantone	Pantone color printing method
PCL	pocket checklist
pk	probability of kill
p.m.	post meridian (after noon)
PMS	PANTONE® matching system
POS	print order sheet
pph	pounds per hour
pps	pulses per second
psi	pound per square inch
psn	position
q	dynamic pressure; torque
RCR	runway condition reading
RDF	radio direction finder
RF	radio frequency
RGB	red green blue color printing method
rpm	revolutions per minute
S/A	safety and arming device
SATNAV	satellite navingation
shp	shaft horsepower
SL	short landing
SRS	sonobuoy reference system
stbd	starboard
STO	short takeoff
t	tonne (metric ton) (also plural) (do not use MT)
TACCO	tactical coordinating officer
TAS	true airspeed
TIFF	tagged image file format
TOC	table of contents
ton	ton (also plural)
TPK	turns per knot
Tshp	takeoff shaft horsepower
TV	television
UHF	ultrahigh frequency
UTM	Universal Transverse Mercator (grid)
V	volt
Vac	volts alternating current
Vdc	volts direct current
VFR	visual flight rules
VHF	very high frequency
VL	vertical landing
V-n	diagram of aircraft load factor versus calibrated airspeed
VOR/DME	VHF omnidirectional radio-range/distance-measuring equipment
vs.	versus (do not use v.)
VTO	vertical takeoff

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VTOL	vertical takeoff and landing; a flight mode for tiltrotor aircraft
yd	yard (also plural) (spell out in text)
yr	year (also plural) (spell out in text)

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